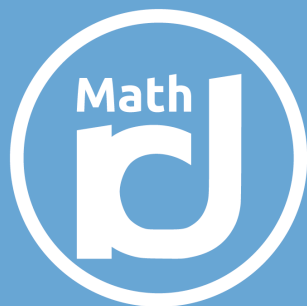




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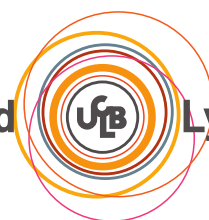
Combinatoire des (p, q) -analogues des polynômes eulériens de type bi-Stirling



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Thèse de doctorat

Université Claude Bernard



Lyon 1

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Thèse de doctorat

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Résumé en français

Cette thèse porte sur les polynômes eulériens généralisés et les statistiques de permutations. Nous introduisons d'abord une généralisation à neuf variables des polynômes eulériens, définie à partir de statistiques de descentes, et nous établissons une formule de connexion avec les polynômes eulériens cycliques (α, t) de Carlitz et Scoville. On en déduit des fonctions génératrices exponentielles explicites ainsi que plusieurs développements γ -positifs, accompagnés d'une interprétation combinatoire des γ -vecteurs normalisés en termes de permutations web, ou, de manière équivalente, de permutations d'André cycliques. Nous établissons ensuite des développements en fractions continues de type Jacobi pour des (p, q) -analogues de polynômes eulériens généralisés, où (p, q) encode soit les motifs vinculaire $(31-2, 2-13)$, soit les statistiques de croisements et d'emboîtements. Ces développements conduisent à de nouveaux résultats sur la positivité totale et la γ -positivité. De plus, nous donnons une interprétation combinatoire d'une sous-famille des nombres eulériens remixés introduits par Nadeau et Tewari, en les réalisant comme des polynômes générateurs suivant les minima de gauche à droite, les minima de droite à gauche, les descentes et un indice majeur mixte. Ces polynômes généralisent simultanément les polynômes bi-Stirling-eulériens de Carlitz-Scoville et les polynômes de Stirling-Euler-Mahoniens de Butler. Enfin, des fonctions génératrices sont obtenues pour des statistiques raffinées de descentes selon la parité sur les permutations colorées $G(r, n)$, ce qui étend et unifie des résultats concernant les types de Coxeter classiques et permet d'évaluer les polynômes de descentes alternées signées en termes de polynômes eulériens.

Mots-clés : polynômes eulériens, γ -positivité, statistiques sur les permutations, fractions continues, (p, q) -analogues, permutations colorées, groupes de Coxeter.

Résumé en anglais

This thesis studies generalized Eulerian-type polynomials and permutation statistics. We first introduce a nine-variable descent-based generalization of Eulerian polynomials and establish a connection formula with the cycle (α, t) -Eulerian polynomials of Carlitz and Scoville. As consequences, explicit exponential generating functions and several γ -positive expansions are obtained, together with a combinatorial interpretation of normalized γ -vectors in terms of web, equivalently cycle André permutations. Next, we establish Jacobi-type continued fraction expansions for (p, q) -analogs of generalized Eulerian polynomials, where (p, q) record either the vincular patterns $(31-2, 2-13)$ or the crossing and nesting statistics. These expansions yield new results on total positivity and γ -positivity. In addition, we provide a combinatorial interpretation of a subfamily of remixed Eulerian numbers introduced by Nadeau and Tewari as generating polynomials tracking left-to-right minima, right-to-left minima, descents, and a mixed major index. These polynomials simultaneously generalize the bi-Stirling–Eulerian polynomials of Carlitz–Scoville and the Stirling–Euler–Mahonian polynomials of Butler. Finally, generating functions are derived for refined descent-parity statistics on colored permutations $G(r, n)$, extending and unifying results for classical Coxeter types and yielding evaluations of signed alternating descent polynomials in terms of Eulerian polynomials.

Keywords: Eulerian polynomials, γ -positivity, permutation statistics, continued fractions, (p, q) -analogs, colored permutations, Coxeter groups.

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First and foremost, I would like to express my deepest gratitude to my supervisor, Professor Jiang Zeng. After my defense, I decided to give up my earlier version of the acknowledgments in which I had avoided mentioning any names. I have to say that I was wrong. Prof. Zeng is the most important person in my doctoral journey. Without him, this thesis would not have been possible. His guidance, patience, encouragement, and support have accompanied me through these four years. He is an outstanding combinatorialist, with a remarkable gift for telling mathematical stories. I have truly cherished every moment of discussion with him.

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There are too many people to thank. I would like to thank everyone who appeared in the first 10,783 days of my life. I do not intend to list your names one by one. If one day you happen to read these words, then I sincerely thank you. I am grateful to heaven for bringing us together, for letting us get to know each other, and for giving us shared memories.

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I have slowly come to understand that human beings are contradictory. The world is absurd, but love redeems it. In the end, I hope we may all have the courage to look life in the face, always to look life in the face, to know it for what it is; at last, to know it, to love it for what it is, and then, to let it go.

Written at the end of the first 10,783 days of my life
On the evening of May 31, 2026
8 Passage des Antonins, Lyon

Contents

Introduction (version française)	1
0.1 Variations des polynômes eulériens	1
0.1.1 Polynômes eulériens	1
0.1.2 Polynômes eulériens de type bi-Stirling	2
0.1.3 Polynômes eulériens binomiaux	4
0.1.4 q -polynômes eulériens	4
0.1.5 Groupes de Coxeter et polynômes eulériens de type W	5
0.1.6 Produit en couronne $\mathbb{Z}_r \wr \mathfrak{S}_n$ et polynômes eulériens colorés	9
0.2 Gamma-positivité des polynômes eulériens	11
0.3 Combinatoire des fractions continues	12
0.4 Plan de la thèse	14
Introduction (English version)	23
0.5 Variations of Eulerian polynomials	23
0.5.1 Eulerian polynomials	23
0.5.2 Bi-Stirling Eulerian polynomials	24
0.5.3 Binomial-Eulerian polynomials	26
0.5.4 q -Eulerian polynomials	26
0.5.5 Coxeter group and W -Eulerian polynomials	27
0.5.6 Wreath product $\mathbb{Z}_r \wr \mathfrak{S}_n$ and colored Eulerian polynomials	31
0.6 Gamma-positivity of Eulerian polynomials	33
0.7 Combinatorics of continued fractions	33
0.8 Outline of thesis	35
1 Variations of (α, t)-Eulerian polynomials and their gamma positivity	45
1.1 Introduction	45
1.2 Proof of Theorem 1.2	52
1.2.1 Two Foata-type transformations	52
1.2.2 Cyclic valley-hopping	52
1.2.3 Proof of Theorem 1.2	54
1.3 Applications of the main theorems	57
1.3.1 (α, t) -binomial-Eulerian polynomials	57
1.3.2 γ -positivity of (α, t) -Eulerian polynomials	59
1.3.3 γ -vector of (α, t) -Eulerian polynomials and cycle André permutations	65
1.4 Cycle André and André permutations: Proof of Theorem 1.23	68
1.4.1 Counting André permutations of type 1 by des and lmax	68

1.4.2	Counting André cycles by drop	69
1.4.3	Proof of Theorem 1.23 (1.79a)	71
1.4.4	Proof of Theorem 1.23 (1.79b)	71
1.4.5	Proof of Theorem 1.23 (1.79c)	72
1.5	Link to successions	77
1.6	Concluding remarks	79
2	The (p, q)-analogues of the bi-Stirling Eulerian polynomials	81
2.1	Introduction	81
2.2	Main results	83
2.3	Encoding permutations by Laguerre histories	92
2.3.1	Preliminaries	92
2.3.2	Proof of Theorem 2.4 and Theorem 2.7	102
2.4	Two bijections over permutations	104
2.4.1	Bijection Φ_C	106
2.4.2	Bijection Φ_{SZ}	107
2.4.3	Proof of Theorem 2.15	108
2.5	Applications to gamma-positivity	109
2.5.1	Proof of Theorem 2.16 and Theorem 2.18	109
2.5.2	Proof of Theorem 2.22	111
2.5.3	Combinatorics of γ -coefficient factorization	112
2.6	Factorization of γ -coefficients via group actions	117
2.6.1	Proof of Theorem 2.44	117
2.6.2	Proof of Theorem 2.47	119
2.7	Application to Total-positivity	123
2.7.1	S-fraction and coefficientwise-TP	123
2.7.2	J-fraction and coefficientwise-TP	125
2.8	Concluding remarks	128
3	A bi-Stirling-Euler-Mahonian polynomial	131
3.1	Introduction	131
3.2	Proof of main Theorems	135
3.3	Corollaries of main theorems	138
3.4	Concluding remarks	140
4	Enumerating Colored Permutations by the Parity of Descent Positions	143
4.1	Introduction	143
4.2	Definitions and main results	145
4.3	Counting colored permutations by the parity of descents	149
4.4	Counting colored permutations by signed alternating descents	156
4.4.1	Proof of Theorem 4.8 when r is even	157
4.4.2	Proof of Theorem 4.8 when r is odd and $n \neq 4m + 3$	158
4.4.3	Proof of Theorem 4.8 when r is odd and $n = 4m + 3$	159
4.5	Counting permutations of type B by the parity of descents of type D	161
4.5.1	Final remark	164
5	Further Research directions	167
5.1	New Interpretations for Euler Numbers and Reduced Tangent Numbers	167
5.2	Link with quasi-Stirling permutations and Cluster parking functions	171

Bibliography**177**

Introduction (version française)

0.1 Variations des polynômes eulériens

0.1.1 Polynômes eulériens

Au XVIII^e siècle, Euler [Eul55] a étudié la série de Dirichlet alternée

$$\varphi(s) = 1 - \frac{1}{2^s} + \frac{1}{3^s} - \frac{1}{4^s} + \cdots,$$

qui converge pour $\Re(s) > 0$ et qui est reliée à la fonction zêta de Riemann $\zeta(s)$ par

$$\varphi(s) = (1 - 2^{1-s})\zeta(s).$$

Le but d'Euler était de comprendre les valeurs spéciales $\varphi(-n)$ pour $n = 0, 1, 2, \dots$. À cette fin, il a introduit les *polynômes eulériens* $A_n(x)$, caractérisés par

$$\frac{A_n(x)}{(1-x)^{n+1}} = \sum_{k \geq 0} (k+1)^n x^k. \quad (0.1)$$

En multipliant les deux membres de (0.1) par $z^n/n!$ puis en sommant sur tous les entiers $n \geq 0$, on obtient

$$\sum_{n \geq 0} \frac{A_n(x)}{(1-x)^{n+1}} \frac{z^n}{n!} = \sum_{n \geq 0} \frac{z^n}{n!} \sum_{k \geq 0} (k+1)^n x^k = \frac{e^z}{1-xe^z}. \quad (0.2)$$

En multipliant ensuite les deux membres de (0.2) par $(1-x)$, puis en remplaçant z par $z(1-x)$, on obtient la fonction génératrice exponentielle de $A_n(x)$,

$$\sum_{n \geq 0} A_n(x) \frac{z^n}{n!} = \frac{(1-x)e^{z(1-x)}}{1-xe^{z(1-x)}}. \quad (0.3)$$

Dominique Foata a rédigé un bel article de synthèse dans lequel il explique la motivation d'Euler ainsi que sa démarche de dérivation [Foa10]. Les premières valeurs de $A_n(x)$ sont les suivantes:

$$A_n(x) = \begin{cases} 1, & \text{if } n = 1, \\ 1 + x, & \text{if } n = 2, \\ 1 + 4x + x^2, & \text{if } n = 3, \\ 1 + 11x + 11x^2 + x^3, & \text{if } n = 4, \\ 1 + 26x + 66x^2 + 26x^3 + x^4, & \text{if } n = 5. \end{cases}$$

Une *permutation* de $[n] := \{1, 2, \dots, n\}$ est une bijection $\sigma : [n] \rightarrow [n]$. On note $\mathfrak{S}_n$ l'ensemble de toutes les permutations de $[n]$. Une *statistique* sur $\mathfrak{S}_n$ est une application $st : \mathfrak{S}_n \rightarrow \mathbb{N}$. Pour chaque permutation $\sigma = \sigma(1)\sigma(2)\cdots\sigma(n) \in \mathfrak{S}_n$, un entier $i \in [n-1]$ est appelé une *descente* (resp. une *excédance*) de σ si $\sigma(i) > \sigma(i+1)$ (resp. $\sigma(i) > i$). On note $\text{DES}(\sigma)$ (resp. $\text{EXC}(\sigma)$) l'ensemble des descentes (resp. des excédances), et $\text{des}(\sigma)$ (resp. $\text{exc}(\sigma)$) le nombre de descentes (resp. d'excédances) de σ .

MacMahon [Mac60, Mac13] a démontré que le nombre de descentes et le nombre d'excédances sur les permutations sont équirépartis, c'est-à-dire

$$\sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)}. \quad (0.4)$$

Riordan [Rio58] a ensuite découvert l'interprétation combinatoire suivante des polynômes eulériens:

$$A_n(x) = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} = \sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle x^k, \quad (0.5)$$

où $\left\langle n \atop k \right\rangle$ est appelé un *nombre eulérien*, défini par

$$\left\langle n \atop k \right\rangle := |\{\sigma \in \mathfrak{S}_n : \text{des}(\sigma) = k\}|. \quad (0.6)$$

Ces nombres eulériens $\left\langle n \atop k \right\rangle$ satisfont la relation de récurrence suivante:

$$\left\langle n \atop k \right\rangle = (n-k) \left\langle n-1 \atop k-1 \right\rangle + (k+1) \left\langle n-1 \atop k \right\rangle, \quad (n > k \geq 0). \quad (0.7)$$

où $\left\langle 0 \atop 0 \right\rangle = 1$ et $\left\langle n \atop k \right\rangle = 0$ si $k \notin \{0, \dots, n-1\}$.

Avec l'interprétation ci-dessus, toute statistique qui est équirépartie avec le nombre de descentes (ou le nombre d'excédances) sur les permutations est généralement appelée une *statistique eulérienne*. La classique *première transformation fondamentale de Foata* [Foa65] fournit une preuve bijective de (0.4).

Leonard Carlitz et ses collaborateurs ont étudié les nombres eulériens et leurs généralisations dans plusieurs articles au XX^e siècle, par exemple [Car54, Car75, CS74, CS73, CRS66]. Foata et Marcel-Paul Schützenberger ont rédigé dans [FS70] une étude approfondie des nombres eulériens du point de vue combinatoire. L'étude des nombres eulériens joue un rôle important dans de nombreux travaux en arithmétique et en combinatoire.

0.1.2 Polynômes eulériens de type bi-Stirling

On peut aussi considérer une permutation $\sigma \in \mathfrak{S}_n$ comme un mot $\sigma = \sigma_1\sigma_2\cdots\sigma_n$, où $\sigma_i := \sigma(i)$ pour tout $i \in [n]$. Pour $\sigma \in \mathfrak{S}_n$, un indice $i \in [n-1]$ est appelé une *montée* de σ si $\sigma_i < \sigma_{i+1}$, et une lettre σ_i est appelée

- un *maximum de gauche à droite* de σ si $\sigma_i > \sigma_j$ pour tout $j < i$;
- un *maximum de droite à gauche* de σ si $\sigma_i > \sigma_j$ pour tout $j > i$.

On note $\text{asc}(\sigma)$, $\text{lmax}(\sigma)$ et $\text{rmax}(\sigma)$ le nombre de montées, de maxima de gauche à droite et de maxima de droite à gauche de σ , respectivement. Les statistiques lmax et rmax sont toutes deux des *statistiques Stirling*:

$$\sum_{\sigma \in \mathfrak{S}_n} x^{\text{lmax}(\sigma)} = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{rmax}(\sigma)} = \sum_{k=0}^n s(n, k) x^k, \quad (0.8)$$

où $s(n, k)$ désigne les nombres de Stirling de première espèce.

En 1974, Carlitz et Scoville [CS74] ont introduit les *polynômes eulériens de type bi-Stirling* comme suit :

$$A_n(x, y | \alpha, \beta) := \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{asc}(\sigma)} y^{\text{des}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}. \quad (0.9)$$

Les premières valeurs de $A_n(x, y | \alpha, \beta)$ sont les suivantes:

$$A_1(x, y | \alpha, \beta) = \alpha x + \beta y,$$

$$A_2(x, y | \alpha, \beta) = \alpha^2 x^2 + 2\alpha\beta xy + \alpha xy + \beta xy + \beta^2 y^2,$$

$$A_3(x, y | \alpha, \beta) = \alpha^3 x^3 + 3\alpha^2\beta x^2 y + 3\alpha^2 x^2 y + 3\alpha\beta x^2 y + \alpha x^2 y + \beta x^2 y \\ + 3\alpha\beta^2 x y^2 + 3\alpha\beta x y^2 + 3\beta^2 x y^2 + \alpha x y^2 + \beta x y^2 + \beta^3 y^3.$$

Ils ont obtenu la fonction génératrice exponentielle suivante pour $A_n(x, y | \alpha, \beta)$.

Theorem 0.1 (Théorème 9, [CS74]). *On a*

$$\sum_{n \geq 0} A_n(x, y | \alpha, \beta) \frac{z^n}{n!} = (1 + xF(x, y; z))^\alpha (1 + yF(x, y; z))^\beta, \quad (0.10)$$

où $F(x, y; z)$ est donné par

$$F(x, y; z) = \frac{e^{xz} - e^{yz}}{xe^{yz} - ye^{xz}}. \quad (0.11)$$

Remark 0.2. Lorsque $\alpha = 0$, $\beta = 1$ et $x = 1$, les polynômes $A_n(x, y | \alpha, \beta)$ se réduisent aux polynômes eulériens classiques $A_n(y)$ définis en (0.5). En conséquence, on retrouve la fonction génératrice (0.3) de $A_n(y)$ en posant $\alpha = 0$, $\beta = 1$ et $x = 1$ dans (0.10).

Pour $x = y = 1$ et $\beta = 0$, il est facile de voir que

$$A_n(1, 1 | \alpha, 0) = \sum_{\sigma \in \mathfrak{S}_n} \alpha^{\text{lmax}(\sigma)}. \quad (0.12)$$

À partir de (0.10), on obtient

$$\sum_{n \geq 0} A_n(1, 1 | \alpha, 0) \frac{z^n}{n!} = \left(\frac{1}{1-z} \right)^\alpha. \quad (0.13)$$

En identifiant les coefficients de $z^n/n!$ dans (0.13), on obtient

$$\sum_{\sigma \in \mathfrak{S}_n} \alpha^{\text{lmax}(\sigma)} = \alpha(\alpha+1)(\alpha+2) \cdots (\alpha+n-1). \quad (0.14)$$

Dans le même article, Carlitz et Scoville ont défini le *nombre eulérien bi-Stirling* $A(r, s | \alpha, \beta)$ comme le coefficient de $x^r y^s$ dans $A_{r+s+1}(x, y | \alpha, \beta)$, c'est-à-dire

$$A(r, s | \alpha, \beta) := [x^r y^s] A_{r+s+1}(x, y | \alpha, \beta) = \sum_{\substack{\sigma \in \mathfrak{S}_{r+s+1} \\ \text{des} = s}} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}, \quad (0.15)$$

où $[x^r y^s]F(x, y)$ désigne le coefficient de $x^r y^s$ dans $F(x, y)$. Ils ont montré que $A(r, s | \alpha, \beta)$ satisfait la relation de récurrence suivante:

$$A(r, s | \alpha, \beta) = (s + \alpha) A(r-1, s | \alpha, \beta) + (r + \beta) A(r, s-1 | \alpha, \beta). \quad (0.16)$$

Remark 0.3. Lorsque $\alpha = \beta = 1$, $r = n - k - 1$ et $s = k$, ces nombres $A(r, s | \alpha, \beta)$ se réduisent aux nombres eulériens classiques définis en (0.6), c'est-à-dire

$$A(n-k-1, k | 1, 1) = \left\langle n \atop k \right\rangle.$$

En conséquence, la récurrence (0.16) se réduit à (0.7).

0.1.3 Polynômes eulériens binomiaux

Postnikov, Reiner et Williams [PRW08] ont introduit les *polynômes eulériens binomiaux* :

$$\tilde{A}_n(x) := \sum_{\sigma \in \text{PRW}_{n+1}} x^{\text{des}(\sigma)} = 1 + x \sum_{m=1}^n \binom{n}{m} A_m(x), \quad (0.17)$$

où PRW_n désigne l'ensemble des permutations $\sigma \in \mathfrak{S}_n$ telles que la première descente éventuelle de σ se produise à la lettre n . Par exemple,

$$\text{PRW}_2 = \{12, 21\}, \quad \text{PRW}_3 = \{123, 132, 231, 312, 321\}.$$

Pour les premières valeurs de n , on a

$$\tilde{A}_n(x) = \begin{cases} 1 + x, & \text{si } n = 1, \\ 1 + 3x + x^2, & \text{si } n = 2, \\ 1 + 7x + 7x^2 + x^3, & \text{si } n = 3, \\ 1 + 15x + 33x^2 + 15x^3 + x^4, & \text{si } n = 4, \\ 1 + 31x + 131x^2 + 131x^3 + 31x^4 + x^5, & \text{si } n = 5. \end{cases}$$

Ce polynôme est apparu pour la première fois dans un contexte énumérativo-géométrique dans [PRW08, Section 10.4], où l'on a montré qu'il coïncide avec le h -polynôme du stelloèdre de dimension n . Chung, Graham et Knuth [CGK10] ont établi l'identité élégante suivante pour des entiers $a, b \geq 0$:

$$\sum_{k \geq 0} \binom{a+b}{k} \left\langle \begin{matrix} k \\ a-1 \end{matrix} \right\rangle = \sum_{k \geq 0} \binom{a+b}{k} \left\langle \begin{matrix} k \\ b-1 \end{matrix} \right\rangle, \quad (0.18)$$

où $\langle 0 \rangle = 1$. Shareshian et Wachs [SW20] ont observé que cette identité est précisément la traduction algébrique du caractère palindromique des coefficients de $\tilde{A}_n(x)$.

0.1.4 q -polynômes eulériens

Nous définissons le q -analogue du nombre réel x , noté $[x]$, par

$$[x] = \frac{1 - q^x}{1 - q}.$$

D'après la règle de l'Hôpital, on a $[x] \rightarrow x$ lorsque $q \rightarrow 1$. Pour $n \in \mathbb{N}$, nous définissons le q -analogue de $n!$ par

$$[n]! = \prod_{i=1}^n [i] = (1+q)(1+q+q^2) \cdots (1+q+\cdots+q^{n-1}).$$

Soit $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$. Une *inversion* de σ est une paire d'indices (i, j) telle que $1 \leq i < j \leq n$ et $\sigma_i > \sigma_j$. Soit $\text{inv}(\sigma)$ le nombre d'inversions de σ , c'est-à-dire

$$\text{inv}(\sigma) := |\{(i, j) : i < j \text{ et } \sigma_i > \sigma_j\}|. \quad (0.19)$$

L'*indice majeur* d'une permutation $\sigma \in \mathfrak{S}_n$ est défini comme la somme des descentes de σ , c'est-à-dire

$$\text{maj}(\sigma) = \sum_{i \in \text{Des}(\sigma)} i. \quad (0.20)$$

En 1915, MacMahon [Mac60] a montré que l'indice majeur est équidistribué avec la statistique d'inversion en établissant la première égalité suivante :

$$\sum_{\sigma \in \mathfrak{S}_n} q^{\text{maj}(\sigma)} = [n]! = \sum_{\sigma \in \mathfrak{S}_n} q^{\text{inv}(\sigma)}. \quad (0.21)$$

La seconde égalité a été obtenue en 1839 par Rodrigues [Rod39]. Foata [Foa68] a donné une preuve bijective de (0.21), qui fut ensuite appelée la *seconde transformation fondamentale de Foata*. En l'honneur de MacMahon, toute statistique sur les permutations ayant la même distribution que maj est appelée une *statistique Mahonienne*.

Toute paire de statistiques (s_1, s_2) est appelée une paire *Eulérienne–Mahonienne* si

$$\sum_{\sigma \in \mathfrak{S}_n} x^{s_1(\sigma)} = A_n(x) \text{ et } \sum_{\sigma \in \mathfrak{S}_n} q^{s_2(\sigma)} = [n]!. \quad (0.22)$$

En particulier, les paires classiques de statistiques (des, maj) et (des, inv) donnent lieu à deux familles distinctes de *q-polynômes eulériens*, définies respectivement par

$$^m A_n(x, q) := \sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} q^{\text{maj}(\sigma)}, \quad (0.23)$$

$$^i A_n(x, q) := \sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} q^{\text{inv}(\sigma)}. \quad (0.24)$$

Rappelons que le symbole de *q*-Pochhammer est défini par

$$(x; q)_k = (1 - x)(1 - xq) \cdots (1 - xq^{k-1}), \quad (0.25)$$

et que la *fonction q-exponentielle* est donnée par

$$\exp(z; q) = \sum_{n \geq 0} \frac{z^n}{[n]!}. \quad (0.26)$$

Carlitz [Car54] et Stanley [Sta76] ont démontré, respectivement,

$$\frac{^m A_n(x, q)}{(x; q)_{n+1}} = \sum_{j \geq 0} x^j [j+1]^n, \quad (0.27)$$

et

$$1 + \sum_{n \geq 1} x \cdot ^i A_n(x, q) \frac{z^n}{[n]!} = \frac{1 - x}{1 - x \exp((1 - x)z; q)}. \quad (0.28)$$

Remark 0.4. Les équations (0.27) et (0.28) sont des *q*-analogues de (0.1) et (0.3), respectivement. Pour les deux autres paires eulériennes–mahoniennes classiques (exc, inv) et (exc, maj) , des familles analogues de polynômes eulériens en *q* ont également été étudiées par Clarke–Steingrímsson–Zeng [CSZ97] et Shareshian–Wachs [SW07a], respectivement.

0.1.5 Groupes de Coxeter et polynômes eulériens de type *W*

On appelle *système de Coxeter* une paire (W, S) , où *W* est un groupe et *S* est un ensemble minimal de générateurs de *W*, soumis aux relations suivantes. Pour tout $s, t \in S$, on a

- $st = e$ si et seulement si $s = t$;
- $(st)^{m(s,t)} = (ts)^{m(s,t)} = e$ pour un certain entier $m(s, t) > 1$,

où e désigne l'élément neutre de W , avec $m(s, s) = 1$ pour tout $s \in S$, et, pour $s \neq t$, on a

$$m(s, t) = m(t, s) \in \{2, 3, \dots\}.$$

Un tel groupe W est appelé un *groupe de Coxeter* de rang $r = |S|$.

À tout système de Coxeter, on peut associer un *graphe de Coxeter* dont les sommets sont les éléments de S . Si $m(s, t) > 2$, on trace une arête entre s et t , étiquetée par $m(s, t)$ lorsque $m(s, t) > 3$. Dans cette thèse, nous ne considérons que les groupes de Coxeter de types A, B et D. Ces systèmes de Coxeter correspondent à des graphes de Coxeter connexes, représentés aux figures 0.1, 0.2 et 0.3.

On définit la *longueur* d'un élément ω de W comme le nombre de facteurs dans une expression de longueur minimale de ω comme produit d'éléments de S . Autrement dit,

$$\ell_W(\omega) := \min\{k : \omega = s_1 \cdots s_k, s_i \in S\}. \quad (0.29)$$

Nous utilisons maintenant la fonction longueur pour formuler une définition générale des descentes pour les éléments d'un groupe de Coxeter. Pour tout élément ω du groupe de Coxeter W , posons

$$\text{Des}_W(\omega) := \{s \in S : \ell_W(\omega) > \ell_W(\omega s)\}, \quad (0.30)$$

et $\text{des}_W(\omega) = |\text{Des}_W(\omega)|$.

On définit les *polynômes eulériens de W* comme la fonction génératrice des descentes, notée

$$W(x) := \sum_{\omega \in W} x^{\text{des}_W(\omega)} = \sum_{k=0}^n \left\langle \begin{matrix} W \\ k \end{matrix} \right\rangle x^k, \quad (0.31)$$

où $n = |S|$ est le rang de W . Les coefficients de $W(x)$

$$\left\langle \begin{matrix} W \\ k \end{matrix} \right\rangle := |\{\omega \in W : \text{des}_W(\omega) = k\}|, \quad (0.32)$$

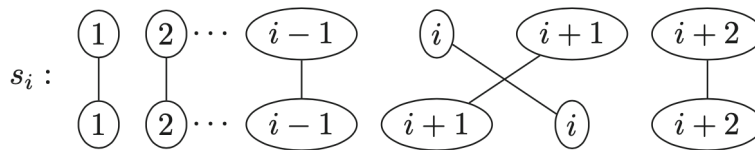
sont appelés les *nombre eulériens de W* .

Par commodité, nous notons la longueur et l'ensemble des descentes du groupe de Coxeter de type A par ℓ_A et Des_A , respectivement. De même, nous utiliserons ℓ_B , Des_B , ℓ_D et Des_D .

L'ensemble $\mathfrak{S}_n$ des permutations est un groupe de Coxeter de type A. En théorie des groupes, le groupe $\mathfrak{S}_n$ est communément appelé le *groupe symétrique*. Pour $n \geq 2$, le groupe $\mathfrak{S}_n$ est engendré par les transpositions adjacentes

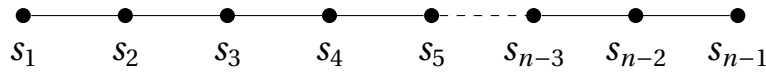
$$s_i = (i \ i+1) \quad (1 \leq i \leq n-1), \quad (0.33)$$

c'est-à-dire que s_i échange les éléments situés aux positions i et $i+1$.



Ces générateurs satisfont les relations suivantes, et leur graphe de Coxeter est donné à la Fig. 0.1.

- $s_i^2 = e$ for $1 \leq i \leq n-1$;

Figure 0.1: Coxeter graph A_n ($n \geq 1$).

- $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$ for $1 \leq i \leq n-2$;
- $s_i s_j = s_j s_i$ for $|i - j| \geq 2$.

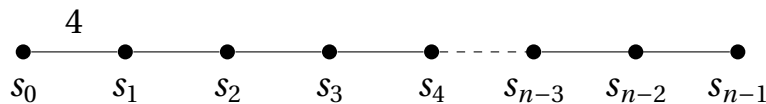
Il est bien connu que, dans le cas d'un groupe de Coxeter de type A, la longueur et l'ensemble des descentes du groupe $\mathfrak{S}_n$ admettent une description combinatoire simple. Plus précisément, pour toute permutation $\sigma \in \mathfrak{S}_n$, on a

$$\ell_A(\sigma) = \text{inv}(\sigma) \text{ et } \text{Des}_A(\sigma) = \{i \in [n-1] : \sigma(i) > \sigma(i+1)\}. \quad (0.34)$$

Dans le cas du groupe symétrique $\mathfrak{S}_n$, les polynômes eulériens de W et les nombres eulériens de W coïncident respectivement avec les polynômes eulériens classiques et les nombres eulériens classiques définis en (0.5) et (0.6).

Soit $\mathfrak{B}_n$ le *groupe hyperoctaédral* sur $[n]$, qui est un groupe de Coxeter de type B engendré par $\{s_0, s_1, \dots, s_{n-1}\}$, où $s_0 = (1 - 1)$ est le changement de signe, et s_i ($1 \leq i \leq n-1$) est la transposition adjacente, comme dans le cas de $\mathfrak{S}_n$. Ces générateurs satisfont les relations suivantes, et leur graphe de Coxeter est donné à la Fig. 0.2.

- $s_i^2 = e$, pour $0 \leq i \leq n-1$;
- $(s_i s_{i+1})^3 = e$ pour $i = 1, \dots, n-2$;
- $(s_0 s_1)^4 = e$; et
- $(s_i s_j)^2 = e$ si $|i - j| > 1$.

Figure 0.2: Coxeter graph B_n ($n \geq 2$).

On peut représenter le groupe de Coxeter de type B comme l'ensemble des *permutations signées*. Ce sont des permutations

$$w : \{-n, \dots, -1, 0, 1, \dots, n\} \rightarrow \{-n, \dots, -1, 0, 1, \dots, n\}, \quad (0.35)$$

telles que $w(-i) = -w(i)$ pour tout i . On remarque que cela impose $w(0) = 0$, et l'élément w est alors entièrement déterminé par $w(1), \dots, w(n)$. En notation en ligne, on écrit $w = w(1) \cdots w(n)$, en utilisant une barre pour indiquer les entiers négatifs. Par exemple, si w est déterminé par $w(1) = -3$, $w(2) = 4$, $w(3) = 5$, $w(4) = -1$ et $w(5) = 2$, on écrit $w = \bar{3}45\bar{1}2$.

L'ensemble des descentes de type B peut être décrit combinatoirement comme suit:

$$\text{Des}_B(w) = \{0 \leq i \leq n-1 : w(i) > w(i+1)\}, \quad (0.36)$$

où $w(0) = 0$. Nous notons par $\text{des}_B(w) = |\text{Des}_B(w)|$ le nombre de descentes de type B.

Brenti [Bre94] a défini les *polynômes eulériens de $\mathfrak{B}_n$* à partir du nombre de descentes de type B, pour $n \geq 1$,

$$B_n(x) := \sum_{w \in \mathfrak{B}_n} x^{\text{des}_B(w)}, \quad (0.37)$$

et a obtenu la fonction génératrice exponentielle de $B_n(x)$,

$$\sum_{n \geq 0} B_n(x) \frac{z^n}{n!} = \frac{(1-x)e^{z(1-x)}}{1-xe^{2z(1-x)}}. \quad (0.38)$$

Rappelons que la longueur ℓ_B de $\mathfrak{B}_n$ est donnée par [BB05, Proposition 8.1.1]

$$\ell_B(w) = \text{inv}(w) - \sum_{w(i) < 0} w(i). \quad (0.39)$$

Pour $n \geq 1$, la distribution de la longueur ℓ_B sur $\mathfrak{B}_n$ est donnée par

$$\sum_{w \in \mathfrak{B}_n} q^{\ell_B(w)} = [2][4] \cdots [2n]. \quad (0.40)$$

En 1995, Reiner [Rei95] a défini le q -analogue des polynômes eulériens de $\mathfrak{B}_n$ par

$$B_n(x, q) := \sum_{w \in \mathfrak{B}_n} x^{\text{des}_B(w)} q^{\ell_B(w)}, \quad (0.41)$$

et a obtenu la fonction génératrice exponentielle en q suivante pour $B_n(x, q)$.

Theorem 0.5. *On a*

$$\sum_{n \geq 0} B_n(x, q) \frac{z^n}{(-q; q)_n [n]!} = \frac{(1-x) \exp_B(z(1-x); q)}{1-x \exp(z(1-x); q)}, \quad (0.42)$$

où $\exp_B(z; q)$ est la fonction q -exponentielle de type B définie par

$$\exp_B(z; q) = \sum_{n \geq 0} \frac{z^n}{[2][4] \cdots [2n]}. \quad (0.43)$$

Le groupe de Coxeter de type $\mathfrak{D}_n$ est le sous-groupe de $\mathfrak{B}_n$ constitué des permutations signées pour lesquelles le nombre de signes moins parmi $w(1), \dots, w(n)$ est pair. Lorsqu'on écrit un élément de $\mathfrak{D}_n$ sous la forme d'un mot $w = w(1)w(2) \cdots w(n)$, on l'appelle une *permutation signée paire*.

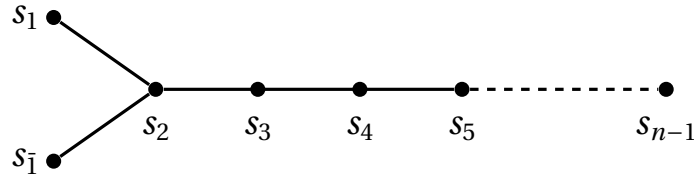


Figure 0.3: Coxeter graph D_n ($n \geq 4$).

L'ensemble des descentes de type D peut être décrit combinatoirement comme suit :

$$\text{Des}_D(w) = \{i \in \{-1, 1, 2, \dots, n-1\} : w(i) > w(|i|+1)\}, \quad (0.44)$$

où $w(0) = 0$. Nous notons par $\text{des}_D(w) = |\text{Des}_D(w)|$ le nombre de descentes de type D.

Brenti [Bre94] a défini les *polynômes eulériens de $\mathfrak{D}_n$* à partir du nombre de descentes de type D, pour $n \geq 1$,

$$D_n(x) := \sum_{w \in \mathfrak{D}_n} x^{\text{des}_D(w)}, \quad (0.45)$$

et a obtenu la fonction génératrice exponentielle de $D_n(x)$,

$$1 + xz + \sum_{n \geq 2} D_n(x) \frac{z^n}{n!} = \frac{(x-1)(z - e^{z(1-x)})}{1 - e^{2z(1-x)}}. \quad (0.46)$$

Rappelons que la longueur ℓ_D de $\mathfrak{D}_n$ est donnée par [BB05, Proposition 8.2.1]

$$\ell_D(w) := \ell_B(w) - \text{neg}(w), \quad (0.47)$$

où $\text{neg}(w)$ désigne le nombre d'entrées négatives dans w .

Pour $n \geq 2$, la distribution de la longueur ℓ_D sur $\mathfrak{D}_n$ est donnée par

$$\sum_{w \in \mathfrak{D}_n} q^{\ell_D(w)} = [2][4] \cdots [2n-2][n]. \quad (0.48)$$

En 1995, Reiner [Rei95] a défini le q -analogue des polynômes eulériens de $\mathfrak{D}_n$ par

$$D_n(x, q) := \sum_{w \in \mathfrak{D}_n} x^{\text{des}_D(w)} q^{\ell_D(w)}, \quad (0.49)$$

et a obtenu la fonction génératrice exponentielle en q suivante pour $D_n(x, q)$.

Theorem 0.6. *On a*

$$2 + \sum_{n \geq 2} D_n(x, q) \frac{z^n}{(-q; q)_{n-1}[n]!} = \frac{(1-x) \exp_D(z(1-x); q) - z(1-x^2 \exp(z(1-x); q))}{1 - x \exp(z(1-x); q)}, \quad (0.50)$$

où $\exp_D(z; q)$ est la fonction q -exponentielle de type D définie par

$$\exp_D(z; q) = 2 + \sum_{n \geq 1} \frac{z^n}{[2][4] \cdots [2n-2][n]}. \quad (0.51)$$

0.1.6 Produit en couronne $\mathbb{Z}_r \wr \mathfrak{S}_n$ et polynômes eulériens colorés

Pour $r, n \in \mathbb{P}$, nous définissons le *produit en couronne* $\mathbb{Z}_r \wr \mathfrak{S}_n$ de $\mathbb{Z}_r$ par $\mathfrak{S}_n$, c'est-à-dire le groupe des permutations colorées $G(r, n)$, par

$$G(r, n) := \{(c_1, \dots, c_n; \sigma) \mid c_i \in [0, r-1], \sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n\}. \quad (0.52)$$

Le produit dans $G(r, n)$ est défini par

$$(c; \sigma) \cdot (c'; \tau) := (c_1 + c'_{\tau^{-1}}, \dots, c_n + c'_{\tau_n}; \sigma \circ \tau),$$

où l'addition $+$ est prise dans $\mathbb{Z}_r$ et la composition $\circ$ dans $\mathfrak{S}_n$. L'entrée c_i est appelée la couleur de σ_i , pour $1 \leq i \leq n$.

Les éléments de $\mathbb{Z}_r \wr \mathfrak{S}_n$ peuvent être vus comme des *permutations r -colorées*, voir Steingrímsson [Ste94] et Bagno et al. [BGM09]. Le produit en couronne $\mathbb{Z}_r \wr \mathfrak{S}_n$ peut être considéré comme une généralisation commune du groupe symétrique $\mathfrak{S}_n$ et du groupe de Coxeter de type B, c'est-à-dire que, pour $r = 1, 2$,

$$\mathbb{Z}_1 \wr \mathfrak{S}_n \cong \mathfrak{S}_n \text{ et } \mathbb{Z}_2 \wr \mathfrak{S}_n \cong \mathfrak{B}_n. \quad (0.53)$$

Nous représenterons un élément $\gamma \in G(r, n)$ en *notation par fenêtre* sous la forme

$$\gamma = [\gamma(1), \dots, \gamma(n)] = [\sigma_1^{c_1}, \dots, \sigma_n^{c_n}], \quad (0.54)$$

et nous appellerons σ_i la *valeur absolue* de $\gamma(i)$, notée $|\gamma(i)|$.

Pour $\gamma \in G(r, n)$, nous définissons l'ensemble des couleurs de γ par

$$\text{Col}_G(\gamma) := \{i \in [n] : c_i \neq 0\}, \quad (0.55)$$

et son cardinal sera noté $\text{col}_G(\gamma)$.

Si $c_i = 0$, il sera omis dans la notation par fenêtre. Par exemple, $\gamma = [3^1, 2, 1^3, 4^2, 6^2, 5^1] \in G(5, 6)$.

Le groupe $G(r, n)$ est engendré par $S_G := \{s_0, s_1, \dots, s_{n-1}\}$, où, pour $i \in [n-1]$, s_i est la transposition adjacente comme dans le cas de $\mathfrak{S}_n$ en (0.146), et

$$s_0 := [1^1, 2, \dots, n].$$

Les relations entre ces générateurs sont décrites par le diagramme de type Dynkin (voir Fig. 0.9).

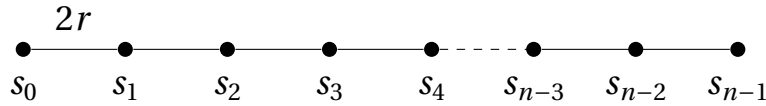


Figure 0.4: The Dynkin-like diagram of $G(r, n)$.

La *longueur* de $\gamma \in G(r, n)$ est le nombre minimal de générateurs de S_G dont le produit est égal à γ ,

$$\ell_G(\gamma) := \min\{r \in \mathbb{N} : \gamma = s_{i_1} \cdots s_{i_r}, \text{ pour un certain choix de } s_{i_j} \in S_G\}. \quad (0.56)$$

L'*ensemble des descentes* de $\gamma \in G(r, n)$ est défini par

$$\text{Des}_G(\gamma) := \{s \in S_G : \ell_G(\gamma s) < \ell_G(\gamma)\}, \quad (0.57)$$

et son cardinal est noté $\text{des}_G(\gamma)$.

Pour donner une description combinatoire de ℓ_G et de $\text{Des}_G(\gamma)$, nous utilisons l'ordre linéaire suivant

$$n^{r-1} < \dots < n^1 < \dots < 1^{r-1} < \dots < 1^1 < 0 < 1 < \dots < n \quad (0.58)$$

sur l'ensemble $\{0, 1, \dots, n, 1^1, \dots, n^1, \dots, 1^{r-1}, \dots, n^{r-1}\}$ des entiers colorés (voir [BZ11]). Pour $\gamma = [\gamma(1), \dots, \gamma(n)] \in G(r, n)$, l'ensemble des descentes de γ admet la définition alternative suivante :

$$\text{Des}_G(\gamma) = \{i \in [0, n-1] : \gamma(i) > \gamma(i+1)\}, \quad (0.59)$$

où $\gamma(0) := 0$.

La longueur de γ est caractérisée par la formule suivante (voir [Bag05, Rei93, Ste94]) :

$$\ell_G(\gamma) = \text{inv}(\gamma) + \sum_{c_i \neq 0} (|\gamma(i)| + c_i - 1). \quad (0.60)$$

Notons que, lorsque $r = 1$ (resp. $r = 2$), l'équation (0.60) se réduit à (0.19) (resp. (0.39)).

Steingrímsson [Ste94] a introduit les *polynômes eulériens colorés* par

$$G_{r,n}(x) := \sum_{\gamma \in G(r,n)} x^{\text{des}_G(\gamma)}, \quad (0.61)$$

et a obtenu la fonction génératrice exponentielle de $G_{r,n}(x)$,

$$\sum_{n \geq 0} G_{r,n}(x) \frac{z^n}{n!} = \frac{(1-x)e^{z(1-x)}}{1 - xe^{rz(1-x)}}. \quad (0.62)$$

Lorsque $r = 1$ et $r = 2$, l'équation (0.62) redonne respectivement (0.3) et (0.38).

Reiner [Rei93] a défini le q -analogue des polynômes eulériens colorés par

$$G_{r,n}(x, q) := \sum_{\gamma \in G(r,n)} x^{\text{des}_G(\gamma)} q^{\ell_G(\gamma)}, \quad (0.63)$$

et a obtenu la fonction génératrice exponentielle en q de $G_{r,n}(x, q)$.

Theorem 0.7. *On a*

$$\sum_{n \geq 0} G_{r,n}(x, q) \frac{z^n}{(-q[r-1]; q)_n [n]!} = \frac{(1-x) \exp_{G(r)}(z(1-x); q)}{1 - x \exp(z(1-x); q)}, \quad (0.64)$$

où $\exp_{G(r)}(z; q)$ est la fonction q -exponentielle colorée définie par

$$\exp_{G(r)}(z; q) = \sum_{n \geq 0} \frac{z^n}{[-q[r-1]; q]_n [n]!}. \quad (0.65)$$

Dans le chapitre 4, nous généraliserons les Théorèmes 0.5, 0.6 et 0.7 en dénombrant les permutations selon la parité des descentes.

0.2 Gamma-positivité des polynômes eulériens

Rappelons qu'un polynôme $f(x) = \sum_i a_i x^i \in \mathbb{R}[x]$ est dit *γ -positif* s'il existe un entier $n \in \mathbb{N}$ et des nombres réels non négatifs $\gamma_0, \gamma_1, \dots, \gamma_{\lfloor n/2 \rfloor}$ tels que

$$f(x) = \sum_{i=0}^{\lfloor n/2 \rfloor} \gamma_i x^i (1+x)^{n-2i}. \quad (0.66)$$

La notion de γ -positivité est apparue pour la première fois dans les travaux de D. Foata et M. Schützenberger [FS70], puis dans ceux de D. Foata et V. Strehl [FS74], consacrés aux polynômes eulériens classiques. La γ -positivité d'une suite combinatoire implique à la fois la unimodalité et la symétrie.

Le polynôme eulérien $A_n(x)$, qui est manifestement symétrique, fournit l'exemple prototypique d'un polynôme γ -positif en combinatoire.

Soit $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$, avec la convention $\sigma_0 = \sigma_{n+1} = 0$. Une lettre $\sigma_i \in [n]$ est appelée une *double descente* de σ si $\sigma_{i-1} > \sigma_i > \sigma_{i+1}$. On note par $\text{dd}(\sigma)$ le nombre de doubles descentes de σ .

Foata et Schützenberger [FS70, Theorem 5.6] ont démontré le développement γ -positif élégant suivant des polynômes eulériens :

$$A_n(x) = \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \gamma_{n,k} x^k (1+x)^{n-1-2k}, \quad (0.67)$$

où

$$\gamma_{n,k} := |\{\sigma \in \mathfrak{S}_n : \text{des}(\sigma) = k, \text{dd}(\sigma) = 0\}|.$$

Les premiers développements en γ de $A_n(x)$ sont les suivants:

$$A_n(x) = \begin{cases} 1+x, & \text{si } n=2, \\ (1+x)^2+2x, & \text{si } n=3, \\ (1+x)^3+8x(1+x), & \text{si } n=4, \\ (1+x)^4+22x(1+x)^2+16x^2, & \text{si } n=5, \\ (1+x)^5+52x(1+x)^3+136x^2(1+x), & \text{si } n=6. \end{cases}$$

D'après les développements ci-dessus, on voit que $\gamma_{n,k}/2^k \in \mathbb{N}^*$ pour $0 \leq k \leq \lfloor \frac{n-1}{2} \rfloor$. Dans un travail ultérieur, Foata et Schützenberger [FS73] ont montré que les coefficients γ normalisés $\gamma_{n,k}/2^k$ dénombrent les permutations d'André ayant k descentes. Voir les Chapitres 1 et 2 pour plus de détails sur les coefficients γ normalisés et les permutations d'André.

Récemment, divers raffinements de (0.67) ainsi que d'autres polynômes γ -positifs apparaissant en combinatoire énumérative et géométrique ont été largement étudiés; le lecteur pourra consulter, pour davantage d'informations, l'article de synthèse d'Athanasiadis [Ath18] ainsi que l'exposé sous forme de livre de Petersen [Pet15].

0.3 Combinatoire des fractions continues

Les travaux pionniers sur les fractions continues analytiques sont dus à Leonhard Euler, qui a introduit plusieurs développements en fractions continues de séries formelles, comme celui de [Eul60]:

$$\sum_{n \geq 0} (n+1)! z^n = \frac{1}{1 - 2z - \frac{1 \cdot 2 z^2}{1 - 4z - \frac{2 \cdot 3 z^2}{1 - 6z - \frac{3 \cdot 4 z^2}{\ddots - \frac{k(k+1) z^2}{1 - 2kz - \ddots}}}}}. \quad (0.68)$$

La théorie générale de telles fractions continues a été principalement développée par Stieltjes dans son mémoire fondamental [Sti94]. Ces fractions continues constituent un cas particulier des *fractions continues de Jacobi*, ou *J-fractions*, c'est-à-dire

$$\mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}) = \frac{1}{1 - b_0 z - \frac{\lambda_1 z^2}{1 - b_1 z - \frac{\lambda_2 z^2}{1 - b_2 z - \frac{\lambda_3 z^2}{\ddots - \frac{\lambda_{k+1} z^2}{1 - b_{k+1} z - \ddots}}}}}, \quad (0.69)$$

où $\{b_k\}_{k \geq 0}$ et $\{\lambda_k\}_{k \geq 1}$ désignent deux suites de « coefficients », qui peuvent être constituées d'entiers, de nombres réels ou complexes, ou, plus généralement, de variables formelles tirées de deux alphabets infinis dénombrables distincts. Dans de nombreux cas concrets, ces coefficients seront des polynômes à coefficients entiers en un nombre fini de variables formelles ou de paramètres.

On définit un *chemin de Motzkin* comme une suite de sommets $(s_0, \dots, s_n)$ du réseau quadrillé $\mathbb{N} \times \mathbb{N}$, satisfaisant les conditions suivantes :

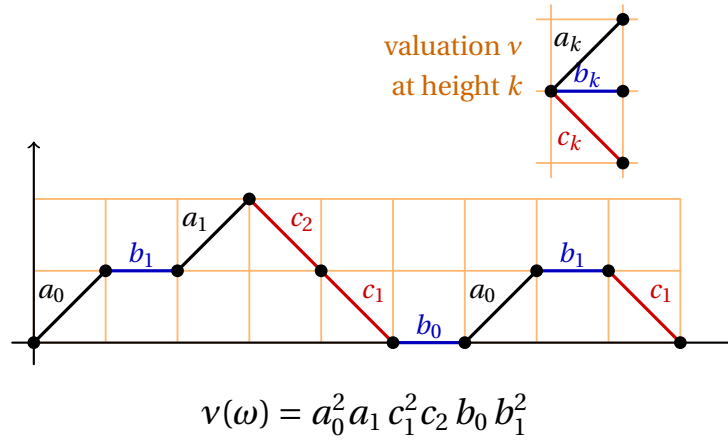


Figure 0.5: Weighted Motzkin path.

- $s_0 = (0, 0)$, $s_n = (n, 0)$.
- Pour $0 \leq i < n$, chaque pas élémentaire (s_i, s_{i+1}) est de l'un des types suivants :
 - montée diagonale (N) : $s_{i+1} = s_i + (1, 1)$;
 - descente diagonale (S) : $s_{i+1} = s_i + (1, -1)$;
 - pas horizontal (E) : $s_{i+1} = s_i + (1, 0)$.

On note $\mathcal{M}_n$ l'ensemble des chemins de Motzkin de longueur n .

Considérons trois suites de "coefficients" ou "lettres" sur trois alphabets (éventuellement distincts) :

$$\{a_k\}_{k \geq 0}, \quad \{b_k\}_{k \geq 0}, \quad \{c_k\}_{k \geq 1}.$$

On définit le poids $v(\omega)$ d'un chemin de Motzkin ω comme le produit des poids de ses pas élémentaires, où le poids d'un pas N (respectivement S, respectivement E) partant de la hauteur k est donné par a_k (respectivement c_k , respectivement b_k), voir Fig. 0.5.

Flajolet [Fla80] a établi une interprétation combinatoire des J -fractions continues générales comme fonctions génératrices de chemins de Motzkin pondérés, résultat que nous appellerons le "lemme fondamental de Flajolet".

Lemma 0.3.1. On a

$$\sum_{n \geq 0} \sum_{\omega \in \mathcal{M}_n} v(\omega) z^n = \frac{1}{1 - b_0 z - \frac{a_0 c_1 z^2}{1 - b_1 z - \frac{a_1 c_2 z^2}{\ddots}}}. \quad (0.70)$$

Dans le cas où $b_k = 0$ pour tout $k \geq 0$, c'est-à-dire lorsque les chemins sont des *chemins de Dyck*, la fraction continue (0.183) est appelée *fraction continue de Stieltjes*, ou *S-fraction*.

Nous supposons maintenant que les suites

$$\{a_k\}_{k \geq 0}, \quad \{b_k\}_{k \geq 0}, \quad \{c_k\}_{k \geq 1} \quad (0.71)$$

sont des suites d'entiers non négatifs. Ces entiers sont appelés *nombres de possibilités*.

Definition 0.8. Une *histoire* h est donnée par une paire

$$(\omega; (p_1, \dots, p_n)),$$

où $\omega = (s_0, \dots, s_n)$ est un chemin de Motzkin de longueur n , et $(p_1, \dots, p_n)$ est une suite d'entiers telle que

$$0 \leq p_i \leq v(s_{i-1}, s_i) - 1, \quad 1 \leq i \leq n, \quad (0.72)$$

où les valeurs possibles de v sont données en (0.71).

De nombreux développements connus de séries formelles en fractions continues peuvent être démontrés en combinant le lemme fondamental de Flajolet avec certaines constructions combinatoires. Les coefficients de la série formelle sont interprétés à l'aide de certains objets combinatoires pondérés. Certaines *histoires pondérées* sont reliées à la fraction continue, et une bijection préservant le poids entre ces objets et ces *histoires* fournit une preuve combinatoire du développement en fraction continue via le lemme fondamental de Flajolet.

Flajolet [Fla80] a fait un usage intensif de la bijection de Françon–Viennot [FV79] entre les permutations et certaines histoires associées à la valuation de chemin suivante :

$$a_k = k + 1, \quad b_k = 2k + 2 \ (k \geq 0), \quad c_k = k + 1 \ (k \geq 1). \quad (0.73)$$

Ces histoires sont souvent appelées *histoires de Laguerre* et fournissent une preuve combinatoire de la fraction continue d'Euler (0.68). Voir le Chapitre 2 pour plus de détails sur la bijection de Françon–Viennot et les histoires de Laguerre.

0.4 Plan de la thèse

Afin de rendre la présentation du plan plus lisible, nous regroupons ici les notations nécessaires à l'énoncé des résultats résumés ci-dessous. Ici comme dans toute la suite, l'abréviation donnée entre parenthèses sera utilisée pour désigner le nombre d'occurrences de la statistique correspondante.

Notations dans les Parties I–III. Pour $\sigma \in \mathfrak{S}_n$, un *cycle* (**cyc**) de σ est une suite $(a_1 a_2 \dots a_k)$ d'éléments distincts de $[n]$ telle que $\sigma(a_i) = a_{i+1}$ pour $1 \leq i < k$ et $\sigma(a_k) = a_1$. Soit $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$,

- pour $i \in [n]$, l'indice i est appelé un *drop* (**drop**) de σ si $i > \sigma_i$, une *excédance* (**exc**) de σ si $i < \sigma_i$, un *point fixe* (**fix**) de σ si $i = \sigma_i$, et une *excédance faible* (**wex**) de σ si $i \leq \sigma_i$;
- pour $i \in [n-1]$, l'indice i est appelé une *descente* (**des**) de σ si $\sigma_i > \sigma_{i+1}$, et une *montée* (**asc**) de σ si $\sigma_i < \sigma_{i+1}$.

Pour $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$, une lettre $\sigma_i \in [n]$ est appelée

- un *maximum de gauche à droite* (**lmax**) de σ si $\sigma_i > \sigma_j$ pour tout $j < i$;
- un *maximum de droite à gauche* (**rmax**) de σ si $\sigma_i > \sigma_j$ pour tout $j > i$;
- un *minimum de droite à gauche* (**rmin**) de σ si $\sigma_i < \sigma_j$ pour tout $j > i$;
- un *minimum de gauche à droite* (**lmin**) de σ si $\sigma_i < \sigma_j$ pour tout $j < i$.

Pour $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$, avec la convention $\sigma_0 = \sigma_{n+1} = 0$, une lettre $\sigma_i \in [n]$ est appelée

- une *vallée* (**val**) de σ si $\sigma_{i-1} > \sigma_i < \sigma_{i+1}$;
- un *pic* (**pk**) de σ si $\sigma_{i-1} < \sigma_i > \sigma_{i+1}$;
- une *double montée* (**da**) de σ si $\sigma_{i-1} < \sigma_i < \sigma_{i+1}$;
- une *double descente* (**dd**) de σ si $\sigma_{i-1} > \sigma_i > \sigma_{i+1}$;
- un *pic maximum de gauche à droite* (**lmaxpk**) de σ si σ_i est à la fois un maximum de gauche à droite et un pic ;
- un *pic maximum de droite à gauche* (**rmaxpk**) de σ si σ_i est à la fois un maximum de droite à gauche et un pic ;
- une *double montée maximum de gauche à droite* (**lmaxda**) de σ si σ_i est à la fois un maximum de gauche à droite et une double montée ;
- une *double descente maximum de droite à gauche* (**rmaxdd**) de σ si σ_i est à la fois un maximum de droite à gauche et une double descente ;
- une *double montée minimum de droite à gauche* (**rminda**) de σ si σ_i est à la fois un minimum de droite à gauche et une double montée.

Pour $\sigma = \sigma(1) \dots \sigma(n) \in \mathfrak{S}_n$, on dit qu'un indice $i \in [n]$ est un

- *pic de cycle* (**cpk**) de σ si $\sigma^{-1}(i) < i > \sigma(i)$;
- une *vallée de cycle* (**cval**) de σ si $\sigma^{-1}(i) > i < \sigma(i)$;
- une *double montée de cycle* (**cda**) de σ si $\sigma^{-1}(i) < i < \sigma(i)$;
- une *double descente de cycle* (**cdd**) de σ si $\sigma^{-1}(i) > i > \sigma(i)$.

Soit $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$. Si $i \in [n-1]$ est une descente, alors l'indice i est appelé une *position de descente*, σ_i est appelé un *sommet de descente*, et σ_{i+1} est appelé un *bas de descente*. Pour $i \in [n]$, l'entrée σ_i est appelée un *non-sommet de descente* ou *sommet de non-descente* (resp. un *bas de non-descente*) de σ , si elle n'est pas un sommet de descente (resp. un bas de descente) de σ . Pour $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$, une lettre $\sigma_i \in [n]$ est appelée

- un *foremaximum* (**fmax**) de σ si σ_i est un sommet de non-descente et aussi un maximum de gauche à droite ;
- un *postminimum* (**pmin**) de σ si σ_i est un bas de non-descente et aussi un minimum de droite à gauche.

Nous notons par $\text{NDtop}(\sigma)$ (resp. $\text{NDbot}(\sigma)$) la somme des sommets de non-descente (resp. des bas de non-descente) de σ , et nous définissons la *différence de non-descente* de σ par

$$\text{NDdif}(\sigma) := \text{NDbot}(\sigma) - \text{NDtop}(\sigma).$$

Le nombre d'occurrences des motifs 31-2, 2-31 et 2-13 dans $\sigma \in \mathfrak{S}_n$ est défini par

$$\begin{aligned} (31-2)\sigma &= |\{(i, j) : i+1 < j \text{ et } \sigma_i > \sigma_j > \sigma_{i+1}\}|; \\ (2-31)\sigma &= |\{(i, j) : j < i-1 \text{ et } \sigma_{i-1} > \sigma_j > \sigma_i\}|; \\ (2-13)\sigma &= |\{(i, j) : j < i-1 \text{ et } \sigma_{i-1} < \sigma_j < \sigma_i\}|. \end{aligned}$$

Pour $\sigma \in \mathfrak{S}_n$, les statistiques MAD et MADL sont définies par

$$\begin{aligned}\text{MAD}(\sigma) &= \text{des}(\sigma) + (31-2)(\sigma) + 2(2-31)(\sigma); \\ \text{MADL}(\sigma) &= \text{des}(\sigma) + (2-31)(\sigma) + 2(31-2)(\sigma).\end{aligned}$$

Le nombre de *croisements* et le nombre de *nichages* de $\sigma \in \mathfrak{S}_n$ sont définis par

$$\begin{aligned}\text{cros}(\sigma) &= |\{(i, j) : i < j \leq \sigma_i < \sigma_j \text{ ou } \sigma_i < \sigma_j < i < j\}|; \\ \text{nest}(\sigma) &= |\{(i, j) : i < j \leq \sigma_j < \sigma_i \text{ ou } \sigma_j < \sigma_i < i < j\}|.\end{aligned}$$

Notations dans la Partie IV. Pour $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$, un indice $i \in [n-1]$ est appelé

- une *descente impaire* (**odes**) de σ si i est impair et est aussi une descente ;
- une *descente paire* (**edes**) de σ si i est pair et est aussi une descente.

Pour $\gamma = [\gamma(1), \dots, \gamma(n)] = [\sigma_1^{c_1}, \dots, \sigma_n^{c_n}] \in G(r, n)$, la *longueur* (ℓ_G) de γ est définie par

$$\ell_G(\gamma) = \text{inv}(\gamma) + \sum_{c_i \neq 0} (|\gamma(i)| + c_i - 1);$$

le nombre de *couleurs* de γ est défini par $\text{col}_G(\gamma) := |\{i \in [n] : c_i \neq 0\}|$; et le nombre de *descentes impaires* et de *descentes paires* de γ est défini respectivement par

$$\begin{aligned}\text{odes}_G(\gamma) &= |\{i \in [0, n-1] : \gamma(i) > \gamma(i+1) \text{ et } i \text{ est impair}\}|; \\ \text{edes}_G(\gamma) &= |\{i \in [0, n-1] : \gamma(i) > \gamma(i+1) \text{ et } i \text{ est pair}\}|,\end{aligned}$$

où $\gamma(0) := 0$.

Chapter 1

Soit $\mathbf{u} = (u_1, u_2, u_3, u_4)$. Nous définissons les *polynômes eulériens bi-Stirling généralisés*

$$A_n(\mathbf{u}, f, g, t | \alpha, \beta) = \sum_{\sigma \in \mathfrak{S}_{n+1}} \mathbf{wt}(\sigma), \quad (0.74)$$

où $\mathbf{wt}$ est une fonction de poids de σ définie par

$$\begin{aligned}\mathbf{wt}(\sigma) &:= (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} \\ &\quad \times t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}.\end{aligned} \quad (0.75)$$

Ces polynômes $A_n(\mathbf{u}, f, g, t | \alpha, \beta)$ raffinent les polynômes eulériens de type bi-Stirling de Carlitz-Scoville définis dans (0.9). Carlitz et Scoville [CS77] ont introduit les *polynômes eulériens cycliques* (α, t) ,

$$A_n^{\text{cyc}}(x, y, t | \alpha) := \sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)} y^{\text{drop}(\sigma)} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)}. \quad (0.76)$$

Le résultat suivant constitue notre premier théorème principal de ce chapitre et fournit une formule de connexion entre les polynômes $A_n(\mathbf{u}, f, g, t | \alpha, \beta)$ et $A_n^{\text{cyc}}(x, y, t | \alpha)$.

Theorem 0.9. *If $xy = u_1 u_2$ and $x + y = u_3 + u_4$, then*

$$A_n(\mathbf{u}, f, g, t | \alpha, \beta) = A_n^{\text{cyc}}\left(x, y, \frac{\alpha u_3 + \beta u_4}{\alpha f + \beta g} t | \alpha f + \beta g\right). \quad (0.77)$$

Nous définissons les *polynômes eulériens* (α, t)

$$A_n(x, y, t | \alpha) = A_n^{\text{cyc}}\left(x, y, \frac{x+y}{2} t | 2\alpha\right). \quad (0.78)$$

En combinant (0.78) et Theorem 0.9 avec $\mathbf{u} = (x, y, x, y)$, $f = g = 1$, and $\alpha = \beta$, nous obtenons

$$A_n(x, y, t | \alpha) = \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{asc}(\sigma)} y^{\text{des}(\sigma)} t^{\text{lmaxda}(\sigma) + \text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2}. \quad (0.79)$$

Le résultat suivant constitue notre deuxième théorème principal de ce chapitre et donne trois interprétations combinatoires différentes de la γ -positivité normalisée de $A_n(x, y, t | \alpha)$.

Theorem 0.10. *Pour $n \geq 0$, on a*

$$A_n(x, y, t | \alpha) = \sum_{j=0}^{\lfloor n/2 \rfloor} 2^j d_{n,j}(\alpha, t) (xy)^j (x+y)^{n-2j}, \quad (0.80a)$$

où pour $0 \leq j \leq \lfloor n/2 \rfloor$,

$$d_{n,j}(\alpha, t) = \sum_{\substack{\sigma \in \mathcal{CA}_n \\ \text{drop}(\sigma) = j}} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} \quad (0.80b)$$

$$= \sum_{\substack{\sigma \in \mathcal{A}_{n+1}^1 \\ \text{des}(\sigma) = j}} t^{\text{rminda}(\sigma)} \alpha^{\text{rmin}(\sigma) - 1} \quad (0.80c)$$

$$= \sum_{\substack{\sigma \in \mathcal{A}_{n+1}^2 \\ \text{des}(\sigma) = j}} t^{\text{rminda}(\sigma)} \alpha^{\text{rmin}(\sigma) - 1}. \quad (0.80d)$$

Les ensembles $\mathcal{CA}_n$, $\mathcal{A}_n^1$ et $\mathcal{A}_n^2$ dans le théorème ci-dessus désignent respectivement l'ensemble des permutations d'André cycliques, l'ensemble des permutations d'André de première espèce et l'ensemble des permutations d'André de seconde espèce. Les définitions précises de ces ensembles seront données dans la Section 1.3.3.

Nous établissons le Théorème 0.9 en construisant une bijection dérivée de deux variantes de la première transformation fondamentale de Foata, puis en intégrant cette construction au valley-hopping cyclique sur les permutations. La preuve du Théorème 0.10 se déroule comme suit : nous établissons d'abord (0.80b) en montrant que les fonctions génératrices exponentielles des deux membres coïncident. Ensuite, nous déduisons (0.80c) de (0.80b) en construisant une bijection $\Phi : \mathcal{CA}_n \rightarrow \mathcal{A}_{n+1}^1$. Enfin, nous déduisons (0.80d) de (0.80c) au moyen d'une bijection $\Psi : \mathcal{A}_{n+1}^1 \rightarrow \mathcal{A}_{n+1}^2$ construite à partir d'un modèle d'arbres d'André.

Grâce à la formule de connexion du Théorème 0.9, nous obtenons des fonctions génératrices exponentielles ainsi que plusieurs développements γ -positifs pour diverses familles de polynômes de type eulérien. Nos résultats unifient et renforcent les travaux récents de Ji [Ji25] et de Ji-Lin [JL24].

Chapitre 2

Dans ce chapitre, nous établissons une J -fraction pour des (p, q) -analogues des polynômes eulériens de type bi-Stirling généralisés définis dans (0.74), avec une statistique supplémentaire D liée à l'aire du chemin de Motzkin associé sous la bijection de Françon-Viennot.

Soient $\text{Val}(\sigma)$ et $\text{Pk}(\sigma)$ les ensembles des vallées et des pics de σ , respectivement. Nous définissons les polynômes eulériens multivariés généraux :

$$A_n(\mathbf{u}, f, g, h, t | \alpha, \beta; p, q) := \sum_{\sigma \in \mathfrak{S}_{n+1}} h^{D(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)} \mathbf{wt}(\sigma), \quad (0.81)$$

où $\mathbf{wt}$ est une fonction de poids de σ définie par (0.75), et D est définie par

$$D(\sigma) := \sum_{i \in \text{Pk}(\sigma) \setminus \{n\}} i - \sum_{j \in \text{Val}(\sigma)} j,$$

Le (p, q) -analogue généralisé de l'entier positif $k \in \mathbb{N}^*$ avec deux paramètres auxiliaires α et β est défini comme suit :

$$[k; \alpha, \beta]_{p,q} := (\alpha - 1)p^{k-1} + (\beta - 1)q^{k-1} + \frac{p^k - q^k}{p - q}.$$

Voici notre premier résultat principal de ce chapitre.

Theorem 0.11. *On a*

$$\sum_{n=0}^{\infty} A_n(\mathbf{u}, f, g, h, t | \alpha, \beta; p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (0.82a)$$

où

$$b_k = h^k \left(u_3 [k+1; 1, \alpha t]_{p,q} + u_4 [k+1; \beta t, 1]_{p,q} \right), \quad (0.82b)$$

$$\lambda_k = u_1 u_2 h^{2k-1} [k]_{p,q} [k+1; \beta g, \alpha f]_{p,q}. \quad (0.82c)$$

Petersen et Tenner [PT15] ont introduit une statistique appelée profondeur pour les groupes de Coxeter, définie en termes de factorisations des éléments du groupe en réflexions. Dans le cas du groupe symétrique $\mathfrak{S}_n$, la *profondeur* d'une permutation $\sigma \in \mathfrak{S}_n$ est définie par

$$\text{depth}(\sigma) = \sum_{\sigma(i) > i} (\sigma(i) - i).$$

Petersen [Pet] a conjecturé une J -fraction pour la distribution jointe des statistiques $(\text{inv}, \text{depth})$, qui a ensuite été établie dans [EFL24, DS23]. Motivés par ces résultats, nous introduisons le (p, q) -analogue de la version cyclique généralisée des polynômes eulériens, incorporant deux statistiques de Stirling,

$$C_n(\mathbf{u}, t, s, h | \alpha, \beta; p, q) := \sum_{\sigma \in \mathfrak{S}_n} (u_1 u_2)^{\text{cval}(\sigma)} u_3^{\text{cda}(\sigma)} u_4^{\text{cdd}(\sigma)} t^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} \\ \times h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma)} \beta^{\text{rmin}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)}. \quad (0.83a)$$

Le résultat suivant constitue notre deuxième résultat de ce chapitre.

Theorem 0.12. *On a*

$$1 + \sum_{n \geq 1} C_n(\mathbf{u}, t, s, h | \alpha, \beta; p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (0.83b)$$

où $b_0 = t\alpha\beta$, et

$$b_k = (sh)^k \left(tq^k s^k + u_4 [k; \beta, 1]_{p,qs} + u_3 p [k; \alpha, 1]_{p,qs} \right), \quad (0.83c)$$

$$\lambda_k = u_1 u_2 (sh)^{2k-1} [k; \alpha, 1]_{p,qs} [k; \beta, 1]_{p,qs}. \quad (0.83d)$$

Nous démontrons le Théorème 0.28 et le Théorème 0.29 à l'aide de la théorie des fractions continues [Fla80], ainsi que de variantes des bijections de Françon-Viennot [FV79] et de Foata-Zeilberger [FZ90] entre permutations et chemins de Motzkin pondérés.

Le troisième résultat principal de ce chapitre montre que la bijection de Corteel Φ_C [Cor07] et la bijection de Shin-Zeng Φ_{SZ} [SZ10] sur $\mathfrak{S}_n$ préservent des propriétés structurelles supplémentaires compatibles avec le codage par chemins de Motzkin.

Theorem 0.13. *Les applications Φ_C et Φ_{SZ} sont des bijections sur $\mathfrak{S}_n$ telles que, pour tout $\sigma \in \mathfrak{S}_n$, on a*

$$(\text{ndes}, \text{pmin}, 31-2, 2-31, \text{MADL}, \text{NDdif}) \sigma = (\text{wex}, \text{fix}, \text{nest}, \text{cros}, \text{inv}, \text{depth}) \Phi_C(\sigma); \quad (0.84)$$

$$(\text{ndes}, \text{fmax}, 31-2, 2-31, \text{MAD}, \text{NDdif}) \sigma = (\text{wex}, \text{fix}, \text{cros}, \text{nest}, \text{inv}, \text{depth}) \Phi_{SZ}(\sigma). \quad (0.85)$$

De plus, si $\Phi^* := \Phi_{SZ} \circ \Phi_C^{-1}$, alors

$$(\text{wex}, \text{nest}, \text{cros}, \text{depth}) \sigma = (\text{wex}, \text{cros}, \text{nest}, \text{depth}) \Phi^*(\sigma). \quad (0.86)$$

En exploitant les fractions continues établies dans les Théorèmes 0.11 et 0.12, nous obtenons des développements γ -positifs pour plusieurs (p, q) -analogues de polynômes eulériens généralisés ; voir les Théorèmes 2.16, 2.18 et 2.22.

Chapitre 3

Soit $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$. Le complément de σ est défini par

$$\sigma^c = (n+1-\sigma_1)(n+1-\sigma_2) \cdots (n+1-\sigma_n). \quad (0.87)$$

Il est clair que le complément définit une bijection sur $\mathfrak{S}_n$ et satisfait les propriétés

$$\text{asc}(\sigma) = \text{des}(\sigma^c), \quad \text{des}(\sigma) = \text{asc}(\sigma^c), \quad (0.88)$$

$$\text{lmax}(\sigma) = \text{lmin}(\sigma^c), \quad \text{rmax}(\sigma) = \text{rmin}(\sigma^c). \quad (0.89)$$

En combinant l'opérateur de complément et (0.15), nous voyons que les nombres eulériens (α, β) $A(r, s | \alpha, \beta)$ admettent une autre interprétation combinatoire

$$A(r, s | \alpha, \beta) = \sum_{\substack{\sigma \in \mathfrak{S}_{r+s+1} \\ \text{des}(\sigma) = r}} \alpha^{\text{lmin}(\sigma)-1} \beta^{\text{rmin}(\sigma)-1}. \quad (0.90)$$

Récemment, Gaudin [Gau24] a observé que les nombres de Carlitz-Scoville $A(r, s | \alpha, \beta)$ forment, à un changement d'échelle près, une sous-famille des nombres eulériens mixtes, et il a proposé un q -analogue $A(r, s | \alpha, \beta)_q$ dans le cadre des nombres eulériens remélangés [NT23]. Il a montré que $A(r, s | \alpha, \beta)_q$ satisfait la récurrence suivante

$$A(r, s | \alpha, \beta)_q = q^{r+\beta-1} [s + \alpha] A(r-1, s | \alpha, \beta)_q + [r + \beta] A(r, s-1 | \alpha, \beta)_q. \quad (0.91)$$

Pour des raisons de commodité, nous introduisons la notation suivante, valable pour $n \geq k \geq 0$:

$$E_{n,k}(\alpha, \beta, q) := A(k, n-k | \alpha, \beta)_q \quad (0.92)$$

avec la convention $E_{n,k}(\alpha, \beta, q) = 0$ si $k \notin \{0, \dots, n\}$. Compte tenu de (0.92), en posant $r + s = n$ et $r = k$ dans (0.91), nous obtenons

$$E_{n,k}(\alpha, \beta, q) = q^{\beta+k-1} [n-k+\alpha] E_{n-1,k-1}(\alpha, \beta, q) + [k+\beta] E_{n-1,k}(\alpha, \beta, q). \quad (0.93)$$

Lorsque $(\alpha, \beta) = (1, 1)$ ou $(0, 1)$, plusieurs q -analogues des nombres eulériens sont déjà connus [Car54, Sta76, Raw81]. Dans le cas $\alpha = 0$, Butler [But04b, But25] a introduit un nouveau q -analogue des nombres eulériens en combinant le paramètre β avec l'indice majeur.

L'objectif de ce chapitre est de fournir une interprétation combinatoire de $E_{n,k}(\alpha, \beta, q)$ en termes de statistiques sur les permutations telles que les minima de gauche à droite, les minima de droite à gauche, les descentes et l'indice majeur (re)mélangé, et d'en déduire plusieurs formules explicites pour ces nombres. Pour $\sigma \in \mathfrak{S}_n$, l'*indice majeur mixte* de σ , noté par $\widetilde{\text{maj}}$, est défini par

$$\widetilde{\text{maj}}(\sigma) = (1 + \text{des}(\sigma) - \text{lmin}(\sigma))(\alpha - 1) + (n - \text{rmin}(\sigma))(\beta - 1) + \text{maj}(\sigma). \quad (0.94)$$

Voici notre premier résultat principal de ce chapitre.

Theorem 0.14. *Pour tout entier $n \geq k \geq 0$, on a*

$$E_{n,k}(\alpha, \beta, q) = \sum_{\sigma \in \mathfrak{S}_{n+1,k}} [\alpha]^{\text{lmin}(\sigma)-1} [\beta]^{\text{rmin}(\sigma)-1} q^{\widetilde{\text{maj}}(\sigma)}. \quad (0.95)$$

où $\mathfrak{S}_{n+1,k}$ désigne l'ensemble des permutations de $\mathfrak{S}_{n+1}$ ayant k descentes.

Nous définissons les *polynômes bi-Stirling-Euler-Mahoniens* $E_n(x | \alpha, \beta, q)$ par

$$E_n(x | \alpha, \beta, q) = \sum_{k=0}^n E_{n,k}(\alpha, \beta, q) x^k. \quad (0.96)$$

Définissons l'opérateur de q -dérivation δ_x par

$$\delta_x p(x) = \frac{p(qx) - p(x)}{(q-1)x}, \quad (0.97)$$

où $p(x) \in \mathbb{R}[[x]]$. Voici notre deuxième résultat principal de ce chapitre.

Theorem 0.15. *Soit $E_n(x; q) := E_n(x | \alpha, \beta, q)$ pour $n \geq 0$. Nous avons les q -identités suivantes:*

(i)

$$E_n(x; q) = ([\beta] + q^\beta [n-1+\alpha]x) E_{n-1}(x; q) + (1-x)xq^\beta \delta_x(E_{n-1}(x; q)). \quad (0.98)$$

(ii)

$$\frac{E_n(x; q)}{(x; q)_{n+\alpha+\beta}} = \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n. \quad (0.99)$$

(iii)

$$\sum_{n \geq 0} \frac{E_n(x; q)}{(x; q)_{n+\alpha+\beta}} \frac{t^n}{n!} = \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} \exp([j + \beta]t). \quad (0.100)$$

(iv)

$$\sum_{k=0}^j \begin{bmatrix} j-k+n+\alpha+\beta-1 \\ j-k \end{bmatrix} E_{n,k}(\alpha, \beta, q) = \begin{bmatrix} j+\alpha+\beta-1 \\ j \end{bmatrix} [j+\beta]^n. \quad (0.101)$$

(v)

$$E_{n,k}(\alpha, \beta, q) = \sum_{j=0}^k \begin{bmatrix} n+\alpha+\beta \\ k-j \end{bmatrix} \begin{bmatrix} j+\alpha+\beta-1 \\ j \end{bmatrix} (-1)^{k-j} q^{\binom{k-j}{2}} [\beta+j]^n. \quad (0.102)$$

Les démonstrations du Théorème 0.14 et du Théorème 0.15 reposent sur une combinaison d'un argument combinatoire par insertion et du q -calcul.

Chapitre 4

En 1973, Carlitz et Scoville [CS73] ont dénombré les permutations selon la parité à la fois des descentes et des montées. Récemment, Pan et Zeng [PZ23] ont considéré le polynôme énumératif des permutations associé à la statistique vectorielle (edes, odes, inv) et ont établi la fonction génératrice exponentielle en q suivante :

$$\sum_{n \geq 1} \frac{t^n}{[n]_q!} \sum_{\sigma \in \mathfrak{S}_n} x^{\text{odes}(\sigma)} y^{\text{edes}(\sigma)} q^{\text{inv}(\sigma)} = \frac{(1+x) \cosh(Mt; q) + M \sinh(Mt; q) - x(\cosh^2(Mt; q) - \sinh^2(Mt; q)) - 1}{1 - (x+y) \cosh(Mt; q) + xy(\cosh^2(Mt; q) - \sinh^2(Mt; q))}, \quad (0.103)$$

où $M = \sqrt{(1-x)(1-y)}$ et

$$\cosh(t; q) = \sum_{n \geq 0} \frac{t^{2n}}{[2n]_q!}, \quad \sinh(t; q) = \sum_{n \geq 1} \frac{t^{2n-1}}{[2n-1]_q!}. \quad (0.104)$$

La formule (0.103) est un q -analogue de la formule de Carlitz-Scoville [CS73, Theorem 3.1] et englobe la formule de Stanley pour (des, inv) dans (0.28). Dans un travail ultérieur, Dey, Shankar et Sivasubramanian [DSS24, Theorems 1.5 and 1.8] ont établi des formules analogues à (0.103) pour les groupes de Coxeter de type B et D.

Dans ce chapitre, nous donnons une formule pour la fonction génératrice de la statistique vectorielle (odes $_G$, edes $_G$, col $_G$, ℓ_G) sur le groupe des permutations colorées $G(r, n)$. Pour $n, r \in \mathbb{P}$, définissons les polynômes

$$G_{(r,n)}(a, q) := \sum_{\gamma \in G(r,n)} a^{\text{col}_G(\gamma)} q^{\ell_G(\gamma)}, \quad (0.105)$$

$$G_{(r,n)}(x, y, a, q) := \sum_{\gamma \in G(r,n)} x^{\text{odes}_G(\gamma)} y^{\text{edes}_G(\gamma)} a^{\text{col}_G(\gamma)} q^{\ell_G(\gamma)}. \quad (0.106)$$

Les fonctions génératrices d'indices pairs et impairs de $G_{(r,n)}(x, y, a, q)$ sont définies par

$$H_0^r := \sum_{n \geq 0} G_{(r,2n)}(x, y, a, q) \frac{t^{2n}}{G_{(r,2n)}(a, q)}, \quad (0.107a)$$

$$H_1^r := \sum_{n \geq 0} G_{(r,2n+1)}(x, y, a, q) \frac{t^{2n+1}}{G_{(r,2n+1)}(a, q)}. \quad (0.107b)$$

Définissons le q -analogue de la série exponentielle sur le produit en couronne

$$\exp_{G(r)}(t; a, q) := \sum_{n \geq 0} \frac{t^n}{G_{(r,n)}(a, q)}, \quad (0.108)$$

ainsi que les séries correspondantes de cosinus hyperbolique et de sinus hyperbolique

$$\cosh_{G(r)}(t; a, q) = \frac{\exp_{G(r)}(t; a, q) + \exp_{G(r)}(-t; a, q)}{2}, \quad (0.109a)$$

$$\sinh_{G(r)}(t; a, q) = \frac{\exp_{G(r)}(t; a, q) - \exp_{G(r)}(-t; a, q)}{2}. \quad (0.109b)$$

Le résultat suivant constitue notre premier résultat principal de ce chapitre.

Theorem 0.16. *Pour tout $r \in \mathbb{P}$, on a*

$$H_0^r = \frac{(1-y) \left((1-x \cosh(Mt; q)) \cosh_{G(r)}(Mt; a, q) + x \sinh(Mt; q) \sinh_{G(r)}(Mt; a, q) \right)}{1 - (x+y) \cosh(Mt; q) + xy \exp(Mt; q) \exp(-Mt; q)}, \quad (0.110a)$$

$$H_1^r = \frac{M \left((1-y \cosh(Mt; q)) \sinh_{G(r)}(Mt; a, q) + y \sinh(Mt; q) \cosh_{G(r)}(Mt; a, q) \right)}{1 - (x+y) \cosh(Mt; q) + xy \exp(Mt; q) \exp(-Mt; q)}, \quad (0.110b)$$

où $M = \sqrt{(1-x)(1-y)}$.

Étant donnée une permutation $\gamma \in G(r, n)$, un indice $i \in \{0, 1, \dots, n-1\}$ est appelé une *descente alternée* si i est une *descente impaire* ou une *montée paire*. Soit $\widehat{Des}_G(\gamma)$ l'ensemble des descentes alternées de γ , c'est-à-dire

$$\widehat{Des}_G(\gamma) = \{2i : \gamma(2i) < \gamma(2i+1)\} \cup \{2i+1 : \gamma(2i+1) > \gamma(2i+2)\},$$

et notons son cardinal par $\widehat{des}_G(\gamma)$. Nous définissons le polynôme des descentes alternées en q sur $G(r, n)$ par

$$\text{Alt}_n^{G(r)}(x, q) := \sum_{\gamma \in G(r, n)} x^{\widehat{des}_G(\gamma)} q^{\ell_G(\gamma)}. \quad (0.111)$$

Comme application du Théorème 0.16, nous allons évaluer $\text{Alt}_n^{G(r)}(x, q)$ lorsque $q = -1$. Le résultat suivant constitue notre deuxième résultat principal de ce chapitre.

Theorem 0.17. *Pour tout entier $n \geq 1$, les identités suivantes sont valables.*

1. *Si r est un entier positif pair, alors*

$$\text{Alt}_n^{G(r)}(x, -1) = (-1)^{\lfloor (n+1)/2 \rfloor} (1-x)^n. \quad (0.112)$$

2. *Si r est un entier positif impair, alors $\text{Alt}_1^{G(r)}(x, -1) = x$ et, pour $n \geq 2$,*

$$\text{Alt}_n^{G(r)}(x, -1) = \begin{cases} x(1-x)^m A_m(x), & \text{si } n = 2m \ (m \in \mathbb{N}^*); \\ \frac{2x^2}{1+x} (1-x)^{2m} A_{2m}(x), & \text{si } n = 4m+1 \ (m \in \mathbb{N}^*); \\ 0, & \text{si } n = 4m+3 \ (m \in \mathbb{N}). \end{cases} \quad (0.113)$$

Les démonstrations du Théorème 0.16 et du Théorème 0.17 reposent sur la méthode des fonctions génératrices et le q -calcul.

Introduction (English version)

0.5 Variations of Eulerian polynomials

0.5.1 Eulerian polynomials

In the 18th century, Euler [Eul55] studied the alternating Dirichlet series

$$\varphi(s) = 1 - \frac{1}{2^s} + \frac{1}{3^s} - \frac{1}{4^s} + \cdots,$$

which converges for $\Re(s) > 0$ and is related to the Riemann zeta function $\zeta(s)$ by

$$\varphi(s) = (1 - 2^{1-s})\zeta(s).$$

Euler's goal was to understand the special values $\varphi(-n)$ for $n = 0, 1, 2, \dots$. To this end, he introduced the *Eulerian polynomials* $A_n(x)$ characterized by

$$\frac{A_n(x)}{(1-x)^{n+1}} = \sum_{k \geq 0} (k+1)^n x^k. \quad (0.114)$$

By multiplying both sides of (0.114) by $z^n/n!$ and summing over all $n \geq 0$, we obtain

$$\sum_{n \geq 0} \frac{A_n(x)}{(1-x)^{n+1}} \frac{z^n}{n!} = \sum_{n \geq 0} \frac{z^n}{n!} \sum_{k \geq 0} (k+1)^n x^k = \frac{e^z}{1-xe^z}. \quad (0.115)$$

Multiplying $(1-x)$ both sides of (0.115) and then substituting z by $z(1-x)$, we derive the exponential generating function of $A_n(x)$,

$$\sum_{n \geq 0} A_n(x) \frac{z^n}{n!} = \frac{(1-x)e^{z(1-x)}}{1-xe^{z(1-x)}}. \quad (0.116)$$

Dominique Foata has a lovely survey in which he explains Euler's motivation and derivation [Foa10]. The first few values of $A_n(x)$ are as follows:

$$A_n(x) = \begin{cases} 1, & \text{if } n = 1, \\ 1 + x, & \text{if } n = 2, \\ 1 + 4x + x^2, & \text{if } n = 3, \\ 1 + 11x + 11x^2 + x^3, & \text{if } n = 4, \\ 1 + 26x + 66x^2 + 26x^3 + x^4, & \text{if } n = 5. \end{cases}$$

A *permutation* of $[n] := \{1, 2, \dots, n\}$ is a bijection $\sigma : [n] \rightarrow [n]$. Let $\mathfrak{S}_n$ denote the set of all permutations of $[n]$. A *statistic* on $\mathfrak{S}_n$ is a mapping $\text{st} : \mathfrak{S}_n \rightarrow \mathbb{N}$. For each permutation

$\sigma = \sigma(1)\sigma(2)\cdots\sigma(n) \in \mathfrak{S}_n$, an integer $i \in [n-1]$ is a *descent* (resp. an *excedance*) of σ if $\sigma(i) > \sigma(i+1)$ (resp. $\sigma(i) > i$). Denote by $\text{DES}(\sigma)$ (resp. $\text{EXC}(\sigma)$) the set of descents (resp. excedances) and denote by $\text{des}(\sigma)$ (resp. $\text{exc}(\sigma)$) the number of descents (resp. excedances) of σ .

MacMahon [Mac60, Mac13] proved that the descent number and excedance number on permutations are equidistributed, i.e.,

$$\sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)}. \quad (0.117)$$

Riordan [Rio58] later discovered the following combinatorial interpretation for the Eulerian polynomials

$$A_n(x) = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} = \sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle x^k, \quad (0.118)$$

where $\left\langle n \atop k \right\rangle$ is called an *Eulerian number*, defined as

$$\left\langle n \atop k \right\rangle := |\{\sigma \in \mathfrak{S}_n : \text{des}(\sigma) = k\}|. \quad (0.119)$$

These Eulerian numbers $\left\langle n \atop k \right\rangle$ satisfy the following recurrence

$$\left\langle n \atop k \right\rangle = (n-k) \left\langle n-1 \atop k-1 \right\rangle + (k+1) \left\langle n-1 \atop k \right\rangle, \quad (n > k \geq 0) \quad (0.120)$$

where $\left\langle 0 \atop 0 \right\rangle = 1$ and $\left\langle n \atop k \right\rangle = 0$ if $k \notin \{0, \dots, n-1\}$.

With the above interpretation, each statistic that is equidistributed with the descent number (or excedance number) on permutations is usually called an *Eulerian statistic*. The classical *Foata's first fundamental transformation* [Foa65] provides a bijective proof of (0.117).

Leonard Carlitz and his collaborators studied Eulerian numbers and their generalizations in several papers in the 20th century, e.g., [Car54, Car75, CS74, CS73, CRS66]. Foata and Marcel-Paul Schützenberger wrote a comprehensive treatment of Eulerian numbers from a combinatorial point of view in [FS70]. The research on the Eulerian numbers plays an important role in numerous arithmetic and combinatorial studies.

0.5.2 Bi-Stirling Eulerian polynomials

We also regard a permutation $\sigma \in \mathfrak{S}_n$ as a word $\sigma = \sigma_1\sigma_2\cdots\sigma_n$, where $\sigma_i := \sigma(i)$ for $i \in [n]$. For $\sigma \in \mathfrak{S}_n$, an index $i \in [n-1]$ is called an *ascent* of σ if $\sigma_i < \sigma_{i+1}$, and a letter σ_i is called a

- *left-to-right maximum* of σ if $\sigma_i > \sigma_j$ for every $j < i$;
- *right-to-left maximum* of σ if $\sigma_i > \sigma_j$ for every $j > i$.

Define $\text{asc}(\sigma)$, $\text{lmax}(\sigma)$, and $\text{rmax}(\sigma)$ to be the number of ascents, left-to-right maxima, and right-to-left maxima of σ , respectively. Both lmax and rmax are *Stirling statistics*:

$$\sum_{\sigma \in \mathfrak{S}_n} x^{\text{lmax}(\sigma)} = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{rmax}(\sigma)} = \sum_{k=0}^n s(n, k) x^k, \quad (0.121)$$

where $s(n, k)$ are the Stirling numbers of the first kind.

In 1974, Carlitz and Scoville [CS74] introduced the *bi-Stirling Eulerian polynomials* as follows,

$$A_n(x, y | \alpha, \beta) := \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{asc}(\sigma)} y^{\text{des}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}. \quad (0.122)$$

The first few values of $A_n(x, y | \alpha, \beta)$ are given as follows:

$$A_1(x, y | \alpha, \beta) = \alpha x + \beta y,$$

$$A_2(x, y | \alpha, \beta) = \alpha^2 x^2 + 2\alpha\beta xy + \alpha xy + \beta xy + \beta^2 y^2,$$

$$A_3(x, y | \alpha, \beta) = \alpha^3 x^3 + 3\alpha^2 \beta x^2 y + 3\alpha^2 x^2 y + 3\alpha \beta x^2 y + \alpha x^2 y + \beta x^2 y \\ + 3\alpha \beta^2 x y^2 + 3\alpha \beta x y^2 + 3\beta^2 x y^2 + \alpha x y^2 + \beta x y^2 + \beta^3 y^3.$$

They derived the following exponential generating function for $A_n(x, y | \alpha, \beta)$.

Theorem 0.18 (Theorem 9 [CS74]). *We have*

$$\sum_{n \geq 0} A_n(x, y | \alpha, \beta) \frac{z^n}{n!} = (1 + xF(x, y; z))^\alpha (1 + yF(x, y; z))^\beta, \quad (0.123)$$

where $F(x, y; z)$ is given by

$$F(x, y; z) = \frac{e^{xz} - e^{yz}}{xe^{yz} - ye^{xz}}. \quad (0.124)$$

Remark 0.19. When $\alpha = 0, \beta = 1$ and $x = 1$, polynomials $A_n(x, y | \alpha, \beta)$ reduce to the classical Eulerian polynomials $A_n(y)$ defined in (0.118). Accordingly, we recover the generating function (0.116) for $A_n(y)$ by setting $\alpha = 0, \beta = 1$ and $x = 1$ in (0.123).

For $x = y = 1$ and $\beta = 0$, it is easy to see that

$$A_n(1, 1 | \alpha, 0) = \sum_{\sigma \in \mathfrak{S}_n} \alpha^{\text{lmax}(\sigma)}. \quad (0.125)$$

By (0.123), we derive

$$\sum_{n \geq 0} A_n(1, 1 | \alpha, 0) \frac{z^n}{n!} = \left(\frac{1}{1-z} \right)^\alpha. \quad (0.126)$$

Equating the coefficients of $z^n/n!$ in (0.126), we obtain

$$\sum_{\sigma \in \mathfrak{S}_n} \alpha^{\text{lmax}(\sigma)} = \alpha(\alpha+1)(\alpha+2) \cdots (\alpha+n-1). \quad (0.127)$$

In the same paper, Carlitz-Scoville defined the *bi-Stirling Eulerian number* $A(r, s | \alpha, \beta)$ as the coefficient $x^r y^s$ in $A_{r+s+1}(x, y | \alpha, \beta)$, i.e.,

$$A(r, s | \alpha, \beta) := [x^r y^s] A_{r+s+1}(x, y | \alpha, \beta) = \sum_{\substack{\sigma \in \mathfrak{S}_{r+s+1} \\ \text{des} = s}} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}, \quad (0.128)$$

where $[x^r y^s] F(x, y)$ denotes the coefficient of $x^r y^s$ in $F(x, y)$. They proved that $A(r, s | \alpha, \beta)$ satisfies the following recurrence

$$A(r, s | \alpha, \beta) = (s + \alpha) A(r-1, s | \alpha, \beta) + (r + \beta) A(r, s-1 | \alpha, \beta) \quad (0.129)$$

Remark 0.20. When $\alpha = \beta = 1, r = n - k - 1$, and $s = k$, these numbers $A(r, s | \alpha, \beta)$ reduce to the classical Eulerian numbers defined in (0.119), i.e., $A(n - k - 1, k | 1, 1) = \langle \frac{n}{k} \rangle$. Accordingly, the recurrence (0.129) reduces to (0.120).

0.5.3 Binomial-Eulerian polynomials

Postnikov, Reiner, and Williams [PRW08] introduced the *binomial-Eulerian polynomials*:

$$\tilde{A}_n(x) := \sum_{\sigma \in \text{PRW}_{n+1}} x^{\text{des}(\sigma)} = 1 + x \sum_{m=1}^n \binom{n}{m} A_m(x), \quad (0.130)$$

where PRW_n denotes the set of permutations $\sigma \in \mathfrak{S}_n$ such that the first descent (if any) of σ occurs at the letter n . For example,

$$\text{PRW}_2 = \{12, 21\}, \quad \text{PRW}_3 = \{123, 132, 231, 312, 321\}.$$

For the first few values of n , we have

$$\tilde{A}_n(x) = \begin{cases} 1 + x, & \text{if } n = 1, \\ 1 + 3x + x^2, & \text{if } n = 2, \\ 1 + 7x + 7x^2 + x^3, & \text{if } n = 3, \\ 1 + 15x + 33x^2 + 15x^3 + x^4, & \text{if } n = 4, \\ 1 + 31x + 131x^2 + 131x^3 + 31x^4 + x^5, & \text{if } n = 5. \end{cases}$$

This polynomial first appeared in an enumerative-geometric context in [PRW08, Section 10.4], where it was shown to equal the h -polynomial of the n -dimensional stellohedron. Chung, Graham, and Knuth [CGK10] established the following elegant identity for integers $a, b \geq 0$:

$$\sum_{k \geq 0} \binom{a+b}{k} \left\langle \begin{matrix} k \\ a-1 \end{matrix} \right\rangle = \sum_{k \geq 0} \binom{a+b}{k} \left\langle \begin{matrix} k \\ b-1 \end{matrix} \right\rangle, \quad (0.131)$$

where $\langle 0 \rangle = 1$. Shareshian and Wachs [SW20] observed that this identity is precisely the algebraic expression of the palindromicity of the coefficients of $\tilde{A}_n(x)$.

0.5.4 q -Eulerian polynomials

We define the q -analogue of the real number x , denoted $[x]$ as

$$[x] = \frac{1 - q^x}{1 - q}.$$

By l'Hôpital's rule, $[x] \rightarrow x$ as $q \rightarrow 1$. For $n \in \mathbb{N}$, we define the q -analogue of $n!$ as

$$[n]! = \prod_{i=1}^n [i] = (1+q)(1+q+q^2) \cdots (1+q+\cdots+q^{n-1}).$$

Let $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$. An *inversion* of σ is a pair of indices (i, j) such that $1 \leq i < j \leq n$ and $\sigma_i > \sigma_j$. Let $\text{inv}(\sigma)$ be the number of inversions of σ , i.e.,

$$\text{inv}(\sigma) := |\{(i, j) : i < j \text{ and } \sigma_i > \sigma_j\}|. \quad (0.132)$$

The *major index* of a permutation $\sigma \in \mathfrak{S}_n$ is defined as the sum of the descents of σ , i.e.

$$\text{maj}(\sigma) = \sum_{i \in \text{Des}(\sigma)} i. \quad (0.133)$$

In 1915, MacMahon [Mac60] showed that the major index is equidistributed with the inversion statistic by establishing the following first equality

$$\sum_{\sigma \in \mathfrak{S}_n} q^{\text{maj}(\sigma)} = [n]! = \sum_{\sigma \in \mathfrak{S}_n} q^{\text{inv}(\sigma)}. \quad (0.134)$$

The second equality was found in 1839 by Rodrigues [Rod39]. Foata [Foa68] provided a bijective proof of (0.134), which was later called *Foata's second fundamental transformation*. In honor of MacMahon, any permutation statistic with the same distribution as the maj is said to be a *Mahonian statistic*.

Any pair of statistics (s_1, s_2) is called an *Euler–Mahonian* pair if

$$\sum_{\sigma \in \mathfrak{S}_n} x^{s_1(\sigma)} = A_n(x) \text{ and } \sum_{\sigma \in \mathfrak{S}_n} q^{s_2(\sigma)} = [n]!. \quad (0.135)$$

In particular, the classical statistic pairs (des, maj) and (des, inv) give rise to two distinct families of *q–Eulerian polynomials*, defined respectively by

$${}^m A_n(x, q) := \sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} q^{\text{maj}(\sigma)}, \quad (0.136)$$

$${}^i A_n(x, q) := \sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} q^{\text{inv}(\sigma)}. \quad (0.137)$$

Recall that the *q*–Pochhammer symbol

$$(x; q)_k = (1 - x)(1 - xq) \cdots (1 - xq^{k-1}), \quad (0.138)$$

and the *q*–exponential function

$$\exp(z; q) = \sum_{n \geq 0} \frac{z^n}{[n]!}. \quad (0.139)$$

Carlitz [Car54] and Stanley [Sta76] proved respectively,

$$\frac{{}^m A_n(x, q)}{(x; q)_{n+1}} = \sum_{j \geq 0} x^j [j + 1]^n, \quad (0.140)$$

and

$$1 + \sum_{n \geq 1} x \cdot {}^i A_n(x, q) \frac{z^n}{[n]!} = \frac{1 - x}{1 - x \exp((1 - x)z; q)}. \quad (0.141)$$

Remark 0.21. Eqs. (0.140) and (0.141) are *q*–analogues of (0.114) and (0.116), respectively. For the other two classical Euler–Mahonian pairs (exc, inv) and (exc, maj) , analogous *q*–Eulerian polynomial families have also been investigated by Clarke–Steingrímsson–Zeng [CSZ97] and Shareshian–Wachs [SW07a], respectively.

0.5.5 Coxeter group and *W*–Eulerian polynomials

Define a *Coxeter system* to be a pair (W, S) , where W is a group and S is a minimal generating set of W , subject to the following relations. For every $s, t \in S$, we have

- $st = e$ if and only if $s = t$;
- $(st)^{m(s,t)} = (ts)^{m(s,t)} = e$ for some integer $m(s, t) > 1$,

where e denotes the identity element of W , and $m(s, s) = 1$ for all $s \in S$, and for $s \neq t$ one has

$$m(s, t) = m(t, s) \in \{2, 3, \dots\}.$$

We say such a group W is a *Coxeter group* of rank $r = |S|$.

To each Coxeter system, we can associate a *Coxeter graph* whose nodes are the elements of S . If $m(s, t) > 2$, then we draw an edge between s and t , labeled with $m(s, t)$ if $m(s, t) > 3$. In this thesis, we only consider the Coxeter groups of types A, B, and D. These Coxeter systems correspond to connected Coxeter graphs, which are shown in Figs. 0.6, 0.7 and 0.8.

Define the *length* of an element ω in W to be the number of terms in a shortest expression for ω as a product of elements of S . That is,

$$\ell_W(\omega) := \min\{k : \omega = s_1 \cdots s_k, s_i \in S\}. \quad (0.142)$$

We now employ the length function to formulate a general definition of descents for elements of a Coxeter group. For any element ω in the Coxeter group W , let

$$\text{Des}_W(\omega) := \{s \in S : \ell_W(\omega) > \ell_W(\omega s)\}, \quad (0.143)$$

and $\text{des}_W(\omega) = |\text{Des}_W(\omega)|$.

Define the *W-Eulerian polynomials* to be the generating function for descents, denoted by

$$W(x) := \sum_{\omega \in W} x^{\text{des}_W(\omega)} = \sum_{k=0}^n \left\langle \begin{matrix} W \\ k \end{matrix} \right\rangle x^k, \quad (0.144)$$

where $n = |S|$ is the rank of W . The coefficients of $W(x)$

$$\left\langle \begin{matrix} W \\ k \end{matrix} \right\rangle := |\{\omega \in W : \text{des}_W(\omega) = k\}|, \quad (0.145)$$

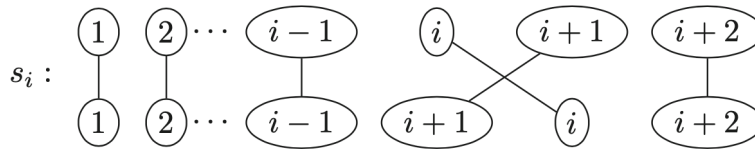
are called the *W-Eulerian numbers*.

For convenience, we denote the length and descent set of the Coxeter group of type A by ℓ_A and Des_A , respectively. Similarly for ℓ_B , Des_B , ℓ_D , and Des_D .

The set $\mathfrak{S}_n$ of permutations is a Coxeter group of type A. In group theory, the group $\mathfrak{S}_n$ is commonly referred to as the *symmetric group*. For $n \geq 2$, the group $\mathfrak{S}_n$ is generated by the adjacent transpositions

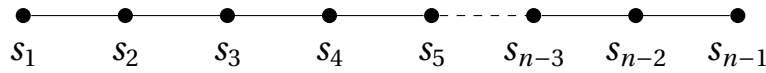
$$s_i = (i \ i+1) \quad (1 \leq i \leq n-1), \quad (0.146)$$

that is, s_i swaps the entries in positions i and $i+1$.



These generators satisfy the following relations and the Coxeter graph in Fig. 0.6.

- $s_i^2 = e$ for $1 \leq i \leq n-1$;
- $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$ for $1 \leq i \leq n-2$;
- $s_i s_j = s_j s_i$ for $|i-j| \geq 2$.

Figure 0.6: Coxeter graph A_n ($n \geq 1$).

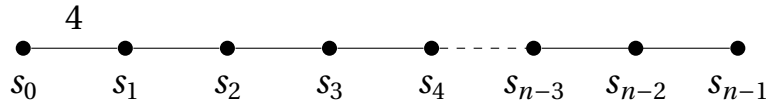
It is well-known that in the case of a Coxeter group of type A, the length and descent set of group $\mathfrak{S}_n$ admit a simple combinatorial description, that is, for any permutation $\sigma \in \mathfrak{S}_n$, we have

$$\ell_A(\sigma) = \text{inv}(\sigma) \text{ and } \text{Des}_A(\sigma) = \{i \in [n-1] : \sigma(i) > \sigma(i+1)\}. \quad (0.147)$$

In the case of symmetric group $\mathfrak{S}_n$, the W -Eulerian polynomials and W -Eulerian numbers are the classical Eulerian polynomials and Eulerian numbers defined in (0.118) and (0.119), respectively.

Let $\mathfrak{B}_n$ be the *hyperoctahedral group* on $[n]$, which is a Coxeter group of type B with generators $\{s_0, s_1, \dots, s_{n-1}\}$, where $s_0 = (1 - 1)$ is the sign change, and s_i ($1 \leq i \leq n-1$) is the adjacent transposition as in the case of $\mathfrak{S}_n$. These generators satisfy the following relations and the Coxeter graph in Fig. 0.7.

- $s_i^2 = e$, for $0 \leq i \leq n-1$;
- $(s_i s_{i+1})^3 = e$ for $i = 1, \dots, n-2$;
- $(s_0 s_1)^4 = e$, and
- $(s_i s_j)^2 = e$ if $|i - j| > 1$.

Figure 0.7: Coxeter graph B_n ($n \geq 2$).

We can represent the Coxeter group of type B as the set of *signed permutations*. These are permutations

$$w : \{-n, \dots, -1, 0, 1, \dots, n\} \rightarrow \{-n, \dots, -1, 0, 1, \dots, n\}, \quad (0.148)$$

such that $w(-i) = -w(i)$ for all i . Notice that this forces $w(0) = 0$ and the element w is completely determined by $w(1), \dots, w(n)$. In one-line notation, we write $w = w(1) \cdots w(n)$ with bars to indicate negative numbers. For example, if w is determined by $w(1) = -3$, $w(2) = 4$, $w(3) = 5$, $w(4) = -1$ and $w(5) = 2$, we write $w = \bar{3}45\bar{1}2$.

The type B descent set can be described combinatorially as follows,

$$\text{Des}_B(w) = \{0 \leq i \leq n-1 : w(i) > w(i+1)\}, \quad (0.149)$$

where $w(0) = 0$. We denote the number of type B descents by $\text{des}_B(w) = |\text{Des}_B(w)|$.

Brenti [Bre94] defined the *$\mathfrak{B}_n$ -Eulerian polynomials* based on the number of descents of type B, for $n \geq 1$,

$$B_n(x) := \sum_{w \in \mathfrak{B}_n} x^{\text{des}_B(w)} \quad (0.150)$$

and derived the exponential generating function for $B_n(x)$,

$$\sum_{n \geq 0} B_n(x) \frac{z^n}{n!} = \frac{(1-x)e^{z(1-x)}}{1-xe^{2z(1-x)}}. \quad (0.151)$$

Recall the length ℓ_B of $\mathfrak{B}_n$ [BB05, Proposition 8.1.1]:

$$\ell_B(w) = \text{inv}(w) - \sum_{w(i) < 0} w(i). \quad (0.152)$$

For $n \geq 1$, the distribution of the length ℓ_B on $\mathfrak{B}_n$ is given by

$$\sum_{w \in \mathfrak{B}_n} q^{\ell_B(w)} = [2][4] \cdots [2n]. \quad (0.153)$$

In 1995, Reiner [Rei95] defined the q -analogue of $\mathfrak{B}_n$ -Eulerian polynomials as

$$B_n(x, q) := \sum_{w \in \mathfrak{B}_n} x^{\text{des}_B(w)} q^{\ell_B(w)}, \quad (0.154)$$

and derived the following q -exponential generating function for $B_n(x, q)$.

Theorem 0.22. *We have*

$$\sum_{n \geq 0} B_n(x, q) \frac{z^n}{(-q; q)_n [n]!} = \frac{(1-x) \exp_B(z(1-x); q)}{1 - x \exp(z(1-x); q)}, \quad (0.155)$$

where $\exp_B(z; q)$ is the q -exponential function of type B defined by

$$\exp_B(z; q) = \sum_{n \geq 0} \frac{z^n}{[2][4] \cdots [2n]}. \quad (0.156)$$

The Coxeter group of type $\mathfrak{D}_n$ is the subgroup of $\mathfrak{B}_n$ consisting of those signed permutations with an even number of minus signs among $w(1), \dots, w(n)$. When we write an element of $\mathfrak{D}_n$ as a word $w = w(1)w(2) \cdots w(n)$, we refer to it as an *even signed permutation*.

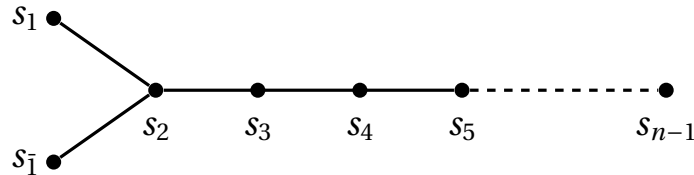


Figure 0.8: Coxeter graph D_n ($n \geq 4$).

The type D descent set can be described combinatorially as follows,

$$\text{Des}_D(w) = \{i \in \{-1, 1, 2, \dots, n-1\} : w(i) > w(|i|+1)\}, \quad (0.157)$$

where $w(0) = 0$. We denote the number of type D descents by $\text{des}_D(w) = |\text{Des}_D(w)|$.

Brenti [Bre94] defined the *$\mathfrak{D}_n$ -Eulerian polynomials* based on the number of descents of type D, for $n \geq 1$,

$$D_n(x) := \sum_{w \in \mathfrak{D}_n} x^{\text{des}_D(w)}, \quad (0.158)$$

and derived the exponential generating function for $D_n(x)$,

$$1 + xz + \sum_{n \geq 2} D_n(x) \frac{z^n}{n!} = \frac{(x-1)(z - e^{z(1-x)})}{1 - e^{2z(1-x)}}. \quad (0.159)$$

Recall the length ℓ_D of $\mathfrak{D}_n$ [BB05, Proposition 8.2.1]:

$$\ell_D(w) := \ell_B(w) - \text{neg}(w), \quad (0.160)$$

where $\text{neg}(w)$ denotes the number of negative entries in w .

For $n \geq 2$, the distribution of the length ℓ_D on $\mathfrak{D}_n$ is given by

$$\sum_{w \in \mathfrak{D}_n} q^{\ell_D(w)} = [2][4] \cdots [2n-2][n]. \quad (0.161)$$

In 1995, Reiner [Rei95] defined the q -analogue of $\mathfrak{D}_n$ -Eulerian polynomials as

$$D_n(x, q) := \sum_{w \in \mathfrak{D}_n} x^{\text{des}_D(w)} q^{\ell_D(w)}, \quad (0.162)$$

and derived the following q -exponential generating function for $D_n(x, q)$.

Theorem 0.23. *We have*

$$2 + \sum_{n \geq 2} D_n(x, q) \frac{z^n}{(-q; q)_{n-1} [n]!} = \frac{(1-x) \exp_D(z(1-x); q) - z(1-x^2 \exp(z(1-x); q))}{1-x \exp(z(1-x); q)}, \quad (0.163)$$

where $\exp_D(z; q)$ is the q -exponential function of type D defined by

$$\exp_D(z; q) = 2 + \sum_{n \geq 1} \frac{z^n}{[2][4] \cdots [2n-2][n]}. \quad (0.164)$$

0.5.6 Wreath product $\mathbb{Z}_r \wr \mathfrak{S}_n$ and colored Eulerian polynomials

For $r, n \in \mathbb{P}$, we define the *wreath product* $\mathbb{Z}_r \wr \mathfrak{S}_n$ of $\mathbb{Z}_r$ by $\mathfrak{S}_n$, i.e., the group of colored permutations $G(r, n)$, by

$$G(r, n) := \{(c_1, \dots, c_n; \sigma) \mid c_i \in [0, r-1], \sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n\}. \quad (0.165)$$

The product in $G(r, n)$ is defined by

$$(c; \sigma) \cdot (c'; \tau) := (c_1 + c'_{\tau^{-1}(1)}, \dots, c_n + c'_{\tau^{-1}(n)}; \sigma \circ \tau),$$

where the addition $+$ is in $\mathbb{Z}_r$ and composition $\circ$ in $\mathfrak{S}_n$. The entry c_i is called the color of the σ_i , for $1 \leq i \leq n$. The elements of $\mathbb{Z}_r \wr \mathfrak{S}_n$ can be viewed as *r -colored permutations*, see Steingrímsson [Ste94] and Bagno et al. [BGM09]. The wreath product $\mathbb{Z}_r \wr \mathfrak{S}_n$ can be viewed as a common generalization of the symmetric group $\mathfrak{S}_n$ and the Coxeter group of type B, that is, for $r = 1, 2$,

$$\mathbb{Z}_1 \wr \mathfrak{S}_n \cong \mathfrak{S}_n \text{ and } \mathbb{Z}_2 \wr \mathfrak{S}_n \cong \mathfrak{B}_n. \quad (0.166)$$

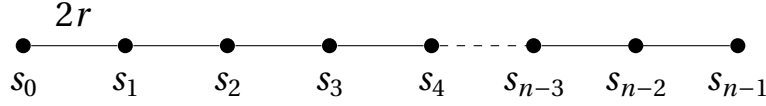
We will represent an element $\gamma \in G(r, n)$ in *window notation* as

$$\gamma = [\gamma(1), \dots, \gamma(n)] = [\sigma_1^{c_1}, \dots, \sigma_n^{c_n}], \quad (0.167)$$

and call σ_i the *absolute value* of $\gamma(i)$, denoted by $|\gamma(i)|$. For $\gamma \in G(r, n)$, we define the color set of γ by

$$\text{Col}_G(\gamma) := \{i \in [n] : c_i \neq 0\} \quad (0.168)$$

and its size will be denoted by $\text{col}_G(\gamma)$. If $c_i = 0$, it will be omitted in the window notation. For example, $\gamma = [3^1, 2, 1^3, 4^2, 6^2, 5^1] \in G(5, 6)$. The group $G(r, n)$ is generated by $S_G := \{s_0, s_1, \dots, s_{n-1}\}$, where for $i \in [n-1]$, s_i is the adjacent transposition in the case of $\mathfrak{S}_n$ in (0.146), and $s_0 := [1^1, 2, \dots, n]$. The relations of these generators are given by the Dynkin-like diagram (see Fig. 0.9).

Figure 0.9: The Dynkin-like diagram of $G(r, n)$.

The *length* of $\gamma \in G(r, n)$ is the minimal number of generators in S_G whose product is γ ,

$$\ell_G(\gamma) := \min\{r \in \mathbb{N} : \gamma = s_{i_1} \cdots s_{i_r}, \text{ for some } s_{i_j} \in S_G\}. \quad (0.169)$$

The *descent set* of $\gamma \in G(r, n)$ is defined as

$$\text{Des}_G(\gamma) := \{s \in S_G : \ell_G(\gamma s) < \ell_G(\gamma)\} \quad (0.170)$$

and its size is denoted by $\text{des}_G(\gamma)$. To give a combinatorial description of ℓ_G and $\text{Des}_G(\gamma)$ we use the following linear order

$$n^{r-1} < \cdots < n^1 < \cdots < 1^{r-1} < \cdots < 1^1 < 0 < 1 < \cdots < n \quad (0.171)$$

on the set $\{0, 1, \dots, n, 1^1, \dots, n^1, \dots, 1^{r-1}, \dots, n^{r-1}\}$ of colored integers (see [BZ11]). For $\gamma = [\gamma(1), \dots, \gamma(n)] \in G(r, n)$, the descent set of $\gamma \in G(r, n)$ has the following alternate definition

$$\text{Des}_G(\gamma) = \{i \in [0, n-1] : \gamma(i) > \gamma(i+1)\}, \quad (0.172)$$

where $\gamma(0) := 0$. The length of γ is characterized by (see [Bag05, Rei93, Ste94])

$$\ell_G(\gamma) = \text{inv}(\gamma) + \sum_{c_i \neq 0} (|\gamma(i)| + c_i - 1). \quad (0.173)$$

Note that when $r = 1$ (resp. $r = 2$) (0.173) reduces to (0.132) (resp. (0.152)).

Steingrímsson [Ste94] introduced the *colored Eulerian polynomials* as

$$G_{r,n}(x) := \sum_{\gamma \in G(r,n)} x^{\text{des}_G(\gamma)}, \quad (0.174)$$

and derived the exponential generating function for $G_{r,n}(x)$,

$$\sum_{n \geq 0} G_{r,n}(x) \frac{z^n}{n!} = \frac{(1-x)e^{z(1-x)}}{1-xe^{rz(1-x)}}. \quad (0.175)$$

When $r = 1$ and $r = 2$, Eq. (0.175) covers (0.116) and (0.151), respectively.

Reiner [Rei93] defined the q -analogue of colored Eulerian polynomials as

$$G_{r,n}(x, q) := \sum_{\gamma \in G(r,n)} x^{\text{des}_G(\gamma)} q^{\ell_G(\gamma)}, \quad (0.176)$$

and derived the q -exponential generating function for $G_{r,n}(x, q)$.

Theorem 0.24. *We have*

$$\sum_{n \geq 0} G_{r,n}(x, q) \frac{z^n}{(-q[r-1]; q)_n [n]!} = \frac{(1-x) \exp_{G(r)}(z(1-x); q)}{1-x \exp(z(1-x); q)}, \quad (0.177)$$

where $\exp_{G(r)}(z; q)$ is the colored q -exponential function defined by

$$\exp_{G(r)}(z; q) = \sum_{n \geq 0} \frac{z^n}{[-q[r-1]; q]_n [n]!}. \quad (0.178)$$

In chapter 4, we will generalize the Theorems 0.22, 0.23, and 0.24 by enumerating permutations according to the parity of the descents.

0.6 Gamma-positivity of Eulerian polynomials

Recall that a polynomial $f(x) = \sum_i a_i x^i \in \mathbb{R}[x]$ is called γ -positive, if

$$f(x) = \sum_{i=0}^{\lfloor n/2 \rfloor} \gamma_i x^i (1+x)^{n-2i} \quad (0.179)$$

for some $n \in \mathbb{N}$ and nonnegative real numbers $\gamma_0, \gamma_1, \dots, \gamma_{\lfloor n/2 \rfloor}$.

The notion of γ -positivity first appeared in the work of D. Foata and M. Schützenberger [FS70] and subsequently of D. Foata and V. Strehl [FS74] on the classic Eulerian polynomials. The gamma positivity of a combinatorial sequence implies both unimodality and symmetry.

The Eulerian polynomial $A_n(x)$, which is clearly symmetric, provides the prototypical example of a γ -positive polynomial in combinatorics. For $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$, with the convention $\sigma_0 = \sigma_{n+1} = 0$, a letter $\sigma_i \in [n]$ is called a *double descent* of σ if $\sigma_{i-1} > \sigma_i > \sigma_{i+1}$. Denote by $\text{dd}(\sigma)$ the number of double descents of σ . Foata and Schützenberger [FS70, Theorem 5.6] proved the following elegant γ -positive expansion of the Eulerian polynomials

$$A_n(x) = \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \gamma_{n,k} x^k (1+x)^{n-1-2k}, \quad (0.180)$$

where $\gamma_{n,k} := |\{\sigma \in \mathfrak{S}_n : \text{des}(\sigma) = k, \text{dd}(\sigma) = 0\}|$. The first γ -expansions of $A_n(x)$ are as follows,

$$A_n(x) = \begin{cases} 1+x, & \text{if } n=2, \\ (1+x)^2+2x, & \text{if } n=3, \\ (1+x)^3+8x(1+x), & \text{if } n=4, \\ (1+x)^4+22x(1+x)^2+16x^2, & \text{if } n=5, \\ (1+x)^5+52x(1+x)^3+136x^2(1+x), & \text{if } n=6. \end{cases}$$

From the above expansions, we see that $\gamma_{n,k}/2^k \in \mathbb{N}^*$ for $0 \leq k \leq \lfloor \frac{n-1}{2} \rfloor$. In a follow-up work, Foata and Schützenberger [FS73] showed that the normalized γ -coefficients $\gamma_{n,k}/2^k$ count the André permutations with k descents. See Chapters 1 and 2 for more details about the normalized γ -coefficients and the André permutations.

Recently, different refinements of (0.180) and other γ -positive polynomials arising in enumerative and geometric combinatorics have been widely studied; the reader is referred to the survey of Athanasiadis [Ath18] and the book exposition by Petersen [Pet15] for more information.

0.7 Combinatorics of continued fractions

The pioneer work on analytic continued fractions is due to Leonhard Euler, who introduced several expansions of power series, such as the one in [Eul60]:

$$\sum_{n \geq 0} (n+1)! z^n = \frac{1}{1-2z - \frac{1 \cdot 2 z^2}{1-4z - \frac{2 \cdot 3 z^2}{\ddots - \frac{k(k+1) z^2}{1-2kz - \ddots}}}}. \quad (0.181)$$

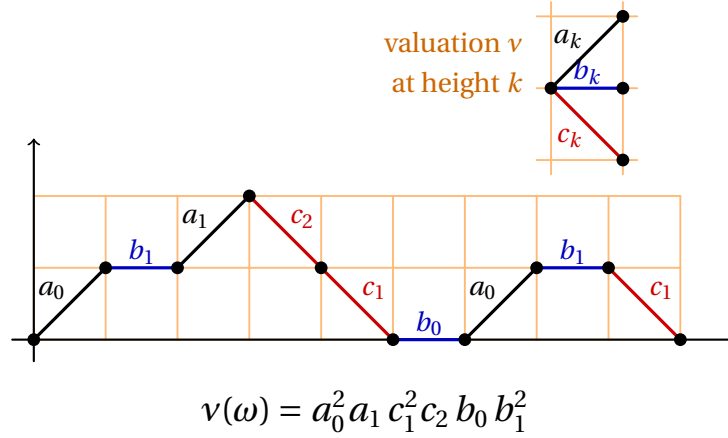


Figure 0.10: Weighted Motzkin path.

The general theory of such continued fractions was primarily developed by Stieltjes in his seminal memoir [Sti94]. These continued fractions are a particular case of the so-called *Jacobi continued fractions*, or *J-fractions*, that is,

$$\mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}) = \frac{1}{1 - b_0 z - \frac{\lambda_1 z^2}{1 - b_1 z - \frac{\lambda_2 z^2}{\ddots 1 - b_k z - \frac{\lambda_{k+1} z^2}{\ddots}}}}, \quad (0.182)$$

where $\{b_k\}_{k \geq 0}$ and $\{\lambda_k\}_{k \geq 1}$ denote two sequences of “coefficients,” which may consist of integers, real or complex numbers, or, more generally, of formal variables drawn from two distinct countably infinite alphabets. In many concrete instances, these coefficients will be taken to be polynomials with integer coefficients in a finite set of formal variables or parameters.

We define a *Motzkin path* as a sequence of vertices $(s_0, \dots, s_n)$ of the square lattice $\mathbb{N} \times \mathbb{N}$, satisfying the following conditions:

- $s_0 = (0, 0)$, $s_n = (0, n)$.
- for $0 \leq i < n$, each elementary step (s_i, s_{i+1}) is one of the following:
 - North-East (N): $s_{i+1} = s_i + (1, 1)$;
 - South-East (S): $s_{i+1} = s_i + (1, -1)$;
 - East (E): $s_{i+1} = s_i + (1, 0)$.

Let $\mathcal{M}_n$ denote the set of Motzkin paths of length n . Consider three sequences of “coefficients” or “letters” over three (possibly distinct) alphabets:

$$\{a_k\}_{k \geq 0}, \quad \{b_k\}_{k \geq 0}, \quad \{c_k\}_{k \geq 1}.$$

We define the weight $v(\omega)$ of a Motzkin path ω as the product of the weights of its elementary steps, where the weight of a N (respectively, S, respectively, E) step starting at height k is given by a_k (respectively, c_k , respectively, b_k), see Fig. 0.10.

Flajolet [Fla80] established a combinatorial interpretation of the general *J*-continued fractions as generating functions of weighted Motzkin paths, a result that we shall refer to as the “Fundamental Lemma of Flajolet”.

Lemma 0.7.1. We have

$$\sum_{n \geq 0} \sum_{\omega \in \mathcal{M}_n} v(\omega) z^n = \frac{1}{1 - b_0 z - \frac{a_0 c_1 z^2}{1 - b_1 z - \frac{a_1 c_2 z^2}{\ddots}}}. \quad (0.183)$$

In the case $b_k = 0$ for all $k \geq 0$, i.e., the paths are *Dyck paths*, the continued fraction (0.183) is called the *Stieltjes continued fraction* or S-fraction.

We now assume that the sequences

$$\{a_k\}_{k \geq 0}, \quad \{b_k\}_{k \geq 0}, \quad \{c_k\}_{k \geq 1} \quad (0.184)$$

are sequences of nonnegative integers. These integers are called *numbers of possibilities*.

Definition 0.25. A *history* h is given by a pair

$$(\omega; (p_1, \dots, p_n)),$$

where $\omega = (s_0, \dots, s_n)$ is a Motzkin path of length n , and $(p_1, \dots, p_n)$ is a sequence of integers such that

$$0 \leq p_i \leq v(s_{i-1}, s_i) - 1, \quad 1 \leq i \leq n, \quad (0.185)$$

where the possible values of v are given in (0.184).

Many known expansions of power series into continued fractions can be proved by combining the Flajolet fundamental Lemma with some combinatorial constructions. The coefficients of the power series are interpreted by some weighted combinatorial objects. Some *weighted histories* are related to the continued fraction, and a weight preserving bijection between these objects and these *histories* will provide a combinatorial proof of the continued fraction expansion using the Flajolet fundamental Lemma.

Flajolet [Fla80] made intensive use of the Françon–Viennot bijection [FV79] between permutations and some histories related to the following path valuation

$$a_k = k + 1, \quad b_k = 2k + 2 \quad (k \geq 0), \quad c_k = k + 1 \quad (k \geq 1). \quad (0.186)$$

These histories are often called *Laguerre histories* and provide a combinatorial proof of Euler's continued fraction (0.181). See Chapter 2 for more details about the Françon–Viennot bijection and Laguerre histories.

0.8 Outline of thesis

For the sake of readability of the outline, we consolidate here the notations required to formulate the results summarized below. Here and throughout, the abbreviation given in parentheses will be used to denote the number of occurrences of the corresponding statistic.

Notations in Parts I–III. For $\sigma \in \mathfrak{S}_n$, a *cycle* (**cyc**) of σ is a sequence $(a_1 a_2 \dots a_k)$ of distinct elements of $[n]$ such that $\sigma(a_i) = a_{i+1}$ for $1 \leq i < k$ and $\sigma(a_k) = a_1$. Let $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$,

- for $i \in [n]$, an index i is called a *drop* (**drop**) of σ if $i > \sigma_i$, an *excedance* (**exc**) of σ if $i < \sigma_i$, a *fixed point* (**fix**) of σ if $i = \sigma_i$, and a *weak excedance* (**wex**) of σ if $i \leq \sigma_i$;

- for $i \in [n-1]$, the index i is called a *descent* (**des**) of σ if $\sigma_i > \sigma_{i+1}$, and an *ascent* (**asc**) of σ if $\sigma_i < \sigma_{i+1}$.

For $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$, a letter $\sigma_i \in [n]$ is called a

- *left-to-right maximum* (**lmax**) of σ if $\sigma_i > \sigma_j$ for every $j < i$;
- *right-to-left maximum* (**rmax**) of σ if $\sigma_i > \sigma_j$ for every $j > i$;
- *right-to-left minimum* (**rmin**) of σ if $\sigma_i < \sigma_j$ for every $j > i$;
- *left-to-right minimum* (**lmin**) of σ if $\sigma_i < \sigma_j$ for every $j < i$.

For $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$, with the boundary $\sigma_0 = \sigma_{n+1} = 0$, a letter $\sigma_i \in [n]$ is called a

- *valley* (**val**) of σ if $\sigma_{i-1} > \sigma_i < \sigma_{i+1}$;
- *peak* (**pk**) of σ if $\sigma_{i-1} < \sigma_i > \sigma_{i+1}$;
- *double ascent* (**da**) of σ if $\sigma_{i-1} < \sigma_i < \sigma_{i+1}$;
- *double descent* (**dd**) of σ if $\sigma_{i-1} > \sigma_i > \sigma_{i+1}$;
- *left-to-right-maximum-peak* (**lmaxpk**) of σ if σ_i is a left-to-right maximum and also a peak;
- *right-to-left-maximum-peak* (**rmaxpk**) of σ if σ_i is a right-to-left maximum and also a peak;
- *left-to-right-maximum-double-ascent* (**lmaxda**) of σ if σ_i is a left-to-right maximum and also a double ascent;
- *right-to-left-maximum-double-descent* (**rmaxdd**) of σ if σ_i is a right-to-left maximum and also a double descent;
- *right-to-left-minimum-double-ascent* (**rminda**) of σ if σ_i is a right-to-left minimum and also a double ascent.

For $\sigma = \sigma(1) \dots \sigma(n) \in \mathfrak{S}_n$, we say that an index $i \in [n]$ is a

- *cycle peak* (**cpk**) of σ if $\sigma^{-1}(i) < i > \sigma(i)$;
- *cycle valley* (**cval**) of σ if $\sigma^{-1}(i) > i < \sigma(i)$;
- *cycle double ascent* (**cda**) of σ if $\sigma^{-1}(i) < i < \sigma(i)$;
- *cycle double descent* (**cdd**) of σ if $\sigma^{-1}(i) > i > \sigma(i)$.

Let $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$. If $i \in [n-1]$ is a descent, then the index i is called a *descent (position)*, σ_i is called a *descent-top*, and σ_{i+1} is called a *descent-bottom*. For $i \in [n]$, the entry σ_i is called a *nondescent* or *nondescent-top* (resp. *nondescent-bottom*) of σ , if it is not a descent-top (resp. descent-bottom) of σ . For $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$, a letter $\sigma_i \in [n]$ is called a

- *foremaximum* (**fmax**) of σ if σ_i is a nondescent top and also a left-to-right maximum;
- *postminimum* (**pmin**) of σ if σ_i is a nondescent bottom and also a right-to-left minimum.

We denote the sum of the nondescent-tops (resp. nondescent-bottoms) of σ by $\text{NDtop}(\sigma)$ (resp. $\text{NDbot}(\sigma)$) and define the *nondescent difference* of σ by

$$\text{NDdif}(\sigma) := \text{NDbot}(\sigma) - \text{NDtop}(\sigma).$$

The number of occurrences of patterns 31-2, 2-31, and 2-13 in $\sigma \in \mathfrak{S}_n$ are defined by

$$(31-2)\sigma = |\{(i, j) : i+1 < j \text{ and } \sigma_i > \sigma_j > \sigma_{i+1}\}|;$$

$$(2-31)\sigma = |\{(i, j) : j < i-1 \text{ and } \sigma_{i-1} > \sigma_j > \sigma_i\}|;$$

$$(2-13)\sigma = |\{(i, j) : j < i-1 \text{ and } \sigma_{i-1} < \sigma_j < \sigma_i\}|.$$

For $\sigma \in \mathfrak{S}_n$, the statistics MAD and MADL are defined by

$$\text{MAD}(\sigma) = \text{des}(\sigma) + (31-2)(\sigma) + 2(2-31)(\sigma);$$

$$\text{MADL}(\sigma) = \text{des}(\sigma) + (2-31)(\sigma) + 2(31-2)(\sigma).$$

The *crossing* number and *nesting* number of $\sigma \in \mathfrak{S}_n$ are defined by

$$\text{cros}(\sigma) = |\{(i, j) : i < j \leq \sigma_i < \sigma_j \text{ or } \sigma_i < \sigma_j < i < j\}|;$$

$$\text{nest}(\sigma) = |\{(i, j) : i < j \leq \sigma_j < \sigma_i \text{ or } \sigma_j < \sigma_i < i < j\}|.$$

Notations in Part IV. For $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$, an index $i \in [n-1]$ is called a

- *odd descent* (**odes**) of σ if i is odd and also a descent;
- *even descent* (**edes**) of σ if i is even and also a descent.

For $\gamma = [\gamma(1), \dots, \gamma(n)] = [\sigma_1^{c_1}, \dots, \sigma_n^{c_n}] \in G(r, n)$, the *length* (ℓ_G) of γ is defined by

$$\ell_G(\gamma) = \text{inv}(\gamma) + \sum_{c_i \neq 0} (|\gamma(i)| + c_i - 1);$$

the number of *colors* of γ is defined by $\text{col}_G(\gamma) := |\{i \in [n] : c_i \neq 0\}|$; and the number of *odd descents* and *even descents* of γ are defined by, respectively,

$$\text{odes}_G(\gamma) = |\{i \in [0, n-1] : \gamma(i) > \gamma(i+1) \text{ and } i \text{ is odd}\}|;$$

$$\text{edes}_G(\gamma) = |\{i \in [0, n-1] : \gamma(i) > \gamma(i+1) \text{ and } i \text{ is even}\}|,$$

where $\gamma(0) := 0$.

Chapter 1

Let $\mathbf{u} = (u_1, u_2, u_3, u_4)$. We define the *generalized bi-Stirling Eulerian polynomials*

$$A_n(\mathbf{u}, f, g, t | \alpha, \beta) = \sum_{\sigma \in \mathfrak{S}_{n+1}} \mathbf{wt}(\sigma), \quad (0.187)$$

where $\mathbf{wt}$ is a weight function of σ defined by

$$\begin{aligned} \mathbf{wt}(\sigma) := & (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} \\ & \times t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}. \end{aligned} \quad (0.188)$$

These polynomials $A_n(\mathbf{u}, f, g, t | \alpha, \beta)$ refine Carlitz-Scoville's bi-Stirling Eulerian polynomials defined in (0.122). Carlitz and Scoville [CS77] introduced the *cycle (α, t) -Eulerian polynomials*,

$$A_n^{\text{cyc}}(x, y, t | \alpha) := \sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)} y^{\text{drop}(\sigma)} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)}. \quad (0.189)$$

The following is our first main result of this chapter, which provides a connection formula between polynomials $A_n(\mathbf{u}, f, g, t | \alpha, \beta)$ and $A_n^{\text{cyc}}(x, y, t | \alpha)$.

Theorem 0.26. *If $xy = u_1 u_2$ and $x + y = u_3 + u_4$, then*

$$A_n(\mathbf{u}, f, g, t | \alpha, \beta) = A_n^{\text{cyc}}\left(x, y, \frac{\alpha u_3 + \beta u_4}{\alpha f + \beta g} t | \alpha f + \beta g\right). \quad (0.190)$$

Define the *(α, t) -Eulerian polynomials*

$$A_n(x, y, t | \alpha) = A_n^{\text{cyc}}\left(x, y, \frac{x + y}{2} t | 2\alpha\right). \quad (0.191)$$

Combining (0.191) and Theorem 0.26 with $\mathbf{u} = (x, y, x, y)$, $f = g = 1$, and $\alpha = \beta$, we obtain

$$A_n(x, y, t | \alpha) = \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{asc}(\sigma)} y^{\text{des}(\sigma)} t^{\text{lmaxda}(\sigma) + \text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2}. \quad (0.192)$$

The following is our second main result of this chapter, which gives three different combinatorial interpretations of the normalised γ -positivity of $A_n(x, y, t | \alpha)$.

Theorem 0.27. *For $n \geq 0$, we have*

$$A_n(x, y, t | \alpha) = \sum_{j=0}^{\lfloor n/2 \rfloor} 2^j d_{n,j}(\alpha, t) (xy)^j (x + y)^{n-2j}, \quad (0.193a)$$

where for $0 \leq j \leq \lfloor n/2 \rfloor$,

$$d_{n,j}(\alpha, t) = \sum_{\substack{\sigma \in \mathcal{CA}_n \\ \text{drop}(\sigma) = j}} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} \quad (0.193b)$$

$$= \sum_{\substack{\sigma \in \mathcal{A}_{n+1}^1 \\ \text{des}(\sigma) = j}} t^{\text{rminda}(\sigma)} \alpha^{\text{rmin}(\sigma) - 1} \quad (0.193c)$$

$$= \sum_{\substack{\sigma \in \mathcal{A}_{n+1}^2 \\ \text{des}(\sigma) = j}} t^{\text{rminda}(\sigma)} \alpha^{\text{rmin}(\sigma) - 1}. \quad (0.193d)$$

The sets $\mathcal{CA}_n$, $\mathcal{A}_n^1$ and $\mathcal{A}_n^2$ in above theorem denote the set of cycle André permutations, André permutations of the first kind and André permutations of the second kind, respectively. The precise definitions of these sets will be given in Section 1.3.3.

We establish Theorem 0.26 by constructing a bijection derived from two variants of Foata's first fundamental transformation, and by integrating this construction with the cyclic valley-hopping on permutations. The proof of Theorem 0.27 proceeds as follows: first, we establish (0.193b) by showing that the exponential generating functions of both sides coincide. Next, we deduce (0.193c) from (0.193b) by constructing a bijection $\Phi : \mathcal{CA}_n \rightarrow \mathcal{A}_{n+1}^1$. Finally, we derive (0.193d) from (0.193c) by means of a bijection $\Psi : \mathcal{A}_{n+1}^1 \rightarrow \mathcal{A}_{n+1}^2$ built upon a model of André trees.

By the connection formula in Theorem 0.26, we derive exponential generating functions and several γ -positive expansions for various Eulerian-type polynomials. Our results unify and strengthen the recent findings of Ji [Ji25] and Ji-Lin [JL24].

Chapter 2

In this chapter, we establish a J -fraction for (p, q) -analogues of the generalized (α, β) -Eulerian polynomials defined in (0.187) with an additional statistic D related to the area of the associated Motzkin path under the Françon–Viennot bijection.

Let $\text{Val}(\sigma)$ and $\text{Pk}(\sigma)$ denote the sets of valleys and peaks of σ , respectively. We define the general multivariable Eulerian polynomials:

$$A_n(\mathbf{u}, f, g, h, t | \alpha, \beta; p, q) := \sum_{\sigma \in \mathfrak{S}_{n+1}} h^{D(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)} \mathbf{wt}(\sigma), \quad (0.194)$$

where $\mathbf{wt}$ is a weight function of σ defined by (0.188), and D is defined by

$$D(\sigma) := \sum_{i \in \text{Pk}(\sigma) \setminus \{n\}} i - \sum_{j \in \text{Val}(\sigma)} j,$$

The generalized (p, q) -analogue of the positive integer $k \in \mathbb{N}^*$ with two auxiliary parameters α and β is as follows:

$$[k; \alpha, \beta]_{p,q} := (\alpha - 1)p^{k-1} + (\beta - 1)q^{k-1} + \frac{p^k - q^k}{p - q}.$$

The following is our first main result of this chapter.

Theorem 0.28. *We have*

$$\sum_{n=0}^{\infty} A_n(\mathbf{u}, f, g, h, t | \alpha, \beta; p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (0.195a)$$

where

$$b_k = h^k \left(u_3 [k+1; 1, \alpha t]_{p,q} + u_4 [k+1; \beta t, 1]_{p,q} \right), \quad (0.195b)$$

$$\lambda_k = u_1 u_2 h^{2k-1} [k]_{p,q} [k+1; \beta g, \alpha f]_{p,q}. \quad (0.195c)$$

Petersen and Tenner [PT15] introduced a statistic called depth for Coxeter groups, defined in terms of factorizations of group elements into reflections. In the case of the symmetric group $\mathfrak{S}_n$, the *depth* of a permutation $\sigma \in \mathfrak{S}_n$ is defined as

$$\text{depth}(\sigma) = \sum_{\sigma(i) > i} (\sigma(i) - i).$$

Petersen [Pet] conjectured a J -fraction for the joint distribution of the statistics $(\text{inv}, \text{depth})$, which was subsequently established in [EFL24, DS23]. Motivated by these results, we introduce the (p, q) -analogue of the generalized cycle version of the Eulerian polynomials, incorporating two Stirling statistics,

$$C_n(\mathbf{u}, t, s, h | \alpha, \beta; p, q) := \sum_{\sigma \in \mathfrak{S}_n} (u_1 u_2)^{\text{cval}(\sigma)} u_3^{\text{cda}(\sigma)} u_4^{\text{cdd}(\sigma)} t^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} \\ \times h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma)} \beta^{\text{rmin}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)}. \quad (0.196a)$$

The following is our second result of this chapter.

Theorem 0.29. *We have*

$$1 + \sum_{n \geq 1} C_n(\mathbf{u}, t, s, h | \alpha, \beta; p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (0.196b)$$

where $b_0 = t\alpha\beta$, and

$$b_k = (sh)^k \left(tq^k s^k + u_4[k; \beta, 1]_{p,qs} + u_3 p[k; \alpha, 1]_{p,qs} \right), \quad (0.196c)$$

$$\lambda_k = u_1 u_2 (sh)^{2k-1} [k; \alpha, 1]_{p,qs} [k; \beta, 1]_{p,qs}. \quad (0.196d)$$

We prove Theorem 0.28 and Theorem 0.29 using the theory of continued fractions [Fla80], together with variants of the bijections of Françon-Viennot [FV79] and Foata-Zeilberger [FZ90] between permutations and weighted Motzkin paths.

The third main result of this chapter shows that Corteel's bijection Φ_C [Cor07] and Shin-Zeng's bijection Φ_{SZ} [SZ10] on $\mathfrak{S}_n$ preserve additional structural properties that are compatible with the Motzkin path encoding.

Theorem 0.30. *The mappings Φ_C and Φ_{SZ} are bijections on $\mathfrak{S}_n$ such that for all $\sigma \in \mathfrak{S}_n$, we have*

$$(\text{ndes}, \text{pmin}, 31-2, 2-31, \text{MADL}, \text{NDdif}) \sigma = (\text{wex}, \text{fix}, \text{nest}, \text{cros}, \text{inv}, \text{depth}) \Phi_C(\sigma); \quad (0.197)$$

$$(\text{ndes}, \text{fmax}, 31-2, 2-31, \text{MAD}, \text{NDdif}) \sigma = (\text{wex}, \text{fix}, \text{cros}, \text{nest}, \text{inv}, \text{depth}) \Phi_{SZ}(\sigma). \quad (0.198)$$

Moreover, if $\Phi^* := \Phi_{SZ} \circ \Phi_C^{-1}$, then

$$(\text{wex}, \text{nest}, \text{cros}, \text{depth}) \sigma = (\text{wex}, \text{cros}, \text{nest}, \text{depth}) \Phi^*(\sigma). \quad (0.199)$$

By employing the continued fractions established in Theorems 0.28 and 0.29, we derive γ -positive expansions for several (p, q) -analogues of generalized Eulerian polynomials; see Theorems 2.16, 2.18, and 2.22.

Chapter 3

Let $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$. The complement of σ is given by

$$\sigma^c = (n+1-\sigma_1)(n+1-\sigma_2) \cdots (n+1-\sigma_n). \quad (0.200)$$

Evidently, the complement establishes a bijection on $\mathfrak{S}_n$ and satisfies the properties

$$\text{asc}(\sigma) = \text{des}(\sigma^c), \quad \text{des}(\sigma) = \text{asc}(\sigma^c), \quad (0.201)$$

$$\text{lmax}(\sigma) = \text{lmin}(\sigma^c), \quad \text{rmax}(\sigma) = \text{rmin}(\sigma^c). \quad (0.202)$$

Combining the complement operator and (0.128), we see that the (α, β) -Eulerian numbers $A(r, s | \alpha, \beta)$ have another combinatorial interpretation

$$A(r, s | \alpha, \beta) = \sum_{\substack{\sigma \in \mathfrak{S}_{r+s+1} \\ \text{des}(\sigma) = r}} \alpha^{\text{lmin}(\sigma)-1} \beta^{\text{rmin}(\sigma)-1}. \quad (0.203)$$

Recently, Gaudin [Gau24] observed that the Carlitz-Scoville numbers $A(r, s | \alpha, \beta)$ are a rescaled subfamily of the mixed Eulerian numbers and proposed a q -analogue $A(r, s | \alpha, \beta)_q$

within the framework of the remixed Eulerian numbers [NT23] and showed that $A(r, s | \alpha, \beta)_q$ satisfies the following recurrence

$$A(r, s | \alpha, \beta)_q = q^{r+\beta-1} [s + \alpha] A(r-1, s | \alpha, \beta)_q + [r + \beta] A(r, s-1 | \alpha, \beta)_q. \quad (0.204)$$

For convenience, we introduce the following notation, valid for $n \geq k \geq 0$:

$$E_{n,k}(\alpha, \beta, q) := A(k, n-k | \alpha, \beta)_q \quad (0.205)$$

with the convention $E_{n,k}(\alpha, \beta, q) = 0$ if $k \notin \{0, \dots, n\}$. In view of (0.205), setting $r + s = n$ and $r = k$ in (0.204) we obtain

$$E_{n,k}(\alpha, \beta, q) = q^{\beta+k-1} [n-k+\alpha] E_{n-1,k-1}(\alpha, \beta, q) + [k+\beta] E_{n-1,k}(\alpha, \beta, q). \quad (0.206)$$

When $(\alpha, \beta) = (1, 1)$ or $(0, 1)$, several q -analogues of Eulerian numbers are already known [Car54, Sta76, Raw81]. In the case $\alpha = 0$, Butler [But04b, But25] introduced a novel q -analogue of Eulerian numbers by combining the parameter β with the major index.

The aim of this chapter is to provide a combinatorial interpretation of $E_{n,k}(\alpha, \beta, q)$ in terms of the permutation statistics such as left-to-right minima, right-to-left minima, descents and (re)mixed major index, and to derive several explicit formulas for these numbers. For $\sigma \in \mathfrak{S}_n$ the *mixed major index* of σ , denoted by $\widetilde{\text{maj}}(\sigma)$, is defined by

$$\widetilde{\text{maj}}(\sigma) = (1 + \text{des}(\sigma) - \text{lmin}(\sigma))(\alpha - 1) + (n - \text{rmin}(\sigma))(\beta - 1) + \text{maj}(\sigma). \quad (0.207)$$

The following is our first main result of this chapter.

Theorem 0.31. *For integer $n \geq k \geq 0$, we have*

$$E_{n,k}(\alpha, \beta, q) = \sum_{\sigma \in \mathfrak{S}_{n+1,k}} [\alpha]^{\text{lmin}(\sigma)-1} [\beta]^{\text{rmin}(\sigma)-1} q^{\widetilde{\text{maj}}(\sigma)}. \quad (0.208)$$

where $\mathfrak{S}_{n+1,k}$ denotes the set of permutations in $\mathfrak{S}_{n+1}$ with k descents.

We define the *bi-Stirling-Euler-Mahonian polynomials* $E_n(x | \alpha, \beta, q)$ by

$$E_n(x | \alpha, \beta, q) = \sum_{k=0}^n E_{n,k}(\alpha, \beta, q) x^k. \quad (0.209)$$

Define the q -derivative operator δ_x by

$$\delta_x p(x) = \frac{p(qx) - p(x)}{(q-1)x}, \quad (0.210)$$

where $p(x) \in \mathbb{R}[[x]]$. The following is our second main result of this chapter.

Theorem 0.32. *Let $E_n(x; q) := E_n(x | \alpha, \beta, q)$ for $n \geq 0$. We have the following q -identities:*

(i)

$$E_n(x; q) = ([\beta] + q^\beta [n-1+\alpha]x) E_{n-1}(x; q) + (1-x)xq^\beta \delta_x(E_{n-1}(x; q)). \quad (0.211)$$

(ii)

$$\frac{E_n(x; q)}{(x; q)_{n+\alpha+\beta}} = \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n. \quad (0.212)$$

(iii)

$$\sum_{n \geq 0} \frac{E_n(x; q)}{(x; q)_{n+\alpha+\beta}} \frac{t^n}{n!} = \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} \exp([j + \beta]t). \quad (0.213)$$

(iv)

$$\sum_{k=0}^j \begin{bmatrix} j - k + n + \alpha + \beta - 1 \\ j - k \end{bmatrix} E_{n,k}(\alpha, \beta, q) = \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n. \quad (0.214)$$

(v)

$$E_{n,k}(\alpha, \beta, q) = \sum_{j=0}^k \begin{bmatrix} n + \alpha + \beta \\ k - j \end{bmatrix} \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} (-1)^{k-j} q^{\binom{k-j}{2}} [\beta + j]^n. \quad (0.215)$$

The proofs of Theorem 0.31 and Theorem 0.32 are based on a combination of a combinatorial insertion argument and q -calculus.

Chapter 4

In 1973, Carlitz and Scoville [CS73] enumerated permutations according to the parity of both descents and ascents. Recently, Pan and Zeng [PZ23] considered the enumeration polynomial of permutations using the vector statistic (edes, odes, inv) and established the following q -exponential generating function,

$$\begin{aligned} & \sum_{n \geq 1} \frac{t^n}{[n]_q!} \sum_{\sigma \in \mathfrak{S}_n} x^{\text{odes}(\sigma)} y^{\text{edes}(\sigma)} q^{\text{inv}(\sigma)} \\ &= \frac{(1+x) \cosh(Mt; q) + M \sinh(Mt; q) - x(\cosh^2(Mt; q) - \sinh^2(Mt; q)) - 1}{1 - (x+y) \cosh(Mt; q) + xy(\cosh^2(Mt; q) - \sinh^2(Mt; q))}, \end{aligned} \quad (0.216)$$

where $M = \sqrt{(1-x)(1-y)}$ and

$$\cosh(t; q) = \sum_{n \geq 0} \frac{t^{2n}}{[2n]_q!}, \quad \sinh(t; q) = \sum_{n \geq 1} \frac{t^{2n-1}}{[2n-1]_q!}. \quad (0.217)$$

Formula (0.216) is a q -analogue of Carlitz-Scoville's formula [CS73, Theorem 3.1] and encompasses Stanley's formula for the (des, inv) in (0.141). In a follow-up work, Dey, Shankar and Sivasubramanian [DSS24, Theorems 1.5 and 1.8] established analogue formulas of (0.216) for Coxeter groups of type B and D.

In this chapter we provide a formula for the generating function of the vector statistic (odes_G, edes_G, col_G, ℓ_G) over the group of colored permutations $G(r, n)$. For $n, r \in \mathbb{P}$, define the polynomials

$$G_{(r,n)}(a, q) := \sum_{\gamma \in G(r,n)} a^{\text{col}_G(\gamma)} q^{\ell_G(\gamma)}, \quad (0.218)$$

$$G_{(r,n)}(x, y, a, q) := \sum_{\gamma \in G(r,n)} x^{\text{odes}_G(\gamma)} y^{\text{edes}_G(\gamma)} a^{\text{col}_G(\gamma)} q^{\ell_G(\gamma)}. \quad (0.219)$$

The even and odd index generating functions of $G_{(r,n)}(x, y, a, q)$ are defined by

$$H_0^r := \sum_{n \geq 0} G_{(r,2n)}(x, y, a, q) \frac{t^{2n}}{G_{(r,2n)}(a, q)}, \quad (0.220a)$$

$$H_1^r := \sum_{n \geq 0} G_{(r,2n+1)}(x, y, a, q) \frac{t^{2n+1}}{G_{(r,2n+1)}(a, q)}. \quad (0.220b)$$

Define the q -analogue of the exponential series over the wreath product

$$\exp_{G(r)}(t; a, q) := \sum_{n \geq 0} \frac{t^n}{G(r, n)(a, q)}, \quad (0.221)$$

and the corresponding hyperbolic cosine and sine series

$$\cosh_{G(r)}(t; a, q) = \frac{\exp_{G(r)}(t; a, q) + \exp_{G(r)}(-t; a, q)}{2}, \quad (0.222a)$$

$$\sinh_{G(r)}(t; a, q) = \frac{\exp_{G(r)}(t; a, q) - \exp_{G(r)}(-t; a, q)}{2}. \quad (0.222b)$$

The following is our first main result of this chapter.

Theorem 0.33. *For any $r \in \mathbb{P}$, we have*

$$H_0^r = \frac{(1-y) \left((1-x \cosh(Mt; q)) \cosh_{G(r)}(Mt; a, q) + x \sinh(Mt; q) \sinh_{G(r)}(Mt; a, q) \right)}{1 - (x+y) \cosh(Mt; q) + xy \exp(Mt; q) \exp(-Mt; q)}, \quad (0.223a)$$

$$H_1^r = \frac{M \left((1-y \cosh(Mt; q)) \sinh_{G(r)}(Mt; a, q) + y \sinh(Mt; q) \cosh_{G(r)}(Mt; a, q) \right)}{1 - (x+y) \cosh(Mt; q) + xy \exp(Mt; q) \exp(-Mt; q)}, \quad (0.223b)$$

where $M = \sqrt{(1-x)(1-y)}$.

Given a permutation $\gamma \in G(r, n)$, an index $i \in \{0, 1, \dots, n-1\}$ is called an *alternating descent* if i is an *odd descent* or *even ascent*. Let $\widehat{Des}_G(\gamma)$ be the set of alternating descents of γ , i.e.,

$$\widehat{Des}_G(\gamma) = \{2i : \gamma(2i) < \gamma(2i+1)\} \cup \{2i+1 : \gamma(2i+1) > \gamma(2i+2)\},$$

and let its cardinality be denoted by $\widehat{des}_G(\gamma)$. We define the q -alternating descent polynomial over $G(r, n)$ as

$$\text{Alt}_n^{G(r)}(x, q) := \sum_{\gamma \in G(r, n)} x^{\widehat{des}_G(\gamma)} q^{\ell_G(\gamma)}. \quad (0.224)$$

As an application of Theorem 0.33 we shall evaluate $\text{Alt}_n^{G(r)}(x, q)$ when $q = -1$. The following is our second main result of this chapter.

Theorem 0.34. *For integer $n \geq 1$ the following identities hold.*

1. *If r is a positive even integer, then*

$$\text{Alt}_n^{G(r)}(x, -1) = (-1)^{\lfloor (n+1)/2 \rfloor} (1-x)^n. \quad (0.225)$$

2. *If r is a positive odd integer, then $\text{Alt}_1^{G(r)}(x, -1) = x$ and for $n \geq 2$,*

$$\text{Alt}_n^{G(r)}(x, -1) = \begin{cases} x(1-x)^m A_m(x), & \text{if } n = 2m \ (m \in \mathbb{N}^*); \\ \frac{2x^2}{1+x} (1-x)^{2m} A_{2m}(x), & \text{if } n = 4m+1 \ (m \in \mathbb{N}^*); \\ 0, & \text{if } n = 4m+3 \ (m \in \mathbb{N}). \end{cases} \quad (0.226)$$

The proofs of Theorem 0.33 and Theorem 0.34 are based on the generating function method and q -calculus.

Chapter 1

Variations of (α, t) -Eulerian polynomials and their gamma positivity²

Abstract

In 1977, Carlitz and Scoville introduced the cycle (α, t) -Eulerian polynomials $A_n^{\text{cyc}}(x, y, t \mid \alpha)$ by enumerating permutations with respect to the number of excedances, drops, fixed points, and cycles. In this chapter, we propose a nine-variable generalization of the Eulerian polynomials $A_n(u_1, u_2, u_3, u_4, f, g, t \mid \alpha, \beta)$ defined in terms of descent-based statistics of permutations, and we establish a connection formula linking these two families of generalized Eulerian polynomials. By means of this connection formula, we derive explicit exponential generating functions and several γ -positive expressions for various Eulerian-type polynomials. In particular, our results unify and strengthen recent findings of Ji and Ji-Lin. In related work on the transition matrix between the Specht and web bases, Hwang, Jang, and Oh recently introduced the web permutations, which can be described in terms of cycle André permutations. We show that enumerating these permutations with respect to the numbers of drops, fixed points, and cycles yields the normalized γ -vectors of the (α, t) -Eulerian polynomials. Our results thus provide a unified framework encompassing and extending several known results in the literature.

Keywords: bi-Stirling Eulerian polynomials, left-to-right maxima, cycle André permutations, cyclic valley-hopping, gamma-positivity

1.1 Introduction

The *Eulerian polynomials* $A_n(x)$ are defined by the exponential generating function [FS70, Pet15]

$$\sum_{n \geq 0} A_n(x) \frac{z^n}{n!} = \frac{1-x}{e^{(x-1)z} - x}. \quad (1.1)$$

²The material presented in this chapter is based on the paper "Variations of (α, t) -Eulerian polynomials and their gamma positivity", accepted for publication in *Combinatorial Theory*, (2026).

These polynomials have a long and rich history. Comprehensive accounts can be found in Foata's survey [Foa10], Stanley's *Enumerative Combinatorics* [Sta12, Sta99], and Petersen's monograph [Pet15]; see also the classic work of Foata and Schützenberger [FS70], which remains a fundamental reference and continues to offer deep insights into the subject.

For a finite set E , let $\mathfrak{S}_E$ denote the set of all permutations of E , and let $\mathfrak{S}_n$ be the symmetric group on $[n] := \{1, 2, \dots, n\}$. For $\sigma \in \mathfrak{S}_n$, an index $1 \leq i \leq n-1$ is called

- a *descent* (**des**) of σ if $\sigma(i) > \sigma(i+1)$;
- an *ascent* (**asc**) of σ if $\sigma(i) < \sigma(i+1)$;
- an *excedance* (**exc**) of σ if $i < \sigma(i)$.

The following combinatorial interpretations of the Eulerian polynomials $A_n(x)$ are well known:

$$A_n(x) = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{asc}(\sigma)} = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)}. \quad (1.2)$$

Postnikov, Reiner, and Williams [PRW08] introduced the *binomial-Eulerian polynomials*:

$$\tilde{A}_n(x) := \sum_{\sigma \in \text{PRW}_{n+1}} x^{\text{des}(\sigma)} = 1 + x \sum_{m=1}^n \binom{n}{m} A_m(x), \quad (1.3)$$

where PRW_n denotes the set of permutations $\sigma \in \mathfrak{S}_n$ such that the first descent (if any) of σ occurs at the letter n . For example,

$$\text{PRW}_2 = \{12, 21\}, \quad \text{PRW}_3 = \{123, 132, 231, 312, 321\}.$$

Any polynomial with real coefficients

$$h(x) = \sum_{i=0}^n h_i x^i$$

satisfying the symmetry condition $h_i = h_{n-i}$ can be written uniquely in the form

$$h(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_k x^k (1+x)^{n-2k}.$$

The coefficients γ_k are called the *γ -coefficients* of $h(x)$. If all γ_k are nonnegative, then $h(x)$ is said to be *γ -positive*. In such cases, it is often of particular interest to find combinatorial interpretations of the γ -coefficients. It is known [FS70, PZ21] that the Eulerian polynomials $A_n(x)$ admit the following γ -positive expansion:

$$A_{n+1}(x) = \sum_{j=0}^{\lfloor n/2 \rfloor} 2^j d_{n,j} x^j (1+x)^{n-2j}, \quad (1.4)$$

where $d_{n,j}$ denotes the number of André permutations in $\mathfrak{S}_{n+1}$ with exactly j descents. For recent developments in this direction, we refer the reader to [LZ15, SW20, PZ21], and for a general overview of γ -positivity phenomena to [Ath18, Pet15].

The γ -positivity of the binomial-Eulerian polynomials $\tilde{A}_n(x)$ follows from a general theorem of Postnikov, Reiner, and Williams [PRW08, Section 10.4], which in particular

implies that these polynomials are symmetric. Moreover, their coefficients can be expressed beautifully as sums of products of Eulerian numbers and binomial coefficients. Writing

$$A_n(x) = \sum_{k=0}^{n-1} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle x^k,$$

Chung, Graham, and Knuth [CGK10] established the following elegant identity (with three different proofs) for integers $a, b \geq 0$:

$$\sum_{k \geq 0} \binom{a+b}{k} \left\langle \begin{matrix} k \\ a-1 \end{matrix} \right\rangle = \sum_{k \geq 0} \binom{a+b}{k} \left\langle \begin{matrix} k \\ b-1 \end{matrix} \right\rangle, \quad (1.5)$$

where $\langle 0 \rangle = 1$. Shareshian and Wachs [SW20] observed that this identity is precisely the algebraic expression of the palindromicity of the coefficients of $\tilde{A}_n(x)$.

For $\sigma \in \mathfrak{S}_n$, an index $i \in [n]$ is called

- a **drop** (**drop**) of σ if $i > \sigma(i)$;
- a **fixed point** (**fix**) of σ if $i = \sigma(i)$.

We also regard a permutation $\sigma \in \mathfrak{S}_n$ as a word $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n$, where $\sigma_i := \sigma(i)$ for $i \in [n]$. A letter σ_i is said to be

- a **left-to-right maximum** (**lmax**) of σ if $\sigma_i > \sigma_j$ for every $j < i$;
- a **right-to-left maximum** (**rmax**) of σ if $\sigma_i > \sigma_j$ for every $j > i$.

In the mid-1970s, Carlitz and Scoville introduced several families of multivariate Eulerian polynomials. Among them are the so-called **bi-Stirling Eulerian polynomials** [CS74], defined by

$$A_n(x, y | \alpha, \beta) := \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{asc}(\sigma)} y^{\text{des}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}, \quad (1.6a)$$

and the **cycle (α, t) -Eulerian polynomials** [CS77],

$$A_n^{\text{cyc}}(x, y, t | \alpha) := \sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)} y^{\text{drop}(\sigma)} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)}, \quad (1.6b)$$

where $\text{cyc}(\sigma)$ denotes the number of cycles of σ . The first values of these polynomials are as follows:

$$\begin{aligned} A_1^{\text{cyc}}(x, y, t | \alpha) &= \alpha t, \\ A_2^{\text{cyc}}(x, y, t | \alpha) &= \alpha x y + \alpha^2 t^2, \\ A_3^{\text{cyc}}(x, y, t | \alpha) &= \alpha (x + y) x y + 3 \alpha^2 x y t + \alpha^3 t^3, \\ A_4^{\text{cyc}}(x, y, t | \alpha) &= \alpha x y (x^2 + (3\alpha + 4) x y + y^2) + 4 \alpha^2 x y (x + y) t + 6 \alpha^3 x y t^2 + \alpha^4 t^4. \end{aligned}$$

Since

$$A_n^{\text{cyc}}(x, y, t | \alpha) = y^n A_n^{\text{cyc}}(x/y, 1, t/y | \alpha),$$

the polynomials $A_n^{\text{cyc}}(x, y, t | \alpha)$ constitute the homogeneous version of $A_n^{\text{cyc}}(x, 1, t | \alpha)$, which were studied in detail in [FS70, Chapter 4].

The exponential generating function of the polynomials $A_n^{\text{cyc}}(x, y, t | \alpha)$ is well known and takes the form

$$\sum_{n \geq 0} A_n^{\text{cyc}}(x, y, t | \alpha) \frac{z^n}{n!} = \left(\frac{(x-y)e^{tz}}{xe^{yz} - ye^{xz}} \right)^\alpha. \quad (1.7a)$$

This identity was implicitly established in Foata–Schützenberger’s monograph [FS70, Chapitre IV, (12)–(13)] via the exponential formula, and later proved analytically by Carlitz and Scoville [CS77, Theorem 1]. A proof using symmetric functions and plethysm was subsequently given by Brenti [Bre00, Proposition 7.3]. As observed in [KZ04], using the exponential formula [FS70, Sta99], Eq. (1.7a) can be rewritten equivalently as either of the following exponential generating functions for derangements.

$$\sum_{n \geq 0} A_n^{\text{cyc}}(x, 1, 0 | 1) \frac{z^n}{n!} = \left(1 - \sum_{n \geq 2} (x + x^2 + \cdots + x^{n-1}) \frac{z^n}{n!} \right)^{-1}, \quad (1.7b)$$

$$\sum_{n \geq 0} A_n^{\text{cyc}}(x, 1, 0 | -1) \frac{z^n}{n!} = 1 - \sum_{n \geq 2} (x + x^2 + \cdots + x^{n-1}) \frac{z^n}{n!}. \quad (1.7c)$$

Gessel [Ges91] gave an elegant combinatorial proof of (1.7b) in the special case $x = 1$. Later, Kim and Zeng [KZ01] provided a bijective proof of (1.7b). Two further combinatorial proofs of (1.7c) were subsequently presented in [KZ04]. More recently, a multivariable extension of (1.7c) was established in [PZ24].

For a permutation $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$, we say that an index $i \in [n]$ is a

- *cycle peak* (**cpk**) of σ if $\sigma^{-1}(i) < i < \sigma(i)$;
- *cycle valley* (**cval**) of σ if $\sigma^{-1}(i) > i > \sigma(i)$;
- *cycle double ascent* (**cda**) of σ if $\sigma^{-1}(i) < i < \sigma(i)$;
- *cycle double descent* (**cdd**) of σ if $\sigma^{-1}(i) > i > \sigma(i)$.

Note that $\text{cpk}(\sigma) = \text{cval}(\sigma)$. The following refinement of (1.7a) was proved in [Zen93, Th. 1]:

$$1 + \sum_{n \geq 1} \frac{z^n}{n!} \sum_{\sigma \in \mathfrak{S}_n} (u_1 u_2)^{\text{cpk}(\sigma)} u_3^{\text{cda}(\sigma)} u_4^{\text{cdd}(\sigma)} t^{\text{fix}(\sigma)} a^{\text{cyc}(\sigma)} = \left(\frac{(x-y)e^{tz}}{xe^{yz} - ye^{xz}} \right)^\alpha, \quad (1.8)$$

where $xy = u_1 u_2$ and $x + y = u_3 + u_4$.

For $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$ with the convention $\sigma_0 = \sigma_{n+1} = 0$, a letter $\sigma_i \in [n]$ is called a

- *valley* (**val**) of σ if $\sigma_{i-1} > \sigma_i < \sigma_{i+1}$;
- *peak* (**pk**) of σ if $\sigma_{i-1} < \sigma_i > \sigma_{i+1}$;
- *double ascent* (**da**) of σ if $\sigma_{i-1} < \sigma_i < \sigma_{i+1}$;
- *double descent* (**dd**) of σ if $\sigma_{i-1} > \sigma_i > \sigma_{i+1}$.

It is clear that the following identities hold

$$\text{val} = \text{pk} - 1, \quad \text{asc} = \text{val} + \text{da}, \quad \text{des} = \text{val} + \text{dd}. \quad (1.9)$$

Recently, Ji [Ji25] introduced the following refinement of the Carlitz–Scoville polynomials $A_n(x, y | \alpha, \beta)$, incorporating six permutation statistics:

$$A_n(u_1, u_2, u_3, u_4 | \alpha, \beta) := \sum_{\sigma \in \mathfrak{S}_{n+1}} (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}. \quad (1.10)$$

Ji's main result [Ji25, Theorem 1.4] gives the following exponential generating function, obtained by means of the context-free grammar calculus.³ Let

$$F(x, y; t) = \frac{e^{xt} - e^{yt}}{xe^{yt} - ye^{xt}}.$$

Theorem 1.1 (Ji). *We have*

$$\sum_{n \geq 0} A_n(u_1, u_2, u_3, u_4 | \alpha, \beta) \frac{z^n}{n!} = (1 + xF(x, y; z))^{\frac{\alpha+\beta}{2}} (1 + yF(x, y; z))^{\frac{\alpha+\beta}{2}} e^{\frac{1}{2}(\beta-\alpha)(u_3-u_4)z}, \quad (1.11)$$

where $x + y = u_3 + u_4$ and $xy = u_1 u_2$.

Ji [Ji25] showed that the generating function (1.11) is related to several families of enumerative polynomials that have appeared in the literature. In a subsequent paper [JL24], using the calculus of context-free grammars together with group actions on permutations, Ji and Lin further established the γ -positivity of two α -analogues of the Eulerian polynomials, as well as an α -analogue of the Chung-Graham-Knuth symmetric Eulerian identity (1.5).

The present chapter originates from the desire to provide a more conceptual approach to Ji's generating function (1.11), in the spirit of Foata-Schützenberger's combinatorial proof of (1.1) in [FS70]. To this end, we first observe that Eq. (1.11) can be rewritten in the following form,

$$\sum_{n \geq 0} A_n(u_1, u_2, u_3, u_4 | \alpha, \beta) \frac{z^n}{n!} = e^{(\alpha u_3 + \beta u_4)z} \left(\frac{x - y}{xe^{yz} - ye^{xz}} \right)^{\alpha + \beta} \quad (1.12)$$

$$= \left(\frac{(x - y)e^{u_3 z}}{xe^{yz} - ye^{xz}} \right)^{\alpha} \cdot \left(\frac{(x - y)e^{u_4 z}}{xe^{yz} - ye^{xz}} \right)^{\beta}, \quad (1.13)$$

where the parameters x and y are related by $x + y = u_3 + u_4$ and $xy = u_1 u_2$. Comparing (1.12) with (1.7a), we see that Eq. (1.11) is equivalent to the following connection formula:

$$A_n(u_1, u_2, u_3, u_4 | \alpha, \beta) = A_n^{\text{cyc}}\left(x, y, \frac{\alpha u_3 + \beta u_4}{\alpha + \beta} | \alpha + \beta\right), \quad (1.14)$$

where $xy = u_1 u_2$ and $x + y = u_3 + u_4$.

Our first main result establishes a generalization of Ji's polynomial (1.10) involving three additional variables. This unified framework subsumes the two families of polynomials studied in [Ji25, JL24] and further incorporates several recent results from [Ji25, JL24]. Before stating the result, we introduce some additional definitions. For $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$ with conventions $\sigma_0 = \sigma_{n+1} = 0$, a letter $\sigma_i \in [n]$ is a

- *left-to-right-maximum-peak* (**lmaxpk**) if σ_i is a left-to-right maximum and also a peak;
- *right-to-left-maximum-peak* (**rmaxpk**) if σ_i is a right-to-left maximum and also a peak;
- *left-to-right-maximum-double-ascent* (**lmaxda**) if σ_i is a left-to-right maximum and also a double ascent;

³Ji [Ji25, (3.33)–(3.35)] assumed that α and β are integers in her proof. In that case, an alternative combinatorial interpretation of the same generating function was given earlier by Carlitz and Scoville [CS74, Theorem 5].

- *right-to-left-maximum-double-descent* (**rmaxdd**) if σ_i is a right-to-left maximum and also a double descent.

Let $\mathbf{u} = (u_1, u_2, u_3, u_4)$ and define the generalized Eulerian polynomial

$$A_n(\mathbf{u}, f, g, t | \alpha, \beta) = \sum_{\sigma \in \mathfrak{S}_{n+1}} (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} \\ \times t^{\text{lmaxda}(\sigma) + \text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}. \quad (1.15)$$

Theorem 1.2. If $xy = u_1 u_2$ and $x + y = u_3 + u_4$, then

$$A_n(\mathbf{u}, f, g, t | \alpha, \beta) = A_n^{\text{cyc}} \left(x, y, \frac{\alpha u_3 + \beta u_4}{\alpha f + \beta g} t | \alpha f + \beta g \right). \quad (1.16)$$

Combining Theorem 1.2 with (1.7a), we derive immediately the exponential generating function of $A_n(\mathbf{u}, f, g, t | \alpha, \beta)$.

Corollary 1.1.1. If $xy = u_1 u_2$ and $x + y = u_3 + u_4$, then

$$\sum_{n \geq 0} A_n(\mathbf{u}, f, g, t | \alpha, \beta) \frac{z^n}{n!} = e^{(\alpha u_3 + \beta u_4)tz} \left(\frac{x - y}{xe^{yz} - ye^{xz}} \right)^{\alpha f + \beta g}. \quad (1.17)$$

In particular, we have

$$\sum_{n \geq 0} A_n(\mathbf{u}, 1, 1, 1 | \alpha, \beta) \frac{z^n}{n!} = e^{(\alpha u_3 + \beta u_4)z} \left(\frac{x - y}{xe^{yz} - ye^{xz}} \right)^{\alpha + \beta}, \quad (1.18)$$

$$\sum_{n \geq 0} A_n(\mathbf{u}, 0, 1, 1 | \alpha, \beta) \frac{z^n}{n!} = e^{(\alpha u_3 + \beta u_4)z} \left(\frac{x - y}{xe^{yz} - ye^{xz}} \right)^{\beta}. \quad (1.19)$$

Remark 1.3. The polynomials $A_n(\mathbf{u}, 1, 1, 1 | \alpha, \beta)$ and $A_n(\mathbf{u}, 0, 1, 1 | \alpha, \beta)$ correspond, respectively, to Ji's generalized Eulerian polynomial (1.10) and Ji-Lin's binomial-Stirling Eulerian polynomial [JL24, Theorem 1.5].

Combining Theorem 1.2 and the continued fraction expansion of the ordinary generating function of $A_n^{\text{cyc}}(x, y, t | \alpha)$ in [Zen93, Th. 3] we obtain the following continued fraction formula for the ordinary generating function of $A_n(\mathbf{u}, f, g, t | \alpha, \beta)$.

Corollary 1.1.2. We have

$$\sum_{n=0}^{\infty} A_n(\mathbf{u}, f, g, t | \alpha, \beta) z^n = \frac{1}{1 - b_0 z - \frac{\lambda_1 z^2}{1 - b_1 z - \frac{\lambda_2 z^2}{1 - b_2 z - \dots}}}, \quad (1.20a)$$

where

$$b_k = k(u_3 + u_4) + (\alpha u_3 + \beta u_4)t, \quad (1.20b)$$

$$\lambda_{k+1} = (k + \alpha f + \beta g)(k + 1)u_1 u_2 \quad (k \geq 0). \quad (1.20c)$$

Define the (α, t) -Eulerian polynomials

$$A_n(x, y, t | \alpha) = A_n^{\text{cyc}} \left(x, y, \frac{x + y}{2} t | 2\alpha \right). \quad (1.21)$$

By Theorem 1.2 with $f = g = 1$, $\alpha = \beta$, we obtain the following t -analogue of Ji-Lin's Theorem 1.6 [JL24].

Corollary 1.1.3. If $xy = u_1 u_2$ and $x + y = u_3 + u_4$, then

$$A_n(x, y, t | \alpha) = \sum_{\sigma \in \mathfrak{S}_{n+1}} (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} t^{\text{lmaxda}(\sigma) + \text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2}. \quad (1.22)$$

In particular, setting $u_1 = u_3 = x$ and $u_2 = u_4 = y$, we have

$$A_n(x, y, t | \alpha) = \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{asc}(\sigma)} y^{\text{des}(\sigma)} t^{\text{lmaxda}(\sigma) + \text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2}. \quad (1.23)$$

The next theorem gives three different combinatorial interpretations of the normalised γ -positivity of $A_n(x, y, t | \alpha)$.

Theorem 1.4. For $n \geq 0$, we have

$$A_n(x, y, t | \alpha) = \sum_{j=0}^{\lfloor n/2 \rfloor} 2^j d_{n,j}(\alpha, t) (xy)^j (x+y)^{n-2j}, \quad (1.24a)$$

where for $0 \leq j \leq \lfloor n/2 \rfloor$,

$$d_{n,j}(\alpha, t) = \sum_{\substack{\sigma \in \mathcal{CA}_n \\ \text{drop}(\sigma) = j}} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} \quad (1.24b)$$

$$= \sum_{\substack{\sigma \in \mathcal{A}_{n+1}^1 \\ \text{des}(\sigma) = j}} t^{\text{rminda}(\sigma)} \alpha^{\text{rmin}(\sigma) - 1} \quad (1.24c)$$

$$= \sum_{\substack{\sigma \in \mathcal{A}_{n+1}^2 \\ \text{des}(\sigma) = j}} t^{\text{rminda}(\sigma)} \alpha^{\text{rmin}(\sigma) - 1}. \quad (1.24d)$$

The sets $\mathcal{CA}_n$, $\mathcal{A}_n^1$ and $\mathcal{A}_n^2$ in above theorem denote the set of cycle André permutations, André permutations of the first kind and André permutations of the second kind, respectively. The precise definitions of these sets will be given in Section 1.3.3.

The rest of this chapter is organized as follows.

In Section 2, we prove Theorem 1.2 by constructing a combinatorial bijection based on two variants of Foata's first fundamental transformation (FFT) [Lot83, Chapter 10], and by combining it with the cyclic valley-hopping on permutations introduced by Cooper, Jones, and Zhuang [CJZ20].

In Section 3, we study (α, t) -binomial-Eulerian polynomials $\tilde{A}_n(x, y, t | \alpha)$ and establish a symmetric (α, t) -Eulerian identity. Then we show that the (α, t) -Eulerian polynomials and the (α, t) -binomial-Eulerian polynomials admit remarkable γ -expansions, including t -analogues of Ji and Lin's results [JL24, Theorems 1.5 and 1.6]. Furthermore, we present an (α, t) -analogue of (1.4), together with three combinatorial interpretations of the corresponding (α, t) -analogue of $d_{n,j}$ in terms of cycle André permutations and André permutations of types 1 and 2 (see Theorem 1.23).

In Section 4, we prove Theorem 1.23, establishing the three combinatorial interpretations of the normalized γ -coefficients of the (α, t) -Eulerian polynomials in the models of cycle André permutations and André permutations of types 1 and 2.

In Section 5, using a suitable variant of the FFT, we relate the polynomials $A_n^{\text{cyc}}(x, y, t | \alpha)$ to the enumeration of successions in permutations and derive the exponential generating function for a multivariate statistic involving the number of successions.

Finally, we conclude the chapter with a discussion of several open problems.

1.2 Proof of Theorem 1.2

1.2.1 Two Foata-type transformations

We recall two variants of the fundamental transformation [Sta12, p. 30], [FS70, Chap. 1]. For a permutation $\sigma \in \mathfrak{S}_n$, the map $\theta_1 : \mathfrak{S}_n \rightarrow \mathfrak{S}_n$ (resp. $\theta_2 : \mathfrak{S}_n \rightarrow \mathfrak{S}_n$) is defined as follows:

1. Write the disjoint cycle decomposition of σ , and in each cycle place its largest element at the *beginning* (resp. *end*).
2. Order the resulting cycles from left to right according to the *increasing* (resp. *decreasing*) order of their largest elements. Denote the concatenation of these ordered cycles by $\tilde{\sigma}$ (resp. $\tilde{\sigma}$).
3. Remove the parentheses of $\tilde{\sigma}$ (resp. $\tilde{\sigma}$); the resulting word is the one-line notation of a permutation, which we denote by $\theta_1(\sigma)$ (resp. $\theta_2(\sigma)$).

Example 1.5. Let $\sigma = (3\ 1\ 4)(6\ 2\ 5) \in \mathfrak{S}_6$.

- After rotating each cycle so that its largest element is first, we obtain $(4\ 3\ 1)$ and $(6\ 2\ 5)$. Ordering the cycles by increasing largest elements ($4 < 6$) gives $\tilde{\sigma} = (4\ 3\ 1)(6\ 2\ 5)$, hence

$$\theta_1(\sigma) = 431625.$$

- After rotating each cycle so that its largest element is last, we obtain $(3\ 1\ 4)$ and $(2\ 5\ 6)$. Ordering the cycles by decreasing largest elements ($6 > 4$) gives $\tilde{\sigma} = (2\ 5\ 6)(3\ 1\ 4)$, hence

$$\theta_2(\sigma) = 256314.$$

Remark 1.6. Under θ_1 , the set of left-to-right maxima of $\theta_1(\sigma)$ coincides with the set of cycle maxima of σ . Thus the number of cycles of σ equals the number of left-to-right maxima of $\theta_1(\sigma)$. Under θ_2 , the set of right-to-left maxima of $\theta_2(\sigma)$ coincides with the set of cycle maxima of σ . Thus the number of cycles of σ equals the number of right-to-left maxima of $\theta_2(\sigma)$.

1.2.2 Cyclic valley-hopping

Fix a letter $x \in [n]$. Each permutation $\sigma \in \mathfrak{S}_n$ can be written as the concatenation $\sigma = w_1 w_2 x w_4 w_5$, where w_2 (resp. w_4) is the maximal contiguous subword immediately to the left (resp. right) of x whose letters are all smaller than x . For example, if $\sigma = 934278516$ and $x = 7$, then $w_1 = 9$, $w_2 = 342$, $w_4 = \emptyset$ and $w_5 = 8516$. Define the involution $\xi_x : \mathfrak{S}_n \rightarrow \mathfrak{S}_n$ by

$$\xi_x(\sigma) := \begin{cases} w_1 w_4 x w_2 w_5, & \text{if } x \text{ is a double ascent or a double descent of } \sigma; \\ \sigma, & \text{if } x \text{ is a valley or a peak of } \sigma. \end{cases} \quad (1.25)$$

Here, we use conventions $\sigma(0) = \sigma(n+1) = \infty$. It is known that the involutions ξ_x and ξ_y commute with each other for all $x, y \in [n]$, see [Brä08]. Hence, for any subset $S \subseteq [n]$, we can define an operator $\xi_S := \prod_{x \in S} \xi_x$. As shown by Bränden [Brä08], the involutions ξ_S ($S \subseteq [n]$) define a $\mathbb{Z}_2^n$ -action on $\mathfrak{S}_n$, which are called *modified Foata and Strehl action* or *MFS-action*, see [FS74]. A geometric version of this action was given by Shapiro et al. [SWG83].

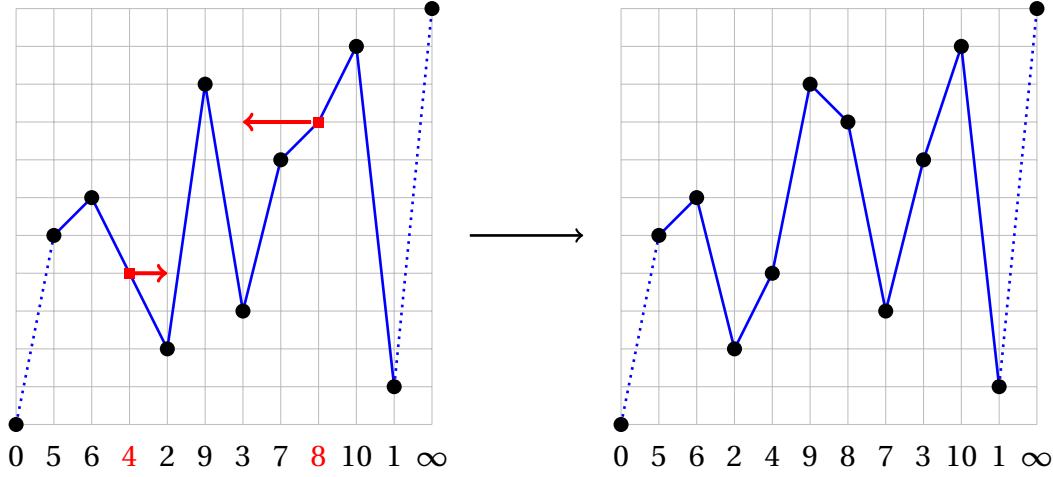


Figure 1.1: Cyclic valley-hopping on $\theta_1(\sigma) = 56429378101$ with $S = \{4, 8\}$ yields $\xi_S \circ \theta_1(\sigma) = 56249873101$.

Athanasiadis-Savvidou [AS12] defined a cyclic analogue of *valley hopping* for derangements by combining Foata's fundamental transformation and a variant of MFS-action with boundary conditions $\sigma(0) = 0$ and $\sigma(n+1) = \infty$, see also Sun-Wang [SW14] and Tirrell-Zhuang [TZ19]. Finally, Cooper, Jones and Zhuang [CJZ20] extended cyclic valley-hopping to permutations by letting the fixed points be unchanged.

Let $\tilde{\xi}_x$ denote the same operation defined in (1.25), but applied with boundary conventions $\sigma(0) = 0$ and $\sigma(n+1) = \infty$. Define $\psi_x : \mathfrak{S}_n \rightarrow \mathfrak{S}_n$ by

$$\psi_x(\sigma) := \begin{cases} \theta_1^{-1} \circ \tilde{\xi}_x \circ \theta_1(\sigma), & \text{if } x \text{ is not a fixed point of } \sigma; \\ \sigma, & \text{if } x \text{ is a fixed point of } \sigma, \end{cases}$$

where we treat the 0-th letter of $\theta_1(\sigma)$ as 0 and the $(n+1)$ -th letter as ∞ . Given a subset $S \subseteq [n]$, define $\psi_S := \prod_{x \in S} \psi_x$. We will call the $\mathbb{Z}_2^n$ -action defined by the involutions $\{\psi_S\}_{S \subseteq [n]}$ *cyclic valley-hopping* or *cyclic MFS-action*.

Example 1.7. Let $\sigma = 106724548931$ and $S = \{4, 8\}$. We have

$$\check{\sigma} = (5)(642)(9378)(101), \quad \theta_1(\sigma) = 56429378101,$$

then $\xi_S \circ \theta_1(\sigma) = 56249873101$, and $\psi_S(\sigma) = 10496523781$, see Fig. 1.1 for the valley hopping on $\theta_1(\sigma)$.

Define $\text{Cval}(\sigma)$, $\text{Cpk}(\sigma)$, $\text{Cda}(\sigma)$, $\text{Cdd}(\sigma)$ and $\text{Fix}(\sigma)$ to be the set of cyclic valleys, cyclic peaks, cyclic double ascents, cyclic double descents and fixed points of σ .

Proposition 1.2.1. [CJZ20, Proposition 3 and 4] For any $S \subseteq [n]$ and $\sigma \in \mathfrak{S}_n$, we have

1. $\text{Cval}(\psi_S(\sigma)) = \text{Cval}(\sigma); \quad \text{Cpk}(\psi_S(\sigma)) = \text{Cpk}(\sigma);$
2. $\text{Cda}(\psi_S(\sigma)) = (\text{Cda}(\sigma) \setminus S) \cup (S \cap \text{Cdd}(\sigma));$
3. $\text{Cdd}(\psi_S(\sigma)) = (\text{Cdd}(\sigma) \setminus S) \cup (S \cap \text{Cda}(\sigma));$
4. $\text{Fix}(\psi_S(\sigma)) = \text{Fix}(\sigma); \quad \text{cyc}(\psi_S(\sigma)) = \text{cyc}(\sigma).$

In other words, the cyclic valley-hopping does not affect cyclic valleys, cyclic peaks and fixed points, the number of cycles, but toggles between cyclic double ascents and cyclic double descents. As an application of cyclic valley-hopping, we give a combinatorial proof of the following result, which can be derived by combining formula (1.8) and (1.7a).

Theorem 1.8. *If $xy = u_1 u_2$ and $x + y = u_3 + u_4$, then*

$$A_n^{\text{cyc}}(x, y, t | \alpha) = \sum_{\sigma \in \mathfrak{S}_n} (u_1 u_2)^{\text{cpk}(\sigma)} u_3^{\text{cda}(\sigma)} u_4^{\text{cdd}(\sigma)} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)}. \quad (1.26)$$

Proof. Given a subset $S \subseteq [n]$, consider cyclic valley action of $\mathbb{Z}_2^n$ on $\mathfrak{S}_n$ via the functions ψ_S . For a permutation $\sigma \in \mathfrak{S}_n$, denote the orbit of σ under cyclic valley-hopping by $\text{Orb}(\sigma) := \{\psi_S(\sigma) | S \subseteq [n]\}$. Let $\bar{\sigma}$ be such a unique element with $\text{cdd}(\bar{\sigma}) = 0$ in $\text{Orb}(\sigma)$. By Proposition 1.2.1, we have

$$\sum_{\pi \in \text{Orb}(\sigma)} (u_1 u_2)^{\text{cpk}(\pi)} u_3^{\text{cda}(\pi)} u_4^{\text{cdd}(\pi)} t^{\text{fix}(\pi)} \alpha^{\text{cyc}(\pi)} = (u_1 u_2)^{\text{cpk}(\bar{\sigma})} (u_3 + u_4)^{\text{cda}(\bar{\sigma})} t^{\text{fix}(\bar{\sigma})} \alpha^{\text{cyc}(\bar{\sigma})}. \quad (1.27)$$

By definition, it is clear that for $\sigma \in \mathfrak{S}_n$,

$$\text{exc}(\sigma) = \text{cda}(\sigma) + \text{cpk}(\sigma) \quad \text{and} \quad \text{drop}(\sigma) = \text{cdd}(\sigma) + \text{cval}(\sigma). \quad (1.28)$$

Setting $u_1 = u_3 = x$ and $u_2 = u_4 = y$ in (1.27) yields

$$\sum_{\pi \in \text{Orb}(\sigma)} x^{\text{exc}(\pi)} y^{\text{drop}(\pi)} t^{\text{fix}(\pi)} \alpha^{\text{cyc}(\pi)} = (xy)^{\text{cpk}(\bar{\sigma})} (x + y)^{\text{cda}(\bar{\sigma})} t^{\text{fix}(\bar{\sigma})} \alpha^{\text{cyc}(\bar{\sigma})}. \quad (1.29)$$

Thus, if $u_1 u_2 = xy$ and $u_3 + u_4 = x + y$, combining (1.27) and (1.29) we have

$$\sum_{\pi \in \text{Orb}(\sigma)} (u_1 u_2)^{\text{cpk}(\pi)} u_3^{\text{cda}(\pi)} u_4^{\text{cdd}(\pi)} t^{\text{fix}(\pi)} \alpha^{\text{cyc}(\pi)} = \sum_{\pi \in \text{Orb}(\sigma)} x^{\text{exc}(\pi)} y^{\text{drop}(\pi)} t^{\text{fix}(\pi)} \alpha^{\text{cyc}(\pi)}. \quad (1.30)$$

Then summing over all the orbits of $\mathfrak{S}_n$ gives Eq. (1.26). $\square$

1.2.3 Proof of Theorem 1.2

We need the following combinatorial formulation of the binomial theorem.

Lemma 1.2.1. Let E be a finite set, we have

$$(\alpha + \beta)^{|E|} = \sum_{h: E \rightarrow \{0,1\}} \alpha^{\#\{x \in E: h(x)=0\}} \beta^{\#\{x \in E: h(x)=1\}}. \quad (1.31)$$

Explanation. The exponent $\#\{x \in E: h(x) = 1\}$ counts the number of elements of E sent to 1 (those contributing a factor β), while the remaining elements, sent to 0, contribute a factor α . Each function $h: E \rightarrow \{0,1\}$ encodes a subset of E , namely the set

$$A_h = \{x \in E: h(x) = 1\}.$$

Summing over all $2^{|E|}$ maps h amounts to summing over all subsets $A_h \subseteq E$, and we obtain the usual binomial expansion

$$(\alpha + \beta)^{|E|} = \sum_{k=0}^{|E|} \binom{|E|}{k} \alpha^{|E|-k} \beta^k.$$

For $\pi \in \mathfrak{S}_n$, let $\text{Fix}(\pi)$ denote the set of fixed points of π (that is, the cycles of length 1), and let $\text{Cyc}_{\geq 2}(\pi)$ denote the set of the remaining cycles of π . Then the set of all cycles of π is

$$C(\pi) := \text{Fix}(\pi) \cup \text{Cyc}_{\geq 2}(\pi).$$

A *cycle-decorated permutation* (π, h) is a permutation π in which each cycle is assigned a label in $\{0, 1\}$. We denote by

$$\mathfrak{S}_n^{\text{cd}} := \{(\pi, h) : \pi \in \mathfrak{S}_n, h : C(\pi) \rightarrow \{0, 1\}\}$$

the set of all such pairs. Let $\mathfrak{S}_n^\star$ denote the set of pairs $(\sigma, \tau) \in \mathfrak{S}_A \times \mathfrak{S}_B$, where (A, B) is an ordered set partition of $[n]$. Given a cycle-decorated permutation $(\pi, h) \in \mathfrak{S}_n^{\text{cd}}$, we associate to it a pair (σ, τ) of permutations acting on two disjoint subsets A and B of $[n]$ such that $A \cup B = [n]$, where A (resp. B) is the underlying support of σ (resp. τ). This defines a bijection

$$\Phi : \mathfrak{S}_n^{\text{cd}} \longrightarrow \mathfrak{S}_n^\star, \quad \Phi(\pi, h) = (\sigma, \tau), \quad (1.32)$$

obtained by grouping together the cycles of π according to their decoration: cycles labeled by 0 (resp. 1) form the permutation σ on A (resp. τ on B).

By Theorem 1.8, under the assumption that $xy = u_1 u_2$, $x + y = u_3 + u_4$, the right-hand side of equality (1.16) can be written as

$$\begin{aligned} A_n^{\text{cyc}}(x, y, t(\alpha u_3 + \beta u_4)(\alpha f + \beta g)^{-1} | \alpha f + \beta g) \\ = \sum_{\pi \in \mathfrak{S}_n} (u_1 u_2)^{\text{cpk}(\pi)} u_3^{\text{cda}(\pi)} u_4^{\text{cdd}(\pi)} (t(\alpha u_3 + \beta u_4))^{\text{fix}(\pi)} (\alpha f + \beta g)^{\text{cyc}(\pi) - \text{fix}(\pi)}. \end{aligned} \quad (1.33)$$

Using (1.31), it is clear that

$$\begin{aligned} (x_0 + x_1)^{\text{fix}(\pi)} (y_0 + y_1)^{\text{cyc}(\pi) - \text{fix}(\pi)} \\ = \sum_{h : C(\pi) \rightarrow \{0, 1\}} \prod_{i=0}^1 x_i^{\#\{x \in \text{Fix}(\pi) : h(x)=i\}} y_i^{\#\{x \in \text{Cyc}_{\geq 2}(\pi) : h(x)=i\}}. \end{aligned} \quad (1.34)$$

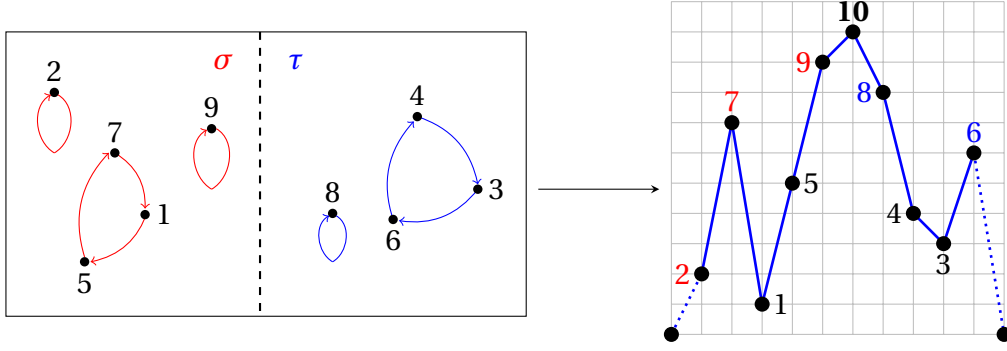
Let E be a finite subset of $\mathbb{Z}_{>0}$. For $\pi \in \mathfrak{S}_E$ define the weight function.

$$w(\pi; \alpha, t, b) = (u_1 u_2)^{\text{cpk}(\pi)} u_3^{\text{cda}(\pi)} u_4^{\text{cdd}(\pi)} t^{\text{fix}(\pi)} b^{\text{cyc}(\pi) - \text{fix}(\pi)} \alpha^{\text{cyc}(\pi)}. \quad (1.35)$$

Since a cycle-decorated permutation (π, h) has the same cycle structure as π and Φ preserves the weight function, applying Eq. (1.34) to (1.33) with $(x_0, x_1) = (t\alpha u_3, t\beta u_4)$ and $(y_0, y_1) = (\alpha f, \beta g)$, we obtain

$$\begin{aligned} A_n^{\text{cyc}}\left(x, y, \frac{t(\alpha u_3 + \beta u_4)}{(\alpha f + \beta g)} | \alpha f + \beta g\right) &= \sum_{(\pi, h) \in \mathfrak{S}_n^{\text{cd}}} w(\pi_0; \alpha, u_3 t, f) w(\pi_1; \beta, u_4 t, g) \\ &= \sum_{(\sigma, \tau) \in \mathfrak{S}_n^\star} w(\sigma; \alpha, u_3 t, f) w(\tau; \beta, u_4 t, g), \end{aligned} \quad (1.36)$$

where π_0 (resp. π_1) the product of cycles in π with label 0 (resp. 1). The second equation follows from (1.32).

Figure 1.2: The mapping $\rho : (\sigma, \tau) \mapsto \tilde{\pi} = 27159108436$.

The mapping ρ . Let $\mathfrak{S}_n^\star$ denote the set of cycle-decorated permutations (π, h) of $[n]$ with a 2-coloring h of the cycles (equivalently, pairs (σ, τ) of disjoint-support permutations whose product is π , with the cycles of σ colored, say, red and those of τ blue). Define

$$\rho : \mathfrak{S}_n^\star \longrightarrow \mathfrak{S}_{n+1}, \quad (\sigma, \tau) \longmapsto \tilde{\pi} := \theta_1(\sigma) (n+1) \theta_2(\tau),$$

where θ_1 and θ_2 are the two transformations introduced in section 1.2.1.

Since σ and τ are permutations on disjoint subsets A and B of $[n]$ with $A \cup B = [n]$, and θ_1 and θ_2 act locally on each cycle, we may apply θ_1 and θ_2 separately to $\sigma \in \mathfrak{S}_A$ and $\tau \in \mathfrak{S}_B$. Clearly the mapping ρ is a bijection satisfying the following properties:

$$\text{cpk}(\sigma) + \text{cpk}(\tau) = \text{pk}(\tilde{\pi}) - 1, \quad (1.37a)$$

$$\text{cda}(\sigma) + \text{fix}(\sigma) + \text{cda}(\tau) = \text{da}(\tilde{\pi}), \quad (1.37b)$$

$$\text{cdd}(\sigma) + \text{cdd}(\tau) + \text{fix}(\tau) = \text{dd}(\tilde{\pi}), \quad (1.37c)$$

$$\text{fix}(\sigma) + \text{fix}(\tau) = \text{lmaxda}(\tilde{\pi}) + \text{rmaxdd}(\tilde{\pi}), \quad (1.37d)$$

$$\text{cyc}(\sigma) - \text{fix}(\sigma) = \text{lmaxpk}(\tilde{\pi}) - 1, \quad (1.37e)$$

$$\text{cyc}(\tau) - \text{fix}(\tau) = \text{rmaxpk}(\tilde{\pi}) - 1, \quad (1.37f)$$

$$\text{cyc}(\sigma) = \text{lmax}(\tilde{\pi}) - 1, \quad (1.37g)$$

$$\text{cyc}(\tau) = \text{rmax}(\tilde{\pi}) - 1. \quad (1.37h)$$

Since $\text{pk} - 1 = \text{val}$, combining (1.37) with (1.36) we obtain

$$\begin{aligned} & \sum_{(\sigma, \tau) \in \mathfrak{S}_n^\star} w(\sigma; \alpha, u_3 t, f) w(\tau; \beta, u_4 t, g) \\ &= \sum_{\tilde{\pi} \in \mathfrak{S}_{n+1}} (u_1 u_2)^{\text{val}(\tilde{\pi})} u_3^{\text{da}(\tilde{\pi})} u_4^{\text{dd}(\tilde{\pi})} f^{\text{lmaxpk}(\tilde{\pi})-1} g^{\text{rmaxpk}(\tilde{\pi})-1} \\ & \quad \times t^{\text{lmaxda}(\tilde{\pi}) + \text{rmaxdd}(\tilde{\pi})} \alpha^{\text{lmax}(\tilde{\pi})-1} \beta^{\text{rmax}(\tilde{\pi})-1}. \end{aligned}$$

This proves (1.16) in view of (1.15). $\square$

We give an example to illustrate the construction of ρ by identifying each element $(\sigma, \tau) \in \mathfrak{S}_n^\star$ with a cycle-decorated permutation (π, h) , where $\pi \in \mathfrak{S}_n$ and h is a 2-coloring of the cycles of π . That is, for each cycle of π , the map h assigns a color in $\{\text{Red}, \text{Blue}\}$.

Example 1.9. Let $(\sigma, \tau) \in \mathfrak{S}_9^\star$ with

$$\sigma = (2)(9)(571), \quad \tau = (643)(8).$$

Then $\theta_1(\sigma) = 27159$ and $\theta_2(\tau) = 8436$. Hence

$$\tilde{\pi} = 27159\mathbf{108436} \in \mathfrak{S}_{10},$$

as illustrated in Fig. 1.2. As a verification of (1.37), we obtain:

- $\text{cyc}(\sigma) = 3, \quad \text{cyc}(\tau) = 2;$
- $\text{cpk}(\sigma) = |\{7\}| = 1, \quad \text{cpk}(\tau) = |\{6\}| = 1;$
- $\text{cda}(\sigma) = |\{5\}| = 1, \quad \text{cdd}(\tau) = |\{4\}| = 1;$
- $\text{fix}(\sigma) = |\{2, 9\}| = 2, \quad \text{fix}(\tau) = |\{8\}| = 1;$
- $\text{pk}(\tilde{\pi}) = |\{6, 7, 10\}| = 3, \quad \text{val}(\tilde{\pi}) = |\{1, 3\}| = 2;$
- $\text{da}(\tilde{\pi}) = |\{2, 5, 9\}| = 3, \quad \text{dd}(\tilde{\pi}) = |\{4, 8\}| = 2;$
- $\text{lmax}(\tilde{\pi}) = |\{2, 7, 9, 10\}| = 4, \quad \text{rmax}(\tilde{\pi}) = |\{6, 8, 10\}| = 3;$
- $\text{lmaxda}(\tilde{\pi}) = |\{2, 9\}| = 2, \quad \text{rmaxdd}(\tilde{\pi}) = |\{8\}| = 1;$
- $\text{lmaxpk}(\tilde{\pi}) = |\{7, 10\}| = 2, \quad \text{rmaxpk}(\tilde{\pi}) = |\{6, 10\}| = 2.$

1.3 Applications of the main theorems

In this section, we investigate a special case of $A_n(\mathbf{u}, f, g, t | \alpha, \beta)$ namely the (α, t) -binomial-Eulerian polynomials $\tilde{A}_n(x, y, t | \alpha)$ and present a symmetric (α, t) -Eulerian identity which is an (α, t) -analogue of Chung-Graham-Knuth's identity (1.5). In the last two subsections, we further show that both the (α, t) -Eulerian polynomials and the (α, t) -binomial-Eulerian polynomials admit various γ -expansions. Moreover, we provide a combinatorial interpretation of the normalized γ -expansions of $A_n(x, y, t | \alpha)$ in terms of cycle André permutations as well as the André permutations of type 1 and type 2.

1.3.1 (α, t) -binomial-Eulerian polynomials

Define the (α, t) -binomial-Eulerian polynomials

$$\tilde{A}_n(x, y, t | \alpha) = A_n^{\text{cyc}}(x, y, (x + y)t | \alpha). \quad (1.38)$$

Proposition 1.3.1. If $xy = u_1 u_2$ and $x + y = u_3 + u_4$, then

$$\tilde{A}_n(x, y, t | \alpha) = \sum_{\sigma \in \text{PRW}_{n+1}} (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} t^{\text{lmaxda}(\sigma) + \text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2}. \quad (1.39)$$

In particular, setting $u_1 = u_3 = x$ and $u_2 = u_4 = y$, we have

$$\tilde{A}_n(x, y, t | \alpha) = \sum_{\sigma \in \text{PRW}_{n+1}} x^{\text{asc}(\sigma)} y^{\text{des}(\sigma)} t^{\text{lmaxda}(\sigma) + \text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2}. \quad (1.40)$$

Proof. By Theorem 1.2 with $f = 0$, $g = 1$, and $\alpha = \beta$, we have

$$A_n^{\text{cyc}}(x, y, (x + y)t | \alpha) = A_n(\mathbf{u}, 0, 1, t | \alpha, \alpha). \quad (1.41)$$

It follows from (1.15) that $A_n(\mathbf{u}, 0, 1, t | \alpha)$ is the enumerative polynomial of permutations σ in $\mathfrak{S}_{n+1}$ such that $\text{lmaxpk}(\sigma) = 1$, this means that $n + 1$ is the unique left-to-right maximum and also a peak of σ , equivalently, $\sigma \in \text{PRW}_{n+1}$. $\square$

Note that Proposition 1.3.1 provides a t -analogue of Ji-Lin's Theorem 1.5 in [JL24]. The first four values of $\tilde{A}_n(x, y, t | \alpha)$ are as follows:

$$\begin{aligned}\tilde{A}_1(x, y, t | \alpha) &= \alpha t(x + y), \\ \tilde{A}_2(x, y, t | \alpha) &= \alpha^2 t^2(x^2 + y^2) + (2\alpha^2 t^2 + \alpha)xy, \\ \tilde{A}_3(x, y, t | \alpha) &= \alpha^3 t^3(x^3 + y^3) + (3\alpha^3 t^3 + 3\alpha^2 t + \alpha)(x^2 y + x y^2), \\ \tilde{A}_4(x, y, t | \alpha) &= \alpha^4 t^4(x^4 + y^4) + (4\alpha^4 t^4 + 6\alpha^3 t^2 + 4\alpha^2 t + \alpha)(x^3 y + x y^3) \\ &\quad + (6\alpha^4 t^4 + 12\alpha^3 t^2 + 8\alpha^2 t + 3\alpha^2 + 4\alpha)x^2 y^2.\end{aligned}$$

We need the following known properties of the fundamental transformation θ_2 in Section 1.2.1, see [Sta12, p. 30].

Lemma 1.3.1. For $\sigma \in \mathfrak{S}_n$, we have

- $\text{exc}(\sigma) = \text{asc}(\theta_2(\sigma))$;
- $\text{fix}(\sigma) = \text{rmaxdd}(\theta_2(\sigma))$;
- $\text{cyc}(\sigma) = \text{rmax}(\theta_2(\sigma))$.

Example 1.10. If $\sigma = 4271365$, then $\tilde{\sigma} = (537)(6)(14)(2)$ and $\sigma' := \theta_2(\sigma) = 5376142$. We verify that

- $\text{exc}(\sigma) = |\{1, 3\}| = 2$, $\text{fix}(\sigma) = |\{2, 6\}| = 2$, $\text{cyc}(\sigma) = 4$;
- $\text{asc}(\sigma') = |\{2, 5\}| = 2$, $\text{rmaxdd}(\sigma') = |\{2, 6\}| = 2$, $\text{rmax}(\sigma') = |\{2, 4, 6, 7\}| = 4$.

Define two kinds of (α, t) -Eulerian numbers as follows:

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{\alpha, t}^{\text{exc}} := \sum_{\substack{\sigma \in \mathfrak{S}_n \\ \text{exc}(\sigma) = n-k}} \alpha^{\text{cyc}(\sigma)} t^{\text{fix}(\sigma)} \quad (1 \leq k \leq n), \quad (1.42a)$$

and

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{\alpha, t}^{\text{asc}} := \sum_{\substack{\sigma \in \mathfrak{S}_n \\ \text{asc}(\sigma) = n-k}} \alpha^{\text{rmax}(\sigma)} t^{\text{rmaxdd}(\sigma)} \quad (1 \leq k \leq n). \quad (1.42b)$$

By Lemma 1.3.1, it is easy to see that the two (α, t) -Eulerian numbers are equal. Let

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{\alpha, t} := \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{\alpha, t}^{\text{exc}} = \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{\alpha, t}^{\text{asc}} \quad (1.43)$$

Theorem 1.11. For integers $a, b \geq 0$, we have

$$\sum_{k \geq 0} (\alpha t)^{a+b-k} \binom{a+b}{k} \left\langle \begin{matrix} k \\ a \end{matrix} \right\rangle_{\alpha, t} = \sum_{k \geq 0} (\alpha t)^{a+b-k} \binom{a+b}{k} \left\langle \begin{matrix} k \\ b \end{matrix} \right\rangle_{\alpha, t}, \quad (1.44)$$

where $\left\langle \begin{matrix} 0 \\ k \end{matrix} \right\rangle_{\alpha, t} = \left\langle \begin{matrix} k \\ 0 \end{matrix} \right\rangle_{\alpha, t} = \delta_{k,0}$.

Proof. We prove (1.44) with $\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{\alpha, t} = \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{\alpha, t}^{\text{exc}}$. As $\text{exc} + \text{drop} + \text{fix} = n$ over $\mathfrak{S}_n$, it follows from (1.6b) that

$$A_n^{\text{cyc}}(x, y, t y | \alpha) = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)} y^{\text{drop}(\sigma) + \text{fix}(\sigma)} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} = \sum_{k=1}^n \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{\alpha, t}^{\text{exc}} x^{n-k} y^k. \quad (1.45)$$

$n \setminus k$	1	2	3	4
1	αt			
2	α	$\alpha^2 t^2$		
3	α	$\alpha + 3\alpha^2 t$	$\alpha^3 t^3$	
4	α	$4\alpha + 3\alpha^2 + 4\alpha^2 t$	$\alpha + 4\alpha^2 t + 6\alpha^3 t^2$	$\alpha^4 t^4$

Table 1.1: The first values of $\langle \frac{n}{k} \rangle_{\alpha, t}$ for $1 \leq k \leq 4$.

By expanding $A_n^{\text{cyc}}(x, y, t(x+y) | \alpha)$ according to a partition of the fixed points of permutations in $\mathfrak{S}_n$ into two types, called *blue* and *red*, we deduce from (1.38) and (1.45) the following binomial identity:

$$\begin{aligned}
 \tilde{A}_n(x, y, t | \alpha) &= A_n^{\text{cyc}}(x, y, t(x+y) | \alpha) \\
 &= \sum_{m=0}^n \binom{n}{m} (\alpha t x)^{n-m} A_m^{\text{cyc}}(x, y, t y | \alpha) \\
 &= \sum_{m=0}^n \binom{n}{m} (\alpha t)^{n-m} \sum_{l=1}^m \langle \frac{m}{l} \rangle_{\alpha, t}^{\text{exc}} x^{n-l} y^l,
 \end{aligned} \tag{1.46}$$

where $\langle \frac{0}{k} \rangle_{\alpha, t}^{\text{exc}} = \langle \frac{k}{0} \rangle_{\alpha, t}^{\text{exc}} = \delta_{k,0}$.

The polynomial $A_n^{\text{cyc}}(x, y, t(x+y) | \alpha)$ is symmetric in x and y , because the inversion map $\vartheta : \sigma \mapsto \sigma^{-1}$ on $\mathfrak{S}_n$ interchanges the numbers of excedances and drops, while leaving the number of fixed points invariant, that is,

$$(\text{exc}, \text{drop}, \text{fix})(\sigma) = (\text{drop}, \text{exc}, \text{fix})(\sigma^{-1}). \tag{1.47}$$

Extracting the coefficients of $x^a y^b$ and $x^b y^a$ with $n = a + b$ in Eq. (1.46) we obtain Eq. (1.44) for $\langle \frac{n}{k} \rangle_{\alpha, t}^{\text{exc}}$. $\square$

Remark 1.12. The equidistribution (1.47) also follows from the γ -formula in Proposition 1.3.2. Identity (1.44) is an (α, t) -analogue of (1.5). When $t = 1$, by complement operation $\sigma \mapsto \sigma^c$, identity (1.44) with $\langle \frac{n}{k} \rangle_{\alpha, 1}^{\text{asc}}$ yields Ji and Lin's Theorem 4.1 in [JL24].

Remark 1.13. If $\sigma \in \mathfrak{S}_n$ with $\sigma_1 = n$, then $\text{lmaxda}(\sigma) = 0$ and $\text{lmax}(\sigma) = 1$. From (1.40), we have

$$\begin{aligned}
 \hat{A}_n(x, y, t | \alpha) &:= \sum_{\substack{\sigma \in \text{PRW}_{n+1} \\ \sigma_1 = n+1}} x^{\text{asc}(\sigma)} y^{\text{des}(\sigma)} t^{\text{rmaxdd}(\sigma)} \alpha^{\text{rmax}(\sigma)-1} \\
 &= \sum_{\sigma \in \mathfrak{S}_n} x^{\text{asc}(\sigma)} y^{\text{des}(\sigma)+1} t^{\text{rmaxdd}(\sigma)} \alpha^{\text{rmax}(\sigma)} \\
 &= \sum_{k=1}^n \langle \frac{n}{k} \rangle_{\alpha, t}^{\text{asc}} x^{n-k+1} y^k.
 \end{aligned} \tag{1.48}$$

A similar proof of (1.44) with $\langle \frac{n}{k} \rangle_{\alpha, t}^{\text{asc}}$ can be given using (1.48).

1.3.2 γ -positivity of (α, t) -Eulerian polynomials

In this section, we investigate the γ -coefficients of the two families of (α, t) -Eulerian polynomials, namely

$$\tilde{A}_n(x, y, t | \alpha) \quad \text{and} \quad A_n(x, y, t | \alpha).$$

We shall show that their γ -coefficients are polynomials in $\mathbb{N}[\alpha, t]$ by providing several combinatorial interpretations. Moreover, we derive explicit expressions for their exponential generating functions.

Lemma 1.3.2. For any variable f , we have

$$A_n(x, y, 0, x + y, f, 1, t | \alpha, \alpha) = \sum_{\substack{\sigma \in \mathfrak{S}_{n+1} \\ \text{da}(\sigma)=0}} (xy)^{\text{asc}(\sigma)} (x + y)^{n-2\text{asc}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} t^{\text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2} \quad (1.49a)$$

$$= \sum_{\substack{\sigma \in \mathfrak{S}_n \\ \text{cda}(\sigma)=0}} (xy)^{\text{exc}(\sigma)} (x + y)^{n-2\text{exc}(\sigma)} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} (f + 1)^{\text{cyc}(\sigma)-\text{fix}(\sigma)}. \quad (1.49b)$$

$$A_n(x, y, x + y, 0, f, 1, t | \alpha, \alpha) = \sum_{\substack{\sigma \in \mathfrak{S}_{n+1} \\ \text{dd}(\sigma)=0}} (xy)^{\text{des}(\sigma)} (x + y)^{n-2\text{des}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} t^{\text{lmaxda}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2} \quad (1.50a)$$

$$= \sum_{\substack{\sigma \in \mathfrak{S}_n \\ \text{cdd}(\sigma)=0}} (xy)^{\text{drop}(\sigma)} (x + y)^{n-2\text{drop}(\sigma)} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} (f + 1)^{\text{cyc}(\sigma)-\text{fix}(\sigma)}. \quad (1.50b)$$

$$A_n\left(1, xy, \frac{x+y}{2}, \frac{x+y}{2}, f, 1, t | \alpha, \alpha\right) = \sum_{\sigma \in \mathfrak{S}_{n+1}} (xy)^{\text{val}(\sigma)} \left(\frac{x+y}{2}\right)^{n-2\text{val}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2} \quad (1.51a)$$

$$= \sum_{\sigma \in \mathfrak{S}_n} (xy)^{\text{cpk}(\sigma)} \left(\frac{x+y}{2}\right)^{n-2\text{cpk}(\sigma)} (2t)^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} (f + 1)^{\text{cyc}(\sigma)-\text{fix}(\sigma)}. \quad (1.51b)$$

Proof. First of all, by definition (see Section 1), it is easy to verify the following identities:

$$\text{val} = \text{pk} - 1, \text{asc} = \text{val} + \text{da}, \text{des} = \text{val} + \text{dd}; \quad (1.52a)$$

$$\text{cval} = \text{cpk}, \text{exc} = \text{cval} + \text{cda}, \text{drop} = \text{cpk} + \text{cdd}; \quad (1.52b)$$

$$\text{fix} + \text{cdd} + \text{cda} = n - \text{cpk} - \text{cval} = n - 2\text{cpk}. \quad (1.52c)$$

When $\alpha = \beta$ and $g = 1$, Theorem 1.2 (with $xy = u_1 u_2$ and $x + y = u_3 + u_4$) becomes

$$A_n(u_1, u_2, u_3, u_4, f, 1, t | \alpha, \alpha) = A_n^{\text{cyc}}\left(x, y, \frac{x+y}{f+1} t | \alpha(f+1)\right). \quad (1.53)$$

- (i) Let $\mathbf{u} = (x, y, 0, x + y)$ in Eq. (1.53), then the left-hand side becomes $A_n(x, y, 0, x + y, f, 1, t | \alpha, \alpha)$, which is equal to (1.49a) by definition (1.15) and (1.52), while the right-hand side of Eq. (1.53) becomes (1.49b) by Theorem 1.8.
- (ii) Let $\mathbf{u} = (x, y, x + y, 0)$ in Eq. (1.53), similarly, we prove Eq. (1.50a) and (1.50b).
- (iii) Let $\mathbf{u} = (1, xy, \frac{x+y}{2}, \frac{x+y}{2})$ in Eq. (1.53), then the left-hand side becomes

$$A_n\left(1, xy, \frac{x+y}{2}, \frac{x+y}{2}, f, 1, t | \alpha, \alpha\right),$$

which is equal to (1.51a) by definition (1.15) and (1.52), while the right-hand side of (1.53) becomes (1.51b) by Theorem 1.8.

□

For convenience of description of the γ -coefficients, we shall define three types of subsets of $\mathfrak{S}_n$ with respect to cyclic statistics and linear statistics of permutations, respectively.

The cycle type subsets of $\mathfrak{S}_n$:

$$\mathfrak{S}_{n,\text{exc}=j}^{\text{cda}=0} := \{\sigma \in \mathfrak{S}_n : \text{cda}(\sigma) = 0 \text{ and } \text{exc}(\sigma) = j\}; \quad (1.54a)$$

$$\mathfrak{S}_{n,\text{drop}=j}^{\text{cdd}=0} := \{\sigma \in \mathfrak{S}_n : \text{cdd}(\sigma) = 0 \text{ and } \text{drop}(\sigma) = j\}; \quad (1.54b)$$

$$\mathfrak{S}_n^{\text{cpk}=j} := \{\sigma \in \mathfrak{S}_n : \text{cpk}(\sigma) = j\}. \quad (1.54c)$$

The linear counterpart subsets of $\mathfrak{S}_n$:

$$\mathfrak{S}_{n,\text{asc}=j}^{\text{da}=0} := \{\sigma \in \mathfrak{S}_n : \text{da}(\sigma) = 0 \text{ and } \text{asc}(\sigma) = j\}; \quad (1.55a)$$

$$\mathfrak{S}_{n,\text{des}=j}^{\text{dd}=0} := \{\sigma \in \mathfrak{S}_n : \text{dd}(\sigma) = 0 \text{ and } \text{des}(\sigma) = j\}; \quad (1.55b)$$

$$\mathfrak{S}_n^{\text{pk}=j} := \{\sigma \in \mathfrak{S}_n : \text{pk}(\sigma) = j\}; \quad (1.55c)$$

and the subsets of PRW_n :

$$\text{PRW}_{n,\text{asc}=j}^{\text{da}=0} := \{\sigma \in \text{PRW}_n : \text{da}(\sigma) = 0 \text{ and } \text{asc}(\sigma) = j\}; \quad (1.56a)$$

$$\text{PRW}_{n,\text{des}=j}^{\text{dd}=0} := \{\sigma \in \text{PRW}_n : \text{dd}(\sigma) = 0 \text{ and } \text{des}(\sigma) = j\}; \quad (1.56b)$$

$$\text{PRW}_n^{\text{pk}=j} := \{\sigma \in \text{PRW}_n : \text{pk}(\sigma) = j\}. \quad (1.56c)$$

In view of (1.38), we see that $\tilde{A}_n(x, y, t | \alpha)$ is symmetric in x and y , and the first few expansions in elementary symmetric polynomials are as follows:

$$\tilde{A}_1(x, y, t | \alpha) = \alpha t(x + y),$$

$$\tilde{A}_2(x, y, t | \alpha) = \alpha^2 t^2(x + y)^2 + \alpha xy,$$

$$\tilde{A}_3(x, y, t | \alpha) = \alpha^3 t^3(x + y)^3 + (\alpha + 3\alpha^2)txy(x + y),$$

$$\tilde{A}_4(x, y, t | \alpha) = \alpha^4 t^4(x + y)^4 + (6\alpha^3 + 4\alpha^2 + \alpha)t^2xy(x + y)^2 + (3\alpha^2 + 2\alpha)(xy)^2.$$

From Lemma 1.3.2, we derive the following combinatorial interpretations of the coefficients in the expansion in elementary symmetric polynomials of $\tilde{A}_n(x, y, t | \alpha)$ in terms of linear and cyclic statistics.

Proposition 1.3.2. For $n \geq 1$, we have

$$\tilde{A}_n(x, y, t | \alpha) = \sum_{j=0}^{\lfloor \frac{n}{2} \rfloor} \tilde{\gamma}_{n,j}(\alpha, t)(xy)^j(x + y)^{n-2j}, \quad (1.57)$$

where for $0 \leq j \leq \lfloor n/2 \rfloor$,

$$\tilde{\gamma}_{n,j}(\alpha, t) = \sum_{\sigma \in \text{PRW}_{n+1, \text{asc}=j}^{\text{da}=0}} \alpha^{\text{rmax}(\sigma)-1} t^{\text{rmaxdd}(\sigma)} \quad (1.58a)$$

$$= \sum_{\sigma \in \mathfrak{S}_{n, \text{exc}=j}^{\text{cda}=0}} \alpha^{\text{cyc}(\sigma)} t^{\text{fix}(\sigma)} \quad (1.58b)$$

$$= \sum_{\sigma \in \text{PRW}_{n+1, \text{des}=j}^{\text{dd}=0}} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2} t^{\text{lmaxda}(\sigma)} \quad (1.58c)$$

$$= \sum_{\sigma \in \mathfrak{S}_{n, \text{drop}=j}^{\text{cdd}=0}} \alpha^{\text{cyc}(\sigma)} t^{\text{fix}(\sigma)} \quad (1.58d)$$

$$= 2^{2j-n} \sum_{\sigma \in \text{PRW}_{n+1}^{\text{pk}=j}} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2} t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)} \quad (1.58e)$$

$$= 2^{2j-n} \sum_{\sigma \in \mathfrak{S}_n^{\text{cpk}=j}} \alpha^{\text{cyc}(\sigma)} (2t)^{\text{fix}(\sigma)}. \quad (1.58f)$$

Proof. Recall (see (1.38)) that

$$\tilde{A}_n(x, y, t | \alpha) = A_n^{\text{cyc}}(x, y, (x+y)t | \alpha). \quad (1.59)$$

From the above formula and Eq. (1.53) with $f = 1$, $u_1 u_2 = xy$ and $u_3 + u_4 = x + y$, we obtain

$$\tilde{A}_n(x, y, t | \alpha) = A_n(u_1, u_2, u_3, u_4, 0, 1, t | \alpha, \alpha). \quad (1.60)$$

Combining (1.60) with $u_3 = 0$ (resp. $u_4 = 0$) and (1.49) (resp. (1.50)) with $f = 0$, we obtain the twin interpretations (1.58a) and (1.58b) (resp. (1.58c) and (1.58d)). Note that we obtain (1.58a) by using the fact that if $\sigma \in \text{PRW}_n$ and $\text{da}(\sigma) = 0$, then $\text{lmax}(\sigma) = 1$.

Similarly, let $u_1 = 1$, $u_2 = xy$ and $u_3 = u_4 = \frac{x+y}{2}$ in (1.60), we obtain

$$\tilde{A}_n(x, y, t | \alpha) = A_n\left(1, xy, \frac{x+y}{2}, \frac{x+y}{2}, 0, 1, t | \alpha, \alpha\right). \quad (1.61)$$

Combining (1.61) and (1.51), we obtain the twin interpretations (1.58e) and (1.58f). $\square$

Remark 1.14. Previously, Shin-Zeng [SZ12, Theorem 11] proved the $t = 0$ case of (1.58b), i.e.,

$$A_n^{\text{cyc}}(x, y, 0 | \alpha) = \sum_{\sigma \in \mathfrak{S}_{n, \text{fix}=0}^{\text{cda}=0}} \alpha^{\text{cyc}(\sigma)} (xy)^{\text{exc}(\sigma)} (x+y)^{n-2\text{exc}(\sigma)}, \quad (1.62)$$

where $\mathfrak{S}_{n, \text{fix}=0}^{\text{cda}=0} := \{\sigma \in \mathfrak{S}_n : \text{cda}(\sigma) = 0 \text{ and } \text{fix}(\sigma) = 0\}$.

In fact, it is easy to derive the general case (1.58b) from (1.62) as in the following. Counting the permutations in $\mathfrak{S}_n$ by the number of fixed points and plugging (1.62), we obtain

$$\begin{aligned} A_n^{\text{cyc}}(x, y, t(x+y) | \alpha) &= \sum_{k=0}^n \binom{n}{k} (\alpha t(x+y))^k A_{n-k}^{\text{cyc}}(x, y, 0 | \alpha) \\ &= \sum_{k=0}^n \binom{n}{k} (\alpha t)^k \sum_{\sigma \in \mathfrak{S}_{n-k, \text{fix}=0}^{\text{cda}=0}} \alpha^{\text{cyc}(\sigma)} (xy)^{\text{exc}(\sigma)} (x+y)^{n-2\text{exc}(\sigma)} \\ &= \sum_{j=0}^{\lfloor n/2 \rfloor} \left(\sum_{\sigma \in \mathfrak{S}_{n, \text{exc}=j}^{\text{cda}=0}} \alpha^{\text{cyc}(\sigma)} t^{\text{fix}(\sigma)} \right) (xy)^j (x+y)^{n-2j}. \end{aligned} \quad (1.63)$$

An equivalent form of (1.63) appeared in [MMYY24, Theorem 3.4] and [PS24, Theorem 7]. One can also derive this result from [CJZ20, Theorem 6]. When $t = 1$, by complement operation $\sigma \mapsto \sigma^c$, the three combinatorial interpretations (1.58a), (1.58c) and (1.58e) yield the three combinatorial interpretations of the γ -coefficients in [JL24, Theorem 2.1].

From Lemma 1.3.2, we obtain the following combinatorial interpretations of the coefficients in the expansion of $A_n(x, y, t | \alpha) = A_n^{\text{cyc}}\left(x, y, \frac{x+y}{2}t | 2\alpha\right)$ in elementary symmetric polynomials, expressed in terms of linear and cyclic statistics.

Proposition 1.3.3. For $n \geq 0$, we have

$$A_n(x, y, t | \alpha) = \sum_{j=0}^{\lfloor n/2 \rfloor} \gamma_{n,j}(\alpha, t) (xy)^j (x+y)^{n-2j}, \quad (1.64)$$

where for $0 \leq j \leq \lfloor n/2 \rfloor$,

$$\gamma_{n,j}(\alpha, t) = \sum_{\sigma \in \mathfrak{S}_{n+1, \text{asc}=j}^{\text{da}=0}} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2} t^{\text{rmaxdd}(\sigma)} \quad (1.65a)$$

$$= \sum_{\sigma \in \mathfrak{S}_{n, \text{exc}=j}^{\text{cda}=0}} 2^{\text{cyc}(\sigma) - \text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} t^{\text{fix}(\sigma)} \quad (1.65b)$$

$$= \sum_{\sigma \in \mathfrak{S}_{n+1, \text{des}=j}^{\text{dd}=0}} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2} t^{\text{lmaxda}(\sigma)} \quad (1.65c)$$

$$= \sum_{\sigma \in \mathfrak{S}_{n, \text{drop}=j}^{\text{cdd}=0}} 2^{\text{cyc}(\sigma) - \text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} t^{\text{fix}(\sigma)} \quad (1.65d)$$

$$= 2^{2j-n} \sum_{\sigma \in \mathfrak{S}_{n+1}^{\text{pk}=j+1}} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2} t^{\text{lmaxda}(\sigma) + \text{rmaxdd}(\sigma)} \quad (1.65e)$$

$$= 2^{2j-n} \sum_{\sigma \in \mathfrak{S}_n^{\text{cpk}=j}} (2\alpha)^{\text{cyc}(\sigma)} t^{\text{fix}(\sigma)}. \quad (1.65f)$$

Proof. Recall (see (1.21)) that

$$A_n(x, y, t | \alpha) = A_n^{\text{cyc}}\left(x, y, \frac{x+y}{2}t | 2\alpha\right).$$

From the above formula and Eq. (1.53) with $f = 1$, $u_1 u_2 = xy$ and $u_3 + u_4 = x + y$, we derive

$$A_n(x, y, t | \alpha) = A_n(u_1, u_2, u_3, u_4, 1, 1, t | \alpha, \alpha). \quad (1.66)$$

Combining (1.66) with $u_3 = 0$ (resp. $u_4 = 0$) and (1.49) (resp. (1.50)) with $f = 1$, we obtain the twin interpretations (1.65a) and (1.65b) (resp. (1.65c) and (1.65d)). Specializing equation (1.66) with $u_1 = 1$, $u_2 = xy$ and $u_3 = u_4 = \frac{x+y}{2}$, we obtain

$$A_n(x, y, t | \alpha) = A_n\left(1, xy, \frac{x+y}{2}, \frac{x+y}{2}, 1, 1, t | \alpha, \alpha\right). \quad (1.67)$$

Combining (1.67) and (1.51), we obtain the twin interpretations (1.65e) and (1.65f). $\square$

The first few values of $\gamma_{n,j}(\alpha, t)$ are given in Table 1.2.

Theorem 1.15. For $n \geq 0$, we have

$$A_n\left(x, y, t | \frac{\alpha + \beta}{2}\right) = \sum_{j=0}^{\lfloor n/2 \rfloor} \gamma_{n,j}\left(\frac{\alpha + \beta}{2}, t\right) (xy)^j (x+y)^{n-2j}, \quad (1.68a)$$

$n \setminus j$	0	1	2
0	1		
1	αt		
2	$\alpha^2 t^2$	2α	
3	$\alpha^3 t^3$	$2\alpha + 6\alpha^2 t$	
4	$\alpha^4 t^4$	$2\alpha + 8\alpha^2 t + 12\alpha^3 t^2$	$4\alpha + 12\alpha^2$
5	$\alpha^5 t^5$	$2\alpha + 10\alpha^2 t + 20\alpha^3 t^2 + 20\alpha^4 t^3$	$16\alpha + 40\alpha^2 + 20\alpha^2 t + 60\alpha^3 t$

Table 1.2: The first values of $\gamma_{n,j}(\alpha, t)$ for $0 \leq 2j \leq n \leq 5$.

where for $0 \leq j \leq \lfloor n/2 \rfloor$,

$$\gamma_{n,j}\left(\frac{\alpha+\beta}{2}, t\right) = 2^{2j-n} \sum_{\sigma \in \mathfrak{S}_{n+1}^{\text{pk}=j+1}} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1} t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)}. \quad (1.68b)$$

Proof. By (1.21), we have

$$A_n\left(x, y, t \mid \frac{\alpha+\beta}{2}\right) = A_n^{\text{cyc}}\left(x, y, \frac{x+y}{2} t \mid \alpha+\beta\right). \quad (1.69)$$

Let $\mathbf{u} = (1, xy, \frac{x+y}{2}, \frac{x+y}{2})$, it follows from Theorem 1.2 and definition (1.15) that

$$\begin{aligned} & A_n^{\text{cyc}}\left(x, y, \frac{x+y}{2} t \mid \alpha+\beta\right) \\ &= A_n\left(1, xy, \frac{x+y}{2}, \frac{x+y}{2}, 1, 1, t \mid \alpha, \beta\right) \\ &= \sum_{\sigma \in \mathfrak{S}_{n+1}} (xy)^{\text{val}(\sigma)} (x+y)^{n-2\text{val}(\sigma)} t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1} 2^{2\text{val}(\sigma)-n}. \end{aligned} \quad (1.70)$$

As $\text{val} = \text{pk} - 1$, combining (1.69) and (1.70) yields (1.68b). $\square$

Remark 1.16. Theorem 1.15 provides a t -analogue of Ji's Theorem 1.8 [Ji25]. Setting $\alpha = \beta$ in equation (1.68b) we obtain (1.65e). Combining (1.65e) with $\alpha \rightarrow \frac{\alpha+\beta}{2}$ and (1.68b) we obtain

$$\begin{aligned} & \sum_{\sigma \in \mathfrak{S}_{n+1}^{\text{pk}=j+1}} (2\alpha)^{\text{lmax}(\sigma)-1} (2\beta)^{\text{rmax}(\sigma)-1} t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)} \\ &= \sum_{\sigma \in \mathfrak{S}_{n+1}^{\text{pk}=j+1}} (\alpha+\beta)^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2} t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)}. \end{aligned} \quad (1.71)$$

This identity is far from trivial. When $t = 1$, Dong, Lin and Pan [DLP24, Theorem 1.5] provided a combinatorial proof of Eq. (1.71). Finding a purely combinatorial explanation of Eq. (1.71) would be of independent interest.

Define the generating polynomials of the two kinds of γ -vectors in (1.57) and (1.64)

$$G_n^{\text{cyc}}(x, t, \alpha) := \sum_{j=0}^{\lfloor \frac{n}{2} \rfloor} \tilde{\gamma}_{n,j}(\alpha, t) x^j = \sum_{\sigma \in \mathfrak{S}_n^{\text{cda}=0}} x^{\text{exc}(\sigma)} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)}, \quad (1.72a)$$

$$G_n(x, t, \alpha) := \sum_{j=0}^{\lfloor \frac{n}{2} \rfloor} \gamma_{n,j}(\alpha, t) x^j = \sum_{\sigma \in \mathfrak{S}_{n+1}^{\text{da}=0}} x^{\text{asc}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2} t^{\text{rmaxdd}(\sigma)}, \quad (1.72b)$$

where

$$\mathfrak{S}_n^{\text{cda}=0} := \{\sigma \in \mathfrak{S}_n : \text{cda}(\sigma) = 0\} \quad \text{and} \quad \mathfrak{S}_n^{\text{da}=0} := \{\sigma \in \mathfrak{S}_n : \text{da}(\sigma) = 0\}. \quad (1.72c)$$

Theorem 1.17. Let $u = \sqrt{1-4x}$. We have

$$1 + \sum_{n \geq 1} G_n^{\text{cyc}}(x, t, \alpha) \frac{z^n}{n!} = \left(\frac{u e^{(t-\frac{1}{2})z}}{u \cosh(uz/2) - \sinh(uz/2)} \right)^\alpha, \quad (1.73a)$$

$$1 + \sum_{n \geq 1} G_n(x, t, \alpha) \frac{z^n}{n!} = \left(\frac{u e^{\frac{1}{2}(t-1)z}}{u \cosh(uz/2) - \sinh(uz/2)} \right)^{2\alpha}. \quad (1.73b)$$

Proof. With $y = 1$ and $s = \frac{x}{(1+x)^2}$, the two γ -expansions (1.57) and (1.64) can be written, respectively, as follows,

$$A_n^{\text{cyc}}(x, 1, t(x+1) | \alpha) = (1+x)^n G_n^{\text{cyc}}(s, t, \alpha), \quad (1.74a)$$

$$A_n(x, 1, t | \alpha) = (1+x)^n G_n(s, t, \alpha). \quad (1.74b)$$

Plugging (1.74a) in (1.7a) yields

$$1 + \sum_{n \geq 1} G_n^{\text{cyc}}(s, t, \alpha) \frac{z^n}{n!} = \left(\frac{(x-1)e^{tz}}{x e^{z/(1+x)} - e^{xz/(1+x)}} \right)^\alpha, \quad (1.74c)$$

After a little computation, the right-hand sides of (1.73a) and (1.74c) agree under substitution $u = \frac{1-x}{1+x} = \sqrt{1-4s}$.

From (1.17) with $u_1 = u_3 = x$, $u_2 = u_4 = f = g = 1$, we derive

$$1 + \sum_{n \geq 1} A_n(x, 1, t | \alpha) \frac{z^n}{n!} = e^{\alpha(x+1)tz} \left(\frac{x-1}{x e^z - e^{xz}} \right)^{2\alpha}. \quad (1.74d)$$

Formula (1.73b) can be proved similarly by plugging (1.74b) in (1.74d). $\square$

Remark 1.18. Formula (1.73a) appeared in [MMYY24, Eq. (13)] with a different proof, and (1.73b) is a t -analogue of Carlitz and Scoville's formula [CS74, Eq. (4.10)].

1.3.3 γ -vector of (α, t) -Eulerian polynomials and cycle André permutations

For $n \geq 0$ and $0 \leq j \leq \lfloor n/2 \rfloor$, let

$$d_{n,j}(\alpha, t) = \gamma_{n,j}(\alpha, t)/2^j. \quad (1.75)$$

Then, Eq. (1.64) reads

$$A_n(x, y, t | \alpha) = \sum_{j=0}^{\lfloor n/2 \rfloor} 2^j d_{n,j}(\alpha, t) (xy)^j (x+y)^{n-2j}. \quad (1.76)$$

From Table 1.2 we derive the first values of $d_{n,j}(\alpha, t)$ in Table 1.3. A glance at these values foretells that the polynomials $d_{n,j}(\alpha, t)$ have nonnegative integral coefficients. Indeed, from Corollary 1.1.2, we derive the continued fraction

$$\sum_{n=0}^{\infty} \sum_{j=0}^{\lfloor n/2 \rfloor} d_{n,j}(\alpha, t) x^j z^n = \frac{1}{1 - b_0 z - \frac{\lambda_1 z^2}{1 - b_1 z - \frac{\lambda_2 z^2}{1 - b_2 z - \dots}}}, \quad (1.77a)$$

$n \setminus j$	0	1	2
0	1		
1	αt		
2	$\alpha^2 t^2$	α	
3	$\alpha^3 t^3$	$\alpha + 3\alpha^2 t$	
4	$\alpha^4 t^4$	$\alpha + 4\alpha^2 t + 6\alpha^3 t^2$	$\alpha + 3\alpha^2$
5	$\alpha^5 t^5$	$\alpha + 5\alpha^2 t + 10\alpha^3 t^2 + 10\alpha^4 t^3$	$4\alpha + 10\alpha^2 + 5\alpha^2 t + 15\alpha^3 t$

Table 1.3: The values of $d_{n,j}(\alpha, t)$ for $0 \leq 2j \leq n \leq 5$.

where

$$b_k = k + \alpha t, \quad (1.77b)$$

$$\lambda_{k+1} = \binom{k+1}{2} x + \alpha(k+1)x \quad (k \geq 0). \quad (1.77c)$$

It follows that the polynomials $d_{n,j}(\alpha, t)$ are elements in $\mathbb{N}[\alpha, t]$. The aim of this section is to provide three combinatorial interpretations in terms of cycle André and André permutations.

Note that when $x = \alpha = t = 1$ Eq.(1.76) is equivalent to the γ -expansion of Eulerian polynomials (1.4) and $d_{n,j}(1, 1)$ is the number of André permutations of $[n+1]$ with j descents.

There are many equivalent ways to define an André permutation. Following Foata-Strehl [FS74] and Heteyi-Reiner [HR98], we define André permutations in terms of *x -factorization*.

Definition 1.19. Let $\sigma = \sigma_1 \sigma_2 \dots \sigma_n \in \mathfrak{S}_n$ and let $x = \sigma_i$ for $i \in [n]$. The x -factorization of σ is given by $\sigma = u \lambda(x) x \rho(x) v$, where

- (1) $\lambda(x) = \sigma_j \dots \sigma_{i-1}$ with $\sigma_l > x$ for $j \leq l \leq i-1$, $u = \sigma_1 \dots \sigma_{j-1}$, and $\sigma_{j-1} < x$.
- (2) $\rho(x) = \sigma_{i+1} \dots \sigma_k$ with $\sigma_l > x$ for $i+1 \leq l \leq k$, $v = \sigma_{k+1} \dots \sigma_n$, and $\sigma_{k+1} < x$.

Here, any of words $u, \lambda(x), \rho(x), v$ may be the empty.

Example 1.20. If $\sigma = 810145113612279 \in \mathfrak{S}_{12}$, then, the 3-factorization of σ is

$$\underbrace{8101}_u \underbrace{4511}_{\lambda(3)} \underbrace{3}_{\rho(3)} \underbrace{612279}_v.$$

Definition 1.21 ([HR98]). Let $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$. Say that σ is an **André permutation of the first kind (resp. second kind)** if σ has no double descents, i.e., $\sigma_{i-1} > \sigma_i > \sigma_{i+1}$, and each x -factorisation $u \lambda(x) x \rho(x) v$ of σ has property

- $\lambda(x) = \emptyset$ if $\rho(x) = \emptyset$,
- $\max(\lambda(x)) < \max(\rho(x))$ (resp. $\min(\rho(x)) < \min(\lambda(x))$) if $\rho(x) \neq \emptyset$ and $\lambda(x) \neq \emptyset$.

Let $\mathcal{A}_n^1$ (resp. $\mathcal{A}_n^2$) be the set of André permutations of the first (resp. second) kind in $\mathfrak{S}_n$. For $n \in [5]$, these permutations are given in Tables 1.4 and 1.5. It is known [FS70] that the cardinality of $\mathcal{A}_n^1$ (resp. $\mathcal{A}_n^2$) is the *Euler number* E_n , which is the n th coefficient in the Taylor expansion of $\sec(z) + \tan(z)$, namely,

$$\sum_{n \geq 0} E_n \frac{z^n}{n!} = \sec(z) + \tan(z). \quad (1.78)$$

$n = 1:$	1;	$n = 2:$	12;
$n = 3:$	123, 213;		
$n = 4:$	1234, 1324, 2314, 2134, 3124;		
$n = 5:$	12345, 12435, 13425, 23415, 13245, 14235, 34125, 24135,		
	23145, 21345, 41235, 31245, 21435, 32415, 41325, 31425.		

Table 1.4: André permutations of the first kind of length ≤ 5 .

$n = 1:$	1;	$n = 2:$	12;
$n = 3:$	123, 312;		
$n = 4:$	1234, 1423, 3412, 4123, 3124;		
$n = 5:$	12345, 12534, 14523, 34512, 15234, 14235, 34125, 45123,		
	35124, 51234, 41235, 31245, 51423, 53412, 41523, 31524.		

Table 1.5: André permutations of the second kind of length ≤ 5 .

The sequence E_n 's starts with 1, 1, 2, 5, 16, 61, 271, ..., see [Slo, A138265].

Let $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$ with $\sigma_0 = \sigma_{n+1} = 0$. A right-to-left minimum (**rmin**) of σ is an element σ_i such that $\sigma_j > \sigma_i$ if $n \geq j > i \geq 1$. A letter $\sigma_i \in [n]$ is a right-to-left-minimum-double-ascent (**rminda**) if it is a double ascent and also a right-to-left minimum. Hwang et al. [HJO23] introduced cycle André permutations to characterise the so-called [Web permutations](#).

Definition 1.22 ([HJO23]). Let $A := \{a_1, \dots, a_k\}$ be a set of k positive integers. Let $C = (a_1 \cdots a_k)$ be a cycle (cyclic permutation) of A with $a_1 = \min\{a_1, \dots, a_k\}$. Then, cycle C is called an **André cycle** if the word $a_2 \cdots a_k$ is an André permutation of the first kind. We say that a permutation is a **cycle André permutation** if it is a product of disjoint André cycles.

Let $\mathcal{CA}_n$ be the set of cycle André permutations of $[n]$. For example, $(1\ 5\ 6)(2\ 7\ 4\ 8)(3)$ is a cycle André permutation, but $(1\ 2\ 5\ 3)(4)$ is not because the $\rho(3) = \emptyset$ and $\lambda(3) = 5$ in 3-factorisation of 253.

Theorem 1.23. For $n \geq 0$ and $0 \leq j \leq \lfloor n/2 \rfloor$, the normalized γ -coefficient $d_{n,j}(\alpha, t)$ in (1.76) has the following combinatorial interpretations

$$d_{n,j}(\alpha, t) = \sum_{\substack{\sigma \in \mathcal{CA}_n \\ \text{drop}(\sigma) = j}} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} \quad (1.79a)$$

$$= \sum_{\substack{\sigma \in \mathcal{A}_{n+1}^1 \\ \text{des}(\sigma) = j}} t^{\text{rminda}(\sigma)} \alpha^{\text{rmin}(\sigma) - 1} \quad (1.79b)$$

$$= \sum_{\substack{\sigma \in \mathcal{A}_{n+1}^2 \\ \text{des}(\sigma) = j}} t^{\text{rminda}(\sigma)} \alpha^{\text{rmin}(\sigma) - 1}. \quad (1.79c)$$

Theorem 1.4 follows from Theorem 1.23 together with (1.76). We shall prove Theorem 1.23 in the next section. Our proof proceeds as follows. First, we establish (1.79a) by showing that the exponential generating functions of both sides coincide. Next, we deduce (1.79b) from (1.79a) by constructing a bijection $\Phi : \mathcal{CA}_n \rightarrow \mathcal{A}_{n+1}^1$. Finally, we derive (1.79c) from (1.79b) by means of a bijection $\Psi : \mathcal{A}_{n+1}^1 \rightarrow \mathcal{A}_{n+1}^2$ built upon André trees.

1.4 Cycle André and André permutations: Proof of Theorem 1.23

1.4.1 Counting André permutations of type 1 by des and lmax

When $x = t = 1$, let $\gamma_{n,j}(\alpha) := \gamma_{n,j}(\alpha, 1)$ and

$$A_{n,k}(\alpha) := \sum_{\substack{\sigma \in \mathfrak{S}_{n+1} \\ \text{des}(\sigma) = k}} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2}, \quad (1.80a)$$

then, Eq. (1.64) can be rephrased as

$$\sum_{k=0}^n A_{n,k}(\alpha) y^k = \sum_{j=0}^{\lfloor n/2 \rfloor} \gamma_{n,j}(\alpha) y^j (1+y)^{n-2j}. \quad (1.80b)$$

By considering the position of the letter 1 in permutations of $\mathfrak{S}_{n+2}$, one readily verifies that the coefficients $A_{n,k}(\alpha)$ obey the following recurrence relation:

$$A_{n+1,k}(\alpha) = (k + \alpha) A_{n,k}(\alpha) + (n + 1 - k + \alpha) A_{n,k-1}(\alpha), \quad (1.80c)$$

where $A_{0,0}(\alpha) = 1$ and $A_{0,-1}(\alpha) = 0$, for $0 \leq k \leq n$.

Lemma 1.4.1. For $n \geq 0$ and $0 \leq j \leq \lfloor n/2 \rfloor$, the γ -coefficients $\gamma_{n,j}(\alpha)$ satisfy the recurrence relation

$$\gamma_{n+1,j}(\alpha) = (\alpha + j) \gamma_{n,j}(\alpha) + 2(n + 2 - 2j) \gamma_{n,j-1}(\alpha), \quad (1.81)$$

with $\gamma_{0,0}(\alpha) = 1$ and $\gamma_{0,-1}(\alpha) = 0$. Equivalently, setting $d_{n+1,j}(\alpha) = \gamma_{n+1,j}(\alpha)/2^j$, then

$$d_{n+1,j}(\alpha) = (\alpha + j) d_{n,j}(\alpha) + (n + 2 - 2j) d_{n,j-1}(\alpha), \quad (1.82)$$

with $d_{0,0}(\alpha) = 1$ and $d_{0,-1}(\alpha) = 0$.

Proof. We verify recurrence (1.81) by induction on n . Since $\gamma_{2,0}(\alpha) = \alpha^2$, $\gamma_{1,0}(\alpha) = \alpha$ and $\gamma_{1,-1}(\alpha) = 0$, this is obvious when $n = 1$. If $n \geq 2$ and $0 \leq k \leq n$, Eq. (1.80b) is equivalent to

$$A_{n,k}(\alpha) = \sum_{0 \leq j \leq k} \binom{n-2j}{k-j} \gamma_{n,j}(\alpha). \quad (1.83)$$

Substituting (1.83) in (1.80c) and using (1.81), we derive

$$\begin{aligned} & \sum_{0 \leq j \leq k} \binom{n+1-2j}{k-j} \left((\alpha + j) \gamma_{n,j}(\alpha) + 2(n + 2 - 2j) \gamma_{n,j-1}(\alpha) \right) \\ &= (k + \alpha) \sum_{0 \leq j \leq k} \binom{n-2j}{k-j} \gamma_{n,j}(\alpha) + (n + 1 - k + \alpha) \sum_{0 \leq j \leq k-1} \binom{n-2j}{k-1-j} \gamma_{n,j}(\alpha). \end{aligned} \quad (1.84)$$

Extracting the coefficients of $\gamma_{n,j}(\alpha)$ we obtain:

$$\begin{aligned} & (\alpha + j) \binom{n+1-2j}{k-j} + 2(n - 2j) \binom{n-1-2j}{k-j-1} \\ &= (k + \alpha) \binom{n-2j}{k-j} + (n + 1 - k + \alpha) \binom{n-2j}{k-1-j}, \end{aligned} \quad (1.85)$$

which can be verified by direct algebraic manipulation. $\square$

Lemma 1.4.2. For $n \geq 0$, we have

$$\sum_{j=0}^{\lfloor n/2 \rfloor} d_{n,j}(\alpha) x^j = \sum_{\sigma \in \mathcal{A}_{n+1}^1} x^{\text{des}(\sigma)} \alpha^{\text{lmax}(\sigma)-1}. \quad (1.86)$$

Proof. Let

$$\sum_{\sigma \in \mathcal{A}_{n+1}^1} x^{\text{des}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} = \sum_{j=0}^{\lfloor n/2 \rfloor} d_{n,j}^*(\alpha) x^j. \quad (1.87)$$

We shall prove that $d_{n,j}^*(\alpha)$ satisfies the same recurrence relation (1.82), i.e.,

$$d_{n+1,j}^*(\alpha) = (\alpha + j) d_{n,j}^*(\alpha) + (n + 2 - 2j) d_{n,j-1}^*(\alpha). \quad (1.88)$$

The left-hand side of equation (1.88) counts permutations $\sigma \in \mathcal{A}_{n+2}^1$ with j descents considering the weights of $\alpha^{\text{lmax}-1}$. Deleting the letter 1 in $\sigma \in \mathcal{A}_{n+2}^1$ yields also an André permutation of $\{2, 3, \dots, n+2\}$. Thus, we can obtain such a permutation uniquely in one of two ways. For the first way, choose a permutation $\sigma = \sigma_1 \cdots \sigma_{n+1} \in \mathcal{A}_E^1$ with j descents where $E = \{2, 3, \dots, n+2\}$, and insert 1 after σ_i if i is descent, or insert 1 at the beginning. Note that letter 1 will contribute a left-to-right maximum at the beginning, while in other positions not. Thus, there are $j+1$ ways to insert 1, so we obtain by this factor $(\alpha + j) d_{n,j}^*(\alpha)$. For the second way, choose $\sigma = \sigma_1 \cdots \sigma_{n+1} \in \mathcal{A}_E^1$ with $j-1$ descents where $E = \{2, 3, \dots, n+2\}$, and insert 1 after σ_i if i is an ascent and $i-1$ is not a descent. Since an André permutation has no double descents, thus there are $n-2(j-1)$ ways to insert 1, so we obtain the second factor $(n+2-2j) d_{n,j-1}^*(\alpha)$. $\square$

Combining Eq. (1.81), (1.82) and Lemma 1.4.2, we derive the following exponential generating function from (1.73b).

Theorem 1.24. We have

$$\sum_{n \geq 0} \sum_{\sigma \in \mathcal{A}_{n+1}^1} x^{\text{des}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \frac{z^n}{n!} = \left(\frac{\sqrt{2x-1} \sec(\frac{z}{2} \sqrt{2x-1})}{\sqrt{2x-1} - \tan(\frac{z}{2} \sqrt{2x-1})} \right)^{2\alpha}. \quad (1.89)$$

Remark 1.25. When $\alpha = 1$, (1.89) was first proved by Foata and Schützenberger [FS73], other proofs are given by Foata and Han [FH01], and Chow and Shiu [CS11, Theorem 2.1].

Remark 1.26. It would be interesting to extend the above approach by incorporating a suitable parameter t .

1.4.2 Counting André cycles by drop

The following is an alternative definition of André permutations of the first kind.

Lemma 1.4.3 ([FH16]). Say that the empty word e and each one-letter word are both André permutations of the first kind. Next, if $w = w_1 w_2 \cdots w_n$ ($n \geq 2$) is a permutation of a set of positive integers $Y = \{y_1 < y_2 < \cdots < y_n\}$, write $w = v \min(w) v'$. Then, w is an André permutation of the first kind, if both v and v' are André permutation of the first kind, with $\max(v) < \max(v')$.

Define the exponential generating function

$$E(x; z) := \sum_{n \geq 0} \left(\sum_{\sigma \in \mathcal{A}_{n+1}^1} x^{\text{des}(\sigma)} \right) \frac{z^n}{n!}. \quad (1.90a)$$

Then, Eq. (1.89) with $\alpha = 1$ yields

$$E(x; z) = \left(\frac{\sqrt{2x-1} \sec(\frac{z}{2} \sqrt{2x-1})}{\sqrt{2x-1} - \tan(\frac{z}{2} \sqrt{2x-1})} \right)^2. \quad (1.90b)$$

We give a combinatorial proof of the following lemma, which can also be derived from (1.90b).

Lemma 1.4.4. We have

$$E(x; z) = \exp \left(z + x \int_0^z \left(\int_0^v E(x; u) du \right) dv \right). \quad (1.90c)$$

Proof. Let

$$E_n(x) := \sum_{\sigma \in \mathcal{A}_n^1} x^{\text{des}(\sigma)}. \quad (1.91a)$$

By Lemma 1.4.3, any permutation $w \in \mathcal{A}_{n+2}^1$ can be written as $w = v \mathbf{1} v'$ where v and v' are André permutations of the first kind. Let $j+1$ be the position of 1. Then, classifying the permutations in $\mathcal{A}_{n+2}^1$ by the index $j \in \{0, 1, \dots, n\}$, we have

$$E_{n+2}(x) = E_{n+1}(x) + x \sum_{j=1}^n \binom{n}{j} E_j(x) E_{n+1-j}(x), \quad (1.91b)$$

in other words,

$$\sum_{n \geq 0} E_{n+2}(x) \frac{z^n}{n!} = \left(\sum_{n \geq 0} E_{n+1}(x) \frac{z^n}{n!} \right) \left(1 + x \sum_{n \geq 1} E_n(x) \frac{z^n}{n!} \right). \quad (1.91c)$$

As $E(x; z) = \sum_{n \geq 0} E_{n+1}(x) \frac{z^n}{n!}$, the above equation can be written as

$$\frac{\partial E(x; z)}{\partial z} = E(x; z) \left(1 + x \int_0^z E(x; u) du \right), \quad (1.91d)$$

which is the equation obtained by differentiating the two sides of (1.90c) with respect to z . As the two sides of (1.90c) agree at $z = 0$, we are done. $\square$

Let $\mathcal{AC}_n$ be the set of André cycles of $[n]$, see Definition 1.22.

Lemma 1.4.5. We have

$$\sum_{n \geq 2} \sum_{\sigma \in \mathcal{AC}_n} x^{\text{drop}(\sigma)} \frac{z^n}{n!} = x \int_0^z \left(\int_0^t E(x; u) du \right) dt. \quad (1.92)$$

Proof. For $\sigma = (1 a_2 \dots a_{n-1} n) \in \mathcal{AC}_n$ with $n \geq 2$, let $\zeta(\sigma) = (a_2 - 1)(a_3 - 1) \cdots (n - 1) \in \mathcal{A}_{n-1}^1$. It is clear that the mapping $\zeta : \mathcal{AC}_n \rightarrow \mathcal{A}_{n-1}^1$ is a bijection satisfying

$$\text{drop}(\sigma) = \text{des}(\zeta(\sigma)) + 1. \quad (1.93)$$

Therefore,

$$\begin{aligned} \sum_{n \geq 2} \sum_{\sigma \in \mathcal{A}_n^c} x^{\text{drop}(\sigma)} \frac{z^n}{n!} &= \sum_{n \geq 2} \sum_{\sigma \in \mathcal{A}_{n-1}^1} x^{\text{des}(\sigma)+1} \frac{z^n}{n!} \\ &= x \sum_{n \geq 0} E_{n+1}(x) \frac{z^{n+2}}{(n+2)!}. \end{aligned} \quad (1.94)$$

By (1.90a), the last series is equal to the right-hand side of (1.92). $\square$

1.4.3 Proof of Theorem 1.23 (1.79a)

By Lemma 1.4.5 we have

$$\begin{aligned} \sum_{n \geq 1} \sum_{\sigma \in \mathcal{A}_n^c} x^{\text{drop}(\sigma)} t^{\text{fix}(\sigma)} \frac{z^n}{n!} &= tz + \sum_{n \geq 2} \sum_{\sigma \in \mathcal{A}_n^c} x^{\text{drop}(\sigma)} \frac{z^n}{n!} \\ &= (t-1)z + \left(z + x \int_0^z \left(\int_0^t E(x; u) du \right) dt \right). \end{aligned} \quad (1.95)$$

Recall that a cycle André permutation is a product of disjoint André cycles, and the set of cycle André permutations of $[n]$ is denoted by $\mathcal{CA}_n$. Combining (1.95), the exponential formula and Lemma 1.4.4 we obtain

$$\sum_{n \geq 0} \sum_{\sigma \in \mathcal{CA}_n} x^{\text{drop}(\sigma)} t^{\text{fix}(\sigma)} a^{\text{cyc}(\sigma)} \frac{z^n}{n!} = e^{\alpha(t-1)z} (E(x; z))^\alpha. \quad (1.96)$$

Substituting (1.90b) in (1.96) yields

$$\sum_{n \geq 0} \sum_{\sigma \in \mathcal{CA}_n} x^{\text{drop}(\sigma)} t^{\text{fix}(\sigma)} a^{\text{cyc}(\sigma)} \frac{z^n}{n!} = e^{(t-1)\alpha z} \left(\frac{\sqrt{2x-1} \sec(\frac{z}{2}\sqrt{2x-1})}{\sqrt{2x-1} - \tan(\frac{z}{2}\sqrt{2x-1})} \right)^{2\alpha}. \quad (1.97)$$

On the other hand, from Proposition 1.3.3 and Theorem 1.17 we derive the generating function

$$\sum_{n \geq 0} \sum_{j=0}^{\lfloor \frac{n}{2} \rfloor} d_{n,j}(\alpha, t) x^j \frac{z^n}{n!} = e^{(t-1)\alpha z} \left(\frac{\sqrt{2x-1} \sec(\frac{z}{2}\sqrt{2x-1})}{\sqrt{2x-1} - \tan(\frac{z}{2}\sqrt{2x-1})} \right)^{2\alpha}. \quad (1.98)$$

Eq. (1.79a) then follows by comparing the above two generating functions. $\square$

1.4.4 Proof of Theorem 1.23 (1.79b)

We define a map $\phi: \mathcal{CA}_n \rightarrow \mathcal{A}_{n+1}^1$ as follows: let $\sigma = C_1 C_2 \dots C_l \in \mathcal{CA}_n$, where $C_1, \dots, C_l$ are André cycles ordered from left to right in increasing order of their smallest letters, which are at the end of each cycle; erasing the parentheses in σ and appending letter $n+1$ to the end results in $\phi(\sigma)$.

It is not difficult to verify that ϕ is a bijection (See, for example, the proof of Theorem 4.1 in Hwang-Jang-Oh [HJO23]). Moreover, similar to Lemma 1.3.1, we have

$$(\text{drop}, \text{fix}, \text{cyc}) \sigma = (\text{des}, \text{rminda}, \text{rmin} - 1) \phi(\sigma). \quad (1.99)$$

Thus, combining (1.99) with (1.79a) we prove (1.79b). $\square$

Example 1.27. If $\sigma = (561)(7482)(3) \in \mathcal{CA}_8$, then $\sigma' := \phi(\sigma) = 561748239 \in \mathcal{A}_9^1$. Moreover,

- $\text{drop}(\sigma) = |\{6, 7, 8\}| = 3$, $\text{fix}(\sigma) = |\{3\}| = 1$ and $\text{cyc}(\sigma) = 3$;
- $\text{des}(\sigma') = |\{2, 4, 6\}| = 3$, $\text{rminda}(\sigma') = |\{3\}| = 1$ and $\text{rmin}(\sigma') = |\{1, 2, 3, 9\}| = 4$.

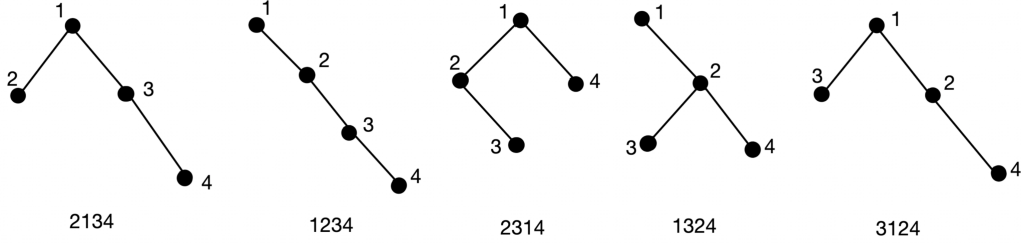


Figure 1.3: The André trees in $\mathcal{T}_4^1$ and associated André permutations in $\mathcal{A}_4^1$.

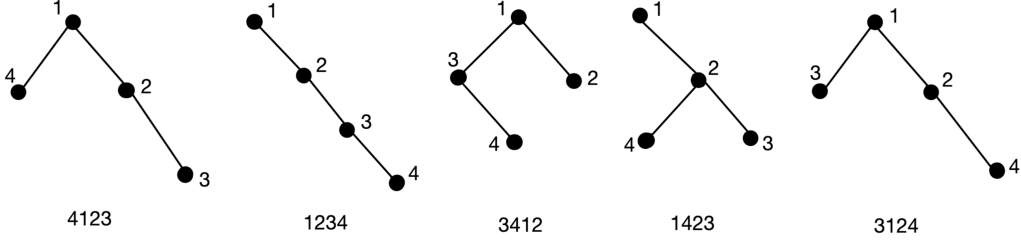


Figure 1.4: The André trees in $\mathcal{T}_4^2$ and associated André permutations in $\mathcal{A}_4^2$.

1.4.5 Proof of Theorem 1.23 (1.79c)

We construct a bijection $\Phi : \mathcal{A}_n^1 \rightarrow \mathcal{A}_n^2$, which preserves the triple statistic $(\text{des}, \text{rmin}, \text{rmin})$.

1.4.5.1 Binary and André trees

A binary tree T on a finite ordered set S is (recursively) defined as follows:

- if $S = \emptyset$, then $T = \emptyset$,
- otherwise, T is a triple (T_1, r, T_2) with $r \in S$ and T_1 and T_2 are binary trees on S_1 and S_2 , respectively such that $(\{r\}, S_1, S_2)$ is a partition of S (possibly with empty S_1 or S_2).

For the binary tree $T = (T_1, r, T_2)$ on S , the element r is called *root* of the tree, S the set of *nodes* or *vertices* of T , T_1 (resp. T_2) is the left (resp. right) subtree of T . A binary tree is called an increasing binary tree, if the vertices along any path from the root are increasing. A vertex of T without successor is called *leaf*, otherwise it is an *interior vertex*. Let $\text{leaf}(T)$ be the number of leaves of T .

Definition 1.28 ([Dis13]). An increasing binary tree T on a finite ordered set is an **André tree of the first (resp. second) kind**, if

- no vertex of T has only left successor;
- and if x is an interior vertex of T with left successor y and right successor z , then

$$\max(T(y)) < \max(T(z)) \text{ (resp. } y > z),$$

where $T(a)$ denotes the subtree rooted at a and $\max(T(a))$ is the largest vertex of $T(a)$.

The set of André trees of the first kind (resp. second kind) on $[n]$ is denoted $\mathcal{T}_n^1$ (resp. $\mathcal{T}_n^2$).

There is a well-known mapping sending permutations to increasing binary trees [Sta12, Section 1.5], which we recall below. Let $\sigma = \sigma_1 \cdots \sigma_n$ be a permutation of $[n]$. We define inductively a binary tree T_σ as follows. Let $x = \sigma_i$ be the smallest letter of σ , and factorize

$$\sigma = u x v, \quad \text{where } u = \sigma_1 \cdots \sigma_{i-1} \quad \text{and} \quad v = \sigma_{i+1} \cdots \sigma_n.$$

Then x is taken as the root of T_σ , the left subtree T_u is constructed recursively from u , and the right subtree T_v from v . If σ is empty, we define T_σ to be the empty tree.

The mapping $\omega : \sigma \mapsto T_\sigma$ is thus a bijection from $\mathfrak{S}_n$ onto the set of increasing binary trees on $[n]$. To recover σ from T_σ , one performs a *symmetric order traversal* (or *inorder traversal*), that is, a recursive visit of the vertices of T_σ defined as follows: first visit in symmetric order the left subtree of the root (if it exists); then visit the root itself; and finally visit in symmetric order the right subtree of the root (if it exists).

By restricting this mapping on André permutations, we obtain a bijection from André permutations to André trees:

$$\omega : \mathcal{A}_n^i \rightarrow \mathcal{T}_n^i, \quad i = 1, 2. \quad (1.100)$$

Example 1.29. The André trees in $\mathcal{T}_4^1$ and $\mathcal{T}_4^2$ and their associated permutations are given in Figs. 1.3 and 1.4, respectively.

Given an increasing binary tree T , the *right face* of T is the path from the root to a leaf by performing only right-steps. Denote by $\text{rface}(T)$ the number of vertices in the right face of T . Let $\text{rface}'(T)$ be the number of interior vertices having only a right successor in the right face of T . For example, in Fig. 1.5, we have $\text{rface}'(T_\sigma) = |\{4, 8\}| = 2$ and $\text{rface}(T_\sigma) = |\{1, 3, 4, 5, 8, 14\}| = 6$.

It is straightforward to verify the following properties, some of which can be found in [Sta12, Section 1.5].

Lemma 1.4.6. Let $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$ and $T_\sigma := \omega(\sigma)$, with $\sigma_0 = \sigma_{n+1} = 0$. Then,

- letter i is a double ascent of σ , iff vertex i of T_σ has only a right successor;
- letter i is a double descent of σ , iff vertex i of T_σ has only a left successor;
- letter i is a valley of σ , iff vertex i of T_σ has left and right successors;
- letter i is a peak of σ , iff vertex i of T_σ is a leaf;
- letter i is a right-to-left minimum of σ , iff vertex i is in the right face of T_σ .

Proposition 1.4.1. For $i = 1, 2$ and $\sigma \in \mathcal{A}_n^i$ with $T_\sigma = \omega(\sigma)$, the following relation holds:

$$(\text{des}, \text{rminda}, \text{rmin}) \sigma = (\text{leaf} - 1, \text{rface}', \text{rface}) T_\sigma. \quad (1.101)$$

Proof. In Definition 1.21, we define André permutations by the so-called x -factorizations of permutations. We only prove the $i = 1$ case. If $\sigma \in \mathcal{A}_n^1$, then $\text{dd}(\sigma) = 0$, so (see (1.9))

$$\text{des}(\sigma) = \text{val}(\sigma) + \text{dd}(\sigma) = \text{pk}(\sigma) - 1 = \text{leaf}(T_\sigma) - 1;$$

besides, it follows from Lemma 1.4.6 that $\text{rmin}(\sigma) = \text{rface}(T_\sigma)$ and $\text{rminda}(\sigma) = \text{rface}'(T_\sigma)$. $\square$

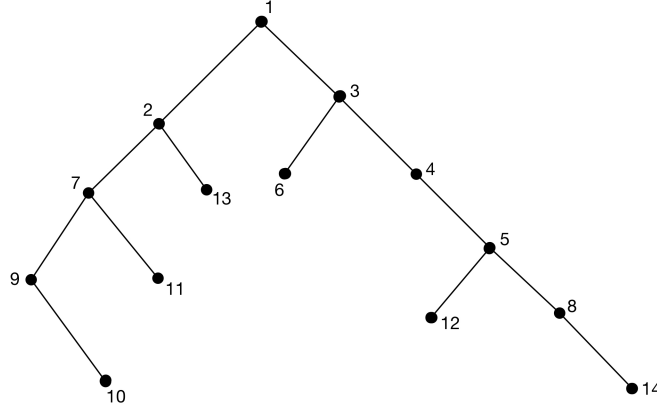


Figure 1.5: The binary tree T_σ with $\sigma = 9107112131634125814 \in \mathcal{A}_{14}^1$.

Example 1.30. If $\sigma = 9107112131634125814 \in \mathcal{A}_{14}^1$, then the binary tree T_σ is given in Fig. 1.5. Moreover,

- $\text{des}(\sigma) = |\{2, 4, 6, 8, 11\}| = 5$, $\text{rminda}(\sigma) = |\{4, 8\}| = 2$, and $\text{rmin}(\sigma) = |\{1, 3, 4, 5, 8, 14\}| = 6$;
- $\text{leaf}(T_\sigma) = |\{6, 10, 11, 12, 13, 14\}| = 6$, $\text{rface}'(T_\sigma) = |\{4, 8\}| = 2$, and $\text{rface}(T_\sigma) = |\{1, 3, 4, 5, 8, 14\}| = 6$.

1.4.5.2 Construction of bijection Φ .

Let $\mathcal{T}_n$ be the set of all increasing binary trees on $[n]$. First, for $\sigma = \sigma_1\sigma_2\cdots\sigma_n \in \mathfrak{S}_n$, let $T_\sigma = \omega(\sigma) \in \mathcal{T}_n$ and define

$$S_{T_\sigma} = \{i : \text{the vertex } \sigma_i \text{ has two successors in } T_\sigma\}. \quad (1.102)$$

Let $i \in S_{T_\sigma}$, and l_{σ_i} and r_{σ_i} denote the left and right successors of σ_i , respectively. Define the operator ϕ_i on T_σ as follows:

- if $l_{\sigma_i} > r_{\sigma_i}$, then $\phi_i T_\sigma = T_\sigma$;
- if $l_{\sigma_i} < r_{\sigma_i}$, then exchange l_{σ_i} with $\max(T(r_{\sigma_i}))$ to obtain a new tree T' , and then rearrange the vertices of $T'(l_{\sigma_i})$ and $T'(r_{\sigma_i})$ so that each subtree preserves its internal relative order.

Example 1.31. If $\sigma = 47281365910 \in \mathfrak{S}_{10}$, then $T_\sigma = \omega(\sigma)$ is shown at the left of Fig. 1.6. As $\sigma_5 = 1$ and $l_1 = 2 < r_1 = 3$, we have $\phi_5(T_\sigma) = T_\tau$ with $\tau = 78410125369$.

Lemma 1.4.7. Let $T_\sigma \in \mathcal{T}_n$ be an increasing tree with $\sigma = \sigma_1\cdots\sigma_n \in \mathfrak{S}_n$ and let $S_{T_\sigma} = \{i : \text{the vertex } \sigma_i \text{ has two successors in } T_\sigma\}$. Then, for any $i, j \in S_{T_\sigma}$, the operators ϕ_i and ϕ_j commute, namely,

$$\phi_i(\phi_j(T_\sigma)) = \phi_j(\phi_i(T_\sigma)). \quad (1.103)$$

Proof. If ϕ_i (or ϕ_j) is an identity, then Eq. (1.103) is trivial. Otherwise, without loss of generality, we assume that $\sigma_i > \sigma_j$. If σ_i is not a descendant of σ_j , then ϕ_i and ϕ_j operate on two disjoint vertex subsets of T_σ and the commutativity is obvious. We suppose that σ_i is a descendant of l_{σ_j} , and

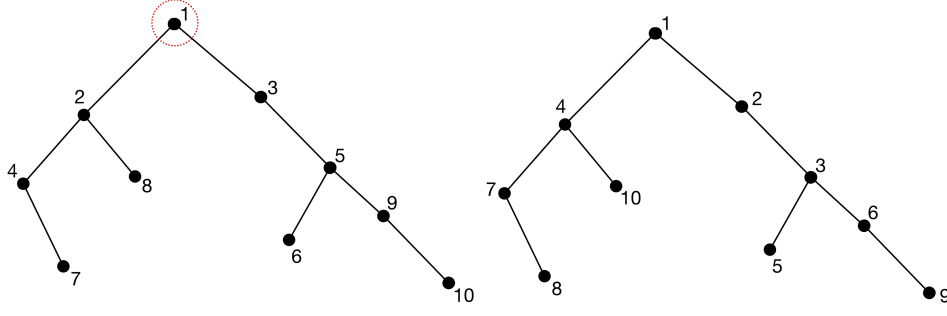


Figure 1.6: Two increasing binary trees T_σ with $\sigma = 47281365910$ (Left) and $\phi_5(T_\sigma) = T_\tau$ with $\tau = 78410125369$.

- the vertex set of $T(l_{\sigma_j})$ (resp. $T(r_{\sigma_j})$) is $A_{l_{\sigma_j}} := \{a_1 < a_2 < \dots < a_m\}$ (resp. $B_{r_{\sigma_j}} := \{b_1 < b_2 < \dots < b_l\}$) with $a_1 = l_{\sigma_j}$ (resp. $b_1 = r_{\sigma_j}$);
- the vertex set of $T(l_{\sigma_i})$ (resp. $T(r_{\sigma_i})$) is $A_{l_{\sigma_i}} := \{a_{t_1} < a_{t_2} < \dots < a_{t_p}\}$ (resp. $A_{r_{\sigma_i}} := \{a_{h_1} < a_{h_2} < \dots < a_{h_q}\}$) with $a_{t_1} = l_{\sigma_i}$ (resp. $a_{h_1} = r_{\sigma_i}$). Note that these two sets are subsets of $A_{l_{\sigma_j}}$.

1. We firstly consider $\phi_i(\phi_j(T_\sigma))$. By definition, the vertices of the $T(l_{\sigma_j})$ and $T(r_{\sigma_j})$ are changed according to the following substitutions,

$$\begin{array}{cccccccccccc} a_1 & a_2 & a_3 & \cdots & a_{m-1} & a_m & & b_1 & b_2 & b_3 & \cdots & b_{l-1} & b_l \\ \downarrow & \downarrow & \downarrow & \cdots & \downarrow & \downarrow & \text{and} & \downarrow & \downarrow & \downarrow & \cdots & \downarrow & \downarrow \\ a_2 & a_3 & a_4 & \cdots & a_m & b_l & & a_1 & b_1 & b_2 & \cdots & b_{l-2} & b_{l-1} \end{array} \quad (1.104)$$

In particular, as $A_{l_{\sigma_i}}$ and $A_{r_{\sigma_i}}$ are subsets of $A_{l_{\sigma_j}}$, under ϕ_j the vertices of $T(l_{\sigma_i})$ and $T(r_{\sigma_i})$ are changed as follows,

$$\begin{array}{cccccccc} a_{t_1} & a_{t_2} & \cdots & a_{t_{p-1}} & a_{t_p} & & a_{h_1} & a_{h_2} & \cdots & a_{h_{q-1}} & a_{h_q} \\ \downarrow & \downarrow & \cdots & \downarrow & \downarrow & \text{and} & \downarrow & \downarrow & \cdots & \downarrow & \downarrow \\ a_{t_1+1} & a_{t_2+1} & \cdots & a_{t_{p-1}+1} & a_{t_p+1} & & a_{h_1+1} & a_{h_2+1} & \cdots & a_{h_{p-1}+1} & a_{h_q+1} \end{array}, \quad (1.105)$$

where if $a_{h_q} = a_m$, then $a_{h_q+1} = b_l$. Next, proceeding by ϕ_i , the vertices of $\phi_j(T(l_{\sigma_i}))$ and $\phi_j(T(r_{\sigma_i}))$ are changed according to

$$\begin{array}{cccccccc} a_{t_1+1} & a_{t_2+1} & \cdots & a_{t_{p-1}+1} & a_{t_p+1} & & a_{h_1+1} & a_{h_2+1} & \cdots & a_{h_{q-1}+1} & a_{h_q+1} \\ \downarrow & \downarrow & \cdots & \downarrow & \downarrow & \text{and} & \downarrow & \downarrow & \cdots & \downarrow & \downarrow \\ a_{t_2+1} & a_{t_3+1} & \cdots & a_{t_p+1} & a_{h_q+1} & & a_{t_1+1} & a_{h_1+1} & \cdots & a_{h_{q-2}+1} & a_{h_{q-1}+1} \end{array} \quad (1.106)$$

2. Now, we consider $\phi_j(\phi_i(T_\sigma))$. Under ϕ_i , the vertices of $T(l_{\sigma_i})$ and $T(r_{\sigma_i})$ are changed according to the following substitutions,

$$\begin{array}{cccccccc} a_{t_1} & a_{t_2} & \cdots & a_{t_{p-1}} & a_{t_p} & & a_{h_1} & a_{h_2} & \cdots & a_{h_{q-1}} & a_{h_q} \\ \downarrow & \downarrow & \cdots & \downarrow & \downarrow & \text{and} & \downarrow & \downarrow & \cdots & \downarrow & \downarrow \\ a_{t_2} & a_{t_3} & \cdots & a_{t_p} & a_{h_q} & & a_{t_1} & a_{h_1} & \cdots & a_{h_{q-2}} & a_{h_{q-1}} \end{array} \quad (1.107)$$

Note that the remaining vertices are not changed under ϕ_i . Next, we proceed by ϕ_j , the vertices of $\phi_i(T(l_{\sigma_j}))$ and $\phi_i(T(r_{\sigma_j}))$ are changed by (1.104), then the vertices of $\phi_i(T(l_{\sigma_i}))$ and $\phi_i(T(r_{\sigma_i}))$ are changed as follows

$$\begin{array}{cccccccc} a_{t_2} & a_{t_3} & \cdots & a_{t_p} & a_{h_q} & & a_{t_1} & a_{h_1} & \cdots & a_{h_{q-2}} & a_{h_{q-1}} \\ \downarrow & \downarrow & \cdots & \downarrow & \downarrow & \text{and} & \downarrow & \downarrow & \cdots & \downarrow & \downarrow \\ a_{t_2+1} & a_{t_3+1} & \cdots & a_{t_p+1} & a_{h_q+1} & & a_{t_1+1} & a_{h_1+1} & \cdots & a_{h_{q-2}+1} & a_{h_{q-1}+1} \end{array} \quad (1.108)$$

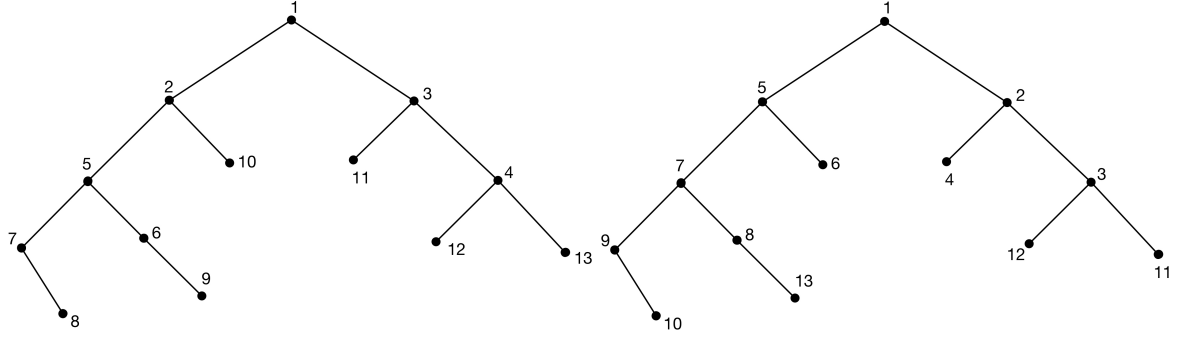


Figure 1.7: (Left) the André tree $T_\sigma \in \mathcal{T}_{13}^1$ with associated $\sigma = 78569210111312413 \in \mathcal{A}_{13}^1$; (Right) the transformed tree $T_\tau \in \mathcal{T}_{13}^2$ under $\Psi = \prod_{i \in S_{T_\sigma}} \phi_i$, where $\tau = 91078135614211312$ and $S_{T_\sigma} = \{3, 6, 8, 10, 12\}$.

Comparing the bottom rows (1.106) and (1.108), we prove (1.103). $\square$

By lemma 1.4.7, we can define the operator $\Psi := \prod_{i \in S_T} \phi_i$ for any $T \in \mathcal{T}_n^1$.

Lemma 1.4.8. The operator Ψ is a bijection from $\mathcal{T}_n^1$ to $\mathcal{T}_n^2$. Moreover, if $T_\sigma = \omega(\sigma)$ with $\sigma \in \mathcal{A}_n^1$, then

$$(\text{leaf}, \text{rfac}'', \text{rfac}) T_\sigma = (\text{leaf}, \text{rfac}'', \text{rfac}) \Psi(T_\sigma). \quad (1.109)$$

Proof. By Definition 1.28 and the construction of Ψ , for each interior vertex of $\Psi(T_\sigma)$ with two successors, its left successor is greater than its right successor. Hence $\Psi(T_\sigma) \in \mathcal{T}_n^2$. For $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$, let $T_\sigma = \omega(\sigma) \in \mathcal{T}_n$ and define

$$S_{T_\sigma} = \{i : \text{the vertex } \sigma_i \text{ has two successors in } T_\sigma\}. \quad (1.110)$$

If $i \in S_{T_\sigma}$, let l_{σ_i} and r_{σ_i} be the left and right successors of σ_i , respectively. We define the operator ϕ_i^{-1} on the vertices of T_σ as follows:

- if $\max(T(l_{\sigma_i})) < \max(T(r_{\sigma_i}))$, then $\phi_i^{-1} T_\sigma = T_\sigma$;
- if $\max(T(l_{\sigma_i})) > \max(T(r_{\sigma_i}))$, then exchange r_{σ_i} with $\max(T(l_{\sigma_i}))$ to obtain a new tree T' , and then rearrange the vertices of $T'(l_{\sigma_i})$ and $T'(r_{\sigma_i})$, so that they keep their same relative orders.

By a similar argument to that in the proof of Lemma 1.4.7, we can show that for any $i, j \in S_{T_\sigma}$ the operators ϕ_i^{-1} and ϕ_j^{-1} commute. Let $T_{\hat{\sigma}} := \Psi(T_\sigma) \in \mathcal{T}_n^2$, with

$$S_{T_{\hat{\sigma}}} = \{i : \hat{\sigma}_i \text{ is a vertex of } T_{\hat{\sigma}} \text{ with 2 successors}\}.$$

Since the operator Ψ merely permutes the labels of vertices of T_σ , we have $S_T = S_{T_{\hat{\sigma}}}$. Define $\Psi^{-1} := \prod_{i \in S_T} \phi_i^{-1}$, then

$$\Psi^{-1}(\Psi(T_\sigma)) = \prod_{i \in S_T} (\phi_i^{-1} \circ \phi_i)(T_\sigma) = \text{id}(T_\sigma) = T_\sigma.$$

Thus, Ψ is an injection from $\mathcal{T}_n^1$ to $\mathcal{T}_n^2$. As $|\mathcal{T}_n^1| = |\mathcal{T}_n^2| = E_n$, see [Dis13, FS73], hence Ψ is a bijection.

Since Ψ does not alter the structure the tree and preserves the relative order within each subtree, Eq. (1.109) follows immediately. $\square$

Proof of Theorem 1.23 (1.79c). It is clear that $\Phi := \omega^{-1} \circ \Psi \circ \omega$ is a bijection from $\mathcal{A}_n^1$ to $\mathcal{A}_n^2$. Combining (1.109) with Proposition 1.4.1, we prove (1.79c). $\square$

Example 1.32. If $\sigma = 78569210111312413 \in \mathcal{A}_{13}^1$, then $T_\sigma \in \mathcal{T}_{13}^1$, see Fig. 1.7. As $S_T = \{3, 6, 8, 10, 12\}$, let $\Psi = \prod_{i \in S_T} \phi_i$ we obtain $\Psi(T_\sigma) = T_\tau \in \mathcal{T}_{13}^2$ and finally

$$\tau = \omega^{-1} \circ \Psi \circ \omega(\sigma) = 91078135614211312 \in \mathcal{A}_{13}^2.$$

Remark 1.33. Foata and Han gave another bijection from $\mathcal{A}_n^1$ to $\mathcal{A}_n^2$ in [FH16, Section 3], but their bijection does not keep the triple statistic (des, rminda, rmin).

1.5 Link to successions

Let $\sigma \in \mathfrak{S}_n$, an index $i \in [n-1]$ is called a

- *succession* (**suc**) of σ if $\sigma(i+1) = \sigma(i) + 1$;
- *big ascent* (**basc**) of σ if $\sigma(i+1) \geq \sigma(i) + 2$.

Let $\text{Suc}(\sigma)$ and $\text{Basc}(\sigma)$ be the sets of successions and big ascents of σ , respectively. Furthermore, we define the sets

- $\text{Basc}_v(\sigma) := \{\sigma(i+1) : \sigma(i) + 1 < \sigma(i+1), i \in [n-1]\}$,
- $\text{Des}_v(\sigma) := \{\sigma(i+1) : \sigma(i) > \sigma(i+1), i \in [n-1]\}$,
- $\text{Suc}_v(\sigma) := \{\sigma(i+1) : \sigma(i) + 1 = \sigma(i+1), i \in [n-1]\}$.

For $\sigma \in \mathfrak{S}_n$ we define

$$\begin{aligned} \widehat{\text{Fix}}(\sigma) &:= \{i : \sigma(i) = i, i \in \{2, \dots, n\}\}, \\ \widehat{\text{Exc}}_v(\sigma) &:= \{\sigma(i) : \sigma(i) > i, i \in \{2, \dots, n\}\}, \\ \text{Drop}_v(\sigma) &:= \{\sigma(i) : \sigma(i) < i, i \in [n]\}, \end{aligned}$$

and denote the corresponding cardinalities by $\widehat{\text{fix}}(\sigma)$, $\widehat{\text{exc}}(\sigma)$ and $\text{drop}(\sigma)$, respectively.

We recall a bijection φ in [CHZ97, Section 5]. For $\sigma = \sigma_1 \sigma_2 \dots \sigma_n \in \mathfrak{S}_n$ we set $\sigma' = \sigma_2 \dots \sigma_n \sigma_1$ and define $\varphi(\sigma) = \theta_2(\sigma')$, where θ_2 is the second variant of the fundamental transformation, see Section 1.2.1.

Example 1.34. If $\sigma = 142836759 \in \mathfrak{S}_9$, then $\sigma' = 428367591$ and

$$\tilde{\sigma}' = (14389)(567)(2), \quad \varphi(\sigma) = 143895672.$$

We have $\widehat{\text{Exc}}_v(\sigma) = \{4, 8\}$, $\text{Drop}_v(\sigma) = \{2, 3, 5\}$, $\widehat{\text{Fix}}(\sigma) = \{6, 7, 9\}$, $\text{Basc}_v(\varphi(\sigma)) = \{4, 8\}$, $\text{Des}_v(\varphi(\sigma)) = \{2, 3, 5\}$, and $\text{Suc}_v(\varphi(\sigma)) = \{6, 7, 9\}$.

A pair of consecutive integers $(i, i+1)$ is called a *cyclic succession* of $\sigma \in \mathfrak{S}_n$ if $\sigma(i) = i+1$ with $i \in [n-1]$.

Theorem 1.35. *The mapping φ is a bijection on $\mathfrak{S}_n$ such that for all $\sigma \in \mathfrak{S}_n$, $\varphi(\sigma)(1) = \sigma(1)$ and*

$$(\widehat{\text{Exc}}_v, \text{Drop}_v, \widehat{\text{Fix}}) \sigma = (\text{Basc}_v, \text{Des}_v, \text{Suc}_v) \varphi(\sigma). \quad (1.111)$$

In particular, we have $(\widehat{\text{exc}}, \text{drop}, \widehat{\text{fix}}) \sigma = (\text{basc}, \text{des}, \text{suc}) \varphi(\sigma)$.

Proof. Clearly the mappings $\sigma \rightarrow \tilde{\sigma}'$ and $\sigma \rightarrow \varphi(\sigma)$ have the following properties:

- $i \in \widehat{\text{Fix}}(\sigma)$ if and only if the pair $(i-1, i)$ is a cyclic succession of $\tilde{\sigma}'$, which is equivalent to $i \in \text{Suc}_\nu(\varphi(\sigma))$.
- If $\sigma(i) \in \text{Drop}_\nu(\sigma)$, then $i-1 \geq \sigma(i)$. So the pair $(i-1, \sigma(i))$ is a cyclic descent or fixed point of $\tilde{\sigma}'$. This is equivalent to say that $\sigma(i) \in \text{Des}_\nu(\varphi(\sigma))$.
- If $\sigma(i) \in \widehat{\text{Exc}}_\nu(\sigma)$, then $(i-1)+1 < \sigma(i)$. This is equivalent to say that the pair $(i-1, \sigma(i))$ is a big ascent of $\varphi(\sigma)$.

Summarising we have (1.111). □

Remark 1.36. The bijection φ provides a combinatorial proof of Theorem 9 in [MQYY25].

Define the polynomials

$$F_n(x, y, t | \alpha) = \sum_{\sigma \in \mathfrak{S}_n} x^{\widehat{\text{exc}}(\sigma)} y^{\text{drop}(\sigma)} t^{\widehat{\text{fix}}(\sigma)} \alpha^{\text{cyc}(\sigma)}.$$

Theorem 1.37. For integer $n \geq 0$, we have

$$xF_{n+1}(x, y, t | \alpha) = A_{n+1}^{\text{cyc}}(x, y, t | \alpha) + \alpha(x-t)A_n^{\text{cyc}}(x, y, t | \alpha), \quad (1.112a)$$

where $A_0(x, y, t | \alpha) = 1$, and

$$\sum_{n \geq 0} F_{n+1}(x, y, t | \alpha) \frac{z^n}{n!} = \alpha e^{(y-t)z} \left(\frac{(x-y)e^{tz}}{xe^{yz} - ye^{xz}} \right)^{1+\alpha}. \quad (1.112b)$$

Proof. Note that

$$\begin{aligned} \sum_{\substack{\sigma \in \mathfrak{S}_{n+1} \\ \sigma(1)=1}} x^{\text{exc}(\sigma)} y^{\text{drop}(\sigma)} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} &= \alpha t A_n^{\text{cyc}}(x, y, t | \alpha), \\ \sum_{\substack{\sigma \in \mathfrak{S}_{n+1} \\ \sigma(1) > 1}} x^{\widehat{\text{exc}}(\sigma)} y^{\text{drop}(\sigma)} t^{\widehat{\text{fix}}(\sigma)} \alpha^{\text{cyc}(\sigma)} &= \alpha A_n^{\text{cyc}}(x, y, t | \alpha). \end{aligned}$$

According to the position of 1 in a permutation, which is 1 or greater than 1, we have

$$\begin{aligned} A_{n+1}^{\text{cyc}}(x, y, t | \alpha) &= \alpha t A_n^{\text{cyc}}(x, y, t | \alpha) + x \sum_{\substack{\sigma \in \mathfrak{S}_{n+1} \\ \sigma(1) > 1}} x^{\widehat{\text{exc}}(\sigma)} y^{\text{drop}(\sigma)} t^{\widehat{\text{fix}}(\sigma)} \alpha^{\text{cyc}(\sigma)}, \\ F_{n+1}(x, y, t | \alpha) &= \alpha A_n^{\text{cyc}}(x, y, t | \alpha) + \sum_{\substack{\sigma \in \mathfrak{S}_{n+1} \\ \sigma(1) > 1}} x^{\widehat{\text{exc}}(\sigma)} y^{\text{drop}(\sigma)} t^{\widehat{\text{fix}}(\sigma)} \alpha^{\text{cyc}(\sigma)}. \end{aligned}$$

Combining the above two equations yields recurrence (1.112a), which is equivalent to

$$x \sum_{n \geq 1} F_n(x, y, t | \alpha) \frac{z^n}{n!} = \sum_{n \geq 1} A_n^{\text{cyc}}(x, y, t | \alpha) \frac{z^n}{n!} + \alpha(x-t) \sum_{n \geq 1} A_{n-1}^{\text{cyc}}(x, y, t | \alpha) \frac{z^n}{n!}. \quad (1.113)$$

Invoking the generating function (1.7a) and applying $\frac{\partial}{\partial z}$ to (1.113) we obtain

$$x \sum_{n \geq 1} F_n(x, y, t | \alpha) \frac{z^{n-1}}{(n-1)!} = \left(\frac{\partial}{\partial z} + \alpha(x-t) \right) \left(\frac{(x-y)e^{tz}}{xe^{yz} - ye^{xz}} \right)^\alpha,$$

which can be shown (or by Maple) to be equivalent to (1.112b). □

Remark 1.38. By Theorem 1.35, when $y = \alpha = 1$, and substitute t by xt , Eq. (1.112b) reduces to Roselle's generating function for the statistic (asc, suc) [Ros68], namely,

$$\sum_{n \geq 0} \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{asc}(\sigma)} t^{\text{suc}(\sigma)} \frac{z^n}{n!} = \frac{(1-x)^2 e^{(xt+1)z}}{(e^{xz} - xe^z)^2}. \quad (1.114)$$

1.6 Concluding remarks

A permutation $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n$ of an ordered set is said to be *(increasing) alternating* (or *up-down*) if $\sigma_1 < \sigma_2 > \sigma_3 < \dots$. We denote by Alt_n the set of all up-down (alternating) permutations in $\mathfrak{S}_n$. A cycle (cyclic permutation) $(\sigma_1 \sigma_2 \dots)$ of an ordered set is said to be *up-down* if $\sigma_1 < \sigma_2 > \sigma_3 < \dots$ and σ_1 is the smallest element. A permutation π is said to be *cycle-up-down* if it is a product of disjoint up-down cycles. Let Δ_n be the set of cycle-up-down permutations of $[n]$. Deutsch and Elizalde [DE11] proved the following bivariate generating function

$$\sum_{n \geq 0} \sum_{\pi \in \Delta_n} t^{\text{fix}(\pi)} \alpha^{\text{cyc}(\pi)} \frac{z^n}{n!} = \frac{e^{\alpha(t-1)z}}{(1 - \sin z)^\alpha}. \quad (1.115)$$

When $x = 1$, Eq. (1.97) reduces to

$$\sum_{n \geq 0} \sum_{\sigma \in \mathcal{CA}_n} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} \frac{z^n}{n!} = \frac{e^{\alpha(t-1)z}}{(1 - \sin z)^\alpha}. \quad (1.116)$$

In [FH16, Section 2], Foata and Han constructed a bijection $\eta : \mathcal{A}_{n+1}^1 \rightarrow \text{Alt}_{n+1}$ preserving the first letter. Applying the mapping η to each cycle of $\sigma \in \mathcal{CA}_n$ immediately yields a bijection $\tilde{\eta} : \mathcal{CA}_n \rightarrow \Delta_n$ such that $(\text{fix}, \text{cyc}) \sigma = (\text{fix}, \text{cyc}) \tilde{\eta}(\sigma)$.

Hwang, Jang and Oh [HJO23, Corollary 4.4] proved the $t = 1$ case of (1.116). By Theorem 1.23, we also have

$$\sum_{n \geq 0} \sum_{\sigma \in \mathcal{A}_{n+1}^i} t^{\text{rminda}(\sigma)} \alpha^{\text{rmin}(\sigma)-1} \frac{z^n}{n!} = \frac{e^{\alpha(t-1)z}}{(1 - \sin z)^\alpha}, \quad (1.117)$$

where $i = 1, 2$.

Disanto [Dis13, Proposition 1] proved the $t = 1$ case of (1.117) and asked for a bijection from $\mathcal{A}_{n+1}^1$ to Δ_n enlightening the correspondence between rmin and cyc . In fact, combining the bijections $\phi^{-1} : \mathcal{A}_{n+1}^1 \rightarrow \mathcal{CA}_n$ in Section 1.4.4 and $\tilde{\eta} : \mathcal{CA}_n \rightarrow \Delta_n$, we obtain a bijection $\Xi := \tilde{\eta} \circ \phi^{-1} : \mathcal{A}_{n+1}^1 \rightarrow \Delta_n$ such that $(\text{rminda}, \text{rmin}) \sigma = (\text{fix}, \text{cyc}) \Xi(\sigma)$ for $\sigma \in \mathcal{A}_{n+1}^1$.

Problem 1. Find a suitable statistic $\widehat{\text{drop}}$ over Δ_n and bijection $\Lambda : \mathcal{CA}_n \rightarrow \Delta_n$ such that $\text{drop}(\sigma) = \widehat{\text{drop}}(\Lambda(\sigma))$ for $\sigma \in \mathcal{CA}_n$.

In view of the work [PZ21], where a (p, q) -analogue of (1.4) is proved, the following question is natural.

Problem 2. Find a combinatorial explanation of the identity $\gamma_{n,j}(\alpha, t) = 2^j d_{n,j}(\alpha, t)$ for $0 \leq j \leq \lfloor n/2 \rfloor$.

Remark 1.39. After the initial preprint of this chapter was posted, Dong, Lin and Pan [DLP25] have resolved the above problems.

Chapter 2

The (p, q) -analogues of the bi-Stirling Eulerian polynomials²

Abstract

We establish Jacobi-type continued fraction expansions for (p, q) -analogues of the bi-Stirling Eulerian polynomials, which refine classical Eulerian statistics on permutations. These polynomials admit equivalent definitions in terms of descents or excedances. The parameters p and q encode refined permutation statistics, including $(31-2, 2-13)$ and $(\text{cros}, \text{nest})$. Our approach is based on weighted Motzkin path models arising from variants of the bijections of Françon–Viennot and Foata–Zeilberger. As applications, we obtain new results on total positivity and γ -positivity for several families of enumerative polynomials. Our results unify and extend a number of recent works in the literature.

Keywords: Eulerian polynomials, continued fractions, Laguerre histories, gamma-positivity, total positivity

2.1 Introduction

The Eulerian polynomials are fundamental objects in enumerative combinatorics, as they encode the distribution of ascents and descents over permutations. Over the years, numerous refinements and generalizations have been introduced, notably the bi-Stirling Eulerian polynomials, which incorporate record-type statistics, and the (p, q) -Eulerian polynomials, which refine Eulerian statistics via generalized permutation patterns. The aim of the present paper is to combine these two directions by introducing (p, q) -analogues of the bi-Stirling Eulerian polynomials.

Our approach is based on continued fractions arising from weighted Motzkin path models. These models provide transparent combinatorial interpretations and lead naturally to explicit Jacobi-type continued fraction expansions for the ordinary generating functions under consideration.

²The material presented in this chapter is based on the preprint "The (p, q) -analogues of bi-Stirling Eulerian Polynomials via continued fractions", submitted for publication, (2026+).

More generally, our work fits naturally into the broad framework connecting permutation statistics, moment sequences, orthogonal polynomials, and continued fractions. In many situations, continued fractions arising from permutation statistics admit an interpretation as moment generating functions of suitable measures, thereby providing a bridge between enumerative combinatorics and the analytic theory of moments. This perspective has been developed, among others, by Blitvić and Steingrímsson [BS21] in their study of permutations, moments, and measures, and more recently by Sokal and Zeng [SZ22] through the introduction of multivariate master polynomials for permutations, set partitions, and perfect matchings together with their continued-fraction expansions. Building on the classical Flajolet–Viennot theory, Pétréolle, Sokal, and Zhu [PSZ23] extended Jacobi-type continued fractions to lattice paths and branched continued fractions, with applications to coefficientwise Hankel-total positivity.

In this context, weighted Motzkin paths and Laguerre histories provide a combinatorial realization of moment sequences, where the step weights encode refined permutation statistics. The Foata–Zeilberger correspondence and its variants therefore furnish a natural mechanism for translating permutation statistics into continued-fraction expansions, and will play a central role in the bijections and J -fraction formulas developed in this paper. The present work offers further evidence for the relevance of this framework by introducing new families of Eulerian-type polynomials whose generating functions admit explicit Jacobi-type continued-fraction expansions.

We now recall the necessary definitions and notation concerning permutation statistics and Eulerian-type polynomials. For a finite subset E of $\mathbb{N}$, let $\mathfrak{S}_E$ denote the set of all permutations of E , and let $\mathfrak{S}_n$ be the symmetric group on $[n] := \{1, 2, \dots, n\}$. For $\sigma \in \mathfrak{S}_n$, regarded as the word $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n$, where $\sigma_i = \sigma(i)$ for $i \in [n]$, an index $1 \leq i \leq n-1$ is called

- a *descent* (**des**) if $\sigma_i > \sigma_{i+1}$;
- an *ascent* (**asc**) if $\sigma_i < \sigma_{i+1}$.

Furthermore, a letter σ_i is said to be

- a *left-to-right maximum* (**lmax**) if $\sigma_i > \sigma_j$ for every $j < i$;
- a *right-to-left maximum* (**rmax**) if $\sigma_i > \sigma_j$ for every $j > i$.

Carlitz and Scoville [CS74] introduced the generalized Eulerian polynomials as the enumerative polynomials of the symmetric group $\mathfrak{S}_n$ with respect to ascent, descent, and record statistics, namely

$$A_n(x, y \mid \alpha, \beta) = \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{asc}(\sigma)} y^{\text{des}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}. \quad (2.1)$$

Note that $\text{asc}(\sigma) + \text{des}(\sigma) = n$ for $\sigma \in \mathfrak{S}_{n+1}$. Motivated by their combinatorial interpretation [XZ26], we refer to these polynomials as the *bi-Stirling Eulerian polynomials*, although they are termed the (α, β) -Eulerian polynomials in [Ji25].

Recently, Ji [Ji25] introduced a refinement of $A_n(x, y \mid \alpha, \beta)$, defined by

$$A_n(u_1, u_2, u_3, u_4 \mid \alpha, \beta) = \sum_{\sigma \in \mathfrak{S}_{n+1}} (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}. \quad (2.2)$$

In the refinement (2.2), the four parameters u_1, u_2, u_3, u_4 distinguish valleys, peaks, double ascents, and double descents, while the parameters α and β record left-to-right and right-to-left maxima. This refined description will later be incorporated into a single weight function

$\mathbf{wt}(\sigma)$, together with additional statistics, including a D-statistic $D(\sigma)$ naturally arising from the Françon–Viennot bijection.

Recall that for $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$, with the convention $\sigma_0 = \sigma_{n+1} = 0$, a letter σ_i with $i \in [n]$ is called a

- *valley* (**val**) if $\sigma_{i-1} > \sigma_i < \sigma_{i+1}$;
- *peak* (**pk**) if $\sigma_{i-1} < \sigma_i > \sigma_{i+1}$;
- *double ascent* (**da**) if $\sigma_{i-1} < \sigma_i < \sigma_{i+1}$;
- *double descent* (**dd**) if $\sigma_{i-1} > \sigma_i > \sigma_{i+1}$.

These statistics satisfy

$$\text{asc} = \text{val} + \text{da}, \quad \text{des} = \text{val} + \text{dd}, \quad \text{val} = \text{pk} - 1. \quad (2.3)$$

The exponential generating function of $A_n(u_1, u_2, u_3, u_4 \mid \alpha, \beta)$ was obtained by Ji [Ji25, Theorem 1.4]. As observed in [XZ26], her formula can be written in the following form

$$\sum_{n \geq 0} A_n(u_1, u_2, u_3, u_4 \mid \alpha, \beta) \frac{z^n}{n!} = e^{(\alpha u_3 + \beta u_4)z} \left(\frac{x - y}{xe^{yz} - ye^{xz}} \right)^{\alpha + \beta}, \quad (2.4)$$

where the parameters x and y are related by $x + y = u_3 + u_4$ and $xy = u_1 u_2$. The sequel [XZ24] established a further generalization (2.4) and derived a J -fraction expansion for the ordinary generating function. In particular, building on earlier work of Zeng [Zen93], we obtained the continued fraction formula

$$\sum_{n=0}^{\infty} A_n(u_1, u_2, u_3, u_4 \mid \alpha, \beta) z^n = \frac{1}{1 - b_0 z - \frac{\lambda_1 z^2}{1 - b_1 z - \frac{\lambda_2 z^2}{1 - b_2 z - \cdots}}}, \quad (2.5a)$$

where $b_0 = \alpha u_3 + \beta u_4$,

$$b_k = k(u_3 + u_4) + (\alpha u_3 + \beta u_4), \quad \lambda_k = k(k - 1 + \alpha + \beta) u_1 u_2 \quad (k \geq 1). \quad (2.5b)$$

In this paper, we will generalize the above continued fraction formula by using the combinatorial theory of continued fractions developed by Flajolet [Fla80] and Viennot [Vie83], together with variants of the bijections of Françon–Viennot [FV79] and Foata–Zeilberger [FZ90] between permutations and weighted Motzkin paths. This approach allows us to derive explicit Jacobi-type continued fraction expansions and to obtain new consequences concerning positivity properties of the associated enumerative polynomials.

2.2 Main results

For $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$ with $\sigma_0 = \sigma_{n+1} = 0$, a letter $\sigma_i \in [n]$ is said to be a

- *left-to-right-maximum-peak* (**lmaxpk**) if σ_i is a left-to-right maximum and also a peak;

- *right-to-left-maximum-peak* (**rmaxpk**) if σ_i is a right-to-left maximum and also a peak;
- *left-to-right-maximum-double-ascent* (**lmaxda**) if σ_i is a left-to-right maximum and also a double ascent;
- *right-to-left-maximum-double-descent* (**rmaxdd**) if σ_i is a right-to-left maximum and also a double descent.

Example 2.1. Let $\sigma = 27159108436 \in \mathfrak{S}_{10}$, we have

$$\text{pk}(\sigma) = |\{6, 7, 10\}| = 3, \text{val}(\sigma) = |\{1, 3\}| = 2, \text{da}(\sigma) = |\{2, 5, 9\}| = 3, \text{dd}(\sigma) = |\{4, 8\}| = 2,$$

$$\text{lmax}(\sigma) = |\{2, 7, 9, 10\}| = 4, \quad \text{rmax}(\sigma) = |\{6, 8, 10\}| = 3,$$

$$\text{lmaxda}(\sigma) = |\{2, 9\}| = 2, \quad \text{rmaxdd}(\sigma) = |\{8\}| = 1,$$

$$\text{lmaxpk}(\sigma) = |\{7, 10\}| = 2, \text{ and } \text{rmaxpk}(\sigma) = |\{6, 10\}| = 2.$$

As a follow-up to [Ji25] the authors [XZ24] generalized (2.2) with three additional variables as follows:

$$A_n(\mathbf{u}, f, g, t | \alpha, \beta) = \sum_{\sigma \in \mathfrak{S}_{n+1}} \mathbf{wt}(\sigma), \quad (2.6a)$$

where $\mathbf{u} = (u_1, u_2, u_3, u_4)$ and

$$\begin{aligned} \mathbf{wt}(\sigma) := & (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1} \\ & \times f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)}. \end{aligned} \quad (2.6b)$$

Let $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$ with $\sigma_0 = \sigma_{n+1} = 0$. Denote by $\text{Val}(\sigma)$ and $\text{Pk}(\sigma)$ the sets of valleys and peaks of σ , respectively. We define the D statistic of σ by

$$D(\sigma) := \sum_{i \in \text{Pk}(\sigma) \setminus \{n\}} i - \sum_{j \in \text{Val}(\sigma)} j. \quad (2.7)$$

This statistic corresponds to the area under the associated Motzkin path via the Françon–Viennot bijection, see Lemma 2.3.4. Note that a shifted version of D statistic appeared in [Fra92].

Example 2.2. Let $\sigma = 42157368 \in \mathfrak{S}_8$. Then $\text{Val}(\sigma) = \{1, 3\}$, $\text{Pk}(\sigma) = \{4, 7, 8\}$, and hence $D(\sigma) = 4 + 7 - 1 - 3 = 7$.

The notion of generalized (dashed) permutation patterns was introduced by Babson and Steingrímsson [BS00]. Recall that the pattern 31–2 (resp. 2–13) occurs in a permutation σ if there exist indices $i < j$ such that the entries in positions i and $i + 1$ (resp. j and $j + 1$) are consecutive in σ , and

$$\sigma(i) > \sigma(j) > \sigma(i + 1) \quad (\text{resp. } \sigma(j) < \sigma(i) < \sigma(j + 1)).$$

We denote by $(31-2)(\sigma)$ (resp. $(2-13)(\sigma)$) the number of occurrences of the pattern 31–2 (resp. 2–13) in σ ; see [SZ12]. Accordingly, we set

$$31-2(\sigma) := (31-2)(\sigma) \quad \text{and} \quad 2-13(\sigma) := (2-13)(\sigma).$$

Similarly we define the pattern 2–31 and the number $(2-31)(\sigma)$.

Example 2.3. Let $\sigma = 4736215 \in \mathfrak{S}_7$. Then

$$\begin{aligned} (31-2)(\sigma) &= |\{73-6, 73-5, 62-5\}| = 3, \\ (2-31)(\sigma) &= |\{4-73, 4-62, 3-62\}| = 3, \\ (2-13)(\sigma) &= |\{4-36, 4-15, 3-15, 2-15\}| = 4. \end{aligned}$$

Shin and one of us [SZ12] proved a continued fraction formula for the ordinary generating function of the following (p, q) -analogue of generalized Eulerian polynomials:

$$A_n(u_1, u_2, u_3, u_4; p, q) = \sum_{\sigma \in \mathfrak{S}_n} (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} p^{2-13(\sigma)} q^{31-2(\sigma)} \quad (2.8)$$

and derived the γ -positivity formula using combinatorics of continued fractions. Combining (2.2) and (2.8), we define the (p, q) -analogue of the generalized Eulerian polynomials as follows:

$$A_n(\mathbf{u}, f, g, h, t | \alpha, \beta; p, q) := \sum_{\sigma \in \mathfrak{S}_{n+1}} \mathbf{wt}(\sigma) h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)}. \quad (2.9)$$

We define the generalized (p, q) -analogue of a positive integer $k \in \mathbb{N}^*$, with auxiliary parameters α and β , as follows:

$$[k; \alpha, \beta]_{p,q} := (\alpha - 1)p^{k-1} + (\beta - 1)q^{k-1} + \frac{p^k - q^k}{p - q}. \quad (2.10a)$$

Note that

$$[1; \alpha, \beta]_{p,q} = \alpha + \beta - 1,$$

and

$$[k]_{p,q} := [k; 1, 1]_{p,q} = \frac{p^k - q^k}{p - q}. \quad (2.10b)$$

Let $\mathbf{b} := (b_k)_{k \geq 0}$ and $\boldsymbol{\lambda} := (\lambda_k)_{k \geq 1}$ be elements of a commutative ring. We define the Jacobi-type continued fraction $\mathcal{J}(t; \mathbf{b}, \boldsymbol{\lambda})$ as a formal power series by

$$\mathcal{J}(t; \mathbf{b}, \boldsymbol{\lambda}) := \frac{1}{1 - b_0 t - \frac{\lambda_1 t^2}{1 - b_1 t - \frac{\lambda_2 t^2}{1 - b_2 t - \ddots}}}. \quad (2.11)$$

Theorem 2.4. We have

$$\sum_{n=0}^{\infty} A_n(\mathbf{u}, f, g, h, t | \alpha, \beta; p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.12a)$$

where

$$b_k = h^k \left(u_3 [k+1; 1, \alpha t]_{p,q} + u_4 [k+1; \beta t, 1]_{p,q} \right), \quad (2.12b)$$

$$\lambda_k = u_1 u_2 h^{2k-1} [k]_{p,q} [k+1; \beta g, \alpha f]_{p,q}. \quad (2.12c)$$

Remark 2.5. The special case $f = g = h = t = p = q = 1$ of (2.12) is Eq. (2.5a). When $\alpha = \beta = h = 1$, other generalizations of the above continued fraction were considered in [SS96, BS21, SZ22].

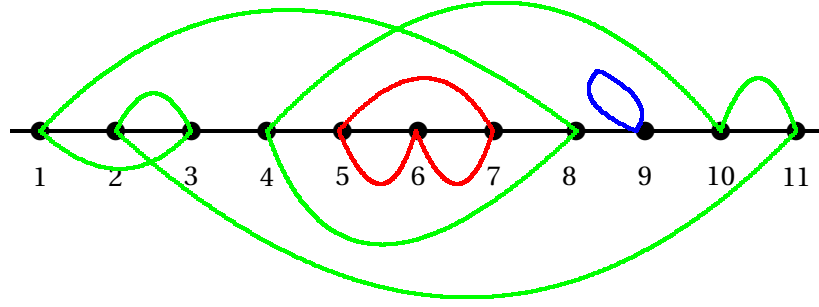


Figure 2.1: The arc-diagram of the permutation $\sigma = 8311075649112$, or as product of disjoint cycles $\sigma = (1, 8, 4, 10, 11, 2, 3) (5, 7, 6) (9)$.

Let $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$. An *inversion* of σ is a pair of indices (i, j) such that $1 \leq i < j \leq n$ and $\sigma_i > \sigma_j$. The *inversion number* of σ , denoted by $\text{inv}(\sigma)$, is the total number of inversions in σ . Petersen and Tenner [PT15] introduced a statistic called *depth* for Coxeter groups, defined in terms of factorizations of group elements into reflections. In the case of the symmetric group $\mathfrak{S}_n$, the reflections are the transpositions $(i j)$ with $1 \leq i < j \leq n$. The *depth* of a permutation $\sigma \in \mathfrak{S}_n$, denoted by $\text{depth}(\sigma)$, is defined as

$$\text{depth}(\sigma) = \min \left\{ \sum_{r=1}^k (j_r - i_r) : \sigma = (i_1 j_1)(i_2 j_2) \cdots (i_k j_k) \right\}. \quad (2.13)$$

They obtained an alternative formula

$$\text{depth}(\sigma) = \sum_{\sigma(i) > i} (\sigma(i) - i). \quad (2.14)$$

Thus, this statistic coincides with the statistic *Edif* over permutations in [CSZ97] and corresponds to the area under the Motzkin path via the bijection of Foata-Zeilberger, see Lemma 2.3.6.

Corteel [Cor07] defined the *crossing* and *nesting* numbers of $\sigma \in \mathfrak{S}_n$ as follows:

$$\text{cros}(\sigma) = \#\{(i, j) \in [n] \times [n] : (i < j \leq \sigma_i < \sigma_j) \vee (i > j > \sigma_i > \sigma_j)\}; \quad (2.15a)$$

$$\text{nest}(\sigma) = \#\{(i, j) \in [n] \times [n] : (i < j \leq \sigma_j < \sigma_i) \vee (i > j > \sigma_j > \sigma_i)\}. \quad (2.15b)$$

These statistics are better illustrated by drawing an arc-diagram as follows: put the numbers from 1 to n on a line and draw an edge from i to σ_i above the line if i is a weak excedance, and below the line otherwise.

Example 2.6. Let $\sigma = 8311075649112 \in \mathfrak{S}_{11}$. Then we have $\text{nest}(\sigma) = 9$ and $\text{cros}(\sigma) = 3$; see Fig. 2.1.

For $\sigma = \sigma(1) \dots \sigma(n) \in \mathfrak{S}_n$, we say that an index $i \in [n]$ is a

- *fixed point* (**fix**) of σ if $i = \sigma(i)$;
- *cycle peak* (**cpk**) of σ if $\sigma^{-1}(i) < i > \sigma(i)$;
- *cycle valley* (**cval**) of σ if $\sigma^{-1}(i) > i < \sigma(i)$;
- *cycle double ascent* (**cda**) of σ if $\sigma^{-1}(i) < i < \sigma(i)$;
- *cycle double descent* (**cdd**) of σ if $\sigma^{-1}(i) > i > \sigma(i)$.

Note that $\text{cpk}(\sigma) = \text{cval}(\sigma)$. A letter $\sigma_i := \sigma(i)$ is said to be a right-to-left minimum (**rmin**) of σ if $\sigma_j > \sigma_i$ and $n \geq j > i \geq 1$. Define the general cycle version of Eulerian polynomials

$$C_n(\mathbf{u}, t, s, h | \alpha, \beta; p, q) := \sum_{\sigma \in \mathfrak{S}_n} (u_1 u_2)^{\text{cval}(\sigma)} u_3^{\text{cda}(\sigma)} u_4^{\text{cdd}(\sigma)} t^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} \\ \times h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma)} \beta^{\text{rmin}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)}. \quad (2.16a)$$

Theorem 2.7. *We have*

$$1 + \sum_{n \geq 1} C_n(\mathbf{u}, t, s, h | \alpha, \beta; p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.16b)$$

where $b_0 = t\alpha\beta$, and

$$b_k = (sh)^k \left(tq^k s^k + u_4[k; \beta, 1]_{p,qs} + u_3 p[k; \alpha, 1]_{p,qs} \right), \quad (2.16c)$$

$$\lambda_k = u_1 u_2 (sh)^{2k-1} [k; \alpha, 1]_{p,qs} [k; \beta, 1]_{p,qs}. \quad (2.16d)$$

Remark 2.8. Theorem 2.2 specializes to Theorem 6.5 of [dMV94] when $\alpha = p = q = h = t = 1$, after a suitable change of variables. When $u_1 = u_4 = s = h = \alpha = \beta = 1$ and $u_2 = u_3 = t = x$, Theorem 2.7 reduces to Eq. (1) in [Cor07].

Remark 2.9. Petersen [Pet] conjectured the special case of the above J -fraction expansion corresponding to the pair of statistics (inv, depth); this conjecture was recently proved in [EFL24, SS24]. In fact, this J -fraction also follows from Theorem 10 or Corollary 11 of [CSZ97] after the successive substitutions $p \rightarrow q$ and then $q \rightarrow zq$.

Remark 2.10. In Theorem 2.7, applying the following substitutions, respectively,

- $\mathbf{u} = (x, 1, x, 1)$ and $(\alpha, \beta, p, q) = (1, 1, 1, 1)$,
- $\mathbf{u} = (x, 1, x, 1)$ and $(t, p, q, \beta) = (tx, 1, 1, 1)$,
- $\mathbf{u} = (x, 1, x, 1)$ and $(t, p, q, \alpha) = (1, 1, 1, 1)$,

we recover Theorem 1.1, (i) and (ii) of Theorem 1.3 in [EFL26].

Definition 2.11. Let $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$. If $i \in [n-1]$ is a descent, then σ_i is called a descent-top, and σ_{i+1} is called a descent-bottom. For $i \in [n]$, the entry σ_i is called a nondescent or a nondescent-top (resp. nondescent-bottom) of σ , if it is not a descent-top (resp. descent-bottom) of σ . The number of nondescents of σ is $\text{ndes}(\sigma) := n - \text{des}(\sigma)$.

Let $\sigma \in \mathfrak{S}_n$. We denote the sum of the nondescent-tops (resp. nondescent-bottoms) of σ by $\text{NDtop}(\sigma)$ (resp. $\text{NDbot}(\sigma)$) and define the *nondescent difference* of σ (see [SZ10]) by

$$\text{NDdif}(\sigma) := \text{NDbot}(\sigma) - \text{NDtop}(\sigma). \quad (2.17)$$

If we impose the boundary condition $\sigma_0 = 0$ and $\sigma_{n+1} = n+1$, then each nondescent-top (resp. nondescent-bottom) of $\sigma \in \mathfrak{S}_n$ is either a valley or a double ascent (resp. a peak or a double descent), it follows that

$$\text{NDdif}(\sigma) = \sum_{i \in \text{Pk}'(\sigma)} i - \sum_{j \in \text{Val}'(\sigma)} j, \quad (2.18)$$

where $\text{Pk}'(\sigma)$ and $\text{Val}'(\sigma)$ denote the sets of peaks and valleys of σ , respectively, with boundary condition $\sigma_0 = 0$ and $\sigma_{n+1} = n+1$.

Example 2.12. Let $\sigma = 42157368 \in \mathfrak{S}_8$. We have

$$\text{NDtop}(\sigma) = 1 + 3 + 5 + 6 + 8 = 23,$$

$$\text{NDbot}(\sigma) = 4 + 5 + 6 + 7 + 8 = 30.$$

Then $\text{NDdif}(\sigma) = 7$. By (2.18), we have $\text{NDdif}(\sigma) = (4 + 7) - (1 + 3) = 7$.

Definition 2.13. Let $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$. A letter $\sigma_i \in [n]$ is called a

- foremaximum of σ if σ_i is a nondescent top and also a left-to-right maximum in σ ;
- postminimum of σ if σ_i is a nondescent bottom and also a right-to-left minimum in σ .

The number of foremaxima (resp. postminima) of σ is denoted by $\text{fmax}(\sigma)$ (resp. $\text{pmin}(\sigma)$).

Example 2.14. For $\sigma = 42157368 \in \mathfrak{S}_8$, we have $\text{fmax}(\sigma) = |\{5, 8\}| = 2$ and $\text{pmin}(\sigma) = |\{6, 8\}| = 2$.

Recall [SZ10] that for $\sigma \in \mathfrak{S}_n$ the statistics MAD and MADL are defined by

$$\text{MAD}(\sigma) := \text{des}(\sigma) + (31-2)(\sigma) + 2(2-31)(\sigma); \quad (2.19a)$$

$$\text{MADL}(\sigma) := \text{des}(\sigma) + (2-31)(\sigma) + 2(31-2)(\sigma). \quad (2.19b)$$

For $\sigma \in \mathfrak{S}_n$, an index $i \in [n]$ is called

- an *excedance* (**exc**) of σ if $i < \sigma(i)$,
- a *weak excedance* (**wex**) of σ if $i \leq \sigma(i)$,
- a *drop* (**drop**) if $\sigma(i) < i$.

Clarke, Steingrímsson and Zeng [CSZ97] constructed a bijection Φ_{CSZ} on $\mathfrak{S}_n$ to characterize the composition $\Phi_{FZ}^{-1} \circ \Phi_{FV}$. Later, Corteel [Cor07] and Shin–Zeng [SZ10] introduced two variants of the bijection Φ_{CSZ} on $\mathfrak{S}_n$, denoted by Φ_C and Φ_{SZ} , respectively. These bijections are known to send the generalized pattern statistic

$$(31-2, 2-31)$$

to the pairs (cros, nest) and (nest, cros) on $\mathfrak{S}_n$. Beyond this correspondence, they admit a natural interpretation in terms of the weighted path model underlying our continued-fraction expansions. In particular, the following result shows that Φ_C and Φ_{SZ} preserve additional structural features compatible with the Motzkin-path encoding and the associated Jacobi-type continued fractions. Related aspects of these constructions have recently been investigated in [HMZ20, CF23, CFZ25].

Theorem 2.15. *The mappings Φ_C and Φ_{SZ} are bijections on $\mathfrak{S}_n$ such that for all $\sigma \in \mathfrak{S}_n$, we have*

$$(\text{ndes}, \text{pmin}, 31-2, 2-31, \text{MADL}, \text{NDdif}) \sigma = (\text{wex}, \text{fix}, \text{nest}, \text{cros}, \text{inv}, \text{depth}) \Phi_C(\sigma); \quad (2.20)$$

$$(\text{ndes}, \text{fmax}, 31-2, 2-31, \text{MAD}, \text{NDdif}) \sigma = (\text{wex}, \text{fix}, \text{cros}, \text{nest}, \text{inv}, \text{depth}) \Phi_{SZ}(\sigma). \quad (2.21)$$

Moreover, if $\Phi^* := \Phi_{SZ} \circ \Phi_C^{-1}$, then

$$(\text{wex}, \text{nest}, \text{cros}, \text{depth}) \sigma = (\text{wex}, \text{cros}, \text{nest}, \text{depth}) \Phi^*(\sigma). \quad (2.22)$$

Any polynomial with real coefficients $h(x) = \sum_{i=0}^n h_i x^i$ satisfying the symmetry condition $h_i = h_{n-i}$ can be written uniquely in the form

$$h(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_k x^k (1+x)^{n-2k}.$$

The coefficients γ_k are called the *γ -coefficients* of $h(x)$. If all γ_k are nonnegative, then $h(x)$ is said to be *γ -positive*. We refer the reader to the general overview of γ -positivity phenomena in [Ath18]. We now consider two specializations of the polynomials $A_n(\mathbf{u}, f, g, h, t | \alpha, \beta; p, q)$:

$$A_n(x, f, g, h | p, q) := A_n((1, x, 1, x), f, g, h, 1 | 1, 1; p, q); \quad (2.23a)$$

$$B_n(x, f, g, h | t, \alpha) := A_n((1, x, 1, x), f, g, h, t | \alpha, \alpha; 1, 1). \quad (2.23b)$$

By (2.9), we obtain

$$A_n(x, f, g, h | p, q) = \sum_{\sigma \in \mathfrak{S}_{n+1}} \mathbf{w}(\sigma) p^{31-2(\sigma)} q^{2-13(\sigma)}, \quad (2.23c)$$

$$B_n(x, f, g, h | t, \alpha) = \sum_{\sigma \in \mathfrak{S}_{n+1}} \mathbf{w}(\sigma) t^{\text{lmaxda}(\sigma) + \text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2}, \quad (2.23d)$$

where the weight $\mathbf{w}(\sigma)$ is defined by

$$\mathbf{w}(\sigma) = x^{\text{des}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)}. \quad (2.23e)$$

By the J -fraction expansion of Theorem 2.4, we show that the polynomials $A_n(x, f, g, h | p, q)$ and $B_n(x, f, g, h | t, \alpha)$ admit natural γ -positive decompositions. The next result provides an explicit combinatorial interpretation of the γ -coefficients in terms of permutations without double descents, refined by generalized pattern statistics.

Theorem 2.16. *For $n \geq 0$, the polynomial $A_n(x, f, g, h | p, q)$ is γ -positive and can be expanded as*

$$A_n(x, f, g, h | p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^a(f) x^k (1+x)^{n-2k}, \quad (2.24)$$

where

$$\gamma_{n,k}^a(f) = \sum_{\sigma \in \mathfrak{S}_{n+1, \text{dd}=0, \text{des}=k}} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)}, \quad (2.25)$$

and

$$\mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0} = \{\sigma \in \mathfrak{S}_n : \text{dd}(\sigma) = 0 \text{ and } \text{des}(\sigma) = k\}.$$

Remark 2.17. When $f = g = h = 1$, this expansion reduces to a classical γ -positive decomposition of Eulerian-type polynomials. An equivalent formulation of (2.24) was obtained by Brändén [Brä08, Eq. (5.1)] via the modified Foata–Strehl action. The present approach derives the γ -coefficients directly from the continued-fraction structure.

Theorem 2.18. *For $n \geq 0$, we have*

$$B_n(x, f, g, h | t, \alpha) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^b(f) x^k (1+x)^{n-2k}, \quad (2.26)$$

where

$$\gamma_{n,k}^b(f) = \sum_{\sigma \in \mathfrak{S}_{n+1, \text{des}=k}^{\text{dd}=0}} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} t^{\text{lmaxda}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2}, \quad (2.27)$$

with $\mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0} = \{\sigma \in \mathfrak{S}_n : \text{dd}(\sigma) = 0 \text{ and } \text{des}(\sigma) = k\}$.

Remark 2.19. When $f = g = h = 1$, we recover (3.28c) of Proposition 3.10 in [XZ24].

Postnikov, Reiner, and Williams [PRW08] proved that the h -polynomials of dual stellohedra has the combinatorial interpretation

$$\tilde{A}_n(x) = \sum_{\sigma \in \text{PRW}_{n+1}} x^{\text{des}(\sigma)},$$

where PRW_n is the set of permutations in $\mathfrak{S}_n$ such that the first descent is at the letter n . For example,

$$\text{PRW}_2 = \{12, 21\} \quad \text{and} \quad \text{PRW}_3 = \{123, 132, 231, 312, 321\}.$$

Clearly, a permutation σ of $[n]$ is in PRW_n if and only if n is simultaneously the unique left-to-right maximum and a peak of σ , namely, $\text{lmaxpk}(\sigma) = 1$. By (2.23), we have

$$A_n(x, 0, g, h \mid p, q) = \sum_{\sigma \in \text{PRW}_{n+1}} x^{\text{des}(\sigma)} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)}, \quad (2.28a)$$

$$B_n(x, 0, g, h \mid t, \alpha) = \sum_{\sigma \in \text{PRW}_{n+1}} x^{\text{des}(\sigma)} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2}. \quad (2.28b)$$

The following result is an immediate consequence of Theorems 2.16 and 2.18.

Theorem 2.20. *We have*

$$A_n(x, 0, g, h \mid p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^a(0) x^k (1+x)^{n-2k}, \quad (2.28c)$$

$$B_n(x, 0, g, h \mid t, \alpha) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^b(0) x^k (1+x)^{n-2k}, \quad (2.28d)$$

where

$$\gamma_{n,k}^a(0) = \sum_{\sigma \in \text{PRW}_{n+1, k}^{\text{dd}=0}} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)}, \quad (2.28e)$$

$$\gamma_{n,k}^b(0) = \sum_{\sigma \in \text{PRW}_{n+1, k}^{\text{dd}=0}} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} t^{\text{lmaxda}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2} \quad (2.28f)$$

with

$$\text{PRW}_{n,k}^{\text{dd}=0} := \{\sigma \in \text{PRW}_n : \text{dd}(\sigma) = 0 \text{ and } \text{des}(\sigma) = k\}. \quad (2.28g)$$

Remark 2.21. Special cases of (2.28d) were obtained in [XZ24, Proposition 3.8] and [JL24, Theorem 2.1].

Consider the two special cases of the polynomials $C_n(\mathbf{u}, t, s, h \mid \alpha, \beta; p, q)$, see (2.16a),

$$\begin{aligned} C_n^1(x, t, s, h \mid \alpha, p, q) &:= C_n((x, 1, x/p, 1), t(1+x), s, h \mid \alpha, \alpha; p, q) \\ &= \sum_{\sigma \in \mathfrak{S}_n} \mathbf{w}^{\mathbf{c}}(\sigma) p^{\text{cros}(\sigma)-\text{cda}(\sigma)} q^{\text{nest}(\sigma)}, \end{aligned} \quad (2.29a)$$

and

$$\begin{aligned} C_n^2(x, t, s, h | \alpha, p, q) &:= C_n((x, p, x, p), t(1+x), s, h | \alpha, \alpha; p, q) \\ &= \sum_{\sigma \in \mathfrak{S}_n} \mathbf{w}^c(\sigma) p^{\text{cros}(\sigma) + \text{drop}(\sigma)} q^{\text{nest}(\sigma)}, \end{aligned} \quad (2.29b)$$

where

$$\mathbf{w}^c(\sigma) = x^{\text{exc}(\sigma)} (t(1+x))^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{rmin}(\sigma) + \text{lmax}(\sigma)}. \quad (2.29c)$$

By the J -fraction of Theorem 2.7, we show that the polynomials $C_n^1(x, t, s, h | \alpha, p, q)$ and $C_n^2(x, t, s, h | \alpha, p, q)$ admit the following γ -expansions.

Theorem 2.22. *For $n \geq 0$, we have*

$$C_n^i(x, t, s, h | \alpha, p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^i(t) x^k (1+x)^{n-2k}, \quad (i = 1, 2) \quad (2.30)$$

where

$$\gamma_{n,k}^1(t) = \sum_{\sigma \in \mathfrak{S}_{n, \text{exc}=k}^{\text{cda}=0}} t^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmin}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)} \quad (2.31)$$

$$\gamma_{n,k}^2(t) = \sum_{\sigma \in \mathfrak{S}_{n, \text{exc}=k}^{\text{cda}=0}} t^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmin}(\sigma)} p^{\text{cros}(\sigma) + \text{drop}(\sigma)} q^{\text{nest}(\sigma)} \quad (2.32)$$

with

$$\mathfrak{S}_{n, \text{exc}=k}^{\text{cda}=0} := \{\sigma \in \mathfrak{S}_n : \text{cda}(\sigma) = 0 \text{ and } \text{exc}(\sigma) = k\}. \quad (2.33)$$

Remark 2.23. The special case $\alpha = s = h = 1$ of Eq. (2.30) with $i = 2$ is equivalent to Eq. (4.14) of Theorem 4.8 in [LWZ19].

The next corollary corresponds to the derangement specialization of the γ -positive decomposition arising from the J -fraction in Theorem 2.4. By setting the fixed-point parameter to zero, the continued-fraction structure naturally restricts to weighted derangements, yielding an explicit combinatorial interpretation of the γ -coefficients. Define

$$C_n^1(x, 0, s, h | \alpha, p, q) = \sum_{\sigma \in \mathcal{D}_n} x^{\text{exc}(\sigma)} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmin}(\sigma)} p^{\text{cros}(\sigma) - \text{cda}(\sigma)} q^{\text{nest}(\sigma)}, \quad (2.34)$$

$$C_n^2(x, 0, s, h | \alpha, p, q) = \sum_{\sigma \in \mathcal{D}_n} x^{\text{exc}(\sigma)} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma) + \text{rmin}(\sigma)} p^{\text{cros}(\sigma) + \text{drop}(\sigma)} q^{\text{nest}(\sigma)}, \quad (2.35)$$

where $\mathcal{D}_n := \{\sigma \in \mathfrak{S}_n : \text{fix}(\sigma) = 0\}$ is the set of derangements of $[n]$.

Corollary 2.2.1. *For $n \geq 0$, we have*

$$C_n^i(x, 0, s, h | \alpha, p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^i(0) x^k (1+x)^{n-2k}, \quad (i = 1, 2) \quad (2.36)$$

where

$$\gamma_{n,k}^1(0) = \sum_{\sigma \in \mathcal{D}_{n,k}^0} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{rmin}(\sigma) + \text{lmax}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)}, \quad (2.37)$$

$$\gamma_{n,k}^2(0) = \sum_{\sigma \in \mathcal{D}_{n,k}^0} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{rmin}(\sigma) + \text{lmax}(\sigma)} p^{\text{cros}(\sigma) + \text{drop}(\sigma)} q^{\text{nest}(\sigma)}, \quad (2.38)$$

with $\mathcal{D}_{n,k}^0 := \{\sigma \in \mathcal{D}_n : \text{cda}(\sigma) = 0 \text{ and } \text{exc}(\sigma) = k\}$.

Remark 2.24. When $h = \alpha = q = 1$, Eq. (2.36) reduces to Theorem 2 of [SZ16].

The remainder of this paper is organized as follows. In Section 2.3, we begin with a detailed review of the Françon–Viennot bijection and the Foata–Zeilberger bijection between permutations and Laguerre histories, a class of weighted Motzkin paths. We then employ these bijections to prove Theorems 2.4 and 2.7 using Flajolet’s fundamental theorem on J -continued fractions.

In Section 2.4, we recall the bijections of Shin–Zeng and Corteel on permutations and use them to establish Theorem 2.15. Section 2.5 is devoted to the γ -positive expansions derived from the continued fractions of Theorems 2.4 and 2.7. In particular, we obtain γ -expansions for the (p, q) -analogues of the generalized Eulerian polynomials (Theorems 2.16, 2.18, and 2.22). We also present two distinct factorizations of the corresponding γ -coefficients together with combinatorial interpretations in terms of André permutations of types 1 and 2; see Theorems 2.44, 2.47, and 2.48.

In Section 2.6, we apply the Pan–Zeng group action to prove Theorem 2.44 and the Dong–Lin–Pan group action to prove Theorem 2.47. Section 2.7 investigates the total positivity of the (p, q) -analogues of generalized Eulerian polynomials via continued fractions. Finally, in Section 2.8, we answer a question of Eu et al. by constructing a bijection that makes explicit the symmetry between two variants of the nest and cros statistics. We also derive several compact closed-form expressions obtained by specializing one of the parameters in the associated continued fractions to -1 .

2.3 Encoding permutations by Laguerre histories

2.3.1 Preliminaries

A *Motzkin path* of length n is a lattice path $\{(x_i, y_i)\}_{0 \leq i \leq n}$ in the plane that

- starts at $(x_0, y_0) = (0, 0)$ and ends at $(x_n, y_n) = (n, 0)$;
- uses only the three types of steps

$$N = (1, 1) \text{ (North-East)}, \quad S = (1, -1) \text{ (South-East)}, \quad E = (1, 0) \text{ (East)};$$

- never goes below the x -axis, that is, the height $h_i := y_{i-1} \geq 0$ for all $i \in [n]$.

Let $\mathcal{M}_n$ denote the set of Motzkin paths of length n .

Following the classical Flajolet–Viennot approach, we consider a *weighted Motzkin-path model*. Each East step at height i is assigned a weight a_i , while each North-East (resp. South-East) step at height i receives a weight b_i (resp. c_i). The weight of a path $P \in \mathcal{M}_n$ is the product of the weights of its steps and is denoted by $\omega(P)$. The generating function of weighted Motzkin paths admits a Jacobi-type continued-fraction expansion, given by Flajolet’s fundamental lemma on continued fractions [Fla80].

Lemma 2.3.1. We have

$$\sum_{n \geq 0} \sum_{P \in \mathcal{M}_n} \omega(P) z^n = \frac{1}{1 - b_0 z - \frac{a_0 c_1 z^2}{1 - b_1 z - \frac{a_1 c_2 z^2}{\ddots}}}. \quad (2.39)$$

For a permutation $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$ and $k = \sigma_j$ with $j \in [n]$, the coordinate generalized pattern statistics are defined by

$$\begin{aligned} (31-2)_k(\sigma) &= \#\{i : i+1 < j \text{ and } \sigma_{i+1} < k < \sigma_i\}; \\ (2-13)_k(\sigma) &= \#\{i : j < i-1 \text{ and } \sigma_{i-1} < k < \sigma_i\}; \\ (2-31)_k(\sigma) &= \#\{i : j < i-1 \text{ and } \sigma_i < k < \sigma_{i-1}\}. \end{aligned} \quad (2.40)$$

Clearly, for $\sigma \in \mathfrak{S}_n$, the statistic associated with the generalized pattern 31-2 decomposes as

$$(31-2)(\sigma) = \sum_{k=1}^n (31-2)_k(\sigma).$$

Similarly,

$$(2-13)(\sigma) = \sum_{k=1}^n (2-13)_k(\sigma), \text{ and } (2-31)(\sigma) = \sum_{k=1}^n (2-31)_k(\sigma).$$

Example 2.25. Let $\sigma = 4723516 \in \mathfrak{S}_7$. We have

$$(31-2)_3(\sigma) = 1, (2-13)_4(\sigma) = 2 \text{ and } (2-31)_4(\sigma) = 2.$$

We can identify a Motzkin path of length n by a Motzkin word $w = w_1 \dots w_n$ such that $w_i = \text{N}$ (resp. S, E) if and only if the i th step is N (resp. S, E). Given a Motzkin path $w = w_1 w_2 \cdots w_n \in \mathcal{M}_n$, the height h_i of the i th step is given by

$$h_i := h_i(w) = \#\{j < i : w_j = \text{N}\} - \#\{j < i : w_j = \text{S}\}. \quad (2.41)$$

2.3.1.1 Françon-Viennot bijection

A *2-Motzkin path* is a Motzkin path whose east steps could come in two colors: red (E_1) or blue (E_2).

Definition 2.26 (Laguerre history). A Laguerre history of length n is a couple (w, p) , where $w = w_1 \cdots w_n$ is a 2-Motzkin path of length n starting at $(0, 0)$ and ending at $(n, 0)$, and $p = (p_1, \dots, p_n)$ is a label sequence such that $0 \leq p_i \leq h_i$ for $1 \leq i \leq n$. Denote by $\mathcal{L}_n$ the set of Laguerre histories of length n .

We describe a variant of Françon-Viennot's bijection $\Psi_{FV} : \mathfrak{S}_{n+1} \rightarrow \mathcal{L}_n$ and its inverse, see [SZ12, Vie83]. For a permutation $\sigma \in \mathfrak{S}_{n+1}$, the Laguerre history $\Psi_{FV}(\sigma) := (w_1 \dots w_n, (p_1, \dots, p_n)) \in \mathcal{L}_n$ is defined by

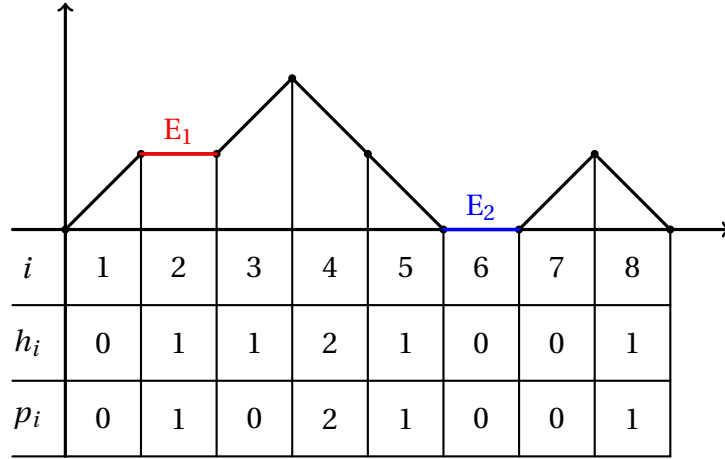
$$w_i = \begin{cases} \text{N (resp. S)} & \text{if } i \text{ is a valley (resp. peak) of } \sigma; \\ \text{E}_1 \text{ (resp. E}_2\text{)} & \text{if } i \text{ is a double descent (resp. double ascent) of } \sigma, \end{cases} \quad (2.42)$$

and $p_i = (31-2)_i(\sigma)$ for each $i \in [n]$.

Inversely, starting from a Laguerre history $(w, p) \in \mathcal{L}_n$ one can recover the corresponding permutation σ by the following algorithm:

- Initialization: $\sigma = \diamond$;
- At the i th ($1 \leq i \leq n$) step of the algorithm, replace the $(p_i + 1)$ th $\diamond$ (from left to right) of σ by

$$\begin{cases} \diamond i \diamond & \text{if } w_i = \text{N}; \\ i \diamond & \text{if } w_i = \text{E}_2; \\ i & \text{if } w_i = \text{S}; \\ \diamond i & \text{if } w_i = \text{E}_1. \end{cases}$$

Figure 2.2: The Laguerre history $\Psi_{FV}(\sigma)$, where $\sigma = 697835142$.

- The final permutation is obtained by putting $n + 1$ at the last remaining $\diamond$.

As a consequence, the cardinality of $\mathcal{L}_n$ is $(n + 1)!$.

Example 2.27. Let $\sigma = 697835142 \in \mathfrak{S}_9$. We have

$$\Psi_{FV}(\sigma) = (\text{NE}_1\text{NSS}\text{E}_2\text{NS}, (0, 1, 0, 2, 1, 0, 0, 1)) \in \mathcal{L}_8,$$

see Fig. 2.2.

Inversely, we have

$$\begin{aligned} \sigma &= \diamond \rightarrow \diamond 1 \diamond \rightarrow \diamond 1 \diamond 2 \rightarrow \diamond 3 \diamond 1 \diamond 2 \rightarrow \diamond 3 \diamond 142 \rightarrow \diamond 35142 \rightarrow 6 \diamond 35142 \\ &\rightarrow 6 \diamond 7 \diamond 35142 \rightarrow 6 \diamond 7835142 \rightarrow 697835142. \end{aligned}$$

There is another more visual description of patterns $(31-2)_x$ and $(2-13)_x$ using the so-called x -decomposition; see [FV79]. For letter $x \in [n]$, the x -decomposition of σ is the unique factorization

$$\sigma = u_1 v_1 \cdots u_k v_k u_{k+1}, \quad k \geq 1, \quad (2.43)$$

in which the words u_i and v_j , except possibly u_1 and u_{k+1} , are nonempty. Moreover, each u_i ($1 \leq i \leq k + 1$) consists solely of letters $< x$, while each v_j ($1 \leq j \leq k$) consists solely of letters $\geq x$. Assume that x lies inside v_j . Then every adjacent pair (v_i, u_{i+1}) with $i \leq j - 1$ contributes to a $(31-2)_x(\sigma)$, while every adjacent pair (u_i, v_i) with $i \geq j$ contributes to a $(2-13)_x(\sigma)$. In Fig 2.3, we see that

$$(31-2)_x(\sigma) = j - 1 \quad \text{and} \quad (2-13)_x(\sigma) = k - j.$$

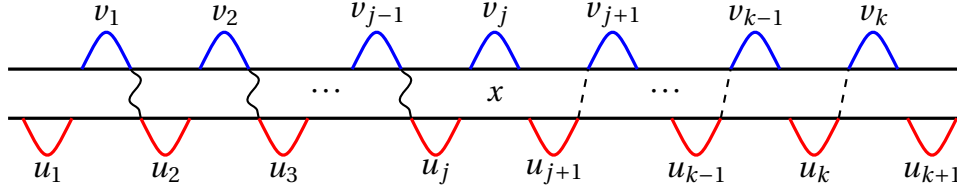
Example 2.28. If $\sigma = 64195381027 \in \mathfrak{S}_{10}$ and $x = 5$. Then the 5-decomposition is

$$\sigma = u_1 v_1 u_2 v_2 u_3 v_3 u_4 v_4,$$

with

$$(u_1, u_2, u_3, u_4) = (\varepsilon, 41, 3, 2) \quad \text{and} \quad (v_1, v_2, v_3, v_4) = (6, 95, 810, 7).$$

Thus, we have $(31-2)_5(\sigma) = 1$ and $(2-13)_5(\sigma) = 2$.

Figure 2.3: The x -decomposition of σ .

Lemma 2.3.2. Given a permutation $\sigma \in \mathfrak{S}_{n+1}$ and the corresponding Laguerre history $\Psi_{FV}(\sigma) = (w, p) \in \mathcal{L}_n$. For any $x \in [n]$, if h_x is the height of the x th step of w , then

$$(31-2)_x(\sigma) + (2-13)_x(\sigma) = h_x. \quad (2.44)$$

Proof. Suppose that the x -decomposition of σ is

$$\sigma = u_1 v_1 \cdots u_k v_k u_{k+1}, \quad k \geq 1, \quad (2.45)$$

where x lies inside v_j . According to Definition 2.41 and the construction of Ψ_{FV} in (2.42), it follows that the height of the x th step of w is

$$h_x = \sum_{j=1}^{k+1} (\text{val}(u_j) - \text{pk}(u_j)),$$

where $\text{val}(u_j)$ (resp. $\text{pk}(u_j)$) denotes the number of valleys (resp. peaks) in u_j . It is straightforward to verify that

- $\text{val}(u_j) = \text{pk}(u_j)$ for $j = 1, k+1$;
- $\text{val}(u_j) = \text{pk}(u_j) + 1$ for $2 \leq j \leq k$.

Hence we obtain $h_x(\sigma) = k - 1$. Every adjacent pair (v_i, u_{i+1}) with $i \leq j - 1$ contributes to a $(31-2)_x(\sigma)$, while every adjacent pair (u_i, v_i) with $i \geq j$ contributes to a $(2-13)_x(\sigma)$. Thus, h_x also can be calculated by using the patterns $(31-2)_x(\sigma)$ and $(2-13)_x(\sigma)$, that is (2.44).

Now we prove Eq. (2.44) by induction for all $x \in [n]$. For $x = 1$, it is obvious that $(31-2)_1(\sigma) + (2-13)_1(\sigma) = 0 = h_1$. If $x > 1$, it suffices to show that

$$(31-2)_x(\sigma) + (2-13)_x(\sigma) = (31-2)_{x-1}(\sigma) + (2-13)_{x-1}(\sigma) + v, \quad (2.46)$$

where $v \in \{0, 1, -1\}$. Assume that the $(x-1)$ -decomposition of σ is

$$\sigma = u_1 v_1 \cdots u_k v_k u_{k+1}, \quad k \geq 1,$$

and $(x-1)$ is in v_j . In view of this $(x-1)$ -decomposition, we see that $(31-2)_{x-1}(\sigma) = j - 1$, $(2-13)_{x-1}(\sigma) = k - j$ and $(31-2)_{x-1}(\sigma) + (2-13)_{x-1}(\sigma) = k - 1$.

- For case $v = 1$. If $(x-1)$ is a valley of σ , and x is also a valley,
 - then x is in v_l where $1 \leq l < j$ (resp. $j < l \leq k$). In this case, $(x-1)$ together with its immediate right (resp. left) neighbor contributes 1 to $(2-13)_x(\sigma)$ (resp. $(31-2)_x(\sigma)$). Hence, we have

$$(31-2)_x(\sigma) + (2-13)_x(\sigma) = (l-1) + (k-l) + 1 = k.$$

- Or x is in v_j . Suppose that $v_j = v_j^l(x-1)v_j^r$ where v_j^l and v_j^r are (contiguous) words consisting of entries strictly larger than $x-1$. Both v_j^l and v_j^r are nonempty. If $x \in v_j^l$, then $(x-1)$ together with its immediate right neighbor contributes 1 to $(2-13)_x(\sigma)$; if $x \in v_j^r$, then $x-1$ together with its immediate left neighbor contributes 1 to $(31-2)_x(\sigma)$. Hence, we have

$$(31-2)_x(\sigma) + (2-13)_x(\sigma) = (j-1) + (k-j) + 1 = k.$$

Analogous arguments hold in the cases when x is a peak, a double ascent, or a double descent of σ .

- For case $v = -1$. If $(x-1)$ is a peak of σ , then from the $(x-1)$ -decomposition we see that v_j contains only $(x-1)$. Assume that x is a valley of σ , and that x is in v_l where $1 \leq l < j$ (resp. $j < l \leq k$). In the x -decomposition of σ , components u_j , $v_j = x-1$ and u_{j+1} will form a new u -word. Therefore, compared with the $(x-1)$ -decomposition, the x -decomposition has one fewer u -word. Hence, we have

$$(31-2)_x(\sigma) + (2-13)_x(\sigma) = (l-1) + (k-l) - 1 = k-2.$$

Analogous arguments hold in the cases when x is a peak, a double ascent, or a double descent of σ .

- For case $v = 0$. If $(x-1)$ is a double descent (resp. double ascent) of σ , then $(x-1)$ is the last (resp. first) element of v_j . Assume that x is a valley of σ and that x is in v_l where $1 \leq l \leq k$. In these cases, it is easy to check that there is no change in the total number of v_i in both $(x-1)$ -decomposition and x -decomposition of σ . Hence, we have

$$(31-2)_x(\sigma) + (2-13)_x(\sigma) = (l-1) + (k-l) = k-1.$$

Analogous arguments hold in the cases when x is a peak, a double ascent, or a double descent of σ . Combining these cases, we obtain (2.46). □

For a Motzkin path w , we define $\text{area}(w)$ to be the geometric area of the region between the x -axis and the Motzkin path w , denoted by $\text{area}(w)$. We have the following characterization for area .

Lemma 2.3.3. Let a Motzkin path $w \in \mathcal{M}_n$. The area of w is given by

$$\text{area}(w) = \sum_{i=1}^n h_i, \tag{2.47}$$

where h_i is the height of the i th step of w defined in (2.41).

Proof. We compute $\text{area}(w)$ by the step heights of the Motzkin path $w \in \mathcal{M}_n$. The contribution of the i th step equals the area of the trapezoid of width 1 whose vertical sides have lengths h_i and $h_i + \Delta_i$, where $\Delta_i \in \{1, 0, -1\}$ is the vertical increment of that step. It is clear that the i th step contributes $h_i + \frac{\Delta_i}{2}$ to $\text{area}(w)$, therefore

$$\text{area}(w) = \sum_{i=1}^n \left(h_i + \frac{\Delta_i}{2} \right) = \sum_{i=1}^n h_i + \frac{1}{2} \sum_{i=1}^n \Delta_i.$$

Since w starts and ends on the x -axis, its total vertical increment is $\sum_{i=1}^n \Delta_i = 0$, and the above equation yields (2.47). □

Lemma 2.3.4. For $\sigma \in \mathfrak{S}_{n+1}$, let $\Psi_{FV}(\sigma) := (w, p) \in \mathcal{L}_n$ be the corresponding Laguerre history. Then

$$D(\sigma) = \text{area}(w). \quad (2.48)$$

Proof. Let $w = w_1 \cdots w_n \in \mathcal{M}_n$. According to the construction of Ψ_{FV} in (2.42), we see that

$$D(\sigma) = \sum_{w_i=S} i - \sum_{w_j=N} j. \quad (2.49)$$

For each South-East step $w_i = S$, match it with the unique North-East step $w_j = N$ with $j < i$ such that the path between steps w_j and w_i stays strictly above height h_j , and $h_i = h_j + 1$. The matched pair of steps $w_i = S$ and $w_j = N$ determines a horizontal strip of area $i - j$ lying within the Motzkin path. Summing over all matched pairs gives

$$\text{area}(w) = \sum_{w_i=S} i - \sum_{w_j=N} j. \quad (2.50)$$

Combining this with Eq. (2.49) yields (2.48). $\square$

Example 2.29. Let $\sigma = 697835142 \in \mathfrak{S}_9$. The corresponding Motzkin path is $w = \text{NE}_1\text{NSSSE}_2\text{NS}$; see Fig. 2.2. We have

$$\begin{aligned} D(\sigma) &= (8 + 5 + 4) - (7 + 3 + 1) = 6, \\ \text{area}(w) &= \sum_{i=1}^8 h_i = 4 \cdot 1 + 2 = 6. \end{aligned}$$

2.3.1.2 Foata-Zeilberger bijection

A *3-Motzkin path* is a Motzkin path whose east steps could come in three colors: red (E_1), blue (E_2) and green (E_3).

Definition 2.30 (3-restricted Laguerre history). A 3-restricted Laguerre history of length n is a couple (w, p) , where $w = w_1 \cdots w_n$ is a 3-Motzkin path of length n starting at $(0, 0)$ and ending at $(n, 0)$, and $p = (p_1, \dots, p_n)$ is a label sequence satisfying $1 \leq i \leq n$,

$$0 \leq p_i \leq \begin{cases} h_i, & \text{if } w_i \in \{N, \text{E}_2, \text{E}_3\}; \\ h_i - 1, & \text{if } w_i \in \{S, \text{E}_1\}, \end{cases}$$

where h_i is the height of the i th step of w . Denote by $\mathcal{L}_n^*$ the set of 3-restricted Laguerre histories of length n .

For a permutation $\sigma \in \mathfrak{S}_n$, we describe Foata-Zeilberger's bijection $\Psi_{FZ}(\sigma) := (w, p) \in \mathcal{L}_n^*$ and its inverse. The associated 3-restricted Laguerre history

$$\Psi_{FZ}(\sigma) = ((w_1, \dots, w_n), (p_1, \dots, p_n)) \in \mathcal{L}_n^*$$

is defined as follows. The step w_i is given by

$$w_i = \begin{cases} N \text{ (resp. } S) & \text{if } i \text{ is a cycle valley (resp. cycle peak) of } \sigma; \\ \text{E}_1 \text{ (resp. } \text{E}_2) & \text{if } i \text{ is a cycle double descent (resp. cycle double ascent) of } \sigma; \\ \text{E}_3 & \text{if } i \text{ is a fixed point of } \sigma, \end{cases} \quad (2.51)$$

and the integer p_i is defined by

$$p_i := \begin{cases} \#\{j < i : \sigma(j) > \sigma(i)\}, & \text{if } \sigma(i) \geq i; \\ \#\{j > i : \sigma(j) < \sigma(i)\}, & \text{if } \sigma(i) < i. \end{cases} \quad (2.52)$$

For a permutation $\sigma = \sigma(1) \cdots \sigma(n) \in \mathfrak{S}_n$, we say that i (resp. σ_i) is the excedance bottom (resp. top) of σ if $i < \sigma(i)$, and that i (resp. σ_i) is the nonexcedance top (resp. bottom) of σ if $i \geq \sigma(i)$. Let $\text{Etop}(\sigma)$, $\text{Ebot}(\sigma)$, $\text{NEtop}(\sigma)$ and $\text{NEbot}(\sigma)$ denote the sets of the excedance tops, excedance bottoms, nonexcedance tops and nonexcedance bottoms of σ , respectively. Let $\text{Cval}(\sigma)$, $\text{Cpk}(\sigma)$, $\text{Cda}(\sigma)$, $\text{Cdd}(\sigma)$ and $\text{Fix}(\sigma)$ be the sets of cycle valleys, cycle peaks, cycle double ascents, cycle double descents and fixed points of σ , respectively.

It is easy to check that the following identities hold:

$$\text{Cpk}(\sigma) \cup \text{Cda}(\sigma) = \text{Etop}(\sigma), \quad \text{Cval}(\sigma) \cup \text{Cda}(\sigma) = \text{Ebot}(\sigma), \quad (2.53a)$$

$$\text{Cpk}(\sigma) \cup \text{Cdd}(\sigma) \cup \text{Fix}(\sigma) = \text{NEtop}(\sigma), \quad (2.53b)$$

$$\text{Cval}(\sigma) \cup \text{Cdd}(\sigma) \cup \text{Fix}(\sigma) = \text{NEbot}(\sigma). \quad (2.53c)$$

Definition 2.31. The left-to-right inversion table (or Lehmer code) of a permutation σ is the sequence $(e_1, e_2, \dots, e_n)$, where e_i denotes the number of entries to the left of position i that are bigger than σ_i , i.e.,

$$e_i = \#\{j < i : \sigma_j > \sigma_i\}. \quad (2.54)$$

Dually, the right-to-left inversion table of σ is the sequence $(f_1, \dots, f_n)$, where f_i denotes the number of entries to the right of position i that are smaller than σ_i , i.e.,

$$f_i = \#\{j > i : \sigma_j < \sigma_i\}. \quad (2.55)$$

The inverse algorithm Ψ_{FZ}^{-1} building a permutation σ from a 3-restricted Laguerre history $(w, p) \in \mathfrak{L}_n^*$ can be described as follows:

According to the classification of steps in 3-restricted Laguerre history and Eq. (2.53), we first obtain the four sets $\text{Etop}(\sigma)$, $\text{Ebot}(\sigma)$, $\text{NEtop}(\sigma)$, and $\text{NEbot}(\sigma)$, where σ is the resulting permutation obtained at the end of the inverse bijection. We then construct two biwords,

$$\sigma_{\text{exc}} := \begin{pmatrix} \eta \\ \eta' \end{pmatrix} \text{ and } \sigma_{\text{nexc}} := \begin{pmatrix} \zeta \\ \zeta' \end{pmatrix},$$

where η and ζ are the subwords, written in increasing order, of the sets $\text{Ebot}(\sigma)$ and $\text{NEtop}(\sigma)$, respectively, and η' and ζ' are some subwords of $\text{Etop}(\sigma)$ and $\text{NEtop}(\sigma)$ obtained from the left-to-right and right-to-left tables with entries p_i , respectively. Then we recover the corresponding permutation $\Psi_{FZ}^{-1}((w, p))$.

As a consequence, the cardinality of $\mathfrak{L}_n^*$ is $n!$.

Example 2.32. If $\sigma = 8311075649112 = (1, 8, 4, 10, 11, 2, 3)(5, 7, 6)(9) \in \mathfrak{S}_{11}$, then we have

$$\Psi_{FZ}(\sigma) = (\text{NNSNN}\textcolor{red}{E}_1\text{SS}\textcolor{green}{E}_3\textcolor{blue}{E}_2\text{S}, (0, 1, 0, 0, 2, 2, 2, 1, 1, 0, 0)) \in \mathfrak{L}_{11}^*.$$

Inversely, from the 3-restricted Laguerre history, we have

$$\begin{aligned} \text{Cval}(\sigma) &= \{1, 2, 4, 5\}, \text{Cpk}(\sigma) = \{3, 7, 8, 11\}, \\ \text{Cdd}(\sigma) &= \{6\}, \text{Cda}(\sigma) = \{10\} \text{ and } \text{Fix}(\sigma) = \{9\}. \end{aligned}$$

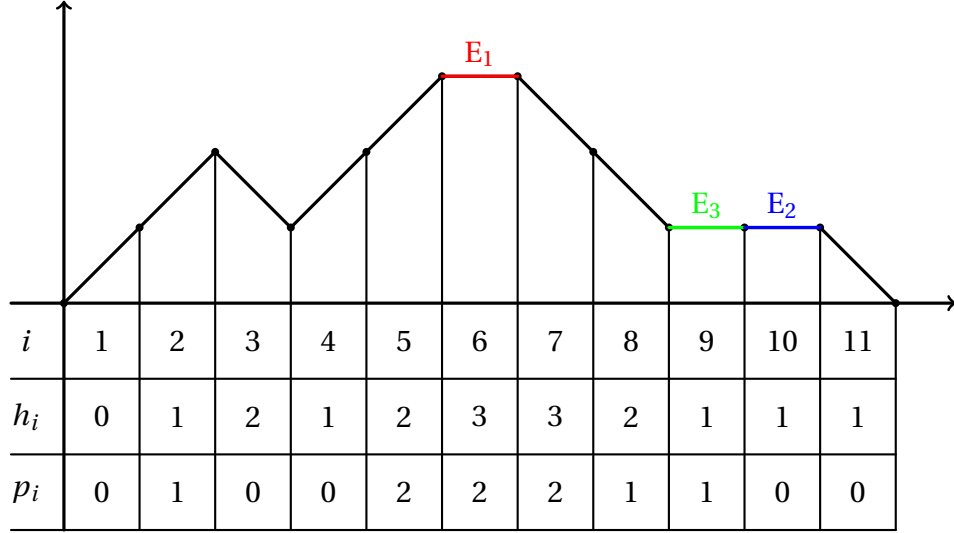


Figure 2.4: The 3-restricted Laguerre history $\Psi_{FZ}(\sigma)$ with $\sigma = 8311075649112 = (1, 8, 4, 10, 11, 2, 3)(5, 7, 6)(9)$.

According to Eq. (2.53), we obtain

$$\begin{aligned} \text{Ebot}(\sigma) &= \{1, 2, 4, 5, 10\}, \text{Etop}(\sigma) = \{3, 7, 8, 10, 11\}; \\ \text{NEbot}(\sigma) &= \{1, 2, 4, 5, 6, 9\} \text{ and } \text{NEtop}(\sigma) = \{3, 6, 7, 8, 9, 11\}. \end{aligned}$$

We now construct the biwords σ_{exc} and σ_{nexc}

$$\sigma_{\text{exc}} := \begin{pmatrix} \eta \\ \eta' \end{pmatrix} = \begin{pmatrix} 1_0 & 2_1 & 4_0 & 5_2 & 10_0 \\ 8 & 3 & 10 & 7 & 11 \end{pmatrix}, \quad \sigma_{\text{nexc}} := \begin{pmatrix} \zeta \\ \zeta' \end{pmatrix} = \begin{pmatrix} 3_0 & 6_2 & 7_2 & 8_1 & 9_1 & 11_0 \\ 1 & 5 & 6 & 4 & 9 & 2 \end{pmatrix}.$$

Combining these two biwords yields the permutation $\sigma = 8311075649112$; see Fig. 2.4.

Definition 2.33 (Indexed crossing). For $k \in [n]$ and $\sigma \in \mathfrak{S}_n$, the upper and lower crossings indexed at k of σ are defined, respectively, by

$$\text{cros}_k^+(\sigma) := \#\{l \mid l < k \leq \sigma_l < \sigma_k\} \text{ and } \text{cros}_k^-(\sigma) = \#\{l \mid \sigma_k < \sigma_l < k < l\}.$$

The number of crossings indexed at k is $\text{cros}_k(\sigma) = \text{cros}_k^+(\sigma) + \text{cros}_k^-(\sigma)$; see Fig 2.5.

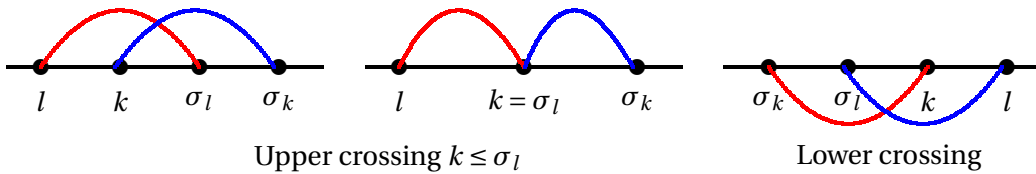
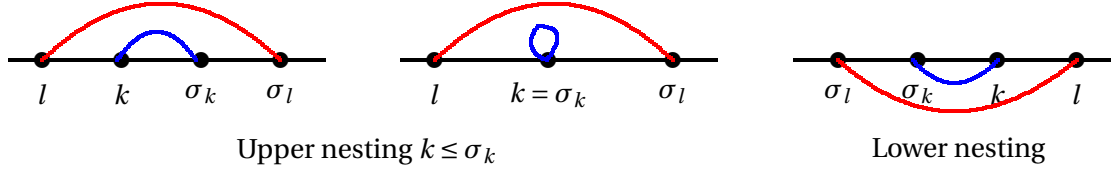


Figure 2.5: Upper crossing and lower crossing indexed at k .

Definition 2.34 (Indexed nesting). For $k \in [n]$ and $\sigma \in \mathfrak{S}_n$, the upper and lower nestings indexed at k of σ are defined, respectively, by

$$\text{nest}_k^+(\sigma) := \#\{l \mid l < k \leq \sigma_k < \sigma_l\} \text{ and } \text{nest}_k^-(\sigma) = \#\{l \mid \sigma_l < \sigma_k < k < l\}.$$

The number of crossings indexed at k is $\text{nest}_k(\sigma) = \text{nest}_k^+(\sigma) + \text{nest}_k^-(\sigma)$; see Fig. 2.6.

Figure 2.6: Upper nesting and lower nesting indexed at k

Note that for any $\sigma \in \mathfrak{S}_n$, we have

$$\text{cros}(\sigma) = \sum_{k=1}^n \text{cros}_k(\sigma) \text{ and } \text{nest}(\sigma) = \sum_{k=1}^n \text{nest}_k(\sigma)$$

The definition of labels p_i in (2.52) admits a straightforward interpretation in terms of the statistic nesting,

$$p_i := \begin{cases} \text{nest}_i^+(\sigma), & \text{if } \sigma(i) \geq i \quad \text{i.e. } i \in \text{Cval}(\sigma) \cup \text{Cda}(\sigma) \cup \text{Fix}(\sigma); \\ \text{nest}_i^-(\sigma), & \text{if } \sigma(i) < i \quad \text{i.e. } i \in \text{Cpk}(\sigma) \cup \text{Cdd}(\sigma). \end{cases}$$

The following Lemma is due to Corteel [Cor07, Lemma 3]. Since the original proof is brief, we provide a visual proof based on arc diagrams of permutations.

Lemma 2.3.5 ([Cor07]). Let $\Psi_{FZ}(\sigma) = (w, p) \in \mathfrak{L}_n^*$ for $\sigma \in \mathfrak{S}_n$. For any $k \in [n]$, if h_k is the height of the k th step of w , then

$$\text{cros}_k(\sigma) + \text{nest}_k(\sigma) = \begin{cases} h_k, & \text{if } k \leq \sigma(k); \\ h_k - 1, & \text{if } k > \sigma(k). \end{cases} \quad (2.56)$$

Proof. It is easy to check that the height of the k th step of w is equal to the number of upper arcs (or lower arcs) that cover node k , that is, more precisely, arcs satisfying one of the following conditions:

- the left endpoint lies to the left of k and the right endpoint lies to the right of k ;
- the left endpoint lies to the left of k and the right endpoint is k ;
- the left endpoint is k and the right endpoint lies to the right of k ;
- both the left and right endpoints are k .

In view of the upper and lower crossing (resp. nesting) indexed at k , see Definition 2.33 and 2.34, it suffices to show that

(1) if $k \in \text{Fix}(\sigma)$, then $\text{cros}_k^+(\sigma) = 0$ and $\text{nest}_k^+(\sigma) = h_k$;

(2) if $k \in \text{Cval}(\sigma)$, then

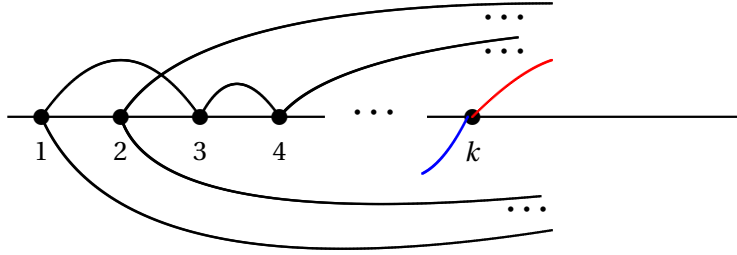
$$\text{cros}_k^+(\sigma) + \text{nest}_k^+(\sigma) = h_k \text{ with } 0 \leq \text{nest}_k^+(\sigma) \leq h_k;$$

(3) if $k \in \text{Cda}(\sigma)$, then

$$\text{cros}_k^+(\sigma) + \text{nest}_k^+(\sigma) = h_k \text{ with } 0 \leq \text{nest}_k^+(\sigma) \leq h_k - 1;$$

(4) and if $k \in \text{Cpk}(\sigma) \cup \text{Cdd}(\sigma)$, then

$$\text{cros}_k^-(\sigma) + \text{nest}_k^-(\sigma) = h_k - 1 \text{ with } 0 \leq \text{nest}_k^-(\sigma) \leq h_k - 1.$$



Case (1) is obvious. If $k \in \text{Cval}(\sigma)$, and there are $h_k = j$ arcs covering k , then the arc from k to $\sigma(k)$ must intersect exactly r of these arcs, for some $0 \leq r \leq j$. We order these j arcs from top to bottom and denote them by $a_1, a_2, \dots, a_j$. Suppose that the arc $(k \rightarrow \sigma(k))$ intersects exactly i of the j arcs above k , namely $a_{j-i+1}, a_{j-i}, \dots, a_j$, where $0 \leq i \leq j$. These intersected arcs together with the arc $(k \rightarrow \sigma(k))$ contribute

$$\text{cros}_k^+(\sigma) = i,$$

while the remaining $j - i$ arcs that do not intersect $(k \rightarrow \sigma(k))$ contribute

$$\text{nest}_k^+(\sigma) = j - i,$$

which proves (2).

If $k \in \text{Cda}(\sigma)$ and there are $h_k = j$ arcs that cover k , exactly one of which has right endpoint k (this arc together with the arc $(k \rightarrow \sigma(k))$ forms an upper crossing), then the arc from k to $\sigma(k)$ must intersect exactly r of the remaining $j - 1$ arcs, for some $0 \leq r \leq j - 1$. We order these $j - 1$ arcs from top to bottom and denote them by $a_1, a_2, \dots, a_{j-1}$. Suppose that the arc $(k \rightarrow \sigma(k))$ intersects exactly i of the $j - 1$ arcs above k , namely $a_{j-i}, a_{j-i-1}, \dots, a_{j-1}$, where $0 \leq i \leq j - 1$. These intersected arcs, together with the arc $(k \rightarrow \sigma(k))$, contribute

$$\text{cros}_k^+(\sigma) = i,$$

while the remaining $j - i - 1$ arcs that do not intersect $(k \rightarrow \sigma(k))$ contribute

$$\text{nest}_k^+(\sigma) = j - i - 1,$$

which proves (3).

If $k \in \text{Cpk}(\sigma) \cup \text{Cdd}(\sigma)$ and there are $h_k = j$ arcs below k , denoted by $b_1, b_2, \dots, b_j$ from bottom to top, exactly one of which has a right endpoint k , namely b_i where $1 \leq i \leq j$, then the arc $(k \rightarrow \sigma(k))$ must intersect exactly $b_{i+1}, b_{i+2}, \dots, b_j$. These intersected arcs, together with the arc $(k \rightarrow \sigma(k))$, contribute

$$\text{cros}_k^-(\sigma) = j - i,$$

while the remaining $i - 1$ arcs that do not intersect the arc $(k \rightarrow \sigma(k))$ contribute

$$\text{nest}_k^-(\sigma) = i - 1,$$

which proves (4). Eq. (2.56) follows by induction on k and can be proved analogously to Lemma 2.3.2; we omit the details. $\square$

Lemma 2.3.6. For $\sigma \in \mathfrak{S}_n$, let $\Psi_{FZ}(\sigma) := (w, p) \in \mathcal{L}_n^*$ be the corresponding 3-restricted Laguerre history. For each $i \in [n]$, denote by h_i the height of the i th step of w . Then

$$\text{depth}(\sigma) = \text{area}(w). \quad (2.57)$$

Proof. By Definition (2.14) and (2.53a), we see that

$$\text{depth}(\sigma) = \text{Etop}(\sigma) - \text{Ebot}(\sigma) = \text{Cpk}(\sigma) - \text{Cval}(\sigma). \quad (2.58)$$

Translating (2.58) through the Ψ_{FZ} yields

$$\text{depth}(\sigma) = \sum_{w_i=S} i - \sum_{w_j=N} j. \quad (2.59)$$

By the same argument in the proof of Lemma 2.3.4 we obtain (2.57). $\square$

Example 2.35. If $\sigma = 8311075649112 \in \mathfrak{S}_{11}$ and the corresponding Motzkin path is $w = \text{NNSNN}\mathbf{E}_1\text{SS}\mathbf{E}_3\mathbf{E}_2\text{S}$, see Fig 2.4, then by Definition (2.14), we have

$$\text{depth}(\sigma) = (8-1) + (3-2) + (10-4) + (7-5) + (11-10) = 17.$$

Moreover, we have

$$\text{area}(w) = \sum_{i=1}^{11} h_i = 5 \cdot 1 + 3 \cdot 2 + 2 \cdot 3 = 17.$$

2.3.2 Proof of Theorem 2.4 and Theorem 2.7

Lemma 2.3.7. Given a permutation $\sigma \in \mathfrak{S}_{n+1}$ and $x \in [n]$, letter x is a

- left-to-right maximum of σ , if and only if p_x is equal to 0 and the x th step is S or $\mathbf{E}_2$ in the Laguerre history $\Psi_{FV}(\sigma)$;
- right-to-left maximum of σ , if and only if p_x is equal to h_x and the x th step is S or $\mathbf{E}_1$ in the Laguerre history $\Psi_{FV}(\sigma)$.

Proof. It is easy to see that if x is a left-to-right (resp. right-to-left) maximum of σ , then x is a peak or a double ascent (resp. a peak or a double descent) of σ . By the construction of Ψ_{FV} , the x th step of $\Psi_{FV}(\sigma)$ is given by S or $\mathbf{E}_2$ (resp. S or $\mathbf{E}_1$) if x is a left-to-right (resp. right-to-left) maximum of σ . If x is a left-to-right maximum of σ , i.e., there is no element to the left of x that is greater than x , which means $p_x = (31-2)_x = 0$. If x is a right-to-left maximum of σ , i.e., there is no element to the right of x that is greater than x , which is $(2-13)_x = 0$. Then by Lemma 2.3.2, we derive that $(31-2)_x = h_x$. $\square$

Proof of Theorem 2.4. For $\sigma \in \mathfrak{S}_{n+1}$, let $\Psi_{FV}(\sigma) = (w, p) := (w_1 \dots w_n, (p_1, \dots, p_n)) \in \mathcal{L}_n$ be the corresponding Laguerre history. Recall (2.6b) that

$$\begin{aligned} \text{wt}(\sigma) = & (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1} \\ & \times f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)}. \end{aligned} \quad (2.60)$$

Combining Lemmas 2.3.2–2.3.4 and 2.3.7, we have

$$\begin{aligned}
\mathbf{wt}(\sigma) h^{D(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)} \\
= \prod_{w_i=N} (u_1 h^{h_i} p^{p_i} q^{h_i-p_i}) \prod_{w_i=S} (u_2 h^{h_i} (\alpha f)^{\chi(p_i=0)} (\beta g)^{\chi(p_i=h_i)} p^{p_i} q^{h_i-p_i}) \\
\times \prod_{w_i=E_1} (u_4 h^{h_i} (\beta t)^{\chi(p_i=h_i)} p^{p_i} q^{h_i-p_i}) \prod_{w_i=E_2} (u_3 h^{h_i} (\alpha t)^{\chi(p_i=0)} p^{p_i} q^{h_i-p_i}). \quad (2.61)
\end{aligned}$$

Summing over $p_i \in \{0, \dots, h_i\}$ for $i \in [n]$ we obtain

$$\begin{aligned}
A_n(\mathbf{u}, f, g, h, t | \alpha, \beta; p, q) \\
= \sum_{w \in \mathcal{M}_n^2} \prod_{w_i=N} u_1 h^{h_i} [h_i + 1]_{p,q} \prod_{w_i=S} u_2 h^{h_i} [h_i + 1; \beta g, \alpha f]_{p,q} \\
\times \prod_{w_i=E_1} u_4 h^{h_i} [h_i + 1; \beta t, 1]_{p,q} \prod_{w_i=E_2} u_3 h^{h_i} [h_i + 1; 1, \alpha t]_{p,q}, \quad (2.62)
\end{aligned}$$

where $\mathcal{M}_n^2$ is the set of Motzkin paths of length n with two colors in step E. This completes the proof by an application of Flajolet's fundamental Lemma 2.3.1. $\square$

For $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$, we say that an integer $i \in [n]$ is a

- left-to-right maximum index of σ , if $\sigma_j < \sigma_i$ for all $j < i$;
- right-to-left minimum index of σ , if $\sigma_j > \sigma_i$ for all $j > i$.

Since σ is a permutation, the map $i \mapsto \sigma(i) = \sigma_i$ is a bijection between positions and values. Consequently, counting left-to-right maxima (resp. right-to-left minima) by record positions or by record values gives the same cardinality. In particular,

$$\begin{aligned}
\text{lmax}(\sigma) &= \#\{i \in [n] : \sigma_j < \sigma_i \text{ for all } j < i\}; \\
\text{rmin}(\sigma) &= \#\{i \in [n] : \sigma_j > \sigma_i \text{ for all } j > i\}.
\end{aligned}$$

The following result provides a characterization of rmin and lmax through the bijection Ψ_{FZ} .

Lemma 2.3.8. Given a permutation $\sigma \in \mathfrak{S}_n$ and $x \in [n]$, an integer x is a

- left-to-right maximum index of σ if and only if p_x is equal to 0 and the x th step is S, E_2 , or E_3 in the 3-restricted Laguerre history $\Psi_{FZ}(\sigma)$;
- right-to-left minimum index of σ if and only if p_x is equal to 0 and the x th step is N, E_1 , or E_3 in the 3-restricted Laguerre history $\Psi_{FZ}(\sigma)$.

Proof. One can readily show that if x is a left-to-right maximum (resp. right-to-left minimum) index of σ , then $\sigma(x) \geq x$ (resp. $\sigma(x) \leq x$). By the construction of Ψ_{FZ} , the x th step of $\Psi_{FZ}(\sigma)$ is given by S, E_2 or E_3 (resp. N, E_1 or E_3) if x is a left-to-right maximum (resp. right-to-left minimum) index of σ . If x is a left-to-right maximum index of σ , then there is no upper arc above the arc $(x \rightarrow \sigma(x))$ because $(\sigma(i) < \sigma(x))$ for all $i < x$; and if x is a right-to-left minimum index of σ , then there is no lower arc below the arc $(x \rightarrow \sigma(x))$ because $(\sigma(i) > \sigma(x))$ for all $i > x$. $\square$

Lemma 2.3.9 ([dMV94]). Let $\sigma \in \mathfrak{S}_n$ and $\Psi_{FZ}(\sigma) = (w, p) \in \mathcal{L}_n^*$. Then

$$\text{inv}(\sigma) = \text{area}(w) + \text{nest}(\sigma) = \sum_{i=1}^n (h_i + p_i), \quad (2.63)$$

where h_i denotes the height of the i th step of w for $i \in [n]$.

Proof of Theorem 2.7. For $\sigma \in \mathfrak{S}_n$, let $\Psi_{FZ}(\sigma) := (w_1 \dots w_n, (p_1, \dots, p_n)) \in \mathcal{L}_n^*$ be the corresponding 3-restricted Laguerre history. Define the weight function

$$\mathbf{wt}^c(\sigma) := (u_1 u_2)^{\text{cval}(\sigma)} u_3^{\text{cda}(\sigma)} u_4^{\text{cdd}(\sigma)} t^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{lmax}(\sigma)} \beta^{\text{rmin}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)}. \quad (2.64)$$

Combining Lemmas 2.3.6, 2.3.8, and 2.3.9, we have

$$\begin{aligned} \mathbf{wt}^c(\sigma) &= \prod_{w_i=\mathbf{N}} (u_1 \beta^{\chi(p_i=0)} p^{h_i-p_i} q^{p_i} s^{h_i+p_i} h^{h_i}) \prod_{w_i=\mathbf{S}} (u_2 \alpha^{\chi(p_i=0)} p^{h_i-p_i-1} q^{p_i} s^{h_i+p_i} h^{h_i}) \\ &\quad \times \prod_{w_i=\mathbf{E}_1} (u_4 \beta^{\chi(p_i=0)} p^{h_i-p_i-1} q^{p_i} s^{h_i+p_i} h^{h_i}) \prod_{w_i=\mathbf{E}_2} (u_3 \alpha^{\chi(p_i=0)} p^{h_i-p_i} q^{p_i} s^{h_i+p_i} h^{h_i}) \\ &\quad \times \prod_{w_i=\mathbf{E}_3} (t(\alpha\beta)^{\chi(h_i=0)} q^{h_i} s^{2h_i} h^{h_i}). \end{aligned} \quad (2.65)$$

By Definition 2.30, summing over $p_i \in \{0, 1, \dots, h_i\}$ if $w_i \in \{\mathbf{N}, \mathbf{E}_2, \mathbf{E}_3\}$ and $p_i \in \{0, 1, \dots, h_i - 1\}$ if $w_i \in \{\mathbf{S}, \mathbf{E}_1\}$ for $i \in [n]$ we have

$$\begin{aligned} C_n(\mathbf{u}, t, s, h | \alpha, \beta; p, q) &= \sum_{w \in \mathcal{M}_n^3} \prod_{w_i=\mathbf{N}} u_1 (sh)^{h_i} [h_i + 1; \beta, 1]_{p,qs} \prod_{w_i=\mathbf{S}} u_2 (sh)^{h_i} [h_i; \alpha, 1]_{p,qs} \\ &\quad \times \prod_{w_i=\mathbf{E}_1} u_4 (sh)^{h_i} [h_i; \beta, 1]_{p,qs} \prod_{w_i=\mathbf{E}_2} u_3 p(sh)^{h_i} [h_i; \alpha, 1]_{p,qs} \prod_{w_i=\mathbf{E}_3} t(\alpha\beta)^{\chi(h_i=0)} (qs^2 h)^{h_i}, \end{aligned}$$

where $\mathcal{M}_n^3$ denotes the set of all 3-Motzkin paths of length n , that is, Motzkin paths whose horizontal step E is endowed with three possible colors: $\mathbf{E}_1$, $\mathbf{E}_2$, and $\mathbf{E}_3$. By applying Flajolet's fundamental Lemma 2.3.1, the proof is complete. $\square$

2.4 Two bijections over permutations

The bijections Φ_{SZ} and Φ_C actually arise from the composition of a restricted Françon–Viennot bijection and a simplified Foata–Zeilberger bijection. First, we need a variant of the bijection Ψ_{FV} (resp. Ψ_{FZ}) defined in (2.42) (resp. (2.51)) as follows.

Definition 2.36 (2-restricted Laguerre history). A 2-restricted Laguerre history of length n is a couple (w, p) , where $w = w_1 \dots w_n$ is a 2-Motzkin path of length n starting at $(0, 0)$ and ending at $(n, 0)$, and $p = (p_1, \dots, p_n)$ is an integer sequence satisfying $1 \leq i \leq n$,

$$0 \leq p_i \leq \begin{cases} h_i, & \text{if } w_i \in \{\mathbf{N}, \mathbf{E}_2\}; \\ h_i - 1, & \text{if } w_i \in \{\mathbf{S}, \mathbf{E}_1\}, \end{cases}$$

where h_i is the height of the i th step of w . Denote by $\tilde{\mathcal{L}}_n^*$ the set of 2-restricted Laguerre histories of length n .

We describe the restricted version of Françon-Viennot's bijection $\tilde{\Psi}_{FV} : \mathfrak{S}_n \rightarrow \tilde{\mathcal{L}}_n^*$ as follows: For a permutation $\sigma \in \mathfrak{S}_n$ with a boundary condition $\sigma_0 = 0$ and $\sigma_{n+1} = n+1$, the 2-restricted Laguerre history $\tilde{\Psi}_{FV}(\sigma) := (w_1 \dots w_n, (p_1, \dots, p_n)) \in \tilde{\mathcal{L}}_n^*$ is defined by

$$w_i = \begin{cases} \text{N (resp. S)} & \text{if } i \text{ is a valley (resp. peak) of } \sigma; \\ \text{E}_1 \text{ (resp. E}_2\text{)} & \text{if } i \text{ is a double descent (resp. double ascent) of } \sigma, \end{cases} \quad (2.66)$$

and $p_i = (31-2)_i(\sigma)$ for each $i \in [n]$.

Lemma 2.4.1. [Cor07] Given a permutation $\sigma \in \mathfrak{S}_n$ and the corresponding 2-restricted Laguerre history $\tilde{\Psi}_{FV}(\sigma) = (w, p) \in \tilde{\mathcal{L}}_n^*$, for any $x \in [n]$, we have

$$(31-2)_x(\sigma) + (2-31)_x(\sigma) = \begin{cases} h_x, & \text{if } w_x \in \{\text{N}, \text{E}_2\}; \\ h_x - 1, & \text{if } w_x \in \{\text{S}, \text{E}_1\}. \end{cases} \quad (2.67)$$

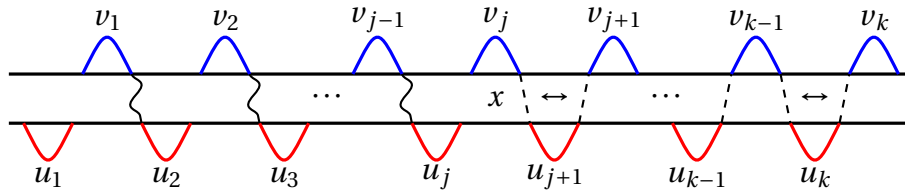
Proof. Let $\sigma \in \mathfrak{S}_n$ and $\sigma' = \sigma \star \sigma_{n+1} \in \mathfrak{S}_{n+1}$ with $\sigma_{n+1} = n+1$, where $\star$ denotes the concatenation operator of two words. Suppose that for $x \in [n]$, the x -decomposition of σ' is

$$\sigma' = u_1 v_1 \cdots u_k v_k, \quad k \geq 1,$$

where x lies inside v_j , see (2.43). It is straightforward to verify the following:

- word v_k has $n+1$ as its last element, possibly together with other entries preceding n .
- If x is a peak or a double descent of σ' , then x cannot lie in v_k .

When x is a valley or a double ascent, the word v_j must contain at least one element larger than x on the right side. Consequently, each contribution to $(2-13)_x(\sigma')$ corresponds to a contribution to $(2-31)_x(\sigma)$. When x is a peak or a double descent, and the word v_j contains x as the last element. In these two cases, one contribution to $(2-31)_x(\sigma)$ is lost. See the corresponding illustration below.



Thus, for any $x \in [n]$, we have

$$(31-2)_x(\sigma) = (31-2)_x(\sigma'), \quad (2.68a)$$

and

$$(2-31)_x(\sigma) = \begin{cases} (2-13)_x(\sigma') & \text{if } x \text{ is a valley or a double ascent;} \\ (2-13)_x(\sigma') - 1 & \text{if } x \text{ is a peak or a double descent.} \end{cases} \quad (2.68b)$$

Combining Eqs. (2.68) and Lemma 2.3.2, we derive (2.67). $\square$

We describe a simplified version of Foata–Zeilberger’s bijection $\tilde{\Psi}_{FZ}$ as follows: For a permutation $\sigma \in \mathfrak{S}_n$, the 2-restricted Laguerre history $\tilde{\Psi}_{FZ}(\sigma) := (w_1 \dots w_n, (p_1, \dots, p_n)) \in \tilde{\mathcal{L}}_n^*$ is defined by

$$w_i = \begin{cases} \mathbf{N} & \text{if } i \text{ is a cycle valley of } \sigma; \\ \mathbf{S} & \text{if } i \text{ is a cycle peak of } \sigma; \\ \mathbf{E}_1 & \text{if } i \text{ is a cycle double descent of } \sigma; \\ \mathbf{E}_2 & \text{if } i \text{ is a cycle double ascent or a fixed point of } \sigma, \end{cases} \quad (2.69)$$

and with $p_i := \text{nest}_i(\sigma)$ for each $i \in [n]$.

The following property follows directly from Lemma 2.3.5 and (2.69).

Lemma 2.4.2 ([Cor07]). Given a permutation $\sigma \in \mathfrak{S}_n$, and the corresponding 2-restricted Laguerre history $\tilde{\Psi}_{FZ}(\sigma) = (w, p) \in \tilde{\mathcal{L}}_n^*$ for $\sigma \in \mathfrak{S}_n$. For any $k \in [n]$, if h_k is the height of the k th step of w , then

$$\text{cros}_k(\sigma) + \text{nest}_k(\sigma) = \begin{cases} h_k, & \text{if } w_x \in \{\mathbf{N}, \mathbf{E}_2\}; \\ h_k - 1, & \text{if } w_x \in \{\mathbf{S}, \mathbf{E}_1\}. \end{cases} \quad (2.70)$$

In particular, if $k \in [n]$ is a fixed point of σ , we have

$$\text{nest}_k(\sigma) = h_k. \quad (2.71)$$

For a 2-restricted Laguerre history $(w, (p_1, \dots, p_n)) \in \tilde{\mathcal{L}}_n^*$, we define the complement operator ρ by

$$\rho((w, (p_1, \dots, p_n))) = (w, (p'_1, \dots, p'_n)), \quad (2.72)$$

where the pairs p_i and p'_i are related by

$$p_i + p'_i = \begin{cases} h_k, & \text{if } w_x \in \{\mathbf{N}, \mathbf{E}_2\}; \\ h_k - 1, & \text{if } w_x \in \{\mathbf{S}, \mathbf{E}_1\}. \end{cases} \quad (2.73)$$

2.4.1 Bijection Φ_C

We recall Corteel’s bijection $\Phi_C := \tilde{\Psi}_{FZ}^{-1} \circ \tilde{\Psi}_{FV}$ in [Cor07, Sec. 5], which is constructed in such a way that if $\tau = \Phi_C(\sigma)$, then

$$(31-2)_k(\sigma) = \text{nest}_k(\tau) \quad \forall k = 1, \dots, n. \quad (2.74)$$

For $k \in [n]$, the pattern $(2-31)_k(\sigma)$ (resp. $(31-2)_k(\sigma)$) indexed by k is called the *right embracing number* (resp. *left embracing number*) of k in σ . Given a permutation σ , one constructs two biwords, $\left(\begin{smallmatrix} f \\ f' \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} g \\ g' \end{smallmatrix}\right)$, and then forms the biword

$$\tau = \left(\begin{array}{cc} f & g \\ f' & g' \end{array} \right)$$

by concatenating f and g , and f' and g' , respectively. These four words are defined as follows:

- The word f (resp. g) is defined as the subword of nondescent (resp. descent) tops in σ , ordered increasingly;

- The word f' (resp. g') is the permutation of nondescent (resp. descent) bottoms in σ obtained from the left-to-right (resp. right-to-left) inversion table (see Definition 2.31) whose entries are the left embracing numbers of σ .

See the following illustration:

$$\left(\begin{array}{c} \text{Nondescent tops (with labels)} \\ \text{Nondescent bottoms} \end{array} \right) \xrightarrow[\text{with entries } (31-2)_i]{\text{left-to-right inversion table}} \left(\begin{array}{c} f \\ f' \end{array} \right); \quad (2.75)$$

$$\left(\begin{array}{c} \text{Descent tops (with labels)} \\ \text{Descent bottoms} \end{array} \right) \xrightarrow[\text{with entries } (31-2)_i]{\text{right-to-left inversion table}} \left(\begin{array}{c} g \\ g' \end{array} \right). \quad (2.76)$$

By rearranging the columns of τ so that the top row is in increasing order, we obtain the permutation $\Phi_C(\sigma)$ as the bottom row of the rearranged biword.

Example 2.37. Let $\sigma = 412796583 \in \mathfrak{S}_9$ with the left embracing numbers

$$((31-2)_1, \dots, (31-2)_8) \sigma = (0, 1, 1, 0, 0, 0, 0, 1, 0).$$

Then

$$\begin{aligned} \left(\begin{array}{c} f \\ f' \end{array} \right) &= \begin{pmatrix} 1_0 & 2_1 & 3_1 & 5_0 & 7_0 \\ 7 & 2 & 4 & 8 & 9 \end{pmatrix}, & \left(\begin{array}{c} g \\ g' \end{array} \right) &= \begin{pmatrix} 4_0 & 6_0 & 8_1 & 9_0 \\ 1 & 3 & 6 & 5 \end{pmatrix}, \\ \tau' = \left(\begin{array}{cc} f & g \\ f' & g' \end{array} \right) &= \begin{pmatrix} 1 & 2 & 3 & 5 & 7 & 4 & 6 & 8 & 9 \\ 7 & 2 & 4 & 8 & 9 & 1 & 3 & 6 & 5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 7 & 2 & 4 & 1 & 8 & 3 & 9 & 6 & 5 \end{pmatrix}. \end{aligned}$$

and thus $\Phi_C(\sigma) = \tau = 724183965$. We verify that

- $\text{ndes}(\sigma) = |\{1, 2, 3, 5, 7\}| = 5$, $\text{pmin}(\sigma) = |\{2\}| = 1$, $(31-2)(\sigma) = |\{41-2, 41-3, 96-8\}| = 3$, $(2-31)(\sigma) = |\{7-96, 4-83, 7-83, 6-83, 5-83\}| = 5$, $\text{MADL}(\sigma) = 15$, and $\text{NDdif}(\sigma) = \text{NDbot}(\sigma) - \text{NDtop}(\sigma) = 30 - 18 = 12$;
- $\text{wex}(\tau) = |\{1, 2, 3, 5, 7\}| = 5$, $\text{fix}(\tau) = |\{2\}| = 1$, $\text{cros}(\tau) = 5$, $\text{nest}(\tau) = 3$, $\text{inv}(\tau) = 15$ and $\text{depth}(\tau) = (7-1) + (4-3) + (8-5) + (9-7) = 12$.

2.4.2 Bijection Φ_{SZ}

We describe a variant of the mapping $\Phi_{SZ} := \tilde{\Psi}_{FZ}^{-1} \circ \rho \circ \tilde{\Psi}_{FV}$ in [SZ10] as in the following.

Define $\tilde{\tau} = \Phi_{SZ}(\sigma)$ for $\sigma \in \mathfrak{S}_n$ such that

$$(2-31)_k(\sigma) = \text{nest}_k(\tilde{\tau}) \quad \forall k = 1, \dots, n. \quad (2.77)$$

Given a permutation $\sigma \in \mathfrak{S}_n$, we first construct two biwords, $\left(\begin{array}{c} f \\ \tilde{f}' \end{array} \right)$ and $\left(\begin{array}{c} g \\ \tilde{g}' \end{array} \right)$, and then form the biword

$$\tilde{\tau} = \left(\begin{array}{cc} f & g \\ \tilde{f}' & \tilde{g}' \end{array} \right)$$

by concatenating f and g , and $\tilde{f}'$ and $\tilde{g}'$, respectively. These four words are defined as follows:

- The word f (resp. g) is defined as the subword of nondescent (resp. descent) tops in σ , ordered increasingly;

- The word $\tilde{f}'$ (resp. $\tilde{g}'$) is the permutation of nondescent (resp. descent) bottoms in σ obtained from the left-to-right (resp. right-to-left) inversion table (see Definition 2.31) whose entries are the right embracing numbers of σ .

See the following illustration:

$$\left(\begin{array}{c} \text{Nondescent tops (with labels)} \\ \text{Nondescent bottoms} \end{array} \right) \xrightarrow[\text{with entries } (2-31)_i]{\text{left-to-right inversion table}} \left(\begin{array}{c} f \\ \tilde{f}' \end{array} \right); \quad (2.78)$$

$$\left(\begin{array}{c} \text{Descent tops (with labels)} \\ \text{Descent bottoms} \end{array} \right) \xrightarrow[\text{with entries } (2-31)_i]{\text{right-to-left inversion table}} \left(\begin{array}{c} g \\ \tilde{g}' \end{array} \right). \quad (2.79)$$

By rearranging the columns of $\tilde{\tau}$ so that the top row is in increasing order, we obtain the permutation $\Phi_{SZ}(\sigma)$ as the bottom row of the rearranged biword.

Example 2.38. Let $\sigma = 412796583 \in \mathfrak{S}_9$ with the right embracing numbers

$$((2-31)_1, \dots, (2-31)_9) \sigma = (0, 0, 0, 1, 1, 1, 2, 0, 0).$$

Then

$$\begin{aligned} \left(\begin{array}{c} f \\ \tilde{f}' \end{array} \right) &= \begin{pmatrix} 1_0 & 2_0 & 3_0 & 5_1 & 7_2 \\ 2 & 4 & 9 & 8 & 7 \end{pmatrix} \quad \left(\begin{array}{c} g \\ \tilde{g}' \end{array} \right) = \begin{pmatrix} 4_1 & 6_1 & 8_0 & 9_0 \\ 3 & 5 & 1 & 6 \end{pmatrix}, \\ \tilde{\tau}' &= \left(\begin{array}{cc} f & g \\ \tilde{f}' & \tilde{g}' \end{array} \right) = \begin{pmatrix} 1 & 2 & 3 & 5 & 7 & 4 & 6 & 8 & 9 \\ 2 & 4 & 9 & 8 & 7 & 3 & 5 & 1 & 6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 4 & 9 & 3 & 8 & 5 & 7 & 1 & 9 \end{pmatrix}. \end{aligned}$$

and thus $\Phi_{SZ}(\sigma) = \tilde{\tau} = 249385716$. We verify that

- $\text{ndes}(\sigma) = |\{1, 2, 3, 5, 7\}| = 5$, $\text{fmax}(\sigma) = |\{7\}| = 1$, $(31-2)(\sigma) = |\{41-2, 41-3, 96-8\}| = 3$, $(2-31)(\sigma) = |\{7-96, 4-83, 7-83, 6-83, 5-83\}| = 5$, $\text{MAD}(\sigma) = 17$, and $\text{NDdif}(\sigma) = \text{NDbot}(\sigma) - \text{NDtop}(\sigma) = 30 - 18 = 12$;
- $\text{wex}(\tilde{\tau}) = |\{1, 2, 3, 5, 7\}| = 5$, $\text{fix}(\tilde{\tau}) = |\{7\}| = 1$, $\text{cros}(\tilde{\tau}) = 3$, $\text{nest}(\tilde{\tau}) = 5$, $\text{inv}(\tilde{\tau}) = 17$ and $\text{depth}(\tilde{\tau}) = (2-1) + (4-2) + (9-3) + (8-5) = 12$.

2.4.3 Proof of Theorem 2.15

We prove only (2.20); the proof of (2.21) is analogous and omitted.

Given $\sigma \in \mathfrak{S}_n$, let $\tau = \Phi_C(\sigma)$. It is easily checked that every column $\binom{i}{j}$ in $\left(\begin{array}{c} f \\ \tilde{f}' \end{array} \right)$ satisfies $i \leq j$ and every column $\binom{i}{j}$ in $\left(\begin{array}{c} g \\ \tilde{g}' \end{array} \right)$ satisfies $i > j$. It follows that each letter σ_i in f corresponds to a nondescent σ_i in π and also a weak excedance σ_i in τ , that is, for all $1 \leq i \leq n$,

$$\sigma_i < \sigma_{i+1} \quad (1 \leq i \leq n-1) \quad \text{or} \quad i = n \iff \sigma_i \leq \tau(\sigma_i). \quad (2.80)$$

Hence, we have $\text{ndes}(\pi) = \text{wex}(\tau)$. By definition, $(31-2)(\sigma)$ is the sum of the left embracing numbers in π and the sum of the inversions in the words f' and g' . If a pair (i, j) is an inversion in f' (resp. g'), then we have $j < i \leq \tau(i) < \tau(j)$ (resp. $\tau(j) < \tau(i) < j < i$) that is a nesting in τ . Thus, we have

$$(31-2)_k(\sigma) = \text{nest}_k(\tau), \quad \forall k \in [n]. \quad (2.81)$$

Combining (2.67) and (2.70), we derive

$$(2-31)_k(\sigma) = \text{cros}_k(\tau), \quad \forall k \in [n]. \quad (2.82)$$

By the definition of pmin (see Definition 2.13), if k is a postminimum in σ , then k is a nondescent bottom. Moreover, since k is also a right-to-left minimum, it follows that k is a nondescent top. Consequently, k belongs to both subwords f and f' . One can check that k is a postminimum of σ , if and only if $p_k = h_k$ and the k th step is E_2 in the 2-restricted Laguerre history $\tilde{\Psi}_{FV}(\sigma)$. Then, combining (2.81) and (2.71), we obtain $\tau(k) = k$.

As shown in [SZ10], the inversion number can be characterized in terms of the crossing and nesting numbers by

$$\text{inv}(\sigma) = \text{drop}(\sigma) + \text{cros}(\sigma) + 2 \text{nest}(\sigma). \quad (2.83)$$

Combining the above discussion with Eqs. (2.19b) and (2.83), we see that

$$\begin{aligned} \text{MADL}(\sigma) &= \text{des}(\sigma) + (2-31)(\sigma) + 2(31-2)(\sigma) \\ &= \text{drop}(\tau) + \text{cros}(\tau) + 2 \text{nest}(\tau) = \text{inv}(\tau). \end{aligned} \quad (2.84)$$

If $i \in [n]$ is a weak-excedance of τ , viz, $\tau(i) \geq i$, we say that $\tau(i)$ is a weak-excedance top and i a weak-excedance bottom. By (2.80), every nondescent top (resp. nondescent bottom) of σ corresponds naturally to a weak-excedance bottom (resp. weak-excedance top) of τ . Therefore, by Definitions (2.14) and (2.17), we have $\text{NDdif}(\sigma) = \text{depth}(\tau)$. $\square$

2.5 Applications to gamma-positivity

2.5.1 Proof of Theorem 2.16 and Theorem 2.18

Consider the following specialization of (2.9)

$$\mathbf{u} = (1, xu_2, u_3, xu_4) \quad \text{and} \quad \beta = \alpha, \quad (2.85)$$

we obtain

$$\begin{aligned} A_n(\mathbf{u}, f, g, h, t | \alpha, \alpha; p, q) &= \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{des}(\sigma)} u_2^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} \\ &\quad \times t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2} p^{31-2(\sigma)} q^{2-13(\sigma)}. \end{aligned} \quad (2.86)$$

For $0 \leq k \leq \lfloor n/2 \rfloor$, let $\gamma_{n,k}(x, u_3, u_4, f, g, h | t, \alpha | p, q)$ be the coefficient of u_2^k in the expansion

$$A_n(\mathbf{u}, f, g, h, t | \alpha, \alpha; p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}(x, u_3, u_4, f, g, h | t, \alpha; p, q) u_2^k, \quad (2.87)$$

where $\mathbf{u} = (1, xu_2, u_3, xu_4)$.

Define the two special cases of (2.86):

$$A_n(x, u_2, u_3, u_4, f, g, h | p, q) := A_n(\mathbf{u}, f, g, h, 1 | 1, 1; p, q); \quad (2.88)$$

$$B_n(x, u_2, u_3, u_4, f, g, h | t, \alpha) := A_n(\mathbf{u}, f, g, h, t | \alpha, \alpha; 1, 1). \quad (2.89)$$

Lemma 2.5.1. The following expansion formulas hold:

$$A_n(x, u_2, u_3, u_4, f, g, h | p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^a(f, g, h | p, q) (xu_2)^k (u_3 + xu_4)^{n-2k}, \quad (2.90)$$

and

$$B_n(x, u_2, u_3, u_4, f, g, h | t, \alpha) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^b(f, g, h | t, \alpha) (xu_2)^k (u_3 + xu_4)^{n-2k}, \quad (2.91)$$

where

$$\gamma_{n,k}^a(f, g, h | p, q) = \sum_{\sigma \in \mathfrak{S}_{n+1, \text{des}=k}^{\text{dd}=0}} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)}, \quad (2.92)$$

and

$$\gamma_{n,k}^b(f, g, h | t, \alpha) = \sum_{\sigma \in \mathfrak{S}_{n+1, \text{des}=k}^{\text{dd}=0}} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} t^{\text{lmaxda}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2} \quad (2.93)$$

with $\mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0} = \{\sigma \in \mathfrak{S}_n : \text{dd}(\sigma) = 0 \text{ and } \text{des}(\sigma) = k\}$.

Proof. Combining (2.87), (2.86) and Theorem 2.4 with (2.85), then substituting $u_2 \rightarrow \frac{u_2(u_3+xu_4)^2}{x}$ and $z \rightarrow \frac{z}{(u_3+xu_4)}$, we obtain

$$\sum_{n \geq 0} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{\gamma_{n,k}(x, u_3, u_4, f, g, h | t, \alpha; p, q)}{x^k (u_3 + xu_4)^{n-2k}} u_2^k z^n = \mathcal{J} \left(z; \frac{b_k}{u_3 + xu_4}, \lambda_k \right), \quad (2.94)$$

where

$$b_k = h^k (u_3 [k+1; \alpha t, 1]_{p,q} + xu_4 [k+1; 1, \alpha t]_{p,q}), \quad (2.95)$$

$$\lambda_{k+1} = h^{2k+1} u_2 [k+1]_{p,q} [k+2; \alpha g, \alpha f]_{p,q} \quad (k \geq 0). \quad (2.96)$$

When $\alpha = t = 1$, we see that for $k \geq 0$,

$$b_k = h^k (u_3 + xu_4) [k+1]_{p,q}. \quad (2.97)$$

Thus, the coefficient of $u_2^k z^n$ in the left-hand side of (2.94) is a polynomial in f, g, h, p and q with coefficients independent of x, u_3 and u_4 . Setting $x = u_3 = 1$ and $u_4 = 0$ in (2.94), we derive that this coefficient is equal to $\gamma_{n,k}(1, 1, 0, f, g, h | 1, 1; p, q) = \gamma_{n,k}^a(f, g, h | p, q)$ by (2.92). It follows from (2.94) that

$$\frac{\gamma_{n,k}(x, u_3, u_4, f, g, h | 1, 1; p, q)}{x^k (u_3 + xu_4)^{n-2k}} = \gamma_{n,k}^a(f, g, h | p, q), \quad (2.98)$$

which is equivalent to (2.90).

When $p = q = 1$, we see that for $k \geq 0$,

$$b_k = h^k (k + \alpha t) (u_3 + xu_4). \quad (2.99)$$

Thus, the coefficient of $u_2^k z^n$ in the left-hand side of (2.94) is a polynomial in f, g, h, t and α with coefficients independent of x, u_3 and u_4 . Setting $x = u_3 = 1$ and $u_4 = 0$ in (2.94), we derive that this coefficient is equal to $\gamma_{n,k}(1, 1, 0, f, g, h | t, \alpha; 1, 1) = \gamma_{n,k}^b(f, g, h | t, \alpha)$ by (2.93). It follows from (2.94) that

$$\frac{\gamma_{n,k}(x, u_3, u_4, f, g, h | t, \alpha; 1, 1)}{x^k (u_3 + xu_4)^{n-2k}} = \gamma_{n,k}^b(f, g, h | t, \alpha),$$

which is equivalent to (2.91). □

Proof of Theorem 2.16. Setting $u_2 = u_3 = u_4 = 1$ in (2.90) of Lemma 2.5.1 yields (2.24). □

Proof of Theorem 2.18. Setting $u_2 = u_3 = u_4 = 1$ in (2.91) of Lemma 2.5.1 yields (2.26). □

2.5.2 Proof of Theorem 2.22

We consider the following special case of (2.16a)

$$\mathbf{u} = (x, u_2, xu_3, u_4) \quad \text{and} \quad \beta = \alpha \quad (2.100)$$

that is,

$$C_n(\mathbf{u}, t, s, h | \alpha, p, q) := \sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)} u_2^{\text{cval}(\sigma)} u_3^{\text{cda}(\sigma)} u_4^{\text{cdd}(\sigma)} t^{\text{fix}(\sigma)} s^{\text{inv}(\sigma)} \\ \times h^{\text{depth}(\sigma)} \alpha^{\text{rmin}(\sigma) + \text{lmax}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)}, \quad (2.101)$$

where $\mathbf{u} = (x, u_2, xu_3, u_4)$. For $0 \leq k \leq \lfloor (n-d)/2 \rfloor$, let $c_{n,k,d}(x, u_3, u_4, s, h | \alpha, p, q)$ be defined by the expansion

$$C_n(\mathbf{u}, t, s, h | \alpha, p, q) = \sum_{d=0}^n t^d \sum_{k=0}^{\lfloor (n-d)/2 \rfloor} c_{n,k,d}(x, u_3, u_4, s, h, \alpha | p, q) u_2^k. \quad (2.102)$$

Lemma 2.5.2. Let $\mathbf{u} = (x, u_2, xu_3, u_4)$. We have

$$C_n(\mathbf{u}, t, s, h | \alpha, p, q) = \sum_{d=0}^n t^d \sum_{k=0}^{\lfloor (n-d)/2 \rfloor} \gamma_{n,k,d}^c(s, h | \alpha, p, q) (xu_2)^k (u_4 + xu_3p)^{n-2k-d}, \quad (2.103)$$

where

$$\gamma_{n,k,d}^c(s, h | \alpha, p, q) = \sum_{\sigma \in \mathfrak{S}_{n, \text{exc}=k}^{\text{cda}=0, \text{fix}=d}} s^{\text{inv}(\sigma)} h^{\text{depth}(\sigma)} \alpha^{\text{rmin}(\sigma) + \text{lmax}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)}, \quad (2.104)$$

where $\mathfrak{S}_{n, \text{exc}=k}^{\text{cda}=0, \text{fix}=d} = \{\sigma \in \mathfrak{S}_n : \text{cda}(\sigma) = 0, \text{fix}(\sigma) = d \text{ and } \text{exc}(\sigma) = k\}$.

Proof. Combining (2.101), (2.102), and Theorem 2.7 with (2.100), then substituting $u_2 \rightarrow \frac{u_2(u_4 + xu_3p)^2}{x}$, $z \rightarrow \frac{z}{(u_4 + xu_3p)}$, and $t \rightarrow t(u_4 + xu_3p)$, we obtain

$$\sum_{n \geq 0} \sum_{d=0}^n \sum_{k=0}^{\lfloor (n-d)/2 \rfloor} \frac{c_{n,k,d}(x, u_3, u_4, s, h | \alpha, p, q) u_2^k}{x^k (u_4 + xu_3p)^{n-2k-d}} u_2^k t^d z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.105)$$

where

$$b_k = (sh)^k (tq^k s^k + [k; \alpha, 1]_{p,qs}), \quad \lambda_k = u_2 (sh)^{2k+1} [k; \alpha, 1]_{p,qs}^2. \quad (2.106)$$

Thus, the coefficient of $u_2^k t^d z^n$ in the left-hand side of (2.105) is a polynomial in s, h, α, p and q with coefficients independent of x, u_3 and u_4 . Setting $x = u_4 = 1$ and $u_3 = 0$ in (2.105), we derive that this coefficient is equal to $c_{n,k,d}(1, 0, 1, s, h | \alpha, p, q) = \gamma_{n,k,d}^c(s, h | \alpha, p, q)$ by (2.104). It follows from (2.105) that

$$\frac{c_{n,k,d}(x, u_3, u_4, s, h | \alpha, p, q) u_2^k}{x^k (u_4 + xu_3p)^{n-2k-d}} = \gamma_{n,k,d}^c(s, h | \alpha, p, q), \quad (2.107)$$

which is equal to (2.103). $\square$

Proof of Theorem 2.22. Setting $u_2 = u_4 = 1$, $u_3 = 1/p$ and $t = t(1+x)$ (resp. $u_2 = u_4 = p$, $u_3 = 1$ and $t = t(1+x)$) in (2.103) of Lemma 2.5.2 yields Eq. (2.30) with $i = 1, 2$. $\square$

Proposition 2.5.1. For $n \geq 1$, we have

$$\sum_{\sigma \in \mathcal{D}_n} (-1)^{\text{nest}(\sigma)} x^{\text{exc}(\sigma)} h^{\text{depth}(\sigma)} = \sum_{j=1}^{\lfloor n/2 \rfloor} \binom{n-j-1}{j-1} h^{n-j} x^j (1+x)^{n-2j}. \quad (2.108)$$

Proof. Specializing Theorem 2.7 with $u_2 = u_3 = u_4 = \alpha = \beta = s = p = q = 1$, $q = -1$, and $t = 0$, we have

$$b_0 = 0, \quad b_1 = (1+x)h, \quad \lambda_1 = xh, \quad \text{and} \quad \lambda_2 = 0. \quad (2.109)$$

Thus

$$1 + \sum_{n \geq 1} \left(\sum_{\sigma \in \mathcal{D}_n} (-1)^{\text{nest}(\sigma)} x^{\text{exc}(\sigma)} h^{\text{depth}(\sigma)} \right) z^n = \frac{1}{1 - \frac{\lambda_1 z^2}{1 - b_1 z}} = \frac{1 - (1+x)hz}{1 - (1+x)hz - xhz^2}. \quad (2.110)$$

For $n \geq 1$, one can check that

$$[z^n] \frac{1 - az}{1 - az - bz^2} = \sum_{j=1}^{\lfloor n/2 \rfloor} \binom{n-j-1}{j-1} a^{n-2j} b^j. \quad (2.111)$$

Substituting $a = (1+x)h$ and $b = xh$ in (2.111) and combining with (2.110) yields (2.108). $\square$

Remark 2.39. As $\text{inv} = \text{nest} + \text{depth}$, Eq. (2.108) is equivalent to Theorem 5.4 (or Theorem 1.6) in [EFL26].

2.5.3 Combinatorics of γ -coefficient factorization

Proposition 2.5.2. For $0 \leq k \leq \lfloor n/2 \rfloor$, there exist polynomials $E_{n,k}(h | p, q) \in \mathbb{N}[h, p, q]$ and $E_{n,k}(f, h | t, \alpha) \in \mathbb{N}[f, h, t, \alpha]$ such that

$$\gamma_{n,k}^a(1, 1, h | p, q) = (p+q)^k E_{n,k}(h | p, q), \quad (2.112)$$

$$\gamma_{n,k}^b(f, f, h | t, \alpha) = 2^k E_{n,k}(f, h | t, \alpha). \quad (2.113)$$

Proof. For all $k \geq 1$, as

$$(p+q) \mid [k]_{p,q} [k+1]_{p,q}, \quad (2.114)$$

it follows from (2.96) that λ_k must have a factor $(p+q)$ if $\alpha = f = g = 1$. Combining this and (2.90) with $f = g = 1$, we prove (2.112).

When $p = q = 1$ and $g \rightarrow f$, it follows from (2.96) that

$$\lambda_k = h^{2k-1} k((k-1) + 2\alpha f) u_2 = 2 \cdot h^{2k-1} (k(k-1)/2 + k\alpha f) u_2 \quad (2.115)$$

for all $k \geq 1$. Combining (2.115) and (2.91) with $g = f$, we prove (2.113). $\square$

Remark 2.40. Brändén [Brä08] conjectured the divisibility (2.112) for $h = 1$; this conjecture was subsequently confirmed in [SZ12].

For the rest of this section, we focus on the combinatorial interpretations of the two quotients in Proposition 2.5.2. Following Foata-Strehl [FS74] and Heteyi-Reiner [HR98], we define the André permutations in terms of *x -factorization*.

Definition 2.41. Let $\sigma = \sigma_1 \sigma_2 \dots \sigma_n \in \mathfrak{S}_n$ and let $x = \sigma_i$ for $i \in [n]$. The x -factorization of σ is given by $\sigma = u \lambda(x) x \rho(x) v$, where

n	$k = 0$	$k = 1$	$k = 2$
0	1		
1	1		
2	1	$h(p+q)$	
3	1	$h(p+q)(h(p+q)+2)$	
4	1	$h(p+q)(h^2(p+q)^2+2h(p+q)+3)$	$h^2(p+q)^2(h^2(p^2+pq+q^2)+1)$

Table 2.1: The first values of $\gamma_{n,k}^a := \gamma_{n,k}^a(1, 1, h \mid p, q)$ for $0 \leq 2k \leq n \leq 4$.

n	$k = 0$	$k = 1$	$k = 2$
0	1		
1	αt		
2	$\alpha^2 t^2$	$2\alpha h f$	
3	$\alpha^3 t^3$	$2\alpha h f(h + \alpha h t + 2\alpha t)$	
4	$\alpha^4 t^4$	$2\alpha h f((h + \alpha h t + \alpha t)^2 + 2\alpha^2 t^2)$	$4\alpha f h^2(h^2 + \alpha f(2h^2 + 1))$

Table 2.2: The first values of $\gamma_{n,k}^b := \gamma_{n,k}^b(f, f, h \mid t, \alpha)$ for $0 \leq 2k \leq n \leq 4$.

(1) $\lambda(x) = \sigma_j \cdots \sigma_{i-1}$ with $\sigma_l > x$ for $j \leq l \leq i-1$, $u = \sigma_1 \cdots \sigma_{j-1}$, and $\sigma_{j-1} < x$.

(2) $\rho(x) = \sigma_{i+1} \cdots \sigma_k$ with $\sigma_l > x$ for $i+1 \leq l \leq k$, $v = \sigma_{k+1} \cdots \sigma_n$, and $\sigma_{k+1} < x$.

Here, any of the words $u, \lambda(x), \rho(x), v$ may be empty.

Example 2.42. If $\sigma = 810145113612279 \in \mathfrak{S}_{12}$, then the 3-factorization of σ is

$$\underbrace{8101}_u \underbrace{4511}_\lambda \underbrace{3612}_{\rho(3)} \underbrace{279}_v.$$

Definition 2.43 ([HR98]). Let $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$. Say that σ is an **André permutation of the first kind (resp. second kind)** if σ has no double descents, i.e., $\sigma_{i-1} > \sigma_i > \sigma_{i+1}$, and each x -factorisation $u \lambda(x) x \rho(x) v$ of σ has property

- $\lambda(x) = \emptyset$ if $\rho(x) = \emptyset$,
- $\max(\lambda(x)) < \max(\rho(x))$ (resp. $\min(\rho(x)) < \min(\lambda(x))$) if $\rho(x) \neq \emptyset$ and $\lambda(x) \neq \emptyset$.

Let $\mathcal{A}_n^1$ (resp. $\mathcal{A}_n^2$) be the set of André permutations of the first (resp. second) kind in $\mathfrak{S}_n$. It is known that [FS74] the cardinality of $\mathcal{A}_n^1$ (resp. $\mathcal{A}_n^2$) is the *Euler number* E_n , which is the n th coefficient in the Taylor expansion of $\sec(z) + \tan(z)$, namely,

$$\sum_{n \geq 0} E_n \frac{z^n}{n!} = \sec(z) + \tan(z). \quad (2.116)$$

For convenience, we define the subsets of $\mathcal{A}_n^1$ and $\mathcal{A}_n^2$, respectively

$$\mathcal{A}_{n, \text{des}=k}^1 := \{\sigma \in \mathcal{A}_n^1 : \text{des}(\sigma) = k\}; \quad (2.117)$$

$$\mathcal{A}_{n, \text{des}=k}^2 := \{\sigma \in \mathcal{A}_n^2 : \text{des}(\sigma) = k\}. \quad (2.118)$$

n	$k = 0$	$k = 1$	$k = 2$
0	1		
1	1		
2	1	h	
3	1	$2h + h^2p + h^2q$	
4	1	$3h + 2h^2(p + q) + h^3(p + q)^2$	$h^2 + h^4(p^2 + pq + q^2)$

Table 2.3: The first values of $E_{n,k}(h | p, q)$ for $0 \leq 2k \leq n \leq 4$.

Theorem 2.44. For $n \geq 0$ and $0 \leq k \leq \lfloor n/2 \rfloor$, we have

$$E_{n,k}(h | p, q) = \sum_{\sigma \in \mathcal{A}_{n+1, \text{des}=k}^2} h^{\text{D}(\sigma)} p^{31-2(\sigma)-\text{des}(\sigma)} q^{2-13(\sigma)}. \quad (2.119)$$

By using a group action on permutations without double descents, Pan and Zeng [PZ21] proved (2.112) and (2.119) when $h = 1$. We shall extend their proof to the general case in Section 2.6.1.

For $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$ with $\sigma_0 = \sigma_{n+1} = +\infty$, a letter $\sigma_i \in [n]$ is said to be a

- *left-to-right minimum* (**lmin**) if $\sigma_j > \sigma_i$ for every $j < i$;
- *left-to-right-minimum-valley* (**lminval**) if σ_i is a left-to-right minimum and also a valley;
- *right-to-left-minimum-valley* (**rminval**) if σ_i is a right-to-left minimum and also a valley;
- *right-to-left-minimum-double-ascent* (**rlminda**) if it is a double ascent and also a right-to-left minimum.

Definition 2.45. Let $\sigma = \sigma_1 \dots \sigma_n$ be a permutation in $\mathfrak{S}_n$ with $\sigma_0 = \sigma_{n+1} = +\infty$. Denote by $\text{Val}(\sigma)$ and $\text{Pk}(\sigma)$ the sets of valleys and peaks of σ , respectively. We define a variant of D by

$$\tilde{\text{D}}(\sigma) := \sum_{i \in \text{Pk}(\sigma)} i - \sum_{j \in \text{Val}(\sigma) \setminus \{1\}} j. \quad (2.120)$$

Let $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$. The *complement* (resp. the *reverse*) of σ is defined by, respectively

$$\sigma^c = (n+1-\sigma_1)(n+1-\sigma_2) \dots (n+1-\sigma_n), \quad (2.121)$$

and

$$\sigma^r = \sigma_n \sigma_{n-1} \dots \sigma_1. \quad (2.122)$$

Example 2.46. Let $\sigma = 71385624 \in \mathfrak{S}_8$, then we have

$$\sigma^c = 28614375 \text{ and } \sigma^r = 42658317.$$

For convenience, let

$$\text{lrmin}(\sigma) := \text{lmin}(\sigma) + \text{rmin}(\sigma) - 2; \quad (2.123a)$$

$$\text{lrminval}(\sigma) := \text{lminval}(\sigma) + \text{rminval}(\sigma) - 2; \quad (2.123b)$$

$$\text{lrmax}(\sigma) := \text{lmax}(\sigma) + \text{rmax}(\sigma) - 2; \quad (2.123c)$$

$$\text{lrmaxpk}(\sigma) := \text{lmaxpk}(\sigma) + \text{rmaxpk}(\sigma) - 2. \quad (2.123d)$$

n	$k = 0$	$k = 1$	$k = 2$
0	1		
1	αt		
2	$\alpha^2 t^2$	$\alpha h f$	
3	$\alpha^3 t^3$	$\alpha h f(h + \alpha h t + 2\alpha t)$	
4	$\alpha^4 t^4$	$\alpha h f((h + \alpha h t + \alpha t)^2 + 2\alpha^2 t^2)$	$\alpha h^2 f(h^2 + 2\alpha h^2 f + \alpha f)$

Table 2.4: The first values of $E_{n,k}(f, h, t, \alpha)$ for $0 \leq 2k \leq n \leq 4$.

Lemma 2.5.3. The mapping $\eta := \sigma \mapsto (\sigma^c)^r$ is a bijection on $\mathfrak{S}_{n, \text{des}=k}^{dd=0}$ such that

$$(\text{des}, \text{lmaxpk}, D, \text{lmaxda}, \text{lmax}) \sigma = (\text{des}, \text{lminval}, \tilde{D}, \text{rminda}, \text{lmin}) \eta(\sigma). \quad (2.124)$$

Lemma 2.5.3 yields another combinatorial interpretation for $\gamma_{n,k}^b(f, f, h | t, \alpha)$ defined in (2.93).

Proposition 2.5.3. For $n \geq 0$ and $0 \leq k \leq \lfloor n/2 \rfloor$, we have

$$\gamma_{n,k}^b(f, f, h | t, \alpha) = \sum_{\sigma \in \mathfrak{S}_{n+1, \text{des}=k}^{dd=0}} f^{\text{lminval}(\sigma)} h^{\tilde{D}(\sigma)} t^{\text{rminda}(\sigma)} \alpha^{\text{lmin}(\sigma)} \quad (2.125)$$

Theorem 2.47. For $n \geq 0$ and $0 \leq k \leq \lfloor n/2 \rfloor$, we have

$$E_{n,k}(f, h | t, \alpha) = \sum_{\sigma \in \mathcal{A}_{n+1, \text{des}=k}^1} f^{\text{rminval}(\sigma)-1} h^{D(\sigma)} t^{\text{rminda}(\sigma)} \alpha^{\text{rmin}(\sigma)-1}. \quad (2.126)$$

By using a group action on permutations without double descents, Dong et al. [DLP25] proved (2.113) and (2.126) when $f = h = 1$. We shall extend their proof to the general case in Section 2.6.2.

We now derive a second combinatorial interpretation of $E_{n,k}(f, h | t, \alpha)$ from Theorem 2.47.

Theorem 2.48. For $n \geq 0$ and $0 \leq k \leq \lfloor n/2 \rfloor$, we have

$$E_{n,k}(f, h | t, \alpha) = \sum_{\sigma \in \mathcal{A}_{n+1, \text{des}=k}^1} f^{\text{lmaxpk}(\sigma)-1} h^{D(\sigma)} t^{\text{lmaxda}(\sigma)} \alpha^{\text{lmax}(\sigma)-1}. \quad (2.127)$$

Proof. We define a bijection $\rho : \mathcal{A}_{n+1}^1 \rightarrow \mathcal{A}_{n+1}^1$ satisfying, for all $\sigma \in \mathcal{A}_{n+1}^1$

$$(\text{des}, \text{lminval}, D, \text{rminda}, \text{rmin}) \sigma = (\text{des}, \text{lmaxpk}, D, \text{lmaxda}, \text{lmax}) \rho(\sigma). \quad (2.128)$$

Given $\sigma \in \mathcal{A}_{n+1}^1$, we define $\rho(\sigma)$ as follows. First, we decompose σ into cycles according to its right-to-left minima. Rewrite each cycle by placing its maximum element in the first position, then reorder the cycles from left to right in increasing order of their maximum elements. Erasing the cycle notation and concatenating the reordered cycles yields a permutation $\rho(\sigma)$, which is readily verified to be an André permutation of the first kind. It is clear that ρ is a bijection.

In this cycle decomposition, all descents of σ occur within cycles. Therefore, the cyclic rotation that moves the maximum element to the first position in each cycle does not change $\text{des}(\sigma)$. Moreover, ρ maps right-to-left minima to left-to-right maxima and right-to-left-minimum double ascents to left-to-right-maximum double ascents. This proves (2.128).

The desired result now follows from (2.126) and the bijection ρ . $\square$

Example 2.49. Let $\sigma = 561748239 \in \mathcal{A}_9^1$. According to its right-to-left minima, we decompose σ into disjoint cycles as

$$\sigma = (561)(7482)(3)(9).$$

Then we have

$$\sigma' := \rho(\sigma) = (3)(615)(8274)(9) = 361582749.$$

Moreover,

- $\text{des}(\sigma) = |\{2, 4, 6\}| = 3$, $\text{rminda}(\sigma) = |\{3\}| = 1$, and $\text{rmin}(\sigma) = |\{1, 2, 3, 9\}| = 4$;
- $\text{des}(\sigma') = |\{2, 5, 7\}| = 3$, $\text{lmaxda}(\sigma') = |\{3\}| = 1$, and $\text{lmax}(\sigma') = |\{3, 6, 8, 9\}| = 4$.

Define the following enumerative polynomials

$$\begin{aligned} E_n(x, h | p, q) &:= \sum_{k=0}^{\lfloor n/2 \rfloor} E_{n,k}(h | p, q) x^k \\ &= \sum_{\sigma \in \mathcal{A}_{n+1}^2} x^{\text{des}(\sigma)} h^{\text{D}(\sigma)} p^{31-2(\sigma)-\text{des}(\sigma)} q^{2-13(\sigma)}; \end{aligned} \quad (2.129a)$$

$$\begin{aligned} E_n(x, f, h | t, \alpha) &:= \sum_{k=0}^{\lfloor n/2 \rfloor} E_{n,k}(f, h, t, \alpha) x^k \\ &= \sum_{\sigma \in \mathcal{A}_{n+1}^1} x^{\text{des}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} t^{\text{lmaxda}(\sigma)} \alpha^{\text{lmax}(\sigma)-1}. \end{aligned} \quad (2.129b)$$

Theorem 2.50. We have

$$1. \quad \sum_{n=0}^{\infty} E_n(x, h | p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.130)$$

where $b_k = h^k [k+1]_{p,q}$ and $\lambda_{k+1} = x h^{2k+1} [k+1]_{p,q} [k]_{p,q} / (p+q)$ for $k \geq 0$.

$$2. \quad \sum_{n=0}^{\infty} E_n(x, f, h | t, \alpha) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.131)$$

where $b_k = h^k (k + \alpha t)$ and $\lambda_{k+1} = x h^{2k+1} k(k-1+2\alpha f)/2$ for $k \geq 0$.

Proof. Combining Eqs. (2.129a), (2.112), (2.24) and Theorem 2.4 with

$$(\mathbf{u}, f, g, h, t, \alpha, \beta, p, q) = (1, x, 1, 0, 1, 1, h, 1, 1, 1, p, q),$$

we obtain the first J -fraction (2.130).

Similarly, combining Eqs. (2.129b), (2.113), (2.26) and Theorem 2.4 with

$$(\mathbf{u}, f, g, h, t, \alpha, \beta, p, q) = (1, x, 1, 0, f, f, h, t, \alpha, \alpha, 1, 1),$$

we obtain the second J -fraction (2.131). $\square$

Remark 2.51. Han [Han20] asked for a combinatorial interpretation of (2.130) when $x = h = p = 1$, which is a q -analogue of the remarkable J -fraction for Euler numbers E_{n+1} :

$$\sum_{n=0}^{\infty} E_{n+1} z^n = \mathcal{J}\left(z; k+1, \binom{k+1}{2}\right). \quad (2.132)$$

Pan and Zeng [PZ21] established the $h = 1$ case of (2.130) shortly. The $f = h = 1$ case of (2.131) appeared in [XZ24].

2.6 Factorization of γ -coefficients via group actions

2.6.1 Proof of Theorem 2.44

We use Pan-Zeng's group action [PZ21] to prove the following equivalent identity, for $0 \leq k \leq \lfloor n-1/2 \rfloor$,

$$\sum_{\sigma \in \mathfrak{S}_n^{\text{dd}=0}, \text{des}=\sigma=k} h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)} = (p+q)^k \sum_{\sigma \in \mathfrak{S}_n^{\text{dd}=0}, \text{des}=\sigma=k} h^{\text{D}(\sigma)} p^{31-2(\sigma)-k} q^{2-13(\sigma)}. \quad (2.133)$$

For convenience, define the subset of $\mathfrak{S}_n$

$$\mathfrak{S}_n^{\text{dd}=0} = \{\sigma \in \mathfrak{S}_n : \text{dd}(\sigma) = 0\}. \quad (2.134)$$

For any permutation $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$ and $x \in [n]$, we shall identify σ with its x -factorization³, i.e.,

$$\sigma = (w_1, w_2, x, w_4, w_5) = w_1 w_2 x w_4 w_5,$$

where w_2 (resp., w_4) is the maximal contiguous interval (possibly empty) immediately to the left (resp., right) of x whose letters are all greater than x . Let y_1 (resp. y_2) be the smallest element in w_2 (resp. w_4), i.e.,

$$y_1 := \min(w_2) \text{ and } y_2 := \min(w_4).$$

Definition 2.52. A valley x of σ is said to be

- *good* (resp. *bad*) if $y_1 > y_2$ (resp. $y_1 < y_2$);
- of *type I* if $\min(y_1, y_2)$ is a peak or double ascent;
- of *type II* if $\min(y_1, y_2)$ is a valley.

Proposition 2.6.1. Let $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$ and $x \in \text{Val}(\sigma)$ with $y = \min(y_1, y_2)$.

- If x is of type I and y is a peak, then $w_4 = y$ (resp. $w_2 = y$) if $y_1 > y_2$ (resp. $y_1 < y_2$).
- If x is of type I and y is a double ascent, then $w_4 = y w_4''$ (resp. $w_2 = y w_2''$) with $w_2'', w_4'' \neq \epsilon$ if $y_1 > y_2$ (resp. $y_1 < y_2$).
- If x is of type II, then $w_4 = w_4' y w_4''$ (resp. $w_2 = w_2' y w_2''$) with $w_2', w_2'', w_4', w_4'' \neq \epsilon$ if $y_1 > y_2$ (resp. $y_1 < y_2$).

Definition 2.53. For $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$ and each $x \in \text{Val}(\sigma)$ with $y = \min(y_1, y_2)$, we define its transform $\varphi(\sigma, x)$ as follows:

1. If x is of type I and y is a peak, then

$$\varphi(\sigma, x) = \begin{cases} (w_1, y, x, w_2, w_5) & \text{if } y = y_2, \\ (w_1, w_4, x, y, w_5) & \text{if } y = y_1. \end{cases}$$

2. If x is of type I and y is a double ascent, then

$$\varphi(\sigma, x) = \begin{cases} (w_1, y w_2, x, w_4'', w_5) & \text{if } y = y_2 \text{ and } w_4 = y w_4'', \\ (w_1, w_2'', x, y w_4, w_5) & \text{if } y = y_1 \text{ and } w_2 = y w_2''. \end{cases}$$

³For convenience, we adopt a concise form that is equivalent to Definition 2.41.

3. If x is of type II, then

$$\varphi(\sigma, x) = \begin{cases} (w_1, w_2 y w', x, w'', w_5) & \text{if } y = y_2 \text{ and } w_4 = w' y w'', \\ (w_1, w', x, w'' y w_4, w_5) & \text{if } y = y_1 \text{ and } w_2 = w' y w''. \end{cases}$$

with $w', w'' \neq \epsilon$.

The three cases (i), (ii), (iii) of the transformation with $y = y_2$ are depicted in Fig. 2.7.

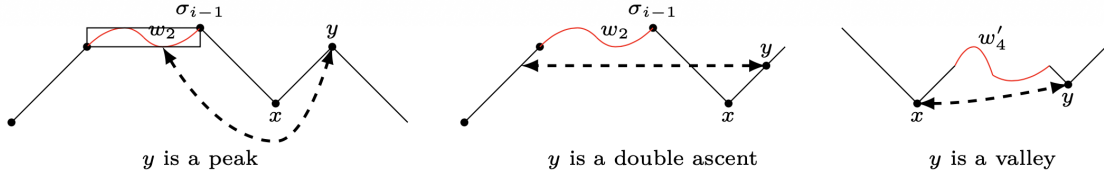


Figure 2.7: The transform $\varphi(\sigma, x)$ according to the type of y .

This transformation satisfies the following proposition.

Proposition 2.6.2. If $\sigma \in \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}$ and $x \in \text{Val}(\sigma)$, then $\varphi(\sigma, x) \in \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}$ and

$$D(\varphi(\sigma, x)) = D(\sigma), \quad (2.135)$$

$$(2-13) \varphi(\sigma, x) = \begin{cases} (2-13)(\sigma) + 1 & \text{if } x \text{ is good,} \\ (2-13)(\sigma) - 1 & \text{if } x \text{ is bad,} \end{cases} \quad (2.136)$$

$$(31-2) \varphi(\sigma, x) = \begin{cases} (31-2)(\sigma) - 1 & \text{if } x \text{ is good,} \\ (31-2)(\sigma) + 1 & \text{if } x \text{ is bad.} \end{cases} \quad (2.137)$$

Proof. Since the transformation $\varphi(\sigma, x)$ preserves the values at the valleys and peaks of σ , it follows directly from (2.7) that identity (2.135) holds. The proofs of (2.136) and (2.137) can be found in [PZ21, Proposition 5]. $\square$

Next, we define the transform $\varphi(\sigma, S)$ for any subset S of $\text{Val}(\sigma)$ with $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$.

Definition 2.54. Let $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$. For any $S \subseteq \text{Val}(\sigma)$, let $\{S_1, S_2\}$ be the partition of S such that

1. S_1 is the subset of S consisting of valleys of type I, say $i_1, \dots, i_\ell$;
2. S_2 is the subset of S consisting of valleys of type II, say $j_k < \dots < j_2 < j_1$.

Define the transforms

$$\varphi(\sigma, S_1) = \varphi(i_\ell, \dots, \varphi(i_2, \varphi(i_1, \sigma))),$$

$$\varphi(\sigma, S_2) = \varphi(j_k, \dots, \varphi(j_2, \varphi(j_1, \sigma))),$$

$$\varphi(\sigma, S) = \varphi(\varphi(\sigma, S_1), S_2).$$

Remark 2.55. Pan-Zeng prove that the image $\varphi(\sigma, S_1)$ is independent of the order of $i_1, \dots, i_\ell$, while $\varphi(\sigma, S_2)$ is defined for the elements of S_2 in the decreasing order $j_1 > j_2 > \dots > j_k$.

Proposition 2.6.3. If $\sigma \in \mathcal{A}_{n, \text{des}=k}^2$ and $S \subseteq \text{Val}(\sigma)$, then $\tau := \varphi(\sigma, S) \in \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}$ is well defined and

$$S = \{x \in \text{Val}(\tau) \mid x \text{ is a bad valley}\}. \quad (2.138)$$

For any set S , we denote by 2^S the set of all subsets of S . In what follows, for $\sigma \in \mathcal{A}_{n, \text{des}=k}^2$ we will identify $\text{Val}(\sigma)$ with $[k]$ under the map $a_i \mapsto i$ for $i \in [k]$ if $\text{Val}(\sigma)$ consists of $a_1 < a_2 < \dots < a_k$, and identify any subset $S \in \text{Val}(\sigma)$ with its image $S' \in 2^{[k]}$. Thus, we will use $2^{[k]}$ instead of $2^{\text{Val}(\sigma)}$.

Proposition 2.6.4. The map $\varphi : \mathcal{A}_{n, \text{des}=k}^2 \times 2^{[k]} \rightarrow \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}$ is a bijection such that for $(\sigma, S') \in \mathcal{A}_{n, k}^2 \times 2^{[k]}$ we have

$$D(\sigma) = D(\varphi(\sigma, S)); \quad (2.139)$$

$$(2-13)(\sigma) + |S'| = (2-13) \varphi(\sigma, S); \quad (2.140)$$

$$(31-2)(\sigma) - |S'| = (31-2) \varphi(\sigma, S). \quad (2.141)$$

Proof. Eq. (2.139) follows directly from (2.135). The proofs of (2.140) and (2.141) can be found in [PZ21, Proposition 7]. $\square$

Continuation of Proof of Eq. (2.133). Since $(p + q)^k = \sum_{S \in 2^{[k]}} p^{|S|} q^{k-|S|}$, it follows from Proposition 2.6.4 that

$$\sum_{(\sigma, S) \in \mathcal{A}_{n, \text{des}=k}^2 \times 2^{[k]}} h^{D(\sigma)} p^{31-2(\sigma)-|S|} q^{2-13(\sigma)+|S|} = \sum_{\sigma \in \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}} h^{D(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)},$$

which is precisely equivalent to (2.6.4). $\square$

Example 2.56. Let $\tau = 11\mathbf{2}13\mathbf{16453897}10 \in \mathfrak{S}_{13, \text{des}=5}^{\text{dd}=0}$.

(i) First, let $(\sigma, S) := (\tau, \emptyset)$. We have $S_2 = \{1, 3\}$ and $\min(S_2) = 1$, so $S = \{1\}$ and

$$\sigma := \varphi(\sigma, 1) = 11\mathbf{1}1213\mathbf{26453897}10.$$

As $S_2 = \{3\}$, we have $\min S_2 = 3$ and $S := \{1, 3\}$, hence

$$\sigma := \varphi(\sigma, 3) = 11\mathbf{1}1213\mathbf{26354897}10.$$

(ii) Now for σ , $S_2 = \emptyset$ and $S_1 = \{4, 7\}$, then $S := \{1, 3, 4, 7\}$ and

$$\sigma := \varphi(\sigma, S_1) = 11\mathbf{1}1213\mathbf{2631078945} \in \mathcal{A}_{13, 5}^2.$$

2.6.2 Proof of Theorem 2.47

We prove the following equivalent identity

$$\begin{aligned} & \sum_{\sigma \in \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}} f^{\text{lrminval}(\sigma)} h^{\tilde{D}(\sigma)} t^{\text{rminda}(\sigma)} a^{\text{lrmin}(\sigma)} \\ &= 2^k \sum_{\sigma \in \mathcal{A}_{n, \text{des}=k}^1} f^{\text{rminval}(\sigma)-1} h^{D(\sigma)} t^{\text{rminda}(\sigma)} a^{\text{rmin}(\sigma)-1}. \end{aligned} \quad (2.142)$$

Dong–Lin–Pan [DLP25] proved the above identity in the special case $f = h = 1$. We show that their approach extends to the general case. Recall that for any $x \in [n]$ and $\sigma \in \mathfrak{S}_n$, the x -factorization of σ is defined by

$$\sigma = w_1 w_2 x w_4 w_5,$$

where w_2 (resp., w_4) is the maximal contiguous interval (possibly empty) immediately to the left (resp., right) of x whose letters are all greater than x . Define the *swap mapping* $\varphi_x: \mathfrak{S}_n \rightarrow \mathfrak{S}_n$ by

$$\varphi_x(\sigma) := w_1 w_4 x w_2 w_5. \quad (2.143)$$

For instance, with $\sigma = 348579162 \in \mathfrak{S}_9$, in the 5-factorization of σ , $w_2 = 8$ and $w_4 = 79$ and so $\varphi_5(\sigma) = 347958162$. It is plain to see that φ_x is an involution acting on $\mathfrak{S}_n$, and for all $x, y \in [n]$, φ_x and φ_y commute. Therefore, for any $S \subseteq [n]$ we can define the function

$$\varphi_S: \mathfrak{S}_n \rightarrow \mathfrak{S}_n \quad \text{by} \quad \varphi_S = \prod_{x \in S} \varphi_x,$$

where multiplication is the composition of mappings. Hence the group $\mathbb{Z}_2^n$ acts on $\mathfrak{S}_n$ via the function φ_S . This action is known as the *Foata–Strehl action* (FS-action for short) on permutations [FS74].

Definition 2.57 (Bi-basic decomposition). For $\sigma \in \mathfrak{S}_n$, with $\text{rmin}(\sigma) - 1 = k$ and $\text{lmin}(\sigma) - 1 = l$, there is a unique decomposition of σ as

$$\sigma = \alpha_1 \alpha_2 \cdots \alpha_k 1 \beta_1 \beta_2 \cdots \beta_l, \quad (2.144)$$

where

- the first letter of each α_i ($1 \leq i \leq k$), which is smallest inside α_i , is a left-to-right minimum of σ ;
- and the last letter of each β_i ($1 \leq i \leq l$), which is smallest inside β_i , is a right-to-left minimum of σ .

This is the bi-basic decomposition of σ and each α_i or β_i is called a block of σ .

Example 2.58. Let $\sigma = 710295181131264 \in \mathfrak{S}_{12}$. Then we have

$$\mathbf{710|295|1|8113|1264}. \quad (2.145)$$

For a word $w = w_1 w_2 \cdots w_n$, let $w^r := w_n w_{n-1} \cdots w_1$ be the reversal of w .

For a given permutation $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$ with bi-basic decomposition as in (2.144) and $x \in [n]$, Dong-Lin-Pan defined the group action $\psi_x(\sigma)$ as follows.

- (1) If x is the first letter of α_i for some i , then x is a valley of σ . Let

$$\tilde{\alpha}_i := \varphi_S(\alpha_i), \quad (2.146)$$

where S is the set of all double ascents of σ inside α_i . Define $\psi_x(\sigma)$ to be the permutation obtained from σ by deleting the block α_i and then inserting $(\tilde{\alpha}_i)^r$ in some proper gap between blocks so that x becomes an additional right-to-left minimum;

- (2) If x is the last letter of β_i for some i , where β_i has length at least two⁴, then x is also a valley of σ . Let

$$\tilde{\beta}_i := \varphi_S(\beta_i),$$

where S is the set of all double ascents of σ inside β_i . Define $\psi_x(\sigma)$ to be the permutation obtained from σ by deleting the block β_i and then inserting $(\tilde{\beta}_i)^r$ in some proper gap between blocks so that x becomes an additional left-to-right minimum.

⁴In case (1), each block α_i has length at least two, as σ has no double descents.

- (3) If $x \neq 1$ is a valley of σ not in the above two cases, then set $\psi_x(\sigma) = \varphi_x(\sigma)$ defined in (2.143);
- (4) Otherwise, set $\psi_x(\sigma) = \sigma$.

As shown by Dong et al. [DLP25], for any $S \subseteq [n]$, the restriction ψ_S over $\mathfrak{S}_n^{\text{dd}=0}$ is stable and defines a $\mathbb{Z}_2^n$ -action on $\mathfrak{S}_n^{\text{dd}=0}$, which is called the *block-Foata-Strehl action* (BFS-action for short). Let $V(\sigma)$ be the set of valleys other than 1 in $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$. It is clear that $\psi_x(\sigma) = \sigma$ if and only if $x \notin V(\sigma)$.

Example 2.59. If $\sigma = 3485710|1|62|9 \in \mathfrak{S}_{10}^{\text{dd}=0}$, then $V(\sigma) = \{2, 3, 5\}$. We compute that

$$\begin{aligned}\psi_2(\sigma) &= 3485710|26|1|9, \\ \psi_3(\sigma) &= 1|62|4710583|9, \\ \psi_5(\sigma) &= 3471058|1|62|9, \\ \psi_{\{2,3\}}(\sigma) &= 26|1|4710583|9, \\ \psi_{\{2,5\}}(\sigma) &= 3471058|26|1|9, \\ \psi_{\{3,5\}}(\sigma) &= 1|62|4857103|9, \\ \psi_V(\sigma) &= 26|1|4857103|9.\end{aligned}$$

Lemma 2.6.1. For any $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$ and $x \in [n]$, the mapping ψ_x satisfies the following identity:

$$(\text{des}, \text{lrminval}, \tilde{D}, \text{rminda}, \text{lrmin}) \sigma = (\text{des}, \text{lrminval}, \tilde{D}, \text{rminda}, \text{lrmin}) \psi_x(\sigma). \quad (2.147)$$

Proof. From the bi-basic decomposition of σ , it is easy to see that

- the first element of each block α_i is a left-to-right-minimum-valley;
- the last element of each block β_i of length at least two is a right-to-left-minimum-valley;
- the (unique) element in each block β_i of length one is a right-to-left-minimum-double-ascent.

In the first two cases, we see that mapping ψ_x transfers a left-to-right-minimum-valley to a right-to-left-minimum-valley, or a right-to-left-minimum-valley to a left-to-right-minimum-valley. In the third case, the right-to-left-minimum-double-ascent is fixed under mapping ψ_x . Thus, the mapping ψ_x preserves the statistics lrminval , lrmin , and rminda . Moreover, since the mapping ψ_x does not change the values of the valleys and peaks of σ , it keeps $\tilde{D}$ unchanged. The reverse operator and the mapping φ_x keep des unchanged as well. Therefore, (2.147) follows. $\square$

Definition 2.60 ([HJO23]). Let $A := \{a_1, \dots, a_k\}$ be a set of k positive integers. Let $C = (a_1 \cdots a_k)$ be a cycle (cyclic permutation) of A with $a_1 = \min\{a_1, \dots, a_k\}$. Then, cycle C is called an *André cycle* if the word $a_2 \cdots a_k$ is an André permutation of the first kind. We say that a permutation is a **cycle André permutation** if it is a product of disjoint André cycles.

Let $\mathcal{CA}_n$ be the set of cycle André permutations of $[n]$. For example, $(1\ 5\ 6)(2\ 7\ 4\ 8)(3)$ is a cycle André permutation.

Lemma 2.6.2 ([HJO23] Theorem 4.1). There exists a bijection $\phi : \mathcal{CA}_n \rightarrow \mathcal{A}_{n+1}^1$ as follows: let $\sigma = C_1 C_2 \dots C_l \in \mathcal{CA}_n$, where $C_1, \dots, C_l$ are André cycles ordered from left to right in increasing order of their smallest letters, which are at the end of each cycle; erasing the parentheses in σ and appending letter $n+1$ to the end results in $\phi(\sigma)$.

Continuation of Proof of Eq. (2.142). For $\sigma \in \mathfrak{S}_n^{\text{dd}=0}$, let $\text{Orb}(\sigma)$ be the orbit of σ under the BSF-action. As $\psi_S(\sigma) = \sigma$ if and only if $S \cap V(\sigma) = \emptyset$, we have $|\text{Orb}(\sigma)| = 2^{V(\sigma)}$. Assuming that $\text{lmin}(\sigma) = l$, then by Definition 2.43 of André permutations of the first kind, there exists a unique $\hat{\sigma} \in \text{Orb}(\sigma)$ with bi-basic decomposition

$$\hat{\sigma} = 1\beta_1\beta_2 \cdots \beta_l \quad (2.148)$$

satisfying that

- ($\star$) each block β_i when removing its last letter becomes an André permutation of the first kind.

By Lemma 2.6.1, we have

$$\sum_{\pi \in \text{Orb}(\sigma)} f^{\text{lrminval}(\pi)} h^{\tilde{D}(\pi)} t^{\text{rminda}(\pi)} \alpha^{\text{lrmin}(\pi)} = 2^{V(\hat{\sigma})} f^{\text{lrminval}(\hat{\sigma})} h^{\tilde{D}(\hat{\sigma})} t^{\text{rminda}(\hat{\sigma})} \alpha^{\text{lrmin}(\hat{\sigma})} \quad (2.149)$$

$$= 2^{\text{des}(\hat{\sigma})} f^{\text{rminval}(\hat{\sigma})-1} h^{\tilde{D}(\hat{\sigma})} t^{\text{rminda}(\hat{\sigma})} \alpha^{\text{rmin}(\hat{\sigma})-1}, \quad (2.150)$$

The second identity follows from $|V(\hat{\sigma})| = \text{des}(\hat{\sigma})$, $\text{lrminval}(\hat{\sigma}) = \text{rminval}(\hat{\sigma}) - 1$ and $\text{lrmin}(\hat{\sigma}) = \text{rmin}(\hat{\sigma}) - 1$. Summing over all orbits of $\mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}$ under the BFS-action gives

$$\sum_{\sigma \in \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0}} f^{\text{lrminval}(\sigma)} h^{\tilde{D}(\sigma)} t^{\text{rminda}(\sigma)} \alpha^{\text{lrmin}(\sigma)} = 2^k \sum_{\hat{\sigma} \in \hat{\mathfrak{S}}_{n, \text{des}=k}^{\text{dd}=0}} f^{\text{rminval}(\hat{\sigma})-1} h^{\tilde{D}(\hat{\sigma})} t^{\text{rminda}(\hat{\sigma})} \alpha^{\text{rmin}(\hat{\sigma})-1}, \quad (2.151)$$

where

$$\hat{\mathfrak{S}}_{n, \text{des}=k}^{\text{dd}=0} := \{\hat{\sigma} \in \mathfrak{S}_{n, \text{des}=k}^{\text{dd}=0} : \text{bi-basic decomposition of } \hat{\sigma} \text{ has the form (2.148) satisfying } (\star)\}.$$

For a word $w = w_1 w_2 \cdots w_n$, let $c(w)$ be the cyclic permutation $(w_1 - 1, w_2 - 1, \dots, w_n - 1)$. There exists a natural one-to-one correspondence

$$\hat{\sigma} = 1\beta_1\beta_2 \cdots \beta_l \mapsto c(\beta_1)c(\beta_2) \cdots c(\beta_l) = \tau. \quad (2.152)$$

between $\hat{\mathfrak{S}}_{n, \text{des}=k}^{\text{dd}=0}$ and $\{\tau \in \mathcal{CA}_{n-1} : \text{drop}(\tau) = k\}$. Then, combining this and the bijection ϕ given in Lemma 2.6.2, we see $\phi(\tau) := \tau' \in \mathcal{A}_n$. One can easily check that

$$(\text{des}, \text{rminval} - 1, \tilde{D}, \text{rminda}, \text{rmin}) \hat{\sigma} = (\text{des}, \text{rminval}, D, \text{rminda}, \text{rmin}) \tau'. \quad (2.153)$$

We obtain Eq. (2.142) by combining (2.151) and (2.153). $\square$

Example 2.61. Let $\sigma = 5107811|46|1|32|9 \in \mathfrak{S}_{11}^{\text{dd}=0}$. Then, we have

$$\begin{aligned} \psi_5(\sigma) &= 46|1|32|8117105|9; \\ \psi_{4,5}(\sigma) &= 1|32|64|8117105|9; \\ \hat{\sigma} := \psi_{4,5,7}(\sigma) &= 1|32|64|1078115|9. \end{aligned}$$

By (2.152), we obtain

$$\hat{\sigma} \rightarrow \tau := (21)(53)(967104)(8) \in \mathcal{CA}_{10}. \quad (2.154)$$

Applying the ϕ , we have

$$\tau' := \phi(\tau) = 2153967104811 \in \mathcal{A}_{11}^1. \quad (2.155)$$

We now verify (2.153).

- On one hand, we have $\text{des}(\hat{\sigma}) = |\{2, 4, 6, 9\}| = 4$, $\text{rminval}(\hat{\sigma}) - 1 = |\{1, 2, 4, 5\}| - 1 = 3$, $\text{rminda}(\hat{\sigma}) = |\{9\}| = 1$, $\text{rmin}(\hat{\sigma}) = |\{1, 2, 4, 5, 9\}| = 5$ and $\tilde{D}(\hat{\sigma}) = 3 + 6 + 10 + 11 - (2 + 4 + 5 + 7) = 12$.
- On the other hand, we have $\text{des}(\tau') = |\{1, 3, 5, 8\}| = 4$, $\text{rminval}(\tau') = |\{1, 3, 4\}| = 3$, $\text{rminda}(\tau') = |\{8\}| = 1$, $\text{rmin}(\tau') = |\{1, 3, 4, 8, 11\}| = 5$, and $D(\tau') = 2 + 5 + 9 + 10 - (1 + 3 + 4 + 6) = 12$.

2.7 Application to Total-positivity

Definition 2.62 (Total positivity). A matrix of real numbers is said to be totally positive (TP in short) if all its minors are nonnegative.

Definition 2.63 (Coefficientwise total positivity). We say that a polynomial in $\mathbb{R}[\mathbf{X}]$, where $\mathbf{X}$ is the set of variables, is *coefficientwise-positive* if all its coefficients are nonnegative. A matrix whose entries are polynomials with real coefficients is said to be *coefficientwise-totally positive* (*coefficientwise-TP*, in short) if all its minors, which are polynomials themselves, are coefficientwise-positive in $\mathbf{X}$.

Let $\mathbf{a} := (a_n)_{n \geq 0}$ denote a sequence of real numbers (resp. polynomials in $\mathbb{R}[\mathbf{X}]$). To this sequence, we associate the corresponding *Hankel matrix*.

$$H_\infty(\mathbf{a}) := (a_{i+j})_{i,j \geq 0} = \begin{pmatrix} a_0 & a_1 & a_2 & \cdots \\ a_1 & a_2 & a_3 & \cdots \\ a_2 & a_3 & a_4 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (2.156)$$

We say that the sequence $\mathbf{a}$ is *Hankel-totally positive* (resp. *coefficientwise Hankel-totally positive*) if the Hankel matrix $H_\infty(\mathbf{a})$ is totally positive (resp. coefficientwise-totally positive in $\mathbf{X}$).

2.7.1 S-fraction and coefficientwise-TP

For a sequence of real numbers $(a_n)_{n \geq 0}$, its Hankel matrix $(a_{i+j})_{i,j \geq 0}$ is *totally positive* if and only if there exist nonnegative real numbers $\alpha := (\alpha_n)_{n \geq 0}$ such that the following identity holds (at the level of formal power series):

$$\mathcal{S}(z; \alpha) := \frac{\alpha_0}{1 - \frac{\alpha_1 z}{1 - \frac{\alpha_2 z}{1 - \frac{\alpha_3 z}{\ddots}}}} = \sum_{n=0}^{\infty} a_n z^n. \quad (2.157)$$

This result is a combination of results due to Stieltjes [Sti94] and Gantmakher–Krein [GK37]. The continued fraction of the form on the right-hand side of equation (2.157) is called a *Stieltjes-type continued fraction* (*S-fraction* in short).

If $(a_n)_{n \geq 0}$ is instead a sequence of polynomials in one or several variables, we no longer have any equivalent condition available to us. However, if an S-fraction such as the one in

equation (2.157) exists, where the $\alpha_0, \alpha_1, \dots$ are all polynomials with nonnegative coefficients, then that is a *sufficient* condition for the Hankel matrix $(a_{i+j})_{i,j \geq 0}$ to be coefficientwise-totally positive. The following fact is folklore.

Lemma 2.7.1. Let $\mathbf{a} = (a_n)_{n \geq 0}$ and $\alpha = (\alpha_n)_{n \geq 0}$ be two sequences of real coefficient polynomials in $\mathbb{R}[\mathbf{X}]$ satisfying the following power series identity (2.157). If the polynomials α_n ($n \geq 0$) are coefficientwise-positive, then $\mathbf{a}$ is *coefficientwise-Hankel totally positive* in $\mathbf{X}$.

This fact was first stated in this form in [Sok14], although it follows straightforwardly from the work of [Fla80] and of [Vie83, Section 3, Chapter 4]. Note that the converse of Lemma 2.7.1 does not hold in general.

A permutation $\sigma = \sigma_1 \sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$ is called an *alternating permutation* (or *zigzag permutation*) if $\sigma_1 > \sigma_2 < \sigma_3 > \sigma_4 < \cdots$. As proved by André [And79], the number of alternating permutations in $\mathfrak{S}_n$ is given by the Euler number E_n . Let $\mathcal{Z}_n$ denote the set of alternating permutations of length n . A permutation of odd length is alternating if and only if it contains neither a double ascent nor a double descent.

Define the polynomials

$$Z_n^o(x, f, g, h, \alpha, \beta | p, q) := \sum_{\sigma \in \mathcal{Z}_{2n+1}} x^{\text{des}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} \times \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1} p^{31-2(\sigma)} q^{2-13(\sigma)}. \quad (2.158)$$

Theorem 2.64. The sequence $\mathcal{Z}_o := (Z_n(x, f, g, h, \alpha, \beta | p, q))_{n \geq 0}$ is coefficientwise-Hankel totally positive in $\mathbf{X} := \{x, f, g, h, \alpha, \beta, p, q\}$.

Proof. By setting $\mathbf{u} = (1, x, 0, 0)$ in Theorem 2.4, we derive

$$\sum_{n=0}^{\infty} Z_n(x, f, g, h, \alpha, \beta | p, q) z^n = \mathcal{J}(\sqrt{z}; \mathbf{0}, \boldsymbol{\lambda}) = \mathcal{S}(z; \boldsymbol{\lambda}), \quad (2.159)$$

where $\lambda_k = x h^{2k-1} [k]_{p,q} [k+1; \beta g, \alpha f]_{p,q}$.

Hence, every λ_k is coefficientwise-positive in the variables $x, f, g, h, \alpha, \beta, p, q$. By Lemma 2.7.1, the sequence $\mathcal{Z}_o$ is therefore coefficientwise Hankel-totally positive. $\square$

A permutation σ of even length $2n$ is an alternating permutation if and only if $\sigma' := \sigma \star (2n+1)$ is an alternating permutation, where $\star$ denotes the concatenation of words.

Lemma 2.7.2. Let $\sigma' := \sigma \star (2n+1) \in \mathcal{Z}_{2n+1}$ with $\sigma \in \mathcal{Z}_{2n}$. We have

$$(\text{des}, \text{lmaxpk}, \text{D} + 2n - \sigma_{2n}, \text{lmax}) \sigma = (\text{des}, \text{lmaxpk} - 1, \text{D}, \text{lmax} - 1) \sigma'. \quad (2.160)$$

For each $x \in [2n]$, we have

$$(31-2)_x(\sigma) = (31-2)_x(\sigma') \text{ and } (2-31)_x(\sigma) = \begin{cases} (2-13)_x(\sigma') & \text{if } \sigma^{-1}(x) \text{ is even;} \\ (2-13)_x(\sigma') - 1 & \text{if } \sigma^{-1}(x) \text{ is odd.} \end{cases}$$

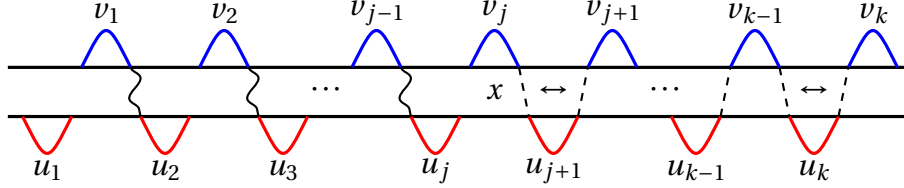
Proof. Eq. (2.160) follows directly from the Definitions of these permutation statistics. Let $\sigma' = \sigma \star (n+1) \in \mathcal{Z}_{2n+1}$, where $\sigma \in \mathcal{Z}_{2n}$. By the x -decomposition of σ' , the first equality is obvious. Suppose that for $x \in [2n]$, the x -decomposition of σ' is

$$\sigma' = u_1 v_1 \cdots u_k v_k, \quad k \geq 1,$$

where x lies in v_j , see (2.43). It is straightforward to verify the following:

- word v_k has $2n+1$ as its last letter, possibly together with other entries preceding $2n+1$.
- If x is a peak of σ' , then x cannot lie in v_k .

When $\sigma^{-1}(x)$ is even, i.e., x is a valley of σ' , the word v_j must contain elements that are larger than x on both sides. Consequently, each contribution to $(2-13)_x(\sigma')$ corresponds to a contribution to $(2-31)_x(\sigma)$. When $\sigma^{-1}(x)$ is odd, the letter x is a peak of σ' , and the word v_j contains only x . In this case, one contribution to $(2-31)_x(\sigma)$ is lost. See the corresponding illustration below.



This completes the proof. $\square$

Define the polynomials

$$Z_n^e(x, f, h, \alpha | p, q) := \sum_{\sigma \in \mathcal{Z}_{2n}} x^{\text{des}(\sigma)} f^{\text{lmaxpk}(\sigma)} h^{\text{D}(\sigma) + 2n - \sigma_{2n}} \alpha^{\text{lmax}(\sigma)} p^{31-2(\sigma)} q^{2-31(\sigma)}. \quad (2.161)$$

Theorem 2.65. *The sequence $\mathcal{Z}_e := (Z_n^e(x, f, h, \alpha | p, q))_{n \geq 0}$ is coefficientwise-Hankel totally positive in $\mathbf{X} := \{x, f, h, \alpha, p, q\}$.*

Proof. Combining Lemma 2.7.2 and Theorem 2.4 with $\mathbf{u} = (1, x, 0, 0)$ and $\beta = 0$ we derive

$$\sum_{n=0}^{\infty} Z_n^e(x, f, h, \alpha | p, q) z^n = \mathcal{J}(\sqrt{z}; \mathbf{0}, \boldsymbol{\lambda}) = \mathcal{S}(z; \boldsymbol{\lambda}), \quad (2.162)$$

where $\lambda_k = x h^{2k-1} [k]_{p,q} [k+1; 0, \alpha f]_{p,q}$. Hence, every λ_k is coefficientwise-positive in the variables x, f, h, α, p, q . By Lemma 2.7.1, the sequence $\mathcal{Z}_e$ is therefore coefficientwise Hankel-totally positive. $\square$

2.7.2 J-fraction and coefficientwise-TP

Let $\mathbf{a} := (a_n)_{n \geq 0}$ be a sequence of polynomials in $\mathbb{R}[\mathbf{X}]$, where $\mathbf{X}$ is the set of variables. Assume that there exists a J -fraction for which the following identity is satisfied:

$$\sum_{n \geq 0} a_n t^n = \mathcal{J}(t; \mathbf{b}, \boldsymbol{\lambda}) = \frac{1}{1 - b_0 t - \frac{\lambda_1 t^2}{1 - b_1 t - \frac{\lambda_2 t^2}{1 - b_2 t - \ddots}}}. \quad (2.163)$$

Define the associated tridiagonal matrix, called *Jacobi matrix* or *production matrix* see [PSZ23],

$$M_{\infty}(\mathbf{b}, \boldsymbol{\lambda}) = \begin{pmatrix} b_0 & 1 & 0 & 0 & \dots \\ \lambda_1 & b_1 & 1 & 0 & \dots \\ 0 & \lambda_2 & b_2 & 1 & \dots \\ 0 & 0 & \lambda_3 & b_3 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (2.164)$$

The following characterization for coefficientwise-TP is due to Sokal [Sok14], see also [PSZ23, Theorem 9.7].

Theorem 2.66 ([Sok14]). *Let $\mathbf{a} := (a_n)_{n \geq 0}$ be a sequence of polynomials in $\mathbb{R}[\mathbf{X}]$ and the associated Hankel matrix $H_\infty(\mathbf{a})$. If $M_\infty(\mathbf{b}, \boldsymbol{\lambda})$ is coefficientwise-TP in $\mathbf{X}$, then so is $H_\infty(\mathbf{a})$.*

The classical criterion in [Pin10, Theorem 4.3] for tridiagonal (Jacobi) matrices over $\mathbb{R}$ extends verbatim to matrices with entries in an arbitrary partially ordered commutative ring; we shall therefore work in this generality throughout.

Lemma 2.7.3. A Jacobi matrix is coefficientwise-totally positive in $\mathbf{X}$ if and only if all its off-diagonal elements and all its principal minors containing consecutive rows and columns are coefficientwise-positive in $\mathbf{X}$.

Proof. Let $A = M_\infty(\mathbf{b}, \boldsymbol{\lambda}) = (a_{i,j})_{i,j \geq 0}$. Then $a_{i,j} = 0$ if $|i - j| > 1$. Fix $r \geq 1$ and index sets $I = (i_1 < \dots < i_r)$ and $J = (j_1 < \dots < j_r)$, and consider the minor

$$A \begin{pmatrix} I \\ J \end{pmatrix} := \det(a_{i_p, j_q})_{1 \leq p, q \leq r}.$$

If $|i_k - j_k| \geq 2$ for some k , then $A \begin{pmatrix} I \\ J \end{pmatrix} = 0$ since A is tridiagonal. Thus assume $|i_k - j_k| \leq 1$ for all k . Whenever $|i_\ell - j_\ell| = 1$, the tridiagonal form forces a block-triangular decomposition of the corresponding submatrix, yielding the factorization

$$A \begin{pmatrix} i_1, \dots, i_r \\ j_1, \dots, j_r \end{pmatrix} = A \begin{pmatrix} i_1, \dots, i_{\ell-1} \\ j_1, \dots, j_{\ell-1} \end{pmatrix} A \begin{pmatrix} i_\ell \\ j_\ell \end{pmatrix} A \begin{pmatrix} i_{\ell+1}, \dots, i_r \\ j_{\ell+1}, \dots, j_r \end{pmatrix}. \quad (2.165)$$

Iterating (2.165) we can factor any minor of A , namely, if

$$i_1 = j_1, \dots, i_{k_1} = j_{k_1}, \quad i_{k_1+1} \neq j_{k_1+1}, \dots, i_{k_2} \neq j_{k_2}, \quad i_{k_2+1} = j_{k_2+1}, \dots, i_{k_3} = j_{k_3}, \dots,$$

then

$$A \begin{pmatrix} i_1, \dots, i_r \\ j_1, \dots, j_r \end{pmatrix} = A \begin{pmatrix} i_1, \dots, i_{k_1} \\ j_1, \dots, j_{k_1} \end{pmatrix} A \begin{pmatrix} i_{k_1+1} \\ j_{k_1+1} \end{pmatrix} \dots A \begin{pmatrix} i_{k_2} \\ j_{k_2} \end{pmatrix} A \begin{pmatrix} i_{k_2+1}, \dots, i_{k_3} \\ j_{k_2+1}, \dots, j_{k_3} \end{pmatrix} \dots, \quad (2.166)$$

where

$$A \begin{pmatrix} i_k \\ j_k \end{pmatrix} = \begin{cases} 1, & \text{if } j_k = i_k + 1, \\ \lambda_{i_k}, & \text{if } j_k = i_k - 1. \end{cases}$$

Finally, if $k_t + 1 < k_{t+1}$ for some $t \in [r - 1]$, then

$$A \begin{pmatrix} k_1, \dots, k_s \\ k_1, \dots, k_s \end{pmatrix} = A \begin{pmatrix} k_1, \dots, k_t \\ k_1, \dots, k_t \end{pmatrix} A \begin{pmatrix} k_{t+1}, \dots, k_s \\ k_{t+1}, \dots, k_s \end{pmatrix}. \quad (2.167)$$

Iterating (2.167), every principal minor factors into a product of consecutive principal minors. Since coefficientwise-positive polynomials are closed under multiplication, all factors on the right-hand side of (2.166) are coefficientwise-positive, and therefore $A \begin{pmatrix} I \\ J \end{pmatrix}$ is coefficientwise-positive for all I, J . Hence A is coefficientwise-totally positive. $\square$

Define the multivariable Eulerian polynomials

$$A_n(x, h | p, q) := \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{des}(\sigma)} h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)}. \quad (2.168)$$

Theorem 2.67. *The sequence $\mathcal{A} := (A_n(x, h \mid p, q))_{n \geq 0}$ is coefficientwise-Hankel totally positive in $\mathbf{X} := \{x, h, p, q\}$.*

Proof. By Theorem 2.4 with $u_2 = u_4 = x$ and $u_1 = u_3 = f = g = t = \alpha = \beta = 1$, we have

$$\sum_{n \geq 0}^{\infty} A_n(x, h \mid p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.169)$$

where $b_k = h^k(1+x)[k+1]_{p,q}$ and $\lambda_k = xh^{2k-1}[k]_{p,q}[k+1]_{p,q}$.

Let $A = M_{\infty}(\mathbf{b}, \boldsymbol{\lambda}) = (a_{i,j})_{i,j \geq 0}$. For $k \geq 0$ and $n \geq 1$, define the contiguous principal minors

$$D_{k,n} := A \begin{pmatrix} I \\ J \end{pmatrix} \text{ with } I = J = \{k, \dots, k+n-1\}. \quad (2.170)$$

Then we have the standard continuant recurrence

$$D_{k,n} = b_{k+n-1}D_{k,n-1} - \lambda_{k+n-1}D_{k,n-2}, \quad (2.171)$$

where $D_{k,0} = 1$ and $D_{k,1} = b_k$.

We prove by induction on $n \geq 1$ that

$$D_{k,n} = h^{nk + \binom{n}{2}} (1+x+x^2+\dots+x^n) \left(\prod_{j=1}^n [k+j]_{p,q} \right). \quad (2.172)$$

Set $[n]_x := 1+x+\dots+x^{n-1}$ and

$$F_{k,n} := h^{nk + \binom{n}{2}} [n+1]_x \left(\prod_{j=1}^n [k+j]_{p,q} \right).$$

Clearly $F_{k,0} = 1 = D_{k,0}$ and $F_{k,1} = (1+x)h^k[k+1]_{p,q} = b_k = D_{k,1}$. By (2.169), for $n \geq 2$, we have

$$\begin{aligned} b_{k+n-1} &= (1+x)h^{k+n-1}[k+n]_{p,q}, \\ \lambda_{k+n-1} &= xh^{2k+2n-3}[k+n-1]_{p,q}[k+n]_{p,q}, \end{aligned}$$

and a direct computation gives

$$b_{k+n-1}F_{k,n-1} - \lambda_{k+n-1}F_{k,n-2} = h^{nk + \binom{n}{2}} ((1+x)[n]_x - x[n-1]_x) \left(\prod_{j=1}^n [k+j]_{p,q} \right).$$

Since $(1+x)[n]_x - x[n-1]_x = [n+1]_x$, we obtain

$$b_{k+n-1}F_{k,n-1} - \lambda_{k+n-1}F_{k,n-2} = F_{k,n}.$$

Thus $F_{k,n}$ satisfies the same recurrence and initial conditions as $D_{k,n}$ for all $k \geq 0$ and $n \geq 1$. This establishes (2.172).

By Lemma 2.7.3 and Theorem 2.66, the sequence $\mathcal{A}$ is therefore coefficientwise Hankel-totally positive.

□

2.8 Concluding remarks

The following identities are direct consequences of the continued fraction: specializing one of the parameters to -1 makes the expression collapse into a simple closed form.

Proposition 2.8.1. The following identities hold.

$$\sum_{\sigma \in \mathfrak{S}_n} (-1)^{\text{inv}(\sigma)} x^{\text{exc}(\sigma)} h^{\text{depth}(\sigma)} = (1 - xh)^{n-1} \quad (2.173a)$$

$$\sum_{\sigma \in \mathfrak{S}_n} (-1)^{\text{nest}(\sigma)} x^{\text{exc}(\sigma)} h^{\text{depth}(\sigma)} = (1 + xh)^{n-1} \quad (2.173b)$$

$$\sum_{\sigma \in \mathfrak{S}_n} (-1)^{\text{rmin}(\sigma)} x^{\text{exc}(\sigma)} h^{\text{depth}(\sigma)} = (-1)^n (1 - xh)^{n-1} \quad (2.173c)$$

$$\sum_{\sigma \in \mathfrak{S}_n} (-1)^{\text{lmax}(\sigma)} h^{\text{depth}(\sigma)} = (-1)^n (1 - h)^{n-1}. \quad (2.173d)$$

Proof. If $\lambda_2 = 0$, the J -fraction (2.11) reduces to the rational fraction:

$$\mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}) = \frac{1}{1 - b_0 z - \frac{\lambda_1 z^2}{1 - b_1 z}} = \frac{1 - b_1 z}{(1 - b_0 z)(1 - b_1 z) - \lambda_1 z^2}. \quad (2.174)$$

Substituting $\mathbf{u} = (x, 1, x, 1)$, $p = q = t = \alpha = \beta = 1$, and $s = -1$ in Theorem 2.7, we obtain $b_0 = 1$, $b_1 = -xh$, $\lambda_1 = -xh$, $\lambda_2 = 0$, and

$$\sum_{n \geq 0} \sum_{\sigma \in \mathfrak{S}_n} (-1)^{\text{inv}(\sigma)} x^{\text{exc}(\sigma)} h^{\text{depth}(\sigma)} z^n = 1 + \frac{z}{1 - (1 - xh)z}, \quad (2.175)$$

which implies (2.173a). The remaining identities follow similarly. $\square$

Remark 2.68. Eqs. (2.173a)–(2.173c) were proved by Eu et al. [EFL26] without using continued fractions.

Eu et al. [EFL26] considered a variant of the notions of crossing and nesting for permutations as follows:

$$\text{cros}^*(\sigma) = \#\{(i, j) \in [n] \times [n] : (i < j < \sigma_i < \sigma_j) \vee (\sigma_i < \sigma_j \leq i < j)\}; \quad (2.176a)$$

$$\text{nest}^*(\sigma) = \#\{(i, j) \in [n] \times [n] : (i < j < \sigma_j < \sigma_i) \vee (\sigma_j < \sigma_i \leq i < j)\}. \quad (2.176b)$$

We note that these statistics are closely related to cros and nest (see (2.15)) by the inverse mapping. More precisely, we have the following result.

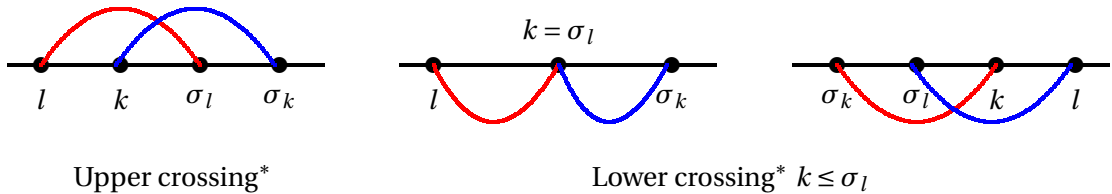
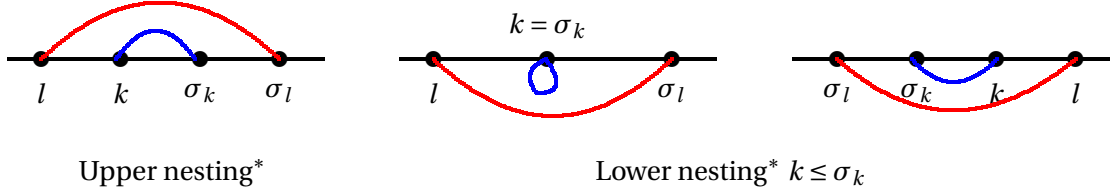


Figure 2.8: Upper crossing* and lower crossing* indexed at k .

Lemma 2.8.1. The inverse mapping $\iota: \mathfrak{S}_n \rightarrow \mathfrak{S}_n$, $\iota(\sigma) = \sigma^{-1}$ is a bijection and satisfies

$$(\text{exc}, \text{drop}, \text{fix}, \text{depth}, \text{cros}, \text{nest}) \sigma = (\text{drop}, \text{exc}, \text{fix}, \text{depth}, \text{cros}^*, \text{nest}^*) \iota(\sigma). \quad (2.177)$$

Figure 2.9: Upper nesting* and lower nesting* indexed at k

Proof. We only verify the statistic depth; the remaining correspondences are straightforward. By Definition 2.14, we see that

$$\text{depth}(\sigma^{-1}) = \sum_{\sigma^{-1}(k) > k} (\sigma^{-1}(k) - k) = \sum_{i > \sigma(i)} (i - \sigma(i)),$$

where the second equality follows by setting $i = \sigma^{-1}(k)$. Moreover,

$$0 = \sum_{i=1}^n (\sigma(i) - i) = \sum_{\sigma(i) > i} (\sigma(i) - i) + \sum_{i > \sigma(i)} (i - \sigma(i)).$$

Combining the above two identities we obtain $\text{depth}(\sigma^{-1}) = \text{depth}(\sigma)$. □

Substituting $\mathbf{u} = (x, 1, x, 1)$ and $s = \alpha = \beta = 1$ in Theorem 2.7 we obtain

$$1 + \sum_{n \geq 1} \left(\sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)} t^{\text{fix}(\sigma)} h^{\text{depth}(\sigma)} p^{\text{cros}(\sigma)} q^{\text{nest}(\sigma)} \right) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.178)$$

where $b_k = h^k(tq^k + (1 + xp)[k]_{p,q})$ and $\lambda_k = xh^{2k-1}[k]_{p,q}^2$.

By Lemma 2.8.1, since $n = \text{exc}(\sigma) + \text{fix}(\sigma) + \text{drop}(\sigma)$ for $\sigma \in \mathfrak{S}_n$, substituting $(x, t) \rightarrow (\frac{1}{x}, \frac{t}{x})$ and then $z \rightarrow xz$ in (2.178), we obtain

$$\sum_{n \geq 0} \left(\sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)} t^{\text{fix}(\sigma)} p^{\text{cros}^*(\sigma)} q^{\text{nest}^*(\sigma)} h^{\text{depth}(\sigma)} \right) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.179)$$

where $b_k = h^k(x[k]_{p,q} + tq^k + p[k]_{p,q})$ and $\lambda_k = xh^{2k-1}[k]_{p,q}^2$.

When $t = 1$, the above J -fraction reduces to

$$\sum_{n \geq 0} \left(\sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)} p^{\text{cros}^*(\sigma)} q^{\text{nest}^*(\sigma)} h^{\text{depth}(\sigma)} \right) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (2.180)$$

where $b_k = h^k(x[k]_{p,q} + [k+1]_{p,q})$ and $\lambda_k = xh^{2k-1}[k]_{p,q}^2$. This J -fraction was obtained by Eu et al. [EFL26, Theorem 1.2]. They further observed that it exhibits a symmetry $\text{nest}^* \leftrightarrow \text{cros}^*$ for fixed values of exc and depth , and asked for a bijective proof of this symmetry. The following theorem provides such a bijection.

Theorem 2.69. *Let $\iota : \mathfrak{S}_n \rightarrow \mathfrak{S}_n$, $\iota(\sigma) = \sigma^{-1}$ and $\Phi^* = \Phi_{SZ} \circ \Phi_C^{-1}$. Then the mapping $\Theta := \iota \circ \Phi^* \circ \iota : \mathfrak{S}_n \rightarrow \mathfrak{S}_n$ is a bijection such that for any $\sigma \in \mathfrak{S}_n$,*

$$(\text{exc}, \text{nest}^*, \text{cros}^*, \text{depth}) \sigma = (\text{exc}, \text{cros}^*, \text{nest}^*, \text{depth}) \Theta(\sigma) \quad (2.181)$$

Proof. By (2.22) of Theorem 2.15 and $n = \text{wex} + \text{drop}$, the bijection Φ^* satisfies

$$(\text{drop}, \text{nest}, \text{cros}, \text{depth}) \sigma = (\text{drop}, \text{cros}, \text{nest}, \text{depth}) \Phi^*(\sigma). \quad (2.182)$$

Combining (2.177) and (2.182), we obtain (2.181). □

From the construction of the bijections Φ_{SZ} and Φ_C , we see that the two bijections satisfy the following identity

$$\begin{aligned}\Phi_{SZ} \circ \Phi_C^{-1} &= \tilde{\Psi}_{FZ}^{-1} \circ \rho \circ \tilde{\Psi}_{FV} \circ (\tilde{\Psi}_{FZ}^{-1} \circ \tilde{\Psi}_{FV})^{-1} \\ &= \tilde{\Psi}_{FZ}^{-1} \circ \rho \circ \tilde{\Psi}_{FZ},\end{aligned}\tag{2.183}$$

where $\tilde{\Psi}_{FZ}$ and ρ are defined in (2.69) and (2.72).

Besides the bijections Φ_C and Φ_{SZ} , Steingrímsson and Williams [SW07b] constructed another variant Φ_{SW} of Φ_{CSZ} satisfying

$$(\text{ndes}, 31-2, 2-31) \sigma = (\text{drop} + 1, \text{nest}, \text{cros}) \Phi_{SW}(\sigma) \quad \text{for } \sigma \in \mathfrak{S}_n.\tag{2.184}$$

Let $\text{Ddif}(\sigma) := \text{Destop}(\sigma) - \text{Desbot}(\sigma)$ for $\sigma \in \mathfrak{S}_n$. It is known [CSZ97, Proposition 3] that the mapping Φ_{CSZ} implies that the two bi-statistics

$$(\text{des}, \text{Ddif}) \text{ and } (\text{exc}, \text{depth})$$

are equidistributed on $\mathfrak{S}_n$. Two alternative proofs of this result were given by Reifegerste [Rei02] and Petersen and Tenner [PT15] using variants of the *fundamental transformation*.

Chapter 3

A bi-Stirling-Euler-Mahonian polynomial²

Abstract

Motivated by recent work on (re)mixed Eulerian numbers, we provide a combinatorial interpretation of a subfamily of the remixed Eulerian numbers introduced by Nadeau and Tewari. More specifically, we show that these numbers can be realized as the generating polynomials of permutations with respect to the statistics of left-to-right minima, right-to-left minima, descents, and the mixed major index. Our results generalize both the bi-Stirling Eulerian polynomials of Carlitz-Scoville and the Stirling-Euler-Mahonian polynomials of Butler.

Keywords: Eulerian polynomial, left-to-right minimum, right-to-left-minimum, descent, mixed major index

3.1 Introduction

The Eulerian numbers $\langle n \rangle_k$ ($n > k \geq 0$) can be defined by the recurrence

$$\langle n \rangle_k = (n - k) \langle n - 1 \rangle_{k-1} + (k + 1) \langle n - 1 \rangle_k, \quad (3.1)$$

with initial condition $\langle 0 \rangle_0 = 1$ and $\langle n \rangle_k = 0$ if $k \notin 0, \dots, n - 1$. This recurrence immediately implies the symmetry $\langle n \rangle_k = \langle n \rangle_{n-k-1}$. It is well-known [FS70, Pet15] that $\langle n \rangle_k$ counts the number of permutations in $\mathfrak{S}_n$ with k descents. The n th Eulerian polynomial, which generates the Eulerian numbers, is defined by

$$A_n(x) = \sum_{k=0}^{n-1} \langle n \rangle_k x^k \quad (3.2)$$

and has the exponential generating function

$$1 + \sum_{n=1}^{\infty} A_n(x) \frac{t^n}{n!} = \frac{1 - x}{e^{t(x-1)} - x}. \quad (3.3)$$

²This chapter is based on the paper "A bi-Stirling-Euler-Mahonian polynomial", published in *Advances in Applied Mathematics*, 173 (2026), Article 102975.

Setting $A(r, s) := \langle r+s+1 \rangle_r$, Carlitz-Scoville [CS74] noticed the following symmetric formula

$$F(x, y) := \sum_{r,s=0}^{\infty} A(r, s) \frac{x^r y^s}{(r+s+1)!} = \frac{e^x - e^y}{xe^y - ye^x}, \quad (3.4)$$

and further introduced the bi-Stirling Eulerian numbers $A(r, s | \alpha, \beta)$ by

$$\sum_{r,s=0}^{\infty} A(r, s | \alpha, \beta) \frac{x^r y^s}{(r+s)!} = (1 + xF(x, y))^{\alpha} (1 + yF(x, y))^{\beta}. \quad (3.5)$$

Note that $A(r, s | \alpha, \beta) = A(s, r | \beta, \alpha)$ and $A(r, s | 1, 1) = A(r, s)$. The first few terms of the above generating function are

$$1 + (\alpha x + \beta y) + \frac{1}{2}(\alpha^2 x^2 + (2\alpha\beta + \alpha + \beta)yx + \beta^2 y^2) + \frac{1}{6}(\alpha^3 x^3 + \beta^3 y^3 + (3\alpha^2\beta + 3\alpha^2 + 3\alpha\beta + \alpha + \beta)yx^2 + (3\alpha\beta^2 + 3\alpha\beta + 3\beta^2 + \alpha + \beta)xy^2) + \dots \quad (3.6)$$

Carlitz-Scoville [CS74, Sec. 8] proved the recurrence

$$A(r, s | \alpha, \beta) = (s + \alpha)A(r-1, s | \alpha, \beta) + (r + \beta)A(r, s-1 | \alpha, \beta) \quad (3.7)$$

and also provided the combinatorial interpretation

$$A(r, s | \alpha, \beta) = \sum_{\substack{\sigma \in \mathfrak{S}_{r+s+1} \\ \text{des}(\sigma)=r}} \alpha^{\text{lmin}(\sigma)-1} \beta^{\text{rmin}(\sigma)-1}, \quad (3.8)$$

where $\text{des}(\sigma)$ is the number of descents of σ , while $\text{lmin}(\sigma)$ (resp. $\text{rmin}(\sigma)$) is the number of *left-to-right* (resp. *right-to-left*) minima of σ , see (3.11).

Let $\lambda_1 \geq \dots \geq \lambda_{r+1}$ be real numbers and let $\lambda = (\lambda_1, \dots, \lambda_{r+1})$. The permutahedron $\text{Perm}(\lambda)$ is the convex hull of points $\lambda_{\sigma} = (\lambda_{\sigma(1)}, \dots, \lambda_{\sigma(r+1)})$ where σ is a permutation in $\mathfrak{S}_{r+1}$. It is known that

$$\text{vol}(\text{Perm}(\lambda)) = \sum_{c=(c_1, \dots, c_r)} A_c \frac{\mu_1^{c_1} \dots \mu_r^{c_r}}{c_1! \dots c_r!}$$

where $c \in \mathcal{W}_r = \{(c_1, \dots, c_r) : c_1 + \dots + c_r = r\}$ and $\mu_i = \lambda_i - \lambda_{i+1}$. The coefficients A_c are positive integers known as the *mixed Eulerian numbers* [Pos09].

In this chapter we adopt the standard q -notations [And76]. For $x \in \mathbb{R}$ the q -real number $[x]$ is defined by

$$[x] = \frac{1 - q^x}{1 - q}.$$

For an integer $r \in \mathbb{N}$, we define $[r]! = 1 \cdot (1 + q) \dots (1 + q + \dots + q^{r-1})$ and the q -binomial coefficient by

$$\begin{bmatrix} r \\ k \end{bmatrix} = \frac{[r]!}{[k]![r-k]!} \quad \text{for } r \geq k \geq 0.$$

Nadeau and Tewari [NT21, NT23] introduced a q -analogue of the mixed Eulerian numbers $A_c(q)$, known as the *remixed Eulerian numbers*. These numbers possess several remarkable properties, among which:

1. $A_c(q)$ is a polynomial with non-negative coefficients and $A_c(1) = A_c$,
2. $A_{(c_1, \dots, c_r)}(q) = q^{\binom{r}{2}} A_{(c_r, \dots, c_1)}(q^{-1})$,

3. $\sum_{c \in \mathcal{W}_r} A_c(q) = [r]! \cdot \frac{1}{r+1} \binom{2r}{r},$
4. $A_{(1, \dots, 1)}(q) = [r]!,$ and
5. $A_{(k, 0, \dots, 0, r-k)}(q) = q^{\binom{k}{2}} \cdot \begin{bmatrix} r \\ k \end{bmatrix}.$

Recently, Gaudin [Gau24] observed that the Carlitz-Scoville numbers $A(r, s | \alpha, \beta)$ are a rescaled subfamily of the mixed Eulerian numbers and proposed the following q -analogue $A(r, s | \alpha, \beta)_q$ within the framework of the remixed Eulerian numbers,

$$A(r, s | \alpha, \beta)_q = \frac{1}{[\alpha + \beta - 1]!} A_{(0^r, 1^{\alpha-1}, r+s+1, 1^{\beta-1}, 0^s)}(q). \quad (3.9)$$

The following q -analogue of the recurrence (3.7) is due to Gaudin [Gau24, Proposition 4.3]:

$$A(r, s | \alpha, \beta)_q = q^{r+\beta-1} [s + \alpha] A(r-1, s | \alpha, \beta)_q + [r + \beta] A(r, s-1 | \alpha, \beta)_q. \quad (3.10)$$

For $\sigma = \sigma_1 \sigma_2 \dots \sigma_n \in \mathfrak{S}_n$, define the following statistics

$$\text{des}(\sigma) = |\{i \in [n-1] \mid \sigma_i > \sigma_{i+1}\}|, \quad (3.11a)$$

$$\text{lmin}(\sigma) = |\{i \in [n] \mid \sigma_i < \sigma_j, \forall j < i\}|, \quad (3.11b)$$

$$\text{rmin}(\sigma) = |\{i \in [n] \mid \sigma_i < \sigma_j, \forall j > i\}|. \quad (3.11c)$$

The statistic des is a well-known Eulerian statistic due to (3.2), both lmin and rmin are Stirling statistics:

$$\sum_{\sigma \in \mathfrak{S}_n} x^{\text{lmin}(\sigma)} = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{rmin}(\sigma)} = \sum_{k=0}^n s(n, k) x^k, \quad (3.12)$$

where $s(n, k)$ are the Stirling numbers of the first kind. The *inversion number* inv and *major index* maj are defined by

$$\text{inv}(\sigma) = |\{(i, j) \mid \sigma_i > \sigma_j, i < j\}|, \quad (3.13)$$

$$\text{maj}(\sigma) = \sum_{\sigma_i > \sigma_{i+1}} i. \quad (3.14)$$

The statistics inv and maj are well-known Mahonian statistics:

$$\sum_{\sigma \in \mathfrak{S}_n} q^{\text{inv}(\sigma)} = \sum_{\sigma \in \mathfrak{S}_n} q^{\text{maj}(\sigma)} = [n]!. \quad (3.15)$$

For convenience, we introduce the following notation, valid for $n \geq k \geq 0$:

$$E_{n,k}(\alpha, \beta, q) := A(k, n-k | \alpha, \beta)_q \quad (3.16)$$

with the convention $E_{n,k}(\alpha, \beta, q) = 0$ if $k \notin \{0, \dots, n\}$. As $A_{(1^r)}(q) = [r]!$ (see [NT23]), by (3.9) we have $E_{0,0}(\alpha, \beta, q) = 1$. In view of (3.16), setting $r + s = n$ and $r = k$ in (3.10) we obtain

$$E_{n,k}(\alpha, \beta, q) = q^{\beta+k-1} [n-k+\alpha] E_{n-1,k-1}(\alpha, \beta, q) + [k+\beta] E_{n-1,k}(\alpha, \beta, q). \quad (3.17)$$

When $(\alpha, \beta) = (1, 1)$ or $(0, 1)$, several q -analogues of Eulerian numbers are already known [Car54, Sta76, Raw81]. In the case $\alpha = 0$, Butler [But04b, But25] introduced a novel q -analogue of Eulerian numbers by combining the parameter β with the major index.

The aim of this chapter is to provide a combinatorial interpretation of $E_{n,k}(\alpha, \beta, q)$ in terms of the permutation statistics such as left-to-right minima, right-to-left minima,

descents and (re)mixed major index, and to derive several explicit formulas for these numbers. Our results generalize and unify those of Carlitz-Scoville [CS74] and Butler [But04b].

For $\sigma \in \mathfrak{S}_n$ the *mixed major index* of σ , denoted by $\widetilde{\text{maj}}$, is defined by

$$\widetilde{\text{maj}}(\sigma) = (1 + \text{des}(\sigma) - \text{lmin}(\sigma))(\alpha - 1) + (n - \text{rmin}(\sigma))(\beta - 1) + \text{maj}(\sigma). \quad (3.18)$$

Theorem 3.1. For integer $n \geq k \geq 0$, we have

$$E_{n,k}(\alpha, \beta, q) = \sum_{\sigma \in \mathfrak{S}_{n+1,k}} [\alpha]^{\text{lmin}(\sigma)-1} [\beta]^{\text{rmin}(\sigma)-1} q^{\widetilde{\text{maj}}(\sigma)}. \quad (3.19)$$

where $\mathfrak{S}_{n+1,k}$ denotes the set of permutations in $\mathfrak{S}_{n+1}$ with k descents.

We define the *bi-Stirling-Euler-Mahonian polynomials* $E_n(x | \alpha, \beta, q)$ by

$$E_n(x | \alpha, \beta, q) = \sum_{k=0}^n E_{n,k}(\alpha, \beta, q) x^k. \quad (3.20)$$

For example, it follows from (3.17) that

$$\begin{aligned} E_0(x | \alpha, \beta, q) &= 1, \\ E_1(x | \alpha, \beta, q) &= [\beta] + q^\beta [\alpha] x, \\ E_2(x | \alpha, \beta, q) &= [\beta]^2 + q^\beta ([1 + \alpha][\beta] + [\alpha][\beta + 1])x + q^{2\beta+1} [\alpha]^2 x^2. \end{aligned}$$

Define the q -derivative operator δ_x by

$$\delta_x p(x) = \frac{p(qx) - p(x)}{(q-1)x}, \quad (3.21)$$

where $p(x) \in \mathbb{R}[[x]]$.

Theorem 3.2. Let $E_n(x; q) := E_n(x | \alpha, \beta, q)$ for $n \geq 0$. We have the following q -identities:

(i)

$$E_n(x; q) = ([\beta] + q^\beta [n-1 + \alpha]x) E_{n-1}(x; q) + (1-x)xq^\beta \delta_x(E_{n-1}(x; q)). \quad (3.22)$$

(ii)

$$\frac{E_n(x; q)}{(x; q)_{n+\alpha+\beta}} = \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n. \quad (3.23)$$

(iii)

$$\sum_{n \geq 0} \frac{E_n(x; q)}{(x; q)_{n+\alpha+\beta}} \frac{t^n}{n!} = \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} \exp([j + \beta]t). \quad (3.24)$$

(iv)

$$\sum_{k=0}^j \begin{bmatrix} j - k + n + \alpha + \beta - 1 \\ j - k \end{bmatrix} E_{n,k}(\alpha, \beta, q) = \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n. \quad (3.25)$$

(v)

$$E_{n,k}(\alpha, \beta, q) = \sum_{j=0}^k \begin{bmatrix} n + \alpha + \beta \\ k - j \end{bmatrix} \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} (-1)^{k-j} q^{\binom{k-j}{2}} [\beta + j]^n. \quad (3.26)$$

We shall prove Theorem 3.1 and Theorem 3.2 in Section 2 and derive some corollaries in Section 3. Barbero et al. [BGSVn15] introduced the ν -order (s, t) -Eulerian numbers. In Section 4 we show that when $\nu = 1$ their numbers coincide with the Carlitz-Scoville numbers $A(r, s | \alpha, \beta)$.

Remark 3.3. Using the inversion statistic on permutations, Dong-Lin-Pan [DLP24] obtained a q -analogue of (3.5) via Gessel's q -compositional formula. For recent developments on Carlitz-Scoville's generalized Eulerian polynomials, we refer the reader to [XZ24].

3.2 Proof of main Theorems

Proof of Theorem 3.1. Consider the enumerative polynomials

$$\hat{E}_{n,k}(\alpha, \beta, q) = \sum_{\sigma \in \mathfrak{S}_{n+1,k}} [\alpha]^{\text{lmin}(\sigma)-1} [\beta]^{\text{rmin}(\sigma)-1} q^{\widetilde{\text{maj}}(\sigma)}. \quad (3.27)$$

It suffices to verify that the polynomials $\hat{E}_{n,k}(\alpha, \beta, q)$ satisfy the recurrence (3.17). Clearly Eq. (3.17) is valid for $n = 0$. It is clear that deleting $n + 1$ from a permutation in $\mathfrak{S}_{n+1,k}$, we obtain a permutation in $\mathfrak{S}_{n,k} \cup \mathfrak{S}_{n,k-1}$. Inversely, any permutation π in $\mathfrak{S}_{n+1,k}$ can be created by adding $n + 1$ to a permutation w in $\mathfrak{S}_{n,k} \cup \mathfrak{S}_{n,k-1}$. We will discuss four types of insertion positions: a) at the beginning of w , b) at the end of w , c) at descent positions of w , d) at ascent positions of w .

(a) If $w = \sigma_1 \dots \sigma_n \in \mathfrak{S}_{n,k-1}$, then $\sigma = (n+1)\sigma_1 \dots \sigma_n \in \mathfrak{S}_{n+1,k}$ with

$$(\text{lmin}, \text{rmin}, \text{des})\pi = (1 + \text{lmin}, \text{rmin}, 1 + \text{des})w.$$

Hence

$$\sum_{\substack{\pi \in \mathfrak{S}_{n+1,k} \\ \pi(1)=n+1}} [\alpha]^{\text{lmin}(\pi)-1} [\beta]^{\text{rmin}(\pi)-1} q^{\widetilde{\text{maj}}(\pi)} = [\alpha] q^{\beta+k-1} \hat{E}_{n-1,k-1}(\alpha, \beta, q).$$

(b) If $w = \sigma_1 \dots \sigma_n \in \mathfrak{S}_{n,k}$, then $\pi = \sigma_1 \dots \sigma_n(n+1) \in \mathfrak{S}_{n+1,k}$ and $(\text{lmin}, \text{rmin})\pi = (\text{lmin}, 1 + \text{rmin})w$. Hence

$$\sum_{\substack{\pi \in \mathfrak{S}_{n+1,k} \\ \pi(n)=n+1}} [\alpha]^{\text{lmin}(\pi)-1} [\beta]^{\text{rmin}(\pi)-1} q^{\widetilde{\text{maj}}(\pi)} = [\beta] \hat{E}_{n-1,k}(\alpha, \beta, q).$$

(c) If $w = \sigma_1 \dots \sigma_n \in \mathfrak{S}_{n,k}$ with the descent set $\{i_1, i_2, \dots, i_k\}$, then we define $\pi = \sigma_1 \dots \sigma_{i_j}(n+1)\sigma_{i_j+1} \dots \sigma_n \in \mathfrak{S}_{n+1,k}$ with $1 \leq j \leq k$. It is clear that $(\text{lmin}, \text{rmin})\pi = (\text{lmin}, \text{rmin})w$ and $\widetilde{\text{maj}}(\pi) = \widetilde{\text{maj}}(w) + k - j$. Thus

$$\sum_{\pi \in \mathfrak{S}'_{n+1,k}} [\alpha]^{\text{lmin}(\pi)-1} [\beta]^{\text{rmin}(\pi)-1} q^{\widetilde{\text{maj}}(\pi)} = q^\beta [k] \hat{E}_{n-1,k}(\alpha, \beta, q),$$

where $\mathfrak{S}'_{n+1,k}$ is the set of permutations $\pi = \sigma_1 \dots \sigma_i(n+1)\sigma_{i+1} \dots \sigma_n$ in $\mathfrak{S}_{n+1,k}$ such that $\sigma_i > \sigma_{i+1}$.

- (d) Let $w = \sigma_1 \dots \sigma_n \in \mathfrak{S}_{n,k-1}$ with descent set $\{i_1, \dots, i_{k-1}\}$. If π is a permutation $\pi = \sigma_1 \dots \sigma_j(n+1)\sigma_{j+1} \dots \sigma_n \in \mathfrak{S}_{n+1,k}$ with $i_l < j < i_{l+1}$, where $0 \leq l \leq k-1$, $i_0 = 0$ and $i_k = n$, then

$$(\text{lmin}, \text{rmin}, \text{des})\pi = (\text{lmin}, \text{rmin}, 1 + \text{des})w, \quad (3.28)$$

$$\widetilde{\text{maj}}(\pi) = (k-1-l) + (j+1) + (\alpha-1) + (\beta-1) + \widetilde{\text{maj}}(w). \quad (3.29)$$

Since

$$\begin{aligned} \bigcup_{l=0}^{k-1} \{k-1-l+j : i_l < j < i_{l+1}\} &= \bigcup_{l=0}^{k-1} \{k-l+i_l, \dots, k-l+i_{l+1}-2\} \\ &= \{k, k+1, \dots, n-1\}, \end{aligned}$$

we have

$$\sum_{\substack{i_l < j < i_{l+1} \\ 0 \leq l \leq k-1}} q^{\alpha+\beta-1+k-1-l+j} = q^{\alpha+\beta+k-1} [n-k].$$

It follows that

$$\sum_{\pi \in \mathfrak{S}_{n+1,k}''} [\alpha]^{\text{lmin}(\pi)-1} [\beta]^{\text{rmin}(\pi)-1} q^{\widetilde{\text{maj}}(\pi)} = q^{\alpha+\beta+k-1} [n-k] \cdot \widehat{E}_{n-1,k-1}(\alpha, \beta, q),$$

where $\mathfrak{S}_{n+1,k}''$ is the set of permutations $\sigma = \sigma_1 \dots \sigma_i(n+1)\sigma_{i+1} \dots \sigma_n$ in $\mathfrak{S}_{n+1,k}$ such that $\sigma_i < \sigma_{i+1}$.

Combining the two cases (a) and (d) (resp. (b) and (c)) we obtain

$$\widehat{E}_{n,k}(\alpha, \beta, q) = q^{\beta+k-1} [n-k+\alpha] \widehat{E}_{n-1,k-1}(\alpha, \beta, q) + [k+\beta] \widehat{E}_{n-1,k}(\alpha, \beta, q). \quad (3.30)$$

Comparing the above equation with (3.17) we obtain $\widehat{E}_{n,k}(\alpha, \beta, q) = E_{n,k}(\alpha, \beta, q)$. $\square$

For non-negative integer $n \in \mathbb{Z}$, we define $(x; q)_0 = 1$ and

$$(x; q)_n = (1-x)(1-xq) \cdots (1-xq^{n-1}) \quad (n \geq 1).$$

The q -binomial coefficients $\begin{bmatrix} x \\ n \end{bmatrix}$ are defined by

$$\begin{bmatrix} x \\ n \end{bmatrix} = \frac{(q^{x-n+1}; q)_n}{(q; q)_n}.$$

Furthermore we define $(x; q)_\infty = \lim_{n \rightarrow \infty} (x; q)_n$ and $(x; q)_n$ for all real numbers n by

$$(x; q)_n = (x; q)_\infty / (xq^n; q)_\infty. \quad (3.31)$$

Recall [And76, Chapter 2] that

$$\sum_{n=0}^{\infty} \frac{(x; q)_n t^n}{(q; q)_n} = \frac{(xt; q)_\infty}{(t; q)_\infty}. \quad (3.32)$$

In particular we have

$$(x; q)_N = \sum_{i=0}^{\infty} \begin{bmatrix} N \\ i \end{bmatrix} (-1)^i x^i q^{i(i-1)/2}, \quad (3.33)$$

$$\frac{1}{(x; q)_N} = \sum_{j=0}^{\infty} x^j \begin{bmatrix} N+j-1 \\ j \end{bmatrix}. \quad (3.34)$$

For $k, n \in \mathbb{N}$ an easy calculation yields

$$\delta_x \frac{x^k}{(x; q)_n} = x^{k-1} \frac{[k] + x[n-k]q^k}{(x; q)_{n+1}} = \begin{cases} [k]x^{k-1}, & \text{if } n = 0; \\ \frac{[n]}{(x; q)_{n+1}}, & \text{if } k = 0. \end{cases} \quad (3.35)$$

Proof of Theorem 3.2. (i) Multiplying Eq. (3.17) or (3.30) by x^k and summing k from 0 to n we obtain

$$\begin{aligned} \sum_{k=0}^n E_{n,k}(\alpha, \beta, q) x^k &= \sum_{k=0}^n \{q^{\beta+k-1} [n-k+\alpha] E_{n-1,k-1}(\alpha, \beta, q) + [k+\beta] E_{n-1,k}(\alpha, \beta, q)\} x^k \\ &= \sum_{k=0}^{n-1} q^{\beta+k} [n-1+\alpha-k] E_{n-1,k}(\alpha, \beta, q) x^{k+1} + \sum_{k=0}^{n-1} [k+\beta] E_{n-1,k}(\alpha, \beta, q) x^k. \end{aligned} \quad (3.36)$$

Substituting $[k+\beta] = [\beta] + q^\beta [k]$ and $q^{\beta+k} [n-1+\alpha-k] = q^\beta [n-1+\alpha] - q^\beta [k]$ in (3.36) and then using $\delta_x x^k = [k]x^{k-1}$, we have

$$\begin{aligned} E_n(x; q) &= q^\beta [n-1+\alpha] x E_{n-1}(x; q) - q^\beta x^2 \delta_x (E_{n-1}(x; q)) \\ &\quad + [\beta] E_{n-1}(x; q) + q^\beta x \delta_x (E_{n-1}(x; q)), \end{aligned} \quad (3.37)$$

which is clearly equal to Eq. (3.22).

(ii) We proceed by induction on $n \geq 0$. When $n = 0$ Eq. (3.23) is (3.34) with $N = \alpha + \beta$. Multiplying (3.23) by x^β and shifting $n \rightarrow n-1$ yields

$$\frac{x^\beta E_{n-1}(x; q)}{(x; q)_{n+\alpha+\beta-1}} = \sum_{j \geq 0} x^{j+\beta} \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^{n-1}. \quad (3.38)$$

Applying the operator $x^{1-\beta} \delta_x$ to (3.38) and using the q -Leibniz formula

$$\delta_x (f(x)g(x)) = \delta_x(f(x))g(qx) + f(x)\delta_x(g(x)), \quad (3.39)$$

we obtain

$$\begin{aligned} &\frac{(1-x)(q^\beta x \delta_x (E_n(x; q)) + [\beta] E_n(x; q)) + [n+\alpha+\beta-1] x E_n(x; q)}{(x; q)_{n+\alpha+\beta}} \\ &= \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n. \end{aligned} \quad (3.40)$$

Combining the above identity with (3.22), we have (3.23).

(iii) Multiplying Eq. (3.23) by $t^n / n!$ followed by summing over $n \geq 0$, we have

$$\begin{aligned} \sum_{n \geq 0} \frac{E_n(x; q)}{(x; q)_{n+\alpha+\beta}} \frac{t^n}{n!} &= \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} \sum_{n \geq 0} \frac{([j + \beta] t)^n}{n!} \\ &= \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} \exp([j + \beta] t). \end{aligned} \quad (3.41)$$

(iv) Eq. (3.25) follows by extracting the coefficient of x^j in both sides of Eq. (3.23) and applying Eq. (3.34).

(v) Eq. (3.26) follows from multiplying both sides of Eq. (3.23) by $(x; q)_{n+\alpha+\beta}$ and extracting the coefficient of x^k in both sides by using Eq. (3.33). $\square$

3.3 Corollaries of main theorems

In this section, we illustrate applications of our main theorem by specializing its parameters.

Corollary 3.4. [But04a, Corollary 4.2.4] For $n \geq 1$, we have

$$\frac{\sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} q^{\widetilde{\text{maj}}_1(\sigma)} [\beta]^{\text{rmin}(\sigma)}}{(x; q)_{n+\beta}} = \sum_{j \geq 0} x^j \begin{bmatrix} j + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n, \quad (3.42)$$

where $\widetilde{\text{maj}}_1(\sigma) = (n - \text{rmin}(\sigma))(\beta - 1) + \text{maj}(\sigma)$.

Proof. When $\alpha = 0$, the summation for $E_n(x | \alpha, \beta, q)$ reduces to the permutations $\sigma \in \mathfrak{S}_{n+1}$ such that $\text{lmin}(\sigma) = 1$, viz, $\sigma_1 = 1$. Define $\sigma'_i = \sigma_{i+1} - 1$ for $i \in [n]$, then $\sigma' \in \mathfrak{S}_n$ and

- $\text{des}(\sigma) = \text{des}(\sigma')$;
- $\text{maj}(\sigma) = \text{maj}(\sigma') + \text{des}(\sigma')$;
- $\text{rmin}(\sigma) = \text{rmin}(\sigma') + 1$.

Hence,

$$\sum_{\substack{\sigma \in \mathfrak{S}_{n+1} \\ \text{lmin}(\sigma)=1}} x^{\text{des}(\sigma)} q^{\widetilde{\text{maj}}_1(\sigma)} [\beta]^{\text{rmin}(\sigma)-1} = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} q^{\widetilde{\text{maj}}_1(\sigma)} [\beta]^{\text{rmin}(\sigma)}. \quad (3.43)$$

The result follows by combining (3.23) and (3.43). $\square$

Corollary 3.5. For $n \geq 1$, we have

$$\frac{\sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} q^{\widetilde{\text{maj}}_2(\sigma)} [\alpha]^{\text{lmin}(\sigma)}}{(x; q)_{n+\alpha}} = \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha \\ j + 1 \end{bmatrix} [j + 1]^n, \quad (3.44)$$

where $\widetilde{\text{maj}}_2(\sigma) = (1 + \text{des}(\sigma) - \text{lmin}(\sigma))(\alpha - 1) + \text{maj}(\sigma)$.

Proof. When $\beta = 0$, the summation for $E_n(x | \alpha, \beta, q)$ reduces to the permutations $\sigma \in \mathfrak{S}_{n+1}$ with $\sigma_{n+1} = 1$. Let $\sigma'_i = \sigma_i - 1$ for $i \in [n]$, then $\sigma' \in \mathfrak{S}_n$ and

- $\text{des}(\sigma) = \text{des}(\sigma') + 1$;
- $\text{maj}(\sigma) = \text{maj}(\sigma') + n$;
- $\text{lmin}(\sigma) = \text{lmin}(\sigma') + 1$.

Hence,

$$\sum_{\substack{\sigma \in \mathfrak{S}_{n+1} \\ \text{rmin}(\sigma)=1}} x^{\text{des}(\sigma)} q^{\widetilde{\text{maj}}_2(\sigma)} [\alpha]^{\text{lmin}(\sigma)-1} = \sum_{\sigma \in \mathfrak{S}_n} x^{1+\text{des}(\sigma)} q^{\widetilde{\text{maj}}_2(\sigma)} [\alpha]^{\text{lmin}(\sigma)}. \quad (3.45)$$

The result follows by combining (3.23) and (3.45). $\square$

Setting $\alpha = 1$ in (3.44) or $\beta = 1$ in (3.42) we recover Carlitz's identity [Car54],

$$\frac{\sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} q^{\text{maj}(\sigma)}}{(x; q)_{n+1}} = \sum_{j \geq 0} x^j [j + 1]^n. \quad (3.46)$$

Corollary 3.6. For $n \geq 1$ we have

$$E_n(1; q) = \prod_{i=0}^{n-1} [\alpha + \beta + i]. \quad (3.47)$$

Proof. Note that

$$\frac{E_n(1, q)}{(q; q)_{n+\alpha+\beta-1}} = \lim_{x \rightarrow 1^-} \frac{E_n(x, q)}{(xq; q)_{n+\alpha+\beta-1}}.$$

Combining (3.23) and Abel's lemma we have

$$\begin{aligned} E_n(1, q) &= (q; q)_{n+\alpha+\beta-1} \lim_{x \rightarrow 1^-} (1-x) \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n \\ &= (q; q)_{n+\alpha+\beta-1} \lim_{j \rightarrow +\infty} \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n \\ &= \frac{(q; q)_{n+\alpha+\beta-1}}{(q; q)_{\alpha+\beta-1} (1-q)^n}, \end{aligned}$$

which is equivalent to (3.47). □

Remark 3.7. Alternatively, setting $x = 1$ in (3.22) yields

$$E_n(1; q) = [n - 1 + \alpha + \beta] E_{n-1}(1; q).$$

As $E_0(1; q) = 1$, we obtain (3.47) by iteration. Moreover, we derive from (3.17) that

$$E_{n,0}(\alpha, \beta, q) = [\beta]^n, \quad E_{n,n}(\alpha, \beta, q) = q^{n\beta + \binom{n}{2}} [\alpha]^n. \quad (3.48)$$

Let $\sigma = \sigma_1 \sigma_2 \dots \sigma_n \in \mathfrak{S}_n$. For an integer $r \geq 1$, as a generalization of descents, the set of *r -descents* of σ is defined by, see [FS70],

$$r\text{Des}(\sigma) := \{i \in [n-1] : \sigma_i \geq \sigma_{i+1} + r\}. \quad (3.49)$$

The cardinality of $r\text{Des}(\sigma)$ is denoted by $r\text{des}(\sigma)$.

Rawlings [Raw81] introduced the *r -major index* as

$$r\text{maj}(\sigma) := \sum_{i \in r\text{Des}(\sigma)} i + |r\text{Inv}(\sigma)|, \quad (3.50)$$

where

$$r\text{Inv}(\sigma) := \{(i, j) : 1 \leq i < j \leq n, \sigma_i > \sigma_j > \sigma_i - r\}.$$

He then defined the (q, r) -Eulerian numbers

$$A[n, k; r] := \sum_{\substack{\sigma \in \mathfrak{S}_n \\ r\text{des}(\sigma) = k}} q^{r\text{maj}(\sigma)}. \quad (3.51)$$

Proposition 3.8. For $r \geq 1$ and $0 \leq k \leq n$, we have

$$A[n + r, k; r] = [r]! \cdot E_{n,k}(1, r, q). \quad (3.52)$$

Proof. It is known [Raw81] that the (q, r) -Eulerian numbers satisfy the recurrence

$$A[n, k; r] = [k + r]A[n - 1, k; r] + q^{k+r-1}[n + 1 - k - r]A[n - 1, k - 1; r], \quad (3.53)$$

for $n \geq 1$, $0 \leq k \leq n - r$ and $A[r, 0; r] = [r]!$. Now, shifting $n \rightarrow n + r$ Eq. (3.53) becomes

$$A[n + r, k; r] = [k + r]A[n + r - 1, k; r] + q^{k+r-1}[n + 1 - k]A[n + r - 1, k - 1; r] \quad (3.54)$$

for $0 \leq k \leq n$. Comparing (3.54) with recurrence (3.17) and their initial values at $(n, k) = (0, 0)$, we derive (3.52). $\square$

Combining Eq. (3.23) and (3.52), we derive

$$\frac{\sum_{0 \leq k \leq n} A[n + r, k; r] x^k}{(x; q)_{n+1+r}} = \sum_{j \geq 0} x^j \frac{[j + r]!}{[j]!} [j + r]^n. \quad (3.55)$$

From the above equality, we see that $A[n + r, k; r]$ for $0 \leq k \leq n$ can be seen as q -hit numbers of Garsia-Remmel [GR86].

3.4 Concluding remarks

In (3.17), let us define

$$E_{n,k}(\alpha, \beta, q) = q^{k\beta+k(k-1)/2} \cdot E_{n,k}^*(\alpha, \beta, q). \quad (3.56)$$

Then

$$E_{n,k}^*(\alpha, \beta, q) = [n - k + \alpha]E_{n-1,k-1}^*(\alpha, \beta, q) + [k + \beta]E_{n-1,k}^*(\alpha, \beta, q). \quad (3.57)$$

It follows that

$$E_{n,k}^*(\alpha, \beta, q) = E_{n,n-k}^*(\beta, \alpha, q) \quad (0 \leq k \leq n). \quad (3.58)$$

Barbero et al. [BGSVn15] introduced a three-parameter generalization of the standard Eulerian numbers $\langle n \rangle_{(s,t)}^{(\nu)}$ for integer $\nu \geq 1$, called the ν -order (s, t) -Eulerian numbers. When $\nu = 1$, the numbers $\langle n \rangle_{(s,t)} := \langle n \rangle_{(s,t)}^{(1)}$ satisfy the following recurrence

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{(s,t)} = (k + s) \left\langle \begin{matrix} n - 1 \\ k \end{matrix} \right\rangle_{(s,t)} + (n - k + t) \left\langle \begin{matrix} n - 1 \\ k - 1 \end{matrix} \right\rangle_{(s,t)} + \delta_{k0} \delta_{n0}, \quad (3.59)$$

with the additional conditions $\langle n \rangle_{(s,t)} = 0$ if $n < 0$ or $k < 0$. Comparing (3.59) with (3.17) (or (3.57)) with $q = 1$, we derive

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{(s,t)} = E_{n,k}(t, s, 1). \quad (3.60)$$

Remark 3.9. From (3.1) and (3.59) we derive

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{(1,0)}, \quad \left\langle \begin{matrix} n + 1 \\ k \end{matrix} \right\rangle = \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{(1,1)}. \quad (3.61)$$

Setting $q = 1$ in Eq. (3.23) we obtain

$$\frac{E_n(x | \alpha, \beta, 1)}{(1 - x)^{n+\alpha+\beta}} = \sum_{j \geq 0} x^j (j + \beta)^n \binom{\alpha + \beta + j - 1}{j}. \quad (3.62)$$

This is Proposition 12 in [BGSVn15] by (3.60). Setting $q = 1$ in (3.24) we obtain

$$\sum_{n \geq 0} E_n(x | \alpha, \beta, 1) \frac{z^n}{n!} = \left(\frac{1-x}{1-xe^{(1-x)z}} \right)^\alpha \left(\frac{1-x}{e^{(x-1)z}-x} \right)^\beta, \quad (3.63)$$

which is equivalent to (3.5).

The *r -Eulerian numbers*, denoted by $A(n, k; r)$, count the number of permutations in $\mathfrak{S}_n$ that have exactly k r -descents (see (3.49)), that is,

$$A(n, k; r) := |\{\sigma \in \mathfrak{S}_n : \text{rdes}(\sigma) = k\}|. \quad (3.64)$$

From (3.54) with $q = 1$, we have

$$A(n+r, k; r) = (k+r)A(n+r-1, k; r) + (n+1-k)A(n+r-1, k-1; r) \quad (3.65)$$

for $0 \leq k \leq n$. Setting $(\alpha, \beta) = (r, 1)$ in (3.59) and combining with (3.65) and their initial value at $(n, k) = (0, 0)$, we derive

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{(r,1)} = \frac{1}{r!} A(n+r, k; r). \quad (3.66)$$

Remark 3.10. In view of (3.60) and (3.66) the 1-order (s, t) -Eulerian numbers coincide with the Carlitz-Scoville numbers and also encompass the r -Eulerian numbers. This fact was overlooked in [BGSVn15, p.2], where it is stated that the Carlitz-Scoville numbers and the r -Eulerian numbers do not fall within their general framework of Eulerian numbers.

The identities (3.23), (3.25) and (3.55) deserve to have combinatorial interpretations. A combinatorial proof of (3.52) with $q = 1$ is given in [FS70]. Moreover, Butler [But04b] originally obtained the permutation interpretation of the q -Eulerian numbers $E_{n,k}(0, \beta, q)$ via the combinatorics of rook configurations (see also Haglund [Hag96] and Garsia and Remmel [GR86]), it would be interesting to extend our results to rook configurations.

Chapter 4

Enumerating Colored Permutations by the Parity of Descent Positions²

Abstract

Motivated by recent works on enumeration of Coxeter groups by the parity of descent positions, we prove a formula for the generating function of the vector statistic $(\text{odes}_G, \text{edes}_G, \text{col}_G, \ell_G)$ over the group of colored permutations $G(r, n)$. Here odes_G , edes_G , col_G and ℓ_G denote the number of odd descent positions, even descent positions, colors, and length of colored permutation, respectively. This generalises and unifies several known results over Coxeter groups of type A and B. Similar results are also given for Coxeter group of type D. In particular, a special case of our formula permits to evaluate the signed alternating descent polynomials over $G(r, n)$ by the usual Eulerian polynomials, which extends Dey and Sivasubramanian's recent results in the special cases when $r = 1, 2$.

Keywords: Euler-Mahonian polynomial, colored permutation, descent, signed alternating descent, inversion

4.1 Introduction

For any positive integer n , let $\mathfrak{S}_n$ be the symmetric group of permutations of $[n] := \{1, 2, \dots, n\}$. Given a permutation $\sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n$, an index i ($1 \leq i \leq n-1$) is a *descent* (resp. *ascent*) of σ if $\sigma_i > \sigma_{i+1}$ (resp. $\sigma_i < \sigma_{i+1}$). The number of descents (resp. ascents) of σ are denoted by $\text{des}(\sigma)$ (resp. $\text{asc}(\sigma)$). A descent i of σ is an *odd descent* (resp. *even descent*) if i is *odd* (resp. *even*). Let $\text{odes}(\sigma)$ (resp. $\text{edes}(\sigma)$) denote the number of odd (resp. even) descents of σ . Similarly, we define the number of even (resp. odd) ascents by easc (resp. oasc). The *inversion number* (inv) of σ is the number of pairs $(i, j) \in [n] \times [n]$ such that $\sigma_i > \sigma_j$ and $i < j$.

²This chapter is adapted from the paper "Enumerating colored permutations by the parity of descent positions", published in *Enumerative Combinatorics and Applications*, 4:3 (2024), Article #S2R22.

It is well known that the enumerative polynomial of permutations of $[n]$ by descents is the Eulerian polynomial $A_n(x)$, which can also be defined by the exponential generating function [Sta12, Pet15, Chap.1]

$$\sum_{n \geq 0} A_n(x) \frac{t^n}{n!} = \frac{x-1}{x - \exp((x-1)t)}. \quad (4.1)$$

In 1973 Carlitz and Scoville [CS73] enumerated permutations according to the parity of both descents and ascents. Recently, Pan and Zeng considered the enumeration polynomial of permutations using the vector statistic (easc, oasc, edes, odes, inv) and established the exponential generating function, [PZ23, Theorem 1.1]. The following is one of the four equivalent forms of their formula [PZ23, Eq. (1.6)].

Theorem A (Pan and Zeng). Let $M = \sqrt{(1-x)(1-y)}$. We have

$$\begin{aligned} \sum_{n \geq 1} \frac{t^n}{[n]_q!} \sum_{\sigma \in \mathfrak{S}_n} x^{\text{odes}(\sigma)} y^{\text{edes}(\sigma)} q^{\text{inv}(\sigma)} \\ = \frac{(1+x) \cosh(Mt; q) + M \sinh(Mt; q) - x(\cosh^2(Mt; q) - \sinh^2(Mt; q)) - 1}{1 - (x+y) \cosh(Mt; q) + xy(\cosh^2(Mt; q) - \sinh^2(Mt; q))}, \end{aligned} \quad (4.2)$$

where

$$\cosh(t; q) = \sum_{n \geq 0} \frac{t^{2n}}{[2n]_q!}, \quad \sinh(t; q) = \sum_{n \geq 1} \frac{t^{2n-1}}{[2n-1]_q!}, \quad (4.3)$$

with $[0]_q! = 1$ and $[n]_q! = \prod_{i=1}^n (1 + q + \dots + q^{i-1})$ for $n \geq 1$.

As shown in [PZ23], formula (4.2) is a q -analogue of Carlitz-Scoville's formula [CS73, Theorem 3.1] and encompasses both Stanley's formula for the bi-statistic (des, inv) [Sta76] and Chebikin's formula for alternating descent polynomials [Che08]. In a follow-up, among other things, Dey, Shankar and Sivasubramanian [DSS24, Theorems 1.5 and 1.8] established analogue formulas of Theorem A for types B and D Coxeter groups. The following is one of their type B formulas [DSS24, Theorems 1.5], which is also a q -analogue of a formula due to Pan-Zeng [PZ23, Theorem 1.4].

Theorem B (Dey, Shankar and Sivasubramanian). Let $M = \sqrt{(1-x)(1-y)}$. We have

$$\begin{aligned} \sum_{n \geq 0} \frac{t^n}{(-q; q)_n [n]_q!} \sum_{\sigma \in \mathcal{B}_n} x^{\text{odes}_B(\sigma)} y^{\text{edes}_B(\sigma)} q^{\text{inv}_B(\sigma)} \\ = \frac{(1-y)((1-x \cosh(Mu; q)) \cosh_B(Mt; q) + x \sinh(Mt; q) \sinh_B(Mt; q))}{1 - (x+y) \cosh(Mt; q) + xy \exp(Mt; q) \exp(-Mt; q)} \\ + \frac{M((1-y \cosh(Mt; q)) \sinh_B(Mt; q) + y \sinh(Mt; q) \cosh_B(Mt; q))}{1 - (x+y) \cosh(Mt; q) + xy \exp(Mt; q) \exp(-Mt; q)}, \end{aligned} \quad (4.4)$$

where

$$\cosh_B(t; q) = \sum_{n \geq 0} \frac{t^{2n}}{(-q; q)_{2n} [2n]_q!}, \quad \sinh_B(t; q) = \sum_{n \geq 1} \frac{t^{2n-1}}{(-q; q)_{2n-1} [2n-1]_q!} \quad (4.5)$$

and odes_B (resp. edes_B and inv_B) denotes the number of odd descent positions (resp. even descent positions and inversions) over type B permutations in $\mathcal{B}_n$, see Remark 4.6.

In this chapter, as a natural continuation of the work done in [PZ23, DSS24], we provide a formula for the generating function of the vector statistic $(\text{odes}_G, \text{edes}_G, \text{col}_G, \ell_G)$ over the group of colored permutations $G(r, n)$, which permits to put Theorems A and B under the same umbrella. Here odes_G (resp. edes_G , col_G and ℓ_G) denotes the number of odd descent positions (resp. even descent positions, colors, and length) of colored permutation. We shall achieve our goal by extending Dey, Shankar and Sivasubramanian's arguments in type B Coxeter groups [DSS24].

The study of signed Eulerian polynomials was initiated by Loday, Désarménien, Foata, Wachs and Reiner in the 1990's and has attracted great attention of researchers [Lod89, DF92, Wac92, Rei93, Rei95], with two recent references being [ELL21, DS23]. The alternating descent statistic on permutations was introduced by Chebikin [Che08] as a variant of the descent statistic. Dey and Sivasubramanian [DS23] further studied the signed enumeration of alternating descents for classical Weyl groups. Applying our generating function formula for colored permutations (see Theorem 4.5), we shall evaluate the signed alternating descent polynomials over $G(r, n)$ by the usual Eulerian polynomials. The resulting formula (see Theorem 4.8) extends Dey and Sivasubramanian's recent results in the special cases when $r = 1, 2$.

The rest of this chapter is organised as follows. We introduce definitions and main results, i.e., Theorem 2.1 and Theorem 2.2 in Section 2 and prove them in Section 3 and Section 4, respectively. In Section 5, we compute the generating function of the quadruple statistic $(\text{odes}_D, \text{edes}_D, \text{col}, \ell_B)$ over the group of signed permutations and make a connection with the corresponding polynomials of type D.

4.2 Definitions and main results

For positive integers m and n with $m \leq n$, we denote by $[m, n]$ the set $\{m, m+1, \dots, n\}$. The cardinality of a set A will be denoted by $|A|$. For $r, n \in \mathbb{P}$, we define the *wreath product* $\mathbb{Z}_r \wr \mathfrak{S}_n$ of $\mathbb{Z}_r$ by $\mathfrak{S}_n$, i.e., the group of colored permutations $G(r, n)$, by

$$G(r, n) := \{(c_1, \dots, c_n; \sigma) \mid c_i \in [0, r-1], \sigma = \sigma_1 \cdots \sigma_n \in \mathfrak{S}_n\}. \quad (4.6)$$

The product in $G(r, n)$ is defined by

$$(c; \sigma) \cdot (c'; \tau) := (c_1 + c'_{\tau^{-1}(1)}, \dots, c_n + c'_{\tau^{-1}(n)}; \sigma \circ \tau),$$

where the addition $+$ is in $\mathbb{Z}_r$ and composition $\circ$ in $\mathfrak{S}_n$. The entry c_i is called the color of the σ_i , for $1 \leq i \leq n$. The elements of $\mathbb{Z}_r \wr \mathfrak{S}_n$ can be viewed as r -colored permutations, see Steingrímsson [Ste94] and Bagno et al. [BGM09]. We will represent an element $\gamma \in G(r, n)$ in *window notation* as

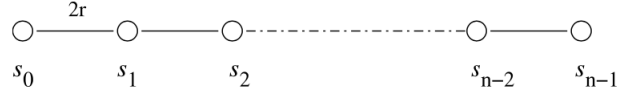
$$\gamma = [\gamma(1), \dots, \gamma(n)] = [\sigma_1^{c_1}, \dots, \sigma_n^{c_n}], \quad (4.7)$$

and call σ_i the *absolute value* of $\gamma(i)$, denoted by $|\gamma(i)|$. For $\gamma \in G(r, n)$, we define the color set of γ by

$$\text{Col}_G(\gamma) := \{i \in [n] : c_i \neq 0\} \quad (4.8)$$

and its size will be denoted by $\text{col}_G(\gamma)$. If $c_i = 0$, it will be omitted in the window notation. For example, $\gamma = [3^1, 2, 1^3, 4^2, 6^2, 5^1] \in G(5, 6)$. The group $G(r, n)$ is generated by $S_G := \{s_0, s_1, \dots, s_{n-1}\}$, where for $i \in [n-1]$

$$s_i := [1, \dots, i-1, i+1, i, i+2, \dots, n] \text{ and } s_0 := [1^1, 2, \dots, n],$$

Figure 4.1: The Dynkin-like diagram of $G(r, n)$.

with relations given by the Dynkin-like diagram (see Fig. 4.1).

The *length* of $\gamma \in G(r, n)$ is the minimal number of generators in S_G whose product is γ ,

$$\ell_G(\gamma) := \min\{r \in \mathbb{N} : \gamma = s_{i_1} \cdots s_{i_r}, \text{ for some } s_{i_j} \in S_G\}. \quad (4.9)$$

Thus, the *descent set* of $\gamma \in G(r, n)$ is

$$\text{Des}_G(\gamma) := \{s \in S_G : \ell_G(\gamma s) < \ell_G(\gamma)\} \quad (4.10)$$

and its size is denoted by $\text{des}_G(\gamma)$. To give a combinatorial description of ℓ_G and $\text{Des}_G(\gamma)$ we use the following linear order

$$n^{r-1} < \cdots < n^1 < \cdots < 1^{r-1} < \cdots < 1^1 < 0 < 1 < \cdots < n \quad (4.11)$$

on the set $\{0, 1, \dots, n, 1^1, \dots, n^1, \dots, 1^{r-1}, \dots, n^{r-1}\}$ of colored integers (see [BZ11]). If $\gamma = [\gamma(1), \dots, \gamma(n)] \in G(r, n)$, the length of γ is then characterized by (see [Bag05, Rei93, Ste94])

$$\ell_G(\gamma) = \text{inv}(\gamma) + \sum_{c_i \neq 0} (|\gamma(i)| + c_i - 1), \quad (4.12)$$

where the inversion number is defined by

$$\text{inv}(\gamma) = |\{(i, j) : 1 \leq i < j \leq n, \text{ and } \gamma(i) > \gamma(j)\}|.$$

The descent set of $\gamma \in G(r, n)$ has the following alternate definition

$$\text{Des}_G(\gamma) = \{i \in [0, n-1] : \gamma(i) > \gamma(i+1)\}, \quad (4.13)$$

where $\gamma(0) := 0$. The number of odd (resp. even) descent positions of γ is denoted by $\text{odes}_G(\gamma)$ (resp. $\text{edes}_G(\gamma)$). The ascent set of $\gamma \in G(r, n)$ is defined by

$$\text{Asc}_G(\gamma) := \{i \in [0, n-1] : \gamma(i) < \gamma(i+1)\}. \quad (4.14)$$

Similarly, we define the statistics $\text{asc}_G(\gamma)$, $\text{easc}_G(\gamma)$ and $\text{oasc}_G(\gamma)$. For any $\gamma \in G(r, n)$ the following identities hold

$$\begin{aligned} \text{edes}_G(\gamma) + \text{easc}_G(\gamma) &= \lfloor (n+1)/2 \rfloor, \\ \text{odes}_G(\gamma) + \text{oasc}_G(\gamma) &= \lfloor n/2 \rfloor. \end{aligned} \quad (4.15)$$

Note that $0 \in \text{Des}_G(\gamma)$ if and only if $c_1 > 0$.

Example 4.1. If $\gamma = [3^1, 2, 1^3, 4^2, 6^2, 5^1] \in G(5, 6)$, then $\text{Des}_G(\gamma) = \{0, 2, 3, 4\}$, $\text{des}_G(\gamma) = 4$, $\text{inv}(\gamma) = 12$, $\sum_{c_i \neq 0} |\gamma(i)| = 19$, $\sum_{c_i \neq 0} (c_i - 1) = 4$, $\ell_G(\gamma) = 35$ and $\text{col}_G(\gamma) = 5$.

For $n, r \in \mathbb{P}$, define the polynomials

$$G_{(r,n)}(a, q) := \sum_{\gamma \in G(r,n)} a^{\text{col}_G(\gamma)} q^{\ell_G(\gamma)}, \quad (4.16)$$

$$G_{(r,n)}(x, y, a, q) := \sum_{\gamma \in G(r,n)} x^{\text{odes}_G(\gamma)} y^{\text{edes}_G(\gamma)} a^{\text{col}_G(\gamma)} q^{\ell_G(\gamma)}, \quad (4.17)$$

and the standard q -factorial notation

$$(a; q)_n := \begin{cases} (1-a)(1-aq) \cdots (1-aq^{n-1}), & \text{if } n \geq 1, \\ 1, & \text{if } n = 0. \end{cases}$$

By an argument similar to the proof of Eq. (4.13) in [BZ11] we can prove the following result.

Proposition 4.2. *For $n, r \in \mathbb{P}$, we have*

$$G_{(r,n)}(a, q) = [n]_q! (-aq[r-1]_q; q)_n. \quad (4.18)$$

Remark 4.3. The case when $r = 1$ is a classical result about the inversion number over $\mathfrak{S}_n$ and the case when $r = 2$ is the counterpart in type B due to Brenti [Bre94, Proposition 3.3].

For convenience, we use the convention

$$G_{(r,0)}(a, q) = 1, \quad G_{(r,0)}(x, y, a, q) = 1. \quad (4.19)$$

The even and odd index generating functions of $G_{(r,n)}(x, y, a, q)$ are defined by

$$H_0^r := \sum_{n \geq 0} G_{(r,2n)}(x, y, a, q) \frac{t^{2n}}{G_{(r,2n)}(a, q)}, \quad (4.20a)$$

$$H_1^r := \sum_{n \geq 0} G_{(r,2n+1)}(x, y, a, q) \frac{t^{2n+1}}{G_{(r,2n+1)}(a, q)}. \quad (4.20b)$$

Define the q -analogue of exponential series over wreath product

$$\exp_{G(r)}(t; a, q) := \sum_{n \geq 0} \frac{t^n}{G_{(r,n)}(a, q)}, \quad (4.21)$$

and the corresponding hyperbolic cosine and sine series

$$\cosh_{G(r)}(t; a, q) = \frac{\exp_{G(r)}(t; a, q) + \exp_{G(r)}(-t; a, q)}{2}, \quad (4.22a)$$

$$\sinh_{G(r)}(t; a, q) = \frac{\exp_{G(r)}(t; a, q) - \exp_{G(r)}(-t; a, q)}{2}. \quad (4.22b)$$

Remark 4.4. When $r = 1, 2$, we recover the classic q -analogue of hyperbolic series in (4.3) and (4.5).

The following is our first main result.

Theorem 4.5. *For any $r \in \mathbb{P}$, we have*

$$H_0^r = \frac{(1-y) \left((1-x \cosh(Mt; q)) \cosh_{G(r)}(Mt; a, q) + x \sinh(Mt; q) \sinh_{G(r)}(Mt; a, q) \right)}{1 - (x+y) \cosh(Mt; q) + xy \exp(Mt; q) \exp(-Mt; q)}, \quad (4.23a)$$

$$H_1^r = \frac{M \left((1-y \cosh(Mt; q)) \sinh_{G(r)}(Mt; a, q) + y \sinh(Mt; q) \cosh_{G(r)}(Mt; a, q) \right)}{1 - (x+y) \cosh(Mt; q) + xy \exp(Mt; q) \exp(-Mt; q)}, \quad (4.23b)$$

where $M = \sqrt{(1-x)(1-y)}$.

Remark 4.6. When $(a, n, r) = (1, n, 1)$ or $(0, n, 2)$ we recover a formula equivalent to Theorem A. When $r = 2$, the length $\ell_G(\gamma)$ coincides with the length in the Coxeter (hyperoctahedral) group $B_n = G(2, n)$ (see [BB05, Bre94]), namely

$$\text{inv}_B(\gamma) := \text{inv}(\gamma) + \sum_{c_i \neq 0} |\gamma(i)|. \quad (4.24)$$

Hence, when $(a, n, r) = (1, n, 2)$, replacing odes_G (resp. edes_G) by odes_B (resp. edes_B), we recover Theorem B.

Remark 4.7. As observed in [PZ23] for permutations of types A and B, among the four statistics easc , oasc , edes , odes over permutations, it is sufficient to consider two of them. This is still valid for colored permutations. Indeed, by (4.15), the distribution of the quadruple statistics $(\text{easc}_G, \text{oasc}_G, \text{edes}_G, \text{odes}_G)$ is completely determined by any pair of the statistics in $\{\text{odes}_G, \text{oasc}_G\} \times \{\text{edes}_G, \text{easc}_G\}$, in particular ³,

$$\sum_{\gamma \in G(r, n)} x_0^{\text{easc}_G(\gamma)} x_1^{\text{oasc}_G(\gamma)} y_0^{\text{edes}_G(\gamma)} y_1^{\text{odes}_G(\gamma)} a^{\text{col}_G(\gamma)} q^{\ell_G(\gamma)} = x_0^{\lfloor (n+1)/2 \rfloor} x_1^{\lfloor n/2 \rfloor} G_{(r, n)}\left(\frac{y_1}{x_1}, \frac{y_0}{x_0}, a, q\right). \quad (4.25)$$

When $r = 2$, $x = y$, $q = 1$, substituting $t \leftarrow 2t$, and adding odd and even indexed generating functions, we obtain Brenti's Theorem 3.4 (iv) in [Bre94],

$$\sum_{n \geq 0} G_{(2, n)}(x, x, a, 1) \frac{t^n}{n!} = \frac{(1-x) \exp(t(1-x))}{1-x \exp(t(1-x)(1+a))}. \quad (4.26)$$

When $x = y$, adding odd and even indexed generating functions, we obtain a formula of Reiner [Rei95, Corollary 4.4, formula (2)],

$$\sum_{n \geq 0} G_{(2, n)}(x, x, a, q) \frac{t^n}{(-aq, q)_n [n]_q!} = \frac{(1-x) \exp_{G(2)}(t(1-x); a, q)}{1-x \exp(t(1-x); q)}. \quad (4.27)$$

Given a permutation $\gamma \in G(r, n)$, an index $i \in \{0, 1, \dots, n-1\}$ is called an *alternating descent* if i is an *odd descent* or *even ascent*. Let $\widehat{\text{Des}}_G(\gamma)$ be the set of alternating descents of γ , i.e.,

$$\widehat{\text{Des}}_G(\gamma) = \{2i : \gamma(2i) < \gamma(2i+1)\} \cup \{2i+1 : \gamma(2i+1) > \gamma(2i+2)\},$$

and let its cardinality be denoted by $\widehat{\text{des}}_G(\gamma)$. We define the q -alternating descent polynomial over $G(r, n)$ by

$$\text{Alt}_n^{G(r)}(x, q) := \sum_{\gamma \in G(r, n)} x^{\widehat{\text{des}}_G(\gamma)} q^{\ell_G(\gamma)}, \quad (4.28)$$

where ℓ_G is the length function (4.12). As an application of Theorem 4.5 we shall evaluate $\text{Alt}_n^{G(r)}(x, q)$ when $q = -1$. The following is our second main result.

Theorem 4.8. *For integer $n \geq 1$ the following identities hold.*

1. *If r is a positive even integer, then*

$$\text{Alt}_n^{G(r)}(x, -1) = (-1)^{\lfloor (n+1)/2 \rfloor} (1-x)^n. \quad (4.29)$$

³Here we count an ascent at the beginning as position 0, which is not counted in [PZ23].

2. If r is a positive odd integer, then $\text{Alt}_1^{G(r)}(x, -1) = x$ and for $n \geq 2$,

$$\text{Alt}_n^{G(r)}(x, -1) = \begin{cases} x(1-x)^m A_m(x), & \text{if } n = 2m \ (m \in \mathbb{N}^*); \\ \frac{2x^2}{1+x} (1-x)^{2m} A_{2m}(x), & \text{if } n = 4m+1 \ (m \in \mathbb{N}^*); \\ 0, & \text{if } n = 4m+3 \ (m \in \mathbb{N}). \end{cases} \quad (4.30)$$

We make the following remarks.

- (i) It is known [Pet15, Chapter 4] that Eulerian polynomial $A_n(x) := \sum_{i=0}^{n-1} A_{n,i} x^i$ is monic, of degree $n-1$ and palindromic, so $A_{2n}(x) = \sum_{i=0}^{n-1} A_{2n,i} x^i (1+x^{2n-2i-1})$, which is clearly divisible by $1+x$.
- (ii) Formula (4.30) reduces to [DS23, Theorem 2] when $r = 1$ ⁴ and formula (4.29) reduces to [DS23, Theorem 13] when $r = 2$.
- (iii) Our proof of the above theorem is à la Désarménien and Foata [DF92] using generating functions and q -calculus. When $r = 1, 2$, Dey and Sivasubramanian [DS23] gave a different proof. It would be interesting to find a combinatorial proof à la Wachs [Wac92].

4.3 Counting colored permutations by the parity of descents

The aim of this section is to prove Theorem 4.5. Throughout this section, we assume that n and r are positive integers. For $0 \leq m \leq n$, let $\binom{[n]}{m}$ be the set of m -subsets of $[n]$, that is, $\binom{[n]}{m} := \{A \subseteq [n] : |A| = m\}$ and let $[n]^r := \{i^{c_i} : i \in [n], c_i \in [0, r-1]\}$ be the set of colored integers. The set of m -subsets A^r of $[n]^r$ such that $A \in \binom{[n]}{m}$ is denoted by $\binom{[n]}{m}^r$.

Let A be a finite ordered set. We write $A = \{a_1, \dots, a_m\}_<$ to mean $a_1 < \dots < a_m$ and denote by $[A] := [a_1, \dots, a_m]$ the increasing sequence of its elements. In particular, if $A^r = \{a_1^{c_1}, a_2^{c_2}, \dots, a_m^{c_m}\}_< \in \binom{[n]}{m}^r$, then $[A^r]$ is the increasing permutation of A^r by the linear order (4.11), that is $[A^r] = [a_1^{c_1}, a_2^{c_2}, \dots, a_m^{c_m}]$.

Observation 4.3.1. Let $\gamma = [\sigma_1^{c_1}, \dots, \sigma_n^{c_n}] \in G(r, n)$. If $i, j \in [n]$ with $i \neq j$, then $\gamma(i) < \gamma(j)$ if and only if one of the following conditions hold,

- (1) $c_i = 0, c_j = 0$ and $|\gamma(i)| < |\gamma(j)|$;
- (2) $c_i > 0, c_j > 0$ and $|\gamma(i)| > |\gamma(j)|$;
- (3) $c_i > 0, c_j = 0$ and $|\gamma(i)| < |\gamma(j)|$ or $|\gamma(i)| > |\gamma(j)|$.

For $\gamma = [\gamma(1), \dots, \gamma(n)] = [\sigma_1^{c_1}, \dots, \sigma_n^{c_n}] \in G(r, n)$, let

$$\text{csum}(\gamma) = \sum_{c_i \neq 0} c_i \quad (4.31)$$

and

$$\text{inv}_c(\gamma) = \sum_{1 \leq i < j \leq n} |\{(i, j) : \gamma_c(i) > \gamma_c(j)\}|, \quad (4.32)$$

⁴When $r = 1$, the position 0 is not counted as an even ascent in [DS23, Theorem 2].

where $\gamma_c(i)$ is defined by

$$\gamma_c(i) = \begin{cases} \sigma_i, & \text{if } c_i \neq 0; \\ \sigma_i^{-1}, & \text{if } c_i = 0. \end{cases} \quad (4.33)$$

We now give an alternative characterization of the length function ℓ_G in (4.9) and (4.12).

Lemma 4.3.1. For $\gamma = [\gamma(1), \dots, \gamma(n)] = [\sigma_1^{c_1}, \dots, \sigma_n^{c_n}] \in G(r, n)$, we have

$$\ell_G(\gamma) = \text{inv}(\gamma) + \text{inv}_c(\gamma) + \text{csum}(\gamma). \quad (4.34)$$

Proof. By definition (4.33) and Observation 4.3.1, we have $(c_i, c_j) \neq (0, 0)$ if $\gamma_c(i) > \gamma_c(j)$. Hence

$$\begin{aligned} \text{inv}_c(\gamma) &= \sum_{1 \leq i < j \leq n} |\{(i, j) : \gamma_c(i) > \gamma_c(j)\}| \\ &= \sum_{i < j} |\{i < j : |\gamma(i)| > |\gamma(j)|, c_i \neq 0\}| + \sum_{i < j} |\{j > i : |\gamma(i)| < |\gamma(j)|, c_j \neq 0\}| \\ &= \sum_{c_k \neq 0} (\sigma_k - 1). \end{aligned}$$

Comparing with (4.12), we are done. $\square$

For $0 \leq m \leq n$, let A^r and B^r be disjoint subsets of $[n]^r$ with $|A^r| = m$ and $|B^r| = n - m$. If π (resp. σ) is a permutation of A^r (resp. B^r), in other words, A^r (resp. B^r) is the set of letters in π (resp. σ), we define the *between-permutation inversion (resp. c-inversion)* as in the following:

$$\text{inv}(\pi, \sigma) = |\{(\pi(i), \sigma(j)) \in A^r \times B^r : \pi(i) > \sigma(j), i \in [m] \text{ and } j \in [n - m]\}|; \quad (4.35a)$$

$$\text{inv}_c(\pi, \sigma) = |\{(\pi(i), \sigma(j)) \in A^r \times B^r : \pi_c(i) > \sigma(j), i \in [m] \text{ and } j \in [n - m]\}|, \quad (4.35b)$$

where $\pi_c(i)$ is defined by (4.33).

For $0 \leq i \leq n - 1$, let $G_i(r, n)$ be the set of colored permutations in $G(r, n)$ with the last $n - i$ elements being increasing from left-to-right, that is,

$$G_i(r, n) := \{\gamma \in G(r, n) : \gamma(i + 1) < \gamma(i + 2) < \dots < \gamma(n - 1) < \gamma(n)\}. \quad (4.36)$$

Note that $G_{n-1}(r, n) = G(r, n)$ and $|G_i(r, n)| = r^n \binom{n}{i} i!$. Define $G(r, 0) = \{\varepsilon\}$, where ε is the empty word. The concatenation operator $*$ of two words u and v is defined by $u * v := uv$ with $\varepsilon * u = u * \varepsilon = u$ for any word u . For $0 \leq i \leq n - 1$, let $\gamma = [\sigma_1^{c_1}, \sigma_2^{c_2}, \dots, \sigma_i^{c_i}] \in G(r, i)$, $A^r \in \binom{[n]}{n-i}^r$ with $[n] \setminus A = \{s_1, s_2, \dots, s_i\}_<$ and

$$\gamma|_{[n] \setminus A} := [s_{\sigma_1}^{c_1}, s_{\sigma_2}^{c_2}, \dots, s_{\sigma_i}^{c_i}]. \quad (4.37)$$

We define

$$f(\gamma, A^r) = \gamma|_{[n] \setminus A} * [A^r]. \quad (4.38)$$

It is easy to see that the mapping $f : G(r, i) \times \binom{[n]}{n-i}^r \rightarrow G_i(r, n)$ is a bijection.

Lemma 4.3.2. For $0 \leq i \leq n - 1$, let $\gamma = [\sigma_1^{c_1}, \sigma_2^{c_2}, \dots, \sigma_i^{c_i}] \in G(r, i)$, $A^r \in \binom{[n]}{n-i}^r$ and $[n] \setminus A = \{s_1, s_2, \dots, s_i\}_<$. The mapping $f : G(r, i) \times \binom{[n]}{n-i}^r \rightarrow G_i(r, n)$ satisfies

$$\text{csum}(f(\gamma, A^r)) = \text{csum}(\gamma) + \text{csum}([s_1, s_2, \dots, s_i] * [A^r]), \quad (4.39a)$$

$$\text{col}(f(\gamma, A^r)) = \text{col}(\gamma) + \text{col}([s_1, s_2, \dots, s_i] * [A^r]), \quad (4.39b)$$

$$\ell_G(f(\gamma, A^r)) = \ell_G(\gamma) + \ell_G([s_1, s_2, \dots, s_i] * [A^r]). \quad (4.39c)$$

Proof. By definition (4.38), we have

$$f(\gamma, A^r) = \gamma|_{[n] \setminus A} * [A^r] = [s_{\sigma_1}^{c_1}, s_{\sigma_2}^{c_2}, \dots, s_{\sigma_i}^{c_i}] * [A^r]. \quad (4.40)$$

So, it is easy to verify the first two identities (4.39a) and (4.39b). By Lemma 4.3.1 we have

$$\ell_G(f(\gamma, A^r)) = \text{inv}(f(\gamma, A^r)) + \text{inv}_c(f(\gamma, A^r)) + \text{csum}(f(\gamma, A^r)).$$

The factorisation (4.40) of $f(\gamma, A^r)$ implies that

$$\text{inv}(f(\gamma, A^r)) = \text{inv}(\gamma|_{[n] \setminus A}) + \text{inv}([A^r]) + \text{inv}(\gamma|_{[n] \setminus A}, [A^r]), \quad (4.41a)$$

$$\text{inv}_c(f(\gamma, A^r)) = \text{inv}_c(\gamma|_{[n] \setminus A}) + \text{inv}_c([A^r]) + \text{inv}_c(\gamma|_{[n] \setminus A}, [A^r]). \quad (4.41b)$$

Note that γ acts on an ordered set (see (4.37)) preserving the inversion (resp. c -inversion) number, i.e.,

$$\text{inv}(\gamma|_{[n] \setminus A}) = \text{inv}(\gamma) \quad \text{and} \quad \text{inv}_c(\gamma|_{[n] \setminus A}) = \text{inv}_c(\gamma). \quad (4.42)$$

By definition (4.35), we have

$$\text{inv}(\gamma|_{[n] \setminus A}, [A^r]) + \text{inv}_c(\gamma|_{[n] \setminus A}, [A^r]) = \text{inv}([s_1, s_2, \dots, s_i], [A^r]) + \text{inv}_c([s_1, s_2, \dots, s_i], [A^r]), \quad (4.43)$$

which is independent from γ . Combining the above results with Lemma 4.3.1 results in

$$\ell_G(f(\gamma, A^r)) - \ell_G(\gamma) - \ell_G([s_1, s_2, \dots, s_i] * [A^r]) = 0,$$

which proves (4.39c). $\square$

Example 4.9. Let $n = 8$, $r = 4$ and $i = 4$. If $\gamma = [2^1, 4, 3^1, 1^3] \in G(4, 4)$, $A^4 = \{6^1, 3^2, 1^1, 5\}_< \in \binom{[8]}{4}$, then $[8] \setminus A = \{s_1, \dots, s_4\}_< = \{2, 4, 7, 8\}$, $\gamma|_{[8] \setminus A} = [4^1, 8, 7^1, 2^3]$, $[A^4] = [6^1, 3^2, 1^1, 5]$ and

$$\gamma' := [s_1, \dots, s_4] * [A^4] = [2, 4, 7, 8] * [6^1, 3^2, 1^1, 5] = [2, 4, 7, 8, 6^1, 3^2, 1^1, 5],$$

and $f(\gamma, A^4) = [4^1, 8, 7^1, 2^3, 6^1, 3^2, 1^1, 5]$. By (4.12) we have

$$\begin{aligned} \ell_G(\gamma) &= \text{inv}(\gamma) + \sum_{c_i \neq 0} (|\gamma(i)| + c_i - 1) \\ &= |\{(2^1, 3^1), (4, 3^1), (4, 1^3)\}| + (2 + 1 - 1) + (3 + 1 - 1) + (1 + 3 - 1) = 11, \\ \ell_G(\gamma') &= \text{inv}(\gamma') + \sum_{c_i \neq 0} (|\gamma'(i)| + c_i - 1) \\ &= |\{(2, 6^1), (2, 3^2), (2, 1^1), (4, 6^1), (4, 3^2), (4, 1^1), (7, 6^1), (7, 3^2), (7, 1^1), \\ &\quad (7, 5), (8, 6^1), (8, 3^2), (8, 1^1), (8, 5)\}| + (6 + 1 - 1) + (3 + 2 - 1) + (1 + 1 - 1) = 25. \end{aligned}$$

In the same manner, we obtain $\ell_G(f(\gamma, A^4)) = 36$.

By convention, for any $n \in \mathbb{P}$ we denote by $\mathbf{1}$ the identity permutation in $G(r, n)$. Thus

$$f(\mathbf{1}, A^r) = [s_1, s_2, \dots, s_i] * [A^r]. \quad (4.44)$$

The q -binomial coefficients are defined by

$$\binom{n}{m}_q := \frac{[n]_q!}{[m]_q! [n-m]_q!} \quad (0 \leq m \leq n). \quad (4.45)$$

Lemma 4.3.3. Let $0 \leq i \leq n-1$. For any $\gamma \in G(r, i)$, we have

$$\sum_{A^r \in \binom{[n]}{n-i}^r} a^{\text{col}(f(\gamma, A^r))} q^{\ell_G(f(\gamma, A^r))} = a^{\text{col}(\gamma)} q^{\ell_G(\gamma)} \binom{n}{n-i}_q (-aq^{i+1}[r-1]_q; q)_{n-i}. \quad (4.46)$$

Proof. By (4.39b) and (4.39c) in Lemma 4.3.2 and (4.44), it suffices to prove the $\gamma = \mathbf{1}$ case of (4.46), which is equivalent to the following identity

$$[i]_q! (-aq[r-1]_q; q)_i \times \sum_{A^r \in \binom{[n]}{n-i}^r} a^{\text{col}(f(\mathbf{1}, A^r))} q^{\ell_G(f(\mathbf{1}, A^r))} \times [n-i]_q! = [n]_q! (-aq[r-1]_q; q)_n. \quad (4.47)$$

To this end, for any $\tau = \tau_1 \tau_2 \cdots \tau_{n-i} \in \mathfrak{S}_{n-i}$, we construct a mapping $f_\tau : G(r, i) \times \binom{[n]}{n-i}^r \rightarrow G(r, n)$ by

$$f_\tau(\gamma, A^r) := \gamma|_{[n] \setminus A} * \tau[A^r], \quad (4.48a)$$

where $[n] \setminus A = \{s_1, s_2, \dots, s_i\}_<$ and

$$\gamma = [\sigma_1^{c_1}, \sigma_2^{c_2}, \dots, \sigma_i^{c_i}], \quad A^r = \{a_1^{c'_1}, a_2^{c'_2}, \dots, a_{n-i}^{c'_{n-i}}\}_<, \quad (4.48b)$$

$$\gamma|_{[n] \setminus A} = [s_{\sigma_1}^{c_1}, s_{\sigma_2}^{c_2}, \dots, s_{\sigma_i}^{c_i}], \quad \tau[A^r] = [a_{\tau_1}^{c'_{\tau_1}}, a_{\tau_2}^{c'_{\tau_2}}, \dots, a_{\tau_{n-i}}^{c'_{\tau_{n-i}}}] . \quad (4.48c)$$

Note that

$$f(\gamma, A^r) = \gamma|_{[n] \setminus A} * [A^r] = [s_{\sigma_1}^{c_1}, s_{\sigma_2}^{c_2}, \dots, s_{\sigma_i}^{c_i}] * [a_1^{c'_1}, a_2^{c'_2}, \dots, a_{n-i}^{c'_{n-i}}], \quad (4.49a)$$

$$f_\tau(\gamma, A^r) = \gamma|_{[n] \setminus A} * \tau[A^r] = [s_{\sigma_1}^{c_1}, s_{\sigma_2}^{c_2}, \dots, s_{\sigma_i}^{c_i}] * [a_{\tau_1}^{c'_{\tau_1}}, a_{\tau_2}^{c'_{\tau_2}}, \dots, a_{\tau_{n-i}}^{c'_{\tau_{n-i}}}] . \quad (4.49b)$$

It is clear that f_τ is a bijection. We show that f_τ satisfies the following properties:

$$\text{col}(f(\mathbf{1}, A^r)) + \text{col}(\gamma) = \text{col}(f_\tau(\gamma, A^r)), \quad (4.50a)$$

$$\ell_G(f(\mathbf{1}, A^r)) + \ell_G(\gamma) + \text{inv}(\tau) = \ell_G(f_\tau(\gamma, A^r)). \quad (4.50b)$$

By (4.44) and (4.48) Eq. (4.50a) is obvious. It remains to prove (4.50b). By definition (4.12) and (4.49), we have

$$\begin{aligned} \ell_G(f(\gamma, A^r)) + \text{inv}(\tau) &= \text{inv}(\gamma|_{[n] \setminus A}) + \text{inv}([A^r]) + \text{inv}(\gamma|_{[n] \setminus A}, [A^r]) + \text{inv}(\tau) \\ &\quad + \sum_{c_j \neq 0} (s_{\sigma_j} + c_j - 1) + \sum_{c'_k \neq 0} (a_k + c'_k - 1), \end{aligned} \quad (4.51)$$

$$\begin{aligned} \ell_G(f_\tau(\gamma, A^r)) &= \text{inv}(\gamma|_{[n] \setminus A}) + \text{inv}(\tau[A^r]) + \text{inv}(\gamma|_{[n] \setminus A}, \tau[A^r]) \\ &\quad + \sum_{c_j \neq 0} (s_{\sigma_j} + c_j - 1) + \sum_{c'_{\tau_k} \neq 0} (a_{\tau_k} + c'_{\tau_k} - 1). \end{aligned} \quad (4.52)$$

We observe the following facts:

- $\text{inv}(\tau) = 0$ and $\text{inv}(\tau[A^r]) = \text{inv}(\tau)$ because $[A^r]$ is an increasing word;
- $\ell_G(f(\mathbf{1}, A^r)) + \ell_G(\gamma) = \ell_G(f(\gamma, A^r))$, see Eq. (4.39c);
- $\text{inv}(\gamma|_{[n] \setminus A}, [A^r]) = \text{inv}(\gamma|_{[n] \setminus A}, \tau[A^r])$, see definition (4.35a).

From (4.51), (4.52) and the above facts, we derive Eq. (4.50b). Finally, combining (4.48a), (4.50) and Proposition 4.2, we prove Eq. (4.47). $\square$

Example 4.10. Let $n = 9$, $r = 4$ and $i = 5$. If $\gamma = [4^1, 5, 1^2, 3^1, 2^3] \in G(4, 5)$, $A^4 = \{6^1, 4^3, 2^1, 1\}_< \in \binom{[9]}{4}^4$ and $\tau = 3412 \in \mathfrak{S}_4$, then $[9] \setminus A = \{3, 5, 7, 8, 9\}$,

$$\gamma_{[9] \setminus A} = [8^1, 9, 3^2, 7^1, 5^3], \quad \tau[A^4] = [2^1, 1, 6^1, 4^3].$$

Hence $f_\tau(\gamma, A^4) = [8^1, 9, 3^2, 7^1, 5^3, 2^1, 1, 6^1, 4^3]$.

Recall the enumerative polynomials see (4.17),

$$G_{(r,n)}(x, y, a, q) = \sum_{\gamma \in G(r,n)} x^{\text{odes}_G(\gamma)} y^{\text{edes}_G(\gamma)} a^{\text{col}(\gamma)} q^{\ell_G(\gamma)}. \quad (4.53)$$

For convenience, we define the weight

$$w(\gamma) = x^{\text{odes}_G(\gamma)} y^{\text{edes}_G(\gamma)} a^{\text{col}(\gamma)} q^{\ell_G(\gamma)}. \quad (4.54)$$

By convention, we set

$$G_{-1}(r, n) = \{[1, 2, \dots, n]\}. \quad (4.55)$$

Lemma 4.3.4. Let $n, r \in \mathbb{P}$ and $0 \leq i \leq n - 1$. For $G_i(r, n)$ in (4.36), the following identities hold.

1. If i is odd,

$$\sum_{\gamma \in G_i(r,n)} w(\gamma) = x \frac{G_{(r,i)}(x, y, a, q) G_{(r,n)}(a, q)}{G_{(r,i)}(a, q) [n-i]_q!} + (1-x) \sum_{\gamma \in G_{i-1}(r,n)} w(\gamma). \quad (4.56)$$

2. If i is even,

$$\sum_{\gamma \in G_i(r,n)} w(\gamma) = y \frac{G_{(r,i)}(x, y, a, q) G_{(r,n)}(a, q)}{G_{(r,i)}(a, q) [n-i]_q!} + (1-y) \sum_{\gamma \in G_{i-1}(r,n)} w(\gamma). \quad (4.57)$$

Proof. Let i be a positive odd integer. Multiplying the two sides of (4.46) by $x^{\text{odes}_G(\gamma)} y^{\text{edes}_G(\gamma)}$ and summing over $\gamma \in G(r, i)$ we obtain the identity

$$\begin{aligned} & \sum_{\gamma \in G(r,i)} \sum_{A^r \in \binom{[n]}{n-i}^r} x^{\text{odes}_G(\gamma)} y^{\text{edes}_G(\gamma)} a^{\text{col}(f(\gamma, A^r))} q^{\ell_G(f(\gamma, A^r))} \\ &= G_{(r,i)}(x, y, a, q) \binom{n}{n-i}_q (-aq^{i+1}[r-1]_q; q)_{n-i} \\ &= G_{(r,i)}(x, y, a, q) \frac{G_{(r,n)}(a, q)}{G_{(r,i)}(a, q) [n-i]_q!}, \end{aligned} \quad (4.58)$$

where the last equality follows from Proposition 4.2.

Let $F_i(r, n)$ denote the subset of colored permutations $\gamma = [\gamma(1), \dots, \gamma(n)]$ in $G_i(r, n)$ such that $\gamma(i) > \gamma(i+1) < \gamma(i+2) < \dots < \gamma(n)$. By definition (4.36), we have $F_i(r, n) = G_i(r, n) \setminus G_{i-1}(r, n)$. For $(\gamma, A^r) \in G(r, i) \times \binom{[n]}{n-i}^r$, let $\gamma' = f(\gamma, A^r)$, see (4.38). Then $f : G(r, i) \times \binom{[n]}{n-i}^r \rightarrow G_i(r, n)$ is a bijection satisfying the following properties:

- if $f(\gamma, A^r) = \gamma' \in F_i(r, n)$, then i is an odd descent of γ' (but i is clearly not a descent of γ), thus $\text{odes}_G(\gamma') = \text{odes}_G(\gamma) + 1$ and $\text{edes}_G(\gamma') = \text{edes}_G(\gamma)$;
- if $f(\gamma, A^r) = \gamma' \in G_{i-1}(r, n)$, then i is not a descent for neither γ' nor γ . Hence $\text{odes}_G(\gamma') = \text{odes}_G(\gamma)$ and $\text{edes}_G(\gamma') = \text{edes}_G(\gamma)$.

By the above arguments, we have

$$\begin{aligned}
\sum_{(\gamma, A^r) \in G(r, i) \times \binom{[n]}{n-i}^r} x^{\text{odes}_G(\gamma)} y^{\text{edes}_G(\gamma)} a^{\text{col}(f(\gamma, A^r))} q^{\ell_G(f(\gamma, A^r))} \\
= \sum_{\gamma' \in G_{i-1}(r, n)} w(\gamma') + \frac{1}{x} \sum_{\gamma' \in G_i(r, n)} w(\gamma') - \frac{1}{x} \sum_{\gamma' \in G_{i-1}(r, n)} w(\gamma') \\
= \left(1 - \frac{1}{x}\right) \sum_{\gamma' \in G_{i-1}(r, n)} w(\gamma') + \frac{1}{x} \sum_{\gamma' \in G_i(r, n)} w(\gamma'). \tag{4.59}
\end{aligned}$$

Equating the right-hand-sides of (4.58) and (4.59) we obtain

$$x \frac{G_{(r, i)}(x, y, a, q) G_{(r, n)}(a, q)}{G_{(r, i)}(a, q) [n-i]_q!} = (x-1) \sum_{\gamma' \in G_{i-1}(r, n)} w(\gamma') + \sum_{\gamma' \in G_i(r, n)} w(\gamma'). \tag{4.60}$$

This completes the proof of (4.56).

For $n \geq 1$, and i is a non negative even integer, formula (4.57) can be proved similarly. We just verify the $i = 0$ case. Clearly we can construct any $\gamma \in G_0(r, n)$ as follows: choose $\gamma(i) = (i, c_i) \in [n] \times \{0, \dots, r-1\}$ for $i \in [n]$ and order $\gamma(1), \dots, \gamma(n)$ increasingly. As $\gamma = \mathbf{1}$ if and only if $c_i = 0$ for all $i \in [n]$, the index 0 is always a descent if $\gamma \neq \mathbf{1}$. Therefore

$$\begin{aligned}
\sum_{\gamma \in G_0(r, n)} w(\gamma) &= 1 + y \left(\prod_{i=1}^n \sum_{c_i=0}^{r-1} a q^{i+c_i-1} - 1 \right) \\
&= 1 - y + y(-a q[r-1]_q; q)_n.
\end{aligned}$$

On the other hand, as $G_{-1}(r, n) = \{\mathbf{1}\}$, Eq. (4.57) holds by Proposition 4.2. $\square$

By definition (4.36), for $0 \leq i \leq n-1$, the elements of $G_i(r, n)$ are colored permutations $\gamma \in G(r, n)$ such that the last $n-i$ elements of γ are increasing. When $i = n-1$, we have $G_{n-1}(r, n) = G(r, n)$. Recall that (see (4.17))

$$G_{(r, n)}(x, y, a, q) = \sum_{\gamma \in G_{n-1}(r, n)} w(\gamma). \tag{4.61}$$

Lemma 4.3.5. For any $n, k \in \mathbb{N}$, $r \in \mathbb{P}$, the polynomials $G_{(r, n)}(x, y, a, q)$ satisfy the following recurrences:

1. If $n = 2k$ is a nonnegative even integer, and $0 \leq j \leq k$, we have

$$\begin{aligned}
\frac{G_{(r, n)}(x, y, a, q)}{G_{(r, n)}(a, q)} &= \frac{(1-x)^j (1-y)^j}{G_{(r, n)}(a, q)} \sum_{\gamma \in G_{n-2j-1}(r, n)} w(\gamma) \\
&+ \sum_{m=0}^{j-1} \frac{x(1-x)^m (1-y)^m}{[2m+1]_q!} \frac{G_{(r, n-2m-1)}(x, y, a, q)}{G_{(r, n-2m-1)}(a, q)} \\
&+ \sum_{m=1}^j \frac{y(1-x)^m (1-y)^{m-1}}{[2m]_q!} \frac{G_{(r, n-2m)}(x, y, a, q)}{G_{(r, n-2m)}(a, q)}. \tag{4.62a}
\end{aligned}$$

2. If $n = 2k + 1$ is a positive odd integer, and $0 \leq j \leq k$, we have

$$\begin{aligned} \frac{G_{(r,n)}(x, y, a, q)}{G_{(r,n)}(a, q)} &= \frac{(1-x)^j(1-y)^{j+1}}{G_{(r,n)}(a, q)} \sum_{\gamma \in G_{n-2j-2}(r,n)} w(\gamma) \\ &+ \sum_{m=0}^j \frac{y(1-x)^m(1-y)^m}{[2m+1]_q!} \frac{G_{(r,n-2m-1)}(x, y, a, q)}{G_{(r,n-2m-1)}(a, q)} \\ &+ \sum_{m=1}^j \frac{x(1-x)^{m-1}(1-y)^m}{[2m]_q!} \frac{G_{(r,n-2m-2)}(x, y, a, q)}{G_{(r,n-2m-2)}(a, q)}. \end{aligned} \quad (4.62b)$$

Proof. It is clear that (4.62a) is valid for $n = 0$. Assuming that $n = 2k$ ($k \geq 1$), we prove (4.62a) by induction on j . The base case when $j = 0$ is obvious by (4.61). Assume that (4.62a) is true for j and show that it holds for $j + 1$, that is,

$$\begin{aligned} G_{(r,n)}(x, y, a, q) &= (1-x)^{j+1}(1-y)^{j+1} \sum_{\gamma \in G_{n-2j-3}(r,n)} w(\gamma) \\ &+ \sum_{m=0}^j \frac{x(1-x)^m(1-y)^m}{[2m+1]_q!} \frac{G_{(r,n-2m-1)}(x, y, a, q)G_{(r,n)}(a, q)}{G_{(r,n-2m-1)}(a, q)} \\ &+ \sum_{m=1}^{j+1} \frac{y(1-x)^m(1-y)^{m-1}}{[2m]_q!} \frac{G_{(r,n-2m)}(x, y, a, q)G_{(r,n)}(a, q)}{G_{(r,n-2m)}(a, q)}. \end{aligned} \quad (4.63)$$

Eq (4.63) is easy to verify by applying (4.56) and (4.57) because

$$\begin{aligned} \sum_{\gamma \in G_{n-2j-1}(r,n)} w(\gamma) &= (1-x) \sum_{\gamma \in G_{n-2j-2}(r,n)} w(\gamma) + x \frac{G_{(r,n-2j-1)}(x, y, a, q)G_{(r,n)}(a, q)}{[2j+1]_q!G_{(r,n-2j-1)}(a, q)} \\ &= (1-x)(1-y) \sum_{\gamma \in G_{n-2j-3}(r,n)} w(\gamma) + x \frac{G_{(r,n-2j-1)}(x, y, a, q)G_{(r,n)}(a, q)}{[2j+1]_q!G_{(r,n-2j-1)}(a, q)} \\ &\quad + (1-x)y \frac{G_{(r,n-2j-2)}(x, y, a, q)G_{(r,n)}(a, q)}{[2j+2]_q!G_{(r,n-2j-2)}(a, q)}. \end{aligned} \quad (4.64)$$

Plugging (4.64) in (4.62a) and dividing both sides by $G_{(r,n)}(a, q)$, we derive Eq. (4.63). Formula (4.62b) can be proved similarly. $\square$

Proof of Theorem 4.5. As $\sum_{\gamma \in G_{-1}(r,n)} w(\gamma) = 1$, multiplying identity (4.62a) (resp. (4.62b)) with $j = k$ by $(1-y)$ (resp. $(1-x)$), and then adding $\frac{yG_{(r,n)}(x, y, a, q)}{G_{(r,n)}(a, q)}$ (resp. $\frac{xG_{(r,n)}(x, y, a, q)}{G_{(r,n)}(a, q)}$) on both sides, we obtain the following recurrence relations:

- if $n = 2k$ is even,

$$\begin{aligned} \frac{G_{(r,n)}(x, y, a, q)}{G_{(r,n)}(a, q)} &= \frac{(1-x)^k(1-y)^{k+1}}{G_{(r,n)}(a, q)} + \sum_{m=0}^{k-1} \frac{x(1-x)^m(1-y)^{m+1}}{[2m+1]_q!} \frac{G_{(r,n-2m-1)}(x, y, a, q)}{G_{(r,n-2m-1)}(a, q)} \\ &+ \sum_{m=0}^k \frac{y(1-x)^m(1-y)^m}{[2m]_q!} \frac{G_{(r,n-2m)}(x, y, a, q)}{G_{(r,n-2m)}(a, q)}, \end{aligned} \quad (4.65)$$

- if $n = 2k + 1$ is odd

$$\begin{aligned} \frac{G_{(r,n)}(x, y, a, q)}{G_{(r,n)}(a, q)} &= \frac{(1-x)^{k+1}(1-y)^{k+1}}{G_{(r,n)}(a, q)} + \sum_{m=0}^k \frac{y(1-x)^{m+1}(1-y)^m}{[2m+1]_q!} \frac{G_{(r,n-2m-1)}(x, y, a, q)}{G_{(r,n-2m-1)}(a, q)} \\ &+ \sum_{m=0}^k \frac{x(1-x)^m(1-y)^m}{[2m]_q!} \frac{G_{(r,n-2m)}(x, y, a, q)}{G_{(r,n-2m)}(a, q)}. \end{aligned} \quad (4.66)$$

Invoking (4.20), multiplying (4.65) (resp. (4.66)) by t^{2k} (resp. t^{2k+1}) and then summing over $k \geq 1$ (resp. $k \geq 0$), we obtain the system

$$\begin{cases} (1 - y \cosh(Mt; q)) H_0^r - xL \sinh(Mt; q) H_1^r &= (1 - y) \cosh_{G(r)}(Mt; a, q), \\ -\frac{y}{L} \sinh(Mt; q) H_0^r + (1 - x \cosh(Mt; q)) H_1^r &= M \sinh_{G(r)}(Mt; a, q), \end{cases} \quad (4.67)$$

where $M = \sqrt{(1-x)(1-y)}$ and $L = \sqrt{(1-y)/(1-x)}$.

Solving the above system using Cramer's rule results in (4.23). $\square$

4.4 Counting colored permutations by signed alternating descents

The aim of this section is to prove Theorem 4.8 by applying Theorem 4.5. By the wreath product analogue of exponential series $\exp_{G(r)}(t; 1, q)$ (see (4.21)) we define the q -trigonometric series over wreath product by

$$\cos_{G(r)}(t; q) := \sum_{n \geq 0} \frac{(-1)^n}{(-q[r-1]_q; q)_{2n}} \cdot \frac{t^{2n}}{[2n]_q!}, \quad (4.68a)$$

$$\sin_{G(r)}(t; q) := \sum_{n \geq 0} \frac{(-1)^n}{(-q[r-1]_q; q)_{2n+1}} \cdot \frac{t^{2n+1}}{[2n+1]_q!}. \quad (4.68b)$$

It follows from (4.17), (4.25) and (4.28) that

$$\text{Alt}_n^{G(r)}(x, q) = x^{[(n+1)/2]} G_{(r,n)}(x, 1/x, 1, q). \quad (4.69)$$

Combining with Theorem 4.5 we obtain the following generating functions.

Lemma 4.4.1. Let $\text{Alt}_0^{G(r)}(x; q) = 1$. We have

$$\begin{aligned} & \sum_{n \geq 0} \frac{\text{Alt}_{2n}^{G(r)}(x, q)}{(-q[r-1]_q; q)_{2n}} \cdot \frac{t^{2n}}{[2n]_q!} \\ &= (x-1) \times \frac{(1 - x \cos((1-x)t; q)) \cos_{G(r)}((1-x)t; q) - x \sin((1-x)t; q) \sin_{G(r)}((1-x)t; q)}{x - (x^2 + 1) \cos((1-x)t; q) + x \exp(i(1-x)t; q) \exp(-i(1-x)t; q)}, \end{aligned} \quad (4.70a)$$

$$\begin{aligned} & \sum_{n \geq 0} \frac{\text{Alt}_{2n+1}^{G(r)}(x, q)}{(-q[r-1]_q; q)_{2n+1}} \cdot \frac{t^{2n+1}}{[2n+1]_q!} \\ &= (x-1) \times \frac{(x - \cos((1-x)t; q)) \sin_{G(r)}((1-x)t; q) + \sin((1-x)t; q) \cos_{G(r)}((1-x)t; q)}{x - (x^2 + 1) \cos((1-x)t; q) + x \exp(i(1-x)t; q) \exp(-i(1-x)t; q)}. \end{aligned} \quad (4.70b)$$

Recall the following q -binomial identity [And76, p. 37]

$$\sum_{j=0}^m (-1)^j \binom{m}{j}_q = \begin{cases} (q; q^2)_n, & \text{if } m = 2n; \\ 0, & \text{if } m \text{ is odd,} \end{cases} \quad (4.71)$$

and the limits of q -binomial coefficients (4.45) when $q \rightarrow -1$:

$$\lim_{q \rightarrow -1} \binom{2n}{2m+1}_q = 0, \quad (4.72a)$$

$$\lim_{q \rightarrow -1} \binom{2n}{2m}_q = \lim_{q \rightarrow -1} \binom{2n+1}{2m}_q = \lim_{q \rightarrow -1} \binom{2n+1}{2m+1}_q = \binom{n}{m}, \quad (4.72b)$$

$$\lim_{q \rightarrow -1} [r-1]_q = \lim_{q \rightarrow -1} \frac{1-q^{r-1}}{1-q} = \begin{cases} 0, & \text{if } r \text{ is odd;} \\ 1, & \text{if } r \text{ is even.} \end{cases} \quad (4.72c)$$

By (4.72c), if r is even and $n-k \geq 2$, then

$$\lim_{q \rightarrow -1} \frac{(-q[r-1]_q; q)_n}{(-q[r-1]_q; q)_k} = ((-1)^k; -1)_{n-k} = 0. \quad (4.72d)$$

Now, we are ready to prove Theorem 4.8 in the next three sections. First we shall prove that for any even integer $r \geq 2$,

$$\text{Alt}_n^{G(r)}(x, -1) = (-1)^{\lfloor (n+1)/2 \rfloor} (1-x)^n \quad \text{for } n \geq 0. \quad (4.73)$$

4.4.1 Proof of Theorem 4.8 when r is even

Multiplying the two sides of Eq. (4.70a) by

$$x - (x^2 + 1) \cos((1-x)t; q) + x \exp(i(1-x)t; q) \cdot \exp(-i(1-x)t; q),$$

and then comparing the coefficients of $\frac{t^{2n}}{[2n]_q!}$ ($n \geq 0$), we derive the recurrence relation, after simplification using (4.71), for even indices:

$$\begin{aligned} & \frac{-(1-x)^2 \text{Alt}_{2n}^{G(r)}(x, q)}{(-q[r-1]_q; q)_{2n}} + \sum_{k=0}^{n-1} \binom{2n}{2k}_q \frac{(-1)^{n-k} \text{Alt}_{2k}^{G(r)}(x, q)}{(-q[r-1]_q; q)_{2k}} (x(q; q^2)_{n-k} - x^2 - 1)(1-x)^{2n-2k} \\ &= \frac{(-1)^{n+1} (1-x)^{2n+2}}{(-q[r-1]_q; q)_{2n}} + x(1-x) \sum_{k=0}^{2n-1} \binom{2n}{k}_q \frac{(-1)^n (1-x)^{2n}}{(-q[r-1]_q; q)_k}. \end{aligned} \quad (4.74)$$

In the same vein, from Eq. (4.70b) we derive the recurrence relation for odd indices ($n \geq 0$):

$$\begin{aligned} & \frac{-(1-x)^2 \text{Alt}_{2n+1}^{G(r)}(x, q)}{(-q[r-1]_q; q)_{2n+1}} + \sum_{k=0}^{n-1} \binom{2n+1}{2k+1}_q \frac{(-1)^{n-k} \text{Alt}_{2k+1}^{G(r)}(x, q)}{(-q[r-1]_q; q)_{2k+1}} (x(q; q^2)_{n-k} - x^2 - 1)(1-x)^{2n-2k} \\ &= \frac{(-1)^n (1-x)^{2n+3}}{(-q[r-1]_q; q)_{2n+1}} + (x-1) \sum_{k=0}^{2n} \binom{2n+1}{k}_q \frac{(-1)^{n-k} (1-x)^{2n+1}}{(-q[r-1]_q; q)_k}. \end{aligned} \quad (4.75)$$

Clearing the fractions in Eq. (4.74) and Eq. (4.75) by multiplying $(-q[r-1]_q; q)_{2n}$ and $(-q[r-1]_q; q)_{2n+1}$, respectively, and then taking the limit $q \rightarrow -1$, we obtain by invoking (4.72),

$$-(1-x)^2 \text{Alt}_{2n}^{G(r)}(x, -1) = (-1)^{n+1} (1-x)^{2n+2}, \quad (4.76a)$$

$$-(1-x)^2 \text{Alt}_{2n+1}^{G(r)}(x, -1) = (-1)^n (1-x)^{2n+3}, \quad (4.76b)$$

which are equivalent to Eq. (4.73).

In the next two sections we shall prove the remaining part of Theorem 4.8, i.e., for $n \geq 2$, if r is a positive odd integer, then

$$\text{Alt}_n^{G(r)}(x, -1) = \begin{cases} x(1-x)^m A_m(x), & \text{if } n = 2m \ (m \in \mathbb{N}^*); \\ \frac{2x^2}{1+x}(1-x)^{2m} A_{2m}(x), & \text{if } n = 4m+1 \ (m \in \mathbb{N}^*); \\ 0, & \text{if } n = 4m+3 \ (m \in \mathbb{N}), \end{cases} \quad (4.77)$$

where $A_m(x)$ are the classical Eulerian polynomials, see (4.1).

4.4.2 Proof of Theorem 4.8 when r is odd and $n \neq 4m+3$

For $n \geq 1$, clearing the fractions in equations (4.74) and (4.75) by multiplying $(-q[r-1]_q; q)_{2n}$ and $(-q[r-1]_q; q)_{2n+1}$, respectively, then taking $q = -1$ results in

$$\begin{aligned} \text{Alt}_{2n}^{G(r)}(x, -1) &= \sum_{k=0}^{n-1} \binom{n}{k} \text{Alt}_{2k}^{G(r)}(x, -1) (-1)^{n-k} (1-x)^{2n-2k-2} (2^{n-k}x - x^2 - 1) \\ &\quad + (-1)^n (1-x)^{2n-1} (1-2^n x), \end{aligned} \quad (4.78a)$$

$$\begin{aligned} \text{Alt}_{2n+1}^{G(r)}(x, -1) &= \sum_{k=0}^{n-1} \binom{n}{k} \text{Alt}_{2k+1}^{G(r)}(x, -1) (-1)^{n-k} (1-x)^{2n-2k-2} (2^{n-k}x - x^2 - 1) \\ &\quad + (-1)^n x(1-x)^{2n}. \end{aligned} \quad (4.78b)$$

Now, we prove that $\text{Alt}_{2m}^{G(r)}(x, -1) = x(1-x)^m A_m(x)$ for $m \geq 1$. By (4.78a), this is clear for $m = 1$. It remains to show that $x(1-x)^m A_m(x)$ satisfies recurrence relation (4.78a), namely,

$$A_m(x) = \sum_{k=0}^{m-1} \binom{m}{k} (1-x)^{m-k-2} A_k(x) (-1)^{m-k} (2^{m-k}x - x^2 - 1) - x(-1)^m (1-x)^{m-1}. \quad (4.78c)$$

Multiplying the above identity by $t^m/m!$ and summing over $m \geq 1$ yields

$$1 + \sum_{m \geq 1} A_m(x) \frac{t^m}{m!} = \frac{x(1-x) \exp((x-1)t) - x}{(x^2+1) \exp((x-1)t) - x - x \exp(2(x-1)t)}, \quad (4.78d)$$

which is equivalent to formula (4.1). Thus, Eq. (4.77) holds true if n is even.

By definition we have $\text{Alt}_1^{G(r)}(x, q) = x(1+q+\dots+q^{r-1})$, hence $\text{Alt}_1^{G(r)}(x, -1) = x$. For the time being, we admit (4.77) for $n = 4m+3$, namely, assume that $\text{Alt}_{4k+3}^{G(r)}(x, -1) = 0$ for $k \in \mathbb{N}$, see the proof in the next section. Thus, replacing n by $2m$ in Eq. (4.78b) results in

$$\text{Alt}_{4m+1}^{G(r)}(x, -1) = \sum_{k=0}^{m-1} \binom{2m}{2k} \text{Alt}_{4k+1}^{G(r)}(x, -1) (1-x)^{4m-4k-2} (2^{2m-2k}x - x^2 - 1) + x(1-x)^{4m}. \quad (4.78e)$$

It remains to show that $\text{Alt}_{4m+1}^{G(r)}(x, -1) = \frac{2x^2(1-x)^{2m}}{1+x} A_{2m}(x)$ for $m \geq 1$. As a check, setting $m = 1$ in (4.78e) yields $\text{Alt}_5^{G(r)}(x, -1) = 2x^2(1-x)^2$, since $A_2(x) = 1+x$, formula (4.77) is valid for $n = 5$. Now we prove that $\frac{2x^2(1-x)^{2m}}{1+x} A_{2m}(x)$ satisfy recurrence relation (4.78e), namely,

$$\begin{aligned} \frac{2x^2(1-x)^{2m}}{1+x} A_{2m}(x) &= \sum_{k=1}^{m-1} \binom{2m}{2k} \frac{2x^2(1-x)^{4m-2k-2} A_{2k}(x)}{1+x} (2^{2m-2k}x - x^2 - 1) \\ &\quad + x(1-x)^{4m-2} (2^{2m}x - 2x). \end{aligned} \quad (4.79)$$

Multiplying the above identity by $t^{2m}/(2m)!$ and summing over $m \geq 1$ yields

$$1 + \sum_{m \geq 1} A_{2m}(x) \frac{t^{2m}}{(2m)!} = \frac{(x-1)(\exp((1-x)t) + \exp((x-1)t))}{2x(\exp((1-x)t) + \exp((x-1)t)) - 2x^2 - 2},$$

which can be verified straightforwardly by (4.1). $\square$

4.4.3 Proof of Theorem 4.8 when r is odd and $n = 4m + 3$

Recall that the elements of the set $\{0, 1, \dots, n, 1^1, \dots, n^1, \dots, 1^{r-1}, \dots, n^{r-1}\}$ are ordered as in the following (see (4.11)),

$$n^{r-1} < \dots < n^1 < \dots < 1^{r-1} < \dots < 1^1 < 0 < 1 < \dots < n. \quad (4.80)$$

For $1 \leq i \leq n$, we define an operator ϕ_i over $G(r, n)$ by

$$\phi_i(\gamma) = \begin{cases} (c_1, \dots, c_i + 1, \dots, c_n; \sigma), & \text{if } c_i \text{ is odd;} \\ (c_1, \dots, c_i - 1, \dots, c_n; \sigma), & \text{if } c_i \neq 0 \text{ is even;} \\ \gamma, & \text{if } c_i = 0, \end{cases} \quad (4.81)$$

where $\gamma = (c_1, \dots, c_n; \sigma) \in G(r, n)$, see (4.6).

Lemma 4.4.2. For $1 \leq i \leq n$, the alternating descent set is invariant under operator ϕ_i , i.e.,

$$\widehat{Des}_G(\phi_i(\gamma)) = \widehat{Des}_G(\gamma) \quad \text{for } \gamma \in G(r, n). \quad (4.82)$$

Hence, we have $\widehat{des}_G(\phi_i(\gamma)) = \widehat{des}_G(\gamma)$.

Proof. Let $\gamma = (c_1, \dots, c_n; \sigma) \in G(r, n)$ and $\phi_i(\gamma) = (c'_1, \dots, c'_n; \sigma)$ for a fixed $i \in [n]$. Then, $c'_j = c_j$ if $j \neq i$ for $j \in [n]$ and $c'_i = c_i \pm 1$. Eq. (4.82) is obvious if $c_i = 0$. Since operator ϕ_i only acts on $\gamma(i)$, we need only to check the nature of positions $i-1$ and i through ϕ_i . In what follows we assume that $c_i > 0$ for some odd index i .

(i) $i \in \widehat{Des}_G(\gamma)$ if and only if (iff) $i \in \widehat{Des}_G(\phi_i(\gamma))$.

Indeed, $i \in \widehat{Des}_G(\gamma)$ (with $c_i > 0$) iff $c_{i+1} > 0$ and $|\gamma(i)| < |\gamma(i+1)|$ iff $c'_i = c_i \pm 1 > 0$, $c'_{i+1} = c_{i+1} > 0$ and $|\gamma(i)| < |\gamma(i+1)|$, which is equivalent to $i \in \widehat{Des}_G(\phi_i(\gamma))$.

(ii) $i-1 \in \widehat{Des}_G(\gamma)$ if and only if $i-1 \in \widehat{Des}_G(\phi_i(\gamma))$.

Indeed, $\gamma(i-1) > \gamma(i)$ (with $c_i > 0$) iff $c_{i-1} = 0$ or $c_{i-1} > 0$ and $|\gamma(i-1)| < |\gamma(i)|$. As $c'_i = c_i \pm 1 > 0$ and $c'_{i-1} = c_{i-1}$, the latter statement is equivalent to $i-1 \in \widehat{Des}_G(\phi_i(\gamma))$.

Thus, Eq. (4.82) is valid for odd $i \in [n]$. The proof for even $i \in [n]$ is similar. $\square$

In what follows, for a permutation $\sigma \in \mathfrak{S}_n$, we write $\ell_G(\sigma) = \text{inv}(\sigma)$ and denote the number of alternating descents of σ by $\widehat{des}(\sigma)$ as in [Che08]. Note that $\text{sgn}(\sigma) = (-1)^{\text{inv}(\sigma)}$ is the signature of σ .

Lemma 4.4.3. Let $n = 4m + 3$ with $m \geq 0$. Then

$$\sum_{\sigma \in \mathfrak{S}_n} (-1)^{\text{inv}(\sigma)} x^{\widehat{des}(\sigma)} = 0. \quad (4.83)$$

Proof. Recall the reversing operator R on $\mathfrak{S}_n$, which maps $\sigma = \sigma_1\sigma_2\cdots\sigma_n \in \mathfrak{S}_n$ to $R(\sigma) = \sigma_n\sigma_{n-1}\cdots\sigma_1$. Clearly, an index $i \in [n-1]$ is an even ascent (resp. odd descent) in σ if and only if $n-i$ is an odd descent (resp. even ascent) in $R(\sigma)$. Also, position 0 is an even ascent in both σ and $R(\sigma)$. Thus

$$\widehat{des}(\sigma) = \widehat{des}(R(\sigma)). \quad (4.84)$$

Since $\text{inv}(\sigma) + \text{inv}(R(\sigma)) = \binom{n}{2}$ and $\binom{n}{2} = (2m+1)(4m+3)$ is odd, we have $\text{sgn}(R(\sigma)) = -\text{sgn}(\sigma)$. Combining with (4.84), we see that R is a *weight preserving and sign reversing involution* or *killing involution* over $\mathfrak{S}_n$, which yields (4.83). $\square$

Let $\mathfrak{S}_{r,n}^c$ be the subset of $G(r, n)$ consisting of permutations $\gamma = (c_1, \dots, c_n; \sigma)$ such that $c_i > 0$ for some index $i \geq 1$.

Lemma 4.4.4. Let r be an odd positive integer and $n \in \mathbb{N}^*$. Then

$$\sum_{\gamma \in \mathfrak{S}_{r,n}^c} (-1)^{\ell_G(\gamma)} x^{\widehat{des}_G(\gamma)} = 0. \quad (4.85)$$

Proof. We construct a *killing involution* Φ on $\mathfrak{S}_{r,n}^c$ such that if $\Phi(\gamma) = \gamma'$ for $\gamma \in \mathfrak{S}_{r,n}^c$, then $\widehat{des}_G(\gamma') = \widehat{des}_G(\gamma)$ and

$$\ell_G(\gamma') = \ell_G(\gamma) \pm 1. \quad (4.86)$$

For $\gamma \in \mathfrak{S}_{r,n}^c$ with $\gamma = (c_1, \dots, c_n; \sigma)$, we define $\Phi(\gamma)$ as follows: let $\Phi(\gamma) = \phi_i(\gamma)$ where i is the smallest index such that $c_i > 0$. It is obvious that Φ is an involution on $\mathfrak{S}_{r,n}^c$.

By Lemma 4.4.2, it is clear that $\widehat{des}_G(\gamma') = \widehat{des}_G(\gamma)$. It remains to verify (4.86). We first show that each inversion pair is invariant through Φ .

- (1) For $1 \leq j < i$, pair (j, i) is an inversion of $\gamma \in \mathfrak{S}_{r,n}^c$ if and only if it is an inversion of $\Phi(\gamma) \in \mathfrak{S}_{r,n}^c$.

Indeed, pair (j, i) is an inversion: $\sigma(j)^{c_j} > \sigma(i)^{c_i}$ with $c_i > 0$ iff $c_j = 0$, or $c_j > 0$ and $|\gamma(j)| < |\gamma(i)|$; as $c'_i = c_i \pm 1 > 0$ and $c'_j = c_j$, the previous statement shows that (j, i) is an inversion of $\Phi(\gamma)$.

- (2) For $i < l \leq n$, pair (i, l) is an inversion of $\gamma \in \mathfrak{S}_{r,n}^c$ if and only if it is an inversion of $\Phi(\gamma) \in \mathfrak{S}_{r,n}^c$.

Indeed, pair (i, l) is an inversion of γ : $\sigma(i)^{c_i} > \sigma(l)^{c_l}$ (with $c_i > 0$) iff $c_l > 0$, and $|\gamma(i)| < |\gamma(l)|$; as $c'_i = c_i \pm 1 > 0$, $c'_l = c_l > 0$, the previous statement means that (i, l) is an inversion of $\Phi(\gamma)$.

As ϕ_i fixes all $\gamma(k)$ for $k \neq i$, it follows that $\text{inv}(\Phi(\gamma)) = \text{inv}(\gamma)$. Therefore, by (4.12),

$$\begin{aligned} \ell_G(\Phi(\gamma)) &= \text{inv}(\Phi(\gamma)) + \sum_{c_k \neq 0} (|\gamma(k)| + c_k - 1) \pm 1 \\ &= \ell_G(\gamma) \pm 1, \end{aligned} \quad (4.87)$$

which is Eq. (4.86). Hence (4.85) is proved. $\square$

4.5 Counting permutations of type B by the parity of descents of type D

When $r = 2$, the group $G(2, n)$ is isomorphic to the group $\mathcal{B}_n$. In this Section, we will compute the generating function of $(\text{odes}_D, \text{edes}_D, \text{col}, \ell_B)$ over type B permutations, see Theorem 4.11. Recall that an element γ of B_n is a permutation of $[\pm n] = \{-n, \dots, -1, 1, \dots, n\}$ such that $\gamma(i) = -\gamma(i)$ for $i \in [n]$. If $\gamma := [\gamma(1), \dots, \gamma(n)] \in \mathcal{B}_n$, we say that $\gamma(i)$ is *negative* if $\gamma(i) < 0$, that is, its color $c_i > 0$, and denote the number of negative terms by $\text{neg}(\gamma)$, see [PZ23, DSS24]. The corresponding q -exponential series and the q -trigonometric series of type B are respectively denoted by

$$\exp_B := \exp_{G(2)}, \quad \cosh_B := \cosh_{G(2)}, \quad \sinh_B := \sinh_{G(2)}.$$

Let D_n be the subgroup of $\mathcal{B}_n$ consisting of permutations having an even number of negative entries, that is,

$$D_n := \{\gamma \in \mathcal{B}_n : \text{neg}(\gamma) \equiv 0 \pmod{2}\}.$$

Note that $\text{col}(\gamma) = \text{neg}(\gamma)$ for $\gamma \in D_n$. We denote -1 by $\bar{1}$ and define $\gamma(\bar{1}) := -\gamma(1)$.

Given a $\gamma = [\gamma(1), \dots, \gamma(n)] \in D_n$, define the type D descent set to be

$$\text{Des}_D(\gamma) := \{i \in \{-1, 1, 2, \dots, n\} : \gamma(i) > \gamma(|i| + 1)\}, \quad (4.88)$$

and its cardinality is denoted by $\text{des}_D(\sigma)$. Similarly we define $\text{odes}_D(\gamma)$ (resp. $\text{edes}_D(\gamma)$) to be the number of odd descent positions (resp. even descent positions) of $\gamma \in D_n$ and also for ascents (resp. $\text{asc}_D(\gamma)$, $\text{easc}_D(\gamma)$ and $\text{easc}_D(\gamma)$), i.e., if $\gamma(i) > \gamma(|i| + 1)$, for $i \in \{-1, 1, 2, \dots, n\}$.

By Proposition 4.2 we have

$$B_n(a, q) := G_{(2,n)}(a, q) = [n]_q!(-aq; q)_n.$$

We consider the twisted type B enumerative polynomials

$$\widehat{B}_n(x, y, a, q) := \sum_{\gamma \in \mathcal{B}_n} x^{\text{odes}_D(\gamma)} y^{\text{edes}_D(\gamma)} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)}, \quad (4.89)$$

and their even index and odd index exponential generating functions,

$$\widehat{B}_0(t; x, y, a, q) := \sum_{n \geq 0} \widehat{B}_{2n}(x, y, a, q) \frac{t^{2n}}{B_{2n}(a, q)}, \quad (4.90)$$

$$\widehat{B}_1(t; x, y, a, q) := \sum_{n \geq 0} \widehat{B}_{2n+1}(x, y, a, q) \frac{t^{2n+1}}{B_{2n+1}(a, q)}. \quad (4.91)$$

Let $M = \sqrt{(1-x)(1-y)}$ and

$$\begin{aligned} \widehat{\text{OD}} &= \frac{(1+aq)(1-\frac{x}{aq})(1-x)}{M} \left(\sinh_B(Mt; a, q) - \frac{Mt}{1+aq} \right) \\ &\quad + \frac{x(1+\frac{1}{aq})(1-x)}{M} (\sinh(Mt; q) - Mt) + x^2 t (\cosh(Mt; q) - 1), \end{aligned} \quad (4.92)$$

$$\begin{aligned} \widehat{\text{ED}} &= (1+aq) \left(1 - \frac{x}{aq}\right) (\cosh_B(Mt; a, q) - 1) \\ &\quad + x \left(1 + \frac{1}{aq}\right) (\cosh(Mt; q) - 1) + x^2 t \frac{M}{1-t} \sinh(Mt; q). \end{aligned} \quad (4.93)$$

Theorem 4.11. Let $M = \sqrt{(1-x)(1-y)}$. For any $n \in \mathbb{P}$, we have

$$\widehat{B}_0(t; x, y, a, q) = \frac{(1-x \cosh(Mt; q))\widehat{ED} + (\frac{t(1-x)}{M} \sinh(Mt; q))\widehat{OD}}{1 - (x+y) \cosh(Mt; q) + xy \exp(Mt; q) \exp(-Mt; q)}, \quad (4.94)$$

$$\widehat{B}_1(t; x, y, a, q) = \frac{(1-y \cosh(Mt; q))\widehat{OD} + (\frac{y(1-x)}{M} \sinh(Mt; q))\widehat{ED}}{1 - (x+y) \cosh(Mt; q) + xy \exp(Mt; q) \exp(-Mt; q)}. \quad (4.95)$$

The proof of Theorem 4.11 is based on the following lemmas.

Lemma 4.5.1. Let $X_{\{1, \bar{1}\}}$ be the set of colored (signed) permutations in $\mathcal{B}_n$ such that the descent set is a subset of $\{1, \bar{1}\}$. Then,

$$\begin{aligned} \sum_{\gamma \in X_{\{1, \bar{1}\}}} x^{\text{odes}_D(\gamma)} y^{\text{edes}_D(\gamma)} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} \\ = x^2 \frac{B_n(a, q)}{[n-1]_q!} + x(1-x) \left(1 + \frac{1}{aq}\right) \left(\frac{B_n(a, q)}{[n]_q!} - 1\right) + (1-x)(1+aq), \end{aligned} \quad (4.96)$$

where $B_n(a, q) = [n]!_q(-aq; q)_n$.

Proof. Let Y_1 be the subset of $\mathcal{B}_n$ consisting permutations whose the last $n-1$ elements in ascending order and Y_0 be the subset of $\mathcal{B}_n$ such that the descent set is $\emptyset$, $\{1\}$ or $\{\bar{1}\}$. By inclusion-exclusion, we derive

$$\begin{aligned} \sum_{\gamma \in X_{\{1, \bar{1}\}}} x^{\text{odes}_D(\gamma)} y^{\text{edes}_D(\gamma)} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} \\ = x^2 \left(\sum_{\gamma \in Y_1} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} - \sum_{\gamma \in Y_0} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} \right) + x \left(\sum_{\gamma \in Y_0} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} - 1 - aq \right) + 1 + aq \\ = x^2 \left(\sum_{\gamma \in Y_1} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} \right) + x(1-x) \left(\sum_{\gamma \in Y_0} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} \right) + (1-x)(1+aq). \end{aligned}$$

The second equality follows because the first term calculates the colored permutations with descent set $\{1, \bar{1}\}$, the second term calculates the colored permutations with descent set $\{1\}$ or $\{\bar{1}\}$, and the last term calculate the colored permutations without descents (i.e. $[1, 2, \dots, n]$ and $[1^1, 2, \dots, n]$). The equation

$$\sum_{\gamma \in Y_1} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} = \frac{B_n(a, q)}{[n-1]_q!}$$

is a special case of Eq. (4.46) when $r = 2$ and $m = n-1$.

Now, it remains to prove,

$$\sum_{\gamma \in Y_0} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} = \left(1 + \frac{1}{aq}\right) \left(\frac{B_n(a, q)}{[n]_q!} - 1\right).$$

Let INB_n be the subset of $\mathcal{B}_n$ consisting all increasing colored permutations without considering the descent position $\bar{1}$, that is, $\text{INB}_n := \{\gamma \in \mathcal{B}_n : \gamma(1) < \gamma(2) < \dots < \gamma(n)\}$. In Eq. (4.46), by letting $r = 2$ and $m = n$, we have

$$\sum_{\gamma \in \text{INB}_n} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} = \frac{B_n(a, q)}{[n]_q!}.$$

There are only two colored permutations contained in INB_n without descents, that is $[1, 2, \dots, n]$ and $[1^1, 2, \dots, n]$. Deleting the color of the first element of the other permutations in INB_n , yields new permutations with a descent position 1 which are denoted by $\overline{\text{INB}}_n := \{\gamma \in \mathcal{B}_n : \gamma(1) > \gamma(2) < \dots < \gamma(n)\}$. Therefore, we have

$$\begin{aligned} \sum_{\gamma \in Y_0} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} &= \sum_{\gamma \in \text{INB}_n} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} + \sum_{\gamma \in \overline{\text{INB}}_n} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} \\ &= \frac{B_n(a, q)}{[n]_q!} + \frac{1}{aq} \left(\frac{B_n(a, q)}{[n]_q!} - 1 - aq \right) \\ &= \left(1 + \frac{1}{aq} \right) \left(\frac{B_n(a, q)}{[n]_q!} - 1 \right). \end{aligned}$$

This completes the proof. $\square$

Let $H_{n,i}$ be the set of colored permutations $\gamma \in \mathcal{B}_n$ such that the last $n-i$ elements of γ are increasing.

Lemma 4.5.2. For any $n \in \mathbb{N}$, $r \in \mathbb{P}$, when i is odd, we have

$$\begin{aligned} \sum_{\gamma' \in H_{n,i}} x^{\text{odes}_D(\gamma')} y^{\text{edes}_D(\gamma')} a^{\text{col}(\gamma')} q^{\ell_B(\gamma')} \\ = x \frac{\widehat{B}_i(x, y, a, q) B_n(a, q)}{B_i(a, q) [n-i]_q!} + (1-x) \sum_{\gamma' \in H_{n,i-1}} x^{\text{odes}_D(\gamma')} y^{\text{edes}_D(\gamma')} a^{\text{col}(\gamma')} q^{\ell_D(\gamma')}. \end{aligned} \quad (4.97)$$

When i is even, we have

$$\begin{aligned} \sum_{\gamma' \in H_{n,i}} x^{\text{odes}_D(\gamma')} y^{\text{edes}_D(\gamma')} a^{\text{col}(\gamma')} q^{\ell_B(\gamma')} \\ = y \frac{\widehat{B}_i(x, y, a, q) B_n(a, q)}{B_i(a, q) [n-i]_q!} + (1-y) \sum_{\gamma' \in H_{n,i-1}} x^{\text{odes}_D(\gamma')} y^{\text{edes}_D(\gamma')} a^{\text{col}(\gamma')} q^{\ell_B(\gamma')}. \end{aligned} \quad (4.98)$$

Proof. The proof is similar to that of Lemma 4.3.4. $\square$

Lemma 4.5.3. When integer $n \geq 3$, the polynomials $\widehat{B}_n(x, y, a, q)$ satisfy the following recurrence:

$$\begin{aligned} \frac{\widehat{B}_n(x, y, a, q)}{B_n(a, q)} &= \frac{(1+aq)(1-\frac{x}{aq})(1-x)^k(1-y)^k}{B_n(a, q)} + \frac{x(1+aq)(1-x)^k(1-y)^k}{aq[n]_q!} \\ &\quad + \frac{x^2(1-x)^{k-1}(1-y)^k}{[n-1]_q!} + \sum_{r=0}^{k-1} x(1-x)^r(1-y)^{r+1} \frac{\widehat{B}_{n-2r-1}(x, y, a, q)}{B_{n-2r-1}(a, q)[2r+1]_q!} \\ &\quad + \sum_{r=0}^{k-1} y(1-x)^r(1-y)^r \frac{\widehat{B}_{n-2r}(x, y, a, q)}{B_{n-2r}(a, q)[2r]_q!} \quad \text{if } n = 2k \text{ is even,} \end{aligned} \quad (4.99)$$

$$\begin{aligned} \frac{\widehat{B}_n(x, y, a, q)}{B_n(a, q)} &= \frac{(1+aq)(1-\frac{x}{aq})(1-x)^{k+1}(1-y)^k}{B_n(a, q)} + \frac{x(1+aq)(1-x)^{k+1}(1-y)^k}{aq[n]_q!} \\ &\quad + \frac{x^2(1-x)^k(1-y)^k}{[n-1]_q!} + \sum_{r=0}^{k-1} y(1-x)^{r+1}(1-y)^r \frac{\widehat{B}_{n-2r-1}(x, y, a, q)}{B_{n-2r-1}(a, q)[2r+1]_q!} \\ &\quad + \sum_{r=0}^{k-1} x(1-x)^r(1-y)^r \frac{\widehat{B}_{n-2r}(x, y, a, q)}{B_{n-2r}(a, q)[2r]_q!} \quad \text{if } n = 2k+1 \text{ is odd.} \end{aligned} \quad (4.100)$$

Proof. When n is even, $H_{n,n-1}$ must be the whole of $\mathcal{B}_n$. By repeatedly applying (4.97) and (4.98), we have

$$\begin{aligned}
\widehat{B}_n(x, y, a, q) &= \sum_{\gamma \in H_{n,n-1}} x^{\text{odes}_D(\gamma)} y^{\text{edes}_D(\gamma)} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} \\
&= x \frac{\widehat{B}_{n-1}(x, y, a, q) B_n(1, q)}{B_{n-1}(1, q) [1]_q!} + (1-x) \left(\sum_{\gamma \in H_{n,n-2}} x^{\text{odes}_D(\gamma)} y^{\text{edes}_D(\gamma)} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} \right) \\
&= x \frac{\widehat{B}_{n-1}(x, y, a, q) B_n(a, q)}{B_{n-1}(a, q) [1]_q!} + y(1-x) \frac{\widehat{B}_{n-2}(x, y, a, q) B_n(a, q)}{B_{n-2}(a, q) [2]_q!} + \dots \\
&\quad + y(1-x)^{\frac{n}{2}-1} (1-y)^{\frac{n}{2}-2} \frac{\widehat{B}_2(x, y, a, q) B_n(a, q)}{B_2(a, q) [n-2]_q!} \\
&\quad + (1-x)^{\frac{n}{2}-1} (1-y)^{\frac{n}{2}-1} \left(\sum_{\gamma \in X_{\{1, \bar{1}\}}} x^{\text{odes}_D(\gamma)} y^{\text{edes}_D(\gamma)} a^{\text{col}(\gamma)} q^{\ell_B(\gamma)} \right).
\end{aligned}$$

This completes the proof of (4.99) by Lemma 4.5.1. The formula (4.100) is proved similarly. $\square$

Proof of Theorem 4.11. We compute the generating functions $\widehat{B}_0$ and $\widehat{B}_1$ (see (4.90) and (4.91)) in terms of $\widehat{\text{ED}}$ and $\widehat{\text{OD}}$ (see (4.93) and (4.92)). For convenience, we introduce the two series

$$\begin{aligned}
U &:= 1 - y \cosh(Mt; q) + \frac{y(1-x) \sinh(Mt; q)}{M}; \\
V &:= 1 - x \cosh(Mt; q) + \frac{x(1-y) \sinh(Mt; q)}{M}.
\end{aligned}$$

The recurrences (4.99) and (4.100) give rise to this following equation:

$$\widehat{\mathcal{B}}_0 U + \widehat{\mathcal{B}}_1 V = \widehat{\text{OD}} + \widehat{\text{ED}}, \quad (4.101)$$

where $\widehat{\mathcal{B}}_0 = \widehat{B}_0(x, y, a, q; t)$ and $\widehat{\mathcal{B}}_1 = \widehat{B}_1(x, y, a, q; t)$. Changing t to $-t$ gives us

$$\widehat{\mathcal{B}}_0 U - \widehat{\mathcal{B}}_1 V = -\widehat{\text{OD}} + \widehat{\text{ED}}. \quad (4.102)$$

Solving the above two equations for $\widehat{\mathcal{B}}_0$ and $\widehat{\mathcal{B}}_1$, we complete the proof. $\square$

4.5.1 Final remark

Recall the length of $\gamma \in D_n$ by

$$\ell_D(\gamma) := \ell_B(\gamma) - \text{col}(\gamma). \quad (4.103)$$

Define the enumerative polynomials of Coxeter group of type D

$$D_n(x, y, a, q) = \sum_{\gamma \in D_n} x^{\text{odes}_D(\gamma)} y^{\text{edes}_D(\gamma)} a^{\text{col}(\gamma)} q^{\ell_D(\gamma)}. \quad (4.104)$$

For a formal power series $f(t) = \sum_{n \geq 0} a_n t^n$, we use $\langle t^n \rangle f(t)$ to denote the coefficient of t^n in $f(t)$, i.e., $\langle t^n \rangle f(t) = a_n$. By definition of length of type B and D (see Eqs. (4.24) and (4.103)), we see that $\ell_B(\gamma) = \ell_D(\gamma) + \text{col}(\gamma)$ and $\text{col}(\gamma) \equiv 0 \pmod{2}$, so we derive the following result from equation (4.94) and (4.95) by replacing a by $-a/q$.

Corollary 4.5.1. For $n \in \mathbb{N}$, we have

$$D_{2n}(x, y, a, q) = \frac{1}{2} \left(B_n(-a, q) \langle t^n \rangle \widehat{B}_0(t; x, y, -\frac{a}{q}, q) + B_n(a, q) \langle t^n \rangle \widehat{B}_0(t; x, y, \frac{a}{q}, q) \right),$$

$$D_{2n+1}(x, y, a, q) = \frac{1}{2} \left(B_n(-a, q) \langle t^n \rangle \widehat{B}_1(t; x, y, -\frac{a}{q}, q) + B_n(a, q) \langle t^n \rangle \widehat{B}_1(t; x, y, \frac{a}{q}, q) \right).$$

When $a = 1$, Dey et al. [DSS24, Theorem 1.8] obtained the type D even and odd index generating functions for the polynomials $D_n(x, y, 1, q)$. It would be interesting to establish a connection between Corollary 4.5.1 and their formulas.

Chapter 5

Further Research directions

The preceding chapters have studied generalized Eulerian-type polynomials from several combinatorial viewpoints, involving refinement permutation statistics, continued-fraction expansions, (p, q) -analogues, and various positivity properties. The research directions presented in this chapter are closely connected with these themes and arise naturally from the results and methods in the thesis. Rather than forming an independent part, they indicate possible continuations of the present work. In particular, we discuss new interpretations for Euler numbers and reduced tangent numbers, as well as possible links with quasi-Stirling permutations and cluster parking functions.

5.1 New Interpretations for Euler Numbers and Reduced Tangent Numbers²

Recall that the *Euler number* E_n is the n -th coefficient in the Taylor expansion of $\sec(z) + \tan(z)$, namely,

$$\sum_{n \geq 0} E_n \frac{z^n}{n!} = \sec(z) + \tan(z). \quad (5.1)$$

It is well-known that the J -fraction for Euler numbers E_{n+1} :

$$\sum_{n=0}^{\infty} E_{n+1} z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (5.2)$$

where $\mathbf{b}$ and $\boldsymbol{\lambda}$ are given by

$$b_k = k + 1 \text{ and } \lambda_{k+1} = \binom{k+2}{2} \text{ for } k \geq 0. \quad (5.3)$$

The continued fraction (5.2) is classical and follows as a specialization of a continued fraction formula due to Stieltjes [Sti94], as pointed out by Zeng in [Sok20, Note 9]. It was rediscovered empirically by P. D. Hanna in a contribution on [Slo, A000111] and later proved by a combinatorial method of Josuat-Vergès [JV14]. Han [Han20] considered a q -version of (5.2) (See Eq. (5.23) with $p = 1$) and asked for a combinatorial interpretation of the

²Part of this section is based on the submitted preprint "New Laguerre History Interpretations of Euler Numbers and Reduced Tangent Numbers", (2026)

corresponding q -Euler numbers, which was later provided by Pan and Zeng [PZ21] using André permutations. Our Theorem 5.1 gives more combinatorial interpretations in terms of partial Laguerre histories.

The odd-indexed Euler numbers E_{2n+1} are called the *tangent numbers*, and the *reduced tangent numbers* are the normalized tangent numbers defined by

$$R_n := \frac{E_{2n+1}}{2^n} \quad (n \geq 0). \quad (5.4)$$

These numbers R_n admit the following J -fraction:

$$\sum_{n \geq 0} R_n z^{2n} = \mathcal{J}(z; \mathbf{0}, \boldsymbol{\lambda}), \quad (5.5)$$

where $\boldsymbol{\lambda}$ is given by $\lambda_{k+1} = \binom{k+2}{2}$ for $k \geq 0$. The above J -fraction follows from a classical formula of Stieltjes, as pointed out by Bala [Bal17].

We introduce two restricted families of Laguerre histories, called r -partial and l -partial Laguerre histories, and show that forbidding suitable colored east steps recovers the two continued fractions above.

Let $w = w_1 \cdots w_n \in \mathcal{M}_n$ be a Motzkin path. For each South-East step $w_j = S$, its matching North-East step is the unique step $w_i = N$, with $i < j$, such that

$$h_j = h_i + 1, \quad (5.6)$$

and every step w_k , for $i < k < j$ satisfies $h_k > h_i$. The pair (i, j) is called a *matched pair*. We denote by $\text{MP}(w)$ the set of all such matched pairs of w . A Laguerre history $(w, p) \in \mathcal{L}_n$ is said to be a *r -partial* (resp. *l -partial*) *Laguerre history* if each matched pair $(i, j) \in \text{MP}(w)$, with $i < j$ satisfies

$$p_i \geq p_j \quad (\text{resp. } p_i < p_j). \quad (5.7)$$

Let $\mathcal{L}_n^r$ (resp. $\mathcal{L}_n^l$) be the set of all r -partial (resp. l -partial) Laguerre histories of length n . Throughout this paper, $|S|$ denotes the cardinality of a finite set S . For a Laguerre history $(w, p) \in \mathcal{L}_n$, we define

$$e_1(w, p) := |\{i \in [n] : w_i = \text{E}_1\}| \text{ and } e_2(w, p) := |\{i \in [n] : w_i = \text{E}_2\}|, \quad (5.8)$$

and the following two enumerative polynomials

$$L_n^l(x, y) := \sum_{\omega := (w, p) \in \mathcal{L}_n^l} x^{e_1(\omega)} y^{e_2(\omega)} \text{ and } L_n^r(x, y) := \sum_{\omega := (w, p) \in \mathcal{L}_n^r} x^{e_1(\omega)} y^{e_2(\omega)}. \quad (5.9)$$

Using Flajolet's fundamental Lemma 0.7.1, we obtain the following result.

Theorem 5.1. *For $s \in \{l, r\}$, we have*

$$\sum_{n \geq 0} L_n^s(x, y) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (5.10)$$

where $\mathbf{b}$ and $\boldsymbol{\lambda}$ are given by $b_k = (x + y)(k + 1)$, and $\lambda_{k+1} = \binom{k+2}{2}$ for $k \geq 0$, respectively.

For $s \in \{r, l\}$ and $I \subseteq \{1, 2\}$, we define

$$\mathcal{L}_n^{s, I} := \{(w, p) \in \mathcal{L}_n^s : e_i((w, p)) = 0 \text{ for all } i \in I\}. \quad (5.11)$$

From the preceding theorem, we immediately deduce the following two results, which give combinatorial interpretations for Euler numbers and reduced tangent numbers, respectively.

Theorem 5.2. For $s \in \{r, l\}$, and $\epsilon \in \{1, 2\}$, we have

$$\sum_{n \geq 0} |\mathcal{L}_n^{s, \{\epsilon\}}| z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (5.12)$$

where $\mathbf{b}$ and $\boldsymbol{\lambda}$ are given by $b_k = k + 1$ and $\lambda_{k+1} = \binom{k+2}{2}$ for $k \geq 0$. Consequently, we have

$$|\mathcal{L}_n^{s, \{\epsilon\}}| = E_{n+1}. \quad (5.13)$$

Theorem 5.3. For $s \in \{r, l\}$, we have

$$\sum_{n \geq 0} |\mathcal{L}_{2n}^{s, \{1, 2\}}| z^{2n} = \mathcal{J}(z; \mathbf{0}, \boldsymbol{\lambda}), \quad (5.14)$$

where $\boldsymbol{\lambda}$ is given by $\lambda_{k+1} = \binom{k+2}{2}$ for $k \geq 0$. Consequently, we have

$$|\mathcal{L}_{2n}^{s, \{1, 2\}}| = R_n. \quad (5.15)$$

Next we consider a natural (p, q) -analogue of Theorem 5.1. Let $(w, p) \in L_n^s$, where $s \in \{r, l\}$. Recall that $\text{MP}(w)$ denotes the set of matched north-east/south-east pairs of the underlying Motzkin path w . If $(i, j) \in \text{MP}(w)$, then $w_i = \text{N}$, $w_j = \text{S}$. For $s \in \{l, r\}$, we define two statistics area_s and coarea_s on two kinds of partial Laguerre histories as follows, respectively. For $(w, p) \in \mathcal{L}_n^r$, define

$$\text{area}_r(w, p) := \sum_{\substack{1 \leq i \leq n \\ w_i = \text{E}_1 \text{ or } \text{E}_2}} p_i + \sum_{(i, j) \in \text{MP}(w)} (p_i + p_j), \quad (5.16)$$

and

$$\text{coarea}_r(w, p) := \sum_{\substack{1 \leq i \leq n \\ w_i = \text{E}_1 \text{ or } \text{E}_2}} (h_i - p_i) + \sum_{(i, j) \in \text{MP}(w)} (2h_i - p_i - p_j). \quad (5.17)$$

For $(w, p) \in \mathcal{L}_n^l$, define

$$\text{area}_l(w, p) := \sum_{\substack{1 \leq i \leq n \\ w_i = \text{E}_1 \text{ or } \text{E}_2}} p_i + \sum_{(i, j) \in \text{MP}(w)} (p_i + p_j - 1), \quad (5.18)$$

and

$$\text{coarea}_l(w, p) := \sum_{\substack{1 \leq i \leq n \\ w_i = \text{E}_1 \text{ or } \text{E}_2}} (h_i - p_i) + \sum_{(i, j) \in \text{MP}(w)} (2h_i + 1 - p_i - p_j). \quad (5.19)$$

For $s \in \{r, l\}$, define the following enumerative polynomials,

$$L_n^s(x, y; p, q) := \sum_{\omega = (w, p) \in \mathcal{L}_n^s} x^{e_1(\omega)} y^{e_2(\omega)} p^{\text{coarea}_s(\omega)} q^{\text{area}_s(\omega)}. \quad (5.20)$$

The following result is a (p, q) -version of Theorem 5.1.

Theorem 5.4. For $s \in \{r, l\}$, we have

$$\sum_{n \geq 0} L_n^s(x, y; p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (5.21)$$

where $\mathbf{b}$ and $\boldsymbol{\lambda}$ are given by

$$b_k = (x + y)[k + 1]_{p, q} \text{ and } \lambda_{k+1} = \left[\begin{matrix} k+2 \\ 2 \end{matrix} \right]_{p, q} \quad (k \geq 0). \quad (5.22)$$

We define (p, q) -analogues $E_n(p, q)$ and $R_n(p, q)$ of the Euler numbers and the reduced tangent numbers by

$$\sum_{n \geq 0} E_{n+1}(p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}) \quad (5.23)$$

and

$$\sum_{n \geq 0} R_n(p, q) z^{2n} = \mathcal{J}(z; 0, \boldsymbol{\lambda}) \quad (5.24)$$

where $\mathbf{b}$ and $\boldsymbol{\lambda}$ are given by $b_k = [k+1]_{p,q}$ and $\lambda_{k+1} = \left[\begin{smallmatrix} k+2 \\ 2 \end{smallmatrix} \right]_{p,q}$ for $k \geq 0$. We now record two specializations of Theorem 5.4. They give combinatorial interpretations of $E_{n+1}(p, q)$ and $R_n(p, q)$ in terms of the statistics area_s and coarea_s on partial Laguerre histories. For $s \in \{r, l\}$, $I \subseteq \{1, 2\}$, and $n \geq 0$, define

$$L_n^{s,I}(p, q) := \sum_{\omega=(w,p) \in \mathcal{L}_n^{s,I}} p^{\text{coarea}_s(\omega)} q^{\text{area}_s(\omega)}. \quad (5.25)$$

The following two results are obtained from Theorem 5.4 by forbidding one of the two colors of east steps, or both. The first follows by comparing (5.23) with the specializations $(x, y) = (0, 1)$ and $(1, 0)$, while the second follows by comparing (5.24) with the specialization $(x, y) = (0, 0)$.

Theorem 5.5. *For $n \geq 0$, $s \in \{r, l\}$, and $\epsilon \in \{1, 2\}$, we have*

$$L_n^{s,\{\epsilon\}}(p, q) = E_{n+1}(p, q). \quad (5.26)$$

Theorem 5.6. *For $n \geq 0$ and $s \in \{r, l\}$, we have*

$$L_{2n}^{s,\{1,2\}}(p, q) = R_n(p, q). \quad (5.27)$$

Remark 5.7. Setting $p = q = 1$ in (5.26) and (5.27) recovers (5.13) and (5.15), respectively. Setting $p = 1$ in Theorem 5.5 gives four Laguerre-history interpretations of the q -Euler numbers $E_{n+1}(q)$ considered by Han [Han20].

We now translate our restrictions on Laguerre histories into conditions on permutations. Recall that, for a Motzkin path w , $\text{MP}(w)$ denotes the set of pairs (i, j) such that $i < j$, $w_i = \text{N}$, $w_j = \text{S}$, $h_j = h_i + 1$, and $h_k > h_i$ for every $i < k < j$.

Let $\sigma \in \mathfrak{S}_{n+1}$, and write $\Psi_{FV}(\sigma) = (w, p) \in \mathcal{L}_n$. We define

$$\text{MP}_{FV}(\sigma) := \text{MP}(w).$$

Thus, if $(i, j) \in \text{MP}_{FV}(\sigma)$, then i is a valley of σ , j is a peak of σ , and the corresponding N-step and S-step are matched in the Motzkin path associated with σ .

For $s \in \{r, l\}$, define $\mathfrak{S}_{n+1}^s$ to be the set of permutations $\sigma \in \mathfrak{S}_{n+1}$ such that, for every $(i, j) \in \text{MP}_{FV}(\sigma)$, one has

$$(31-2)_i(\sigma) \geq (31-2)_j(\sigma) \quad \text{if } s = r, \quad (5.28)$$

and

$$(31-2)_i(\sigma) < (31-2)_j(\sigma) \quad \text{if } s = l. \quad (5.29)$$

Then, by construction of the Françon–Viennot bijection, we have

$$\Psi_{FV}(\mathfrak{S}_{n+1}^s) = \mathcal{L}_n^s. \quad (5.30)$$

Moreover, the colors of the east steps have a simple interpretation in terms of local types of letters. The absence of E_1 -steps is equivalent to the absence of double descents in σ , while the absence of E_2 -steps is equivalent to the absence of double ascents in σ . Hence, Eq. (5.13) in Theorem 5.2 can be restated as

$$|\{\sigma \in \mathfrak{S}_{n+1}^s : \sigma \text{ has no double descents}\}| = E_{n+1}, \quad (5.31)$$

and

$$|\{\sigma \in \mathfrak{S}_{n+1}^s : \sigma \text{ has no double ascents}\}| = E_{n+1}. \quad (5.32)$$

Similarly, forbidding both colors of east steps means that every letter $1, \dots, 2n$ is either a valley or a peak. Therefore, Eq. (5.15) in Theorem 5.3 gives

$$|\{\sigma \in \mathfrak{S}_{2n+1}^s : \text{every letter } 1, \dots, 2n \text{ is a valley or a peak}\}| = R_n. \quad (5.33)$$

Problem 3. Can one find an intrinsic description of the matching $\text{MP}_{FV}(\sigma)$ directly in terms of the permutation σ ? Such a description would turn Theorems 5.2 and 5.3 into purely permutation-theoretic interpretations of Euler numbers and reduced tangent numbers without referring to Laguerre histories.

Pan and Zeng [PZ21] considered the following continued fraction:

$$\sum_{n \geq 0} E_{n+1}(t, p, q) z^n = \mathcal{J}(z; \mathbf{b}, \boldsymbol{\lambda}), \quad (5.34)$$

where $\mathbf{b}$ and $\boldsymbol{\lambda}$ are given by $b_k = [k+1]_{p,q}$ and $\lambda_{k+1} = t \begin{bmatrix} k+2 \\ 2 \end{bmatrix}_{p,q}$ for $k \geq 0$. They proved, by defining a group action on permutations, that polynomials $E_{n+1}(t, p, q)$ are the normalized γ -coefficients of Brändén's (p, q) -Eulerian polynomials [Brä08] in terms of André permutations. More precisely,

$$E_n(t, p, q) = \sum_{\sigma \in \mathcal{D}_n} t^{\text{des}(\sigma)} p^{(2-13)(\sigma)} q^{(31-2)(\sigma) - \text{des}(\sigma)}, \quad (5.35)$$

where $\mathcal{D}_n$ denotes the set of André permutations of the second kind of length n .

The continued fraction defined in (5.34) also arises naturally from our partial Laguerre histories. Indeed, the coefficient $\lambda_{k+1} = t \begin{bmatrix} k+2 \\ 2 \end{bmatrix}_{p,q}$ is obtained as the total weight of admissible labels on a matched pair (i, j) with $h_i = k$, where the additional factor t marks this matched pair. Thus, our model gives a natural weighted-path interpretation of the polynomials $E_{n+1}(t, p, q)$. This suggests that the partial Laguerre history model and the André permutation should be closely related. We leave the following problem for future work.

Problem 4. Find a direct bijection between partial Laguerre histories and André permutations of the second kind. More generally, determine the precise relationship between these two combinatorial interpretations of $E_n(t, p, q)$.

5.2 Link with quasi-Stirling permutations and Cluster parking functions

Archer, Gregory, Pennington and Slayden [AGPS19] introduced the set $\overline{\mathcal{Q}}_n$ of *quasi-Stirling permutations*. These are permutations $\pi_1 \pi_2 \dots \pi_{2n}$ of the multiset $\{1, 1, 2, 2, \dots, n, n\}$ avoiding

patterns 1212 and 2121; that is, those for which there do not exist $i < j < k < l$ such that $\pi_i = \pi_k$ and $\pi_j = \pi_l$. They prove that

$$|\overline{\mathcal{Q}}_n| = n! \cdot C_n, \quad \text{where} \quad C_n = \frac{1}{n+1} \binom{2n}{n}$$

is the n th Catalan number. They also posed the following intriguing formula as a conjecture.

Conjecture 5.8 ([AGPS19]). The number of $\pi \in \overline{\mathcal{Q}}_n$ with $\text{des}(\pi) = n - 1$ is equal to $(n + 1)^{n-1}$.

Elizalde [Eli21] proved Conjecture 5.8³ by using a bijection between quasi-Stirling permutations and edge-labeled plane rooted trees, under which descents become cyclic descents around vertices; the maximal descent case reduces to counting unordered rooted trees with n labeled edges, yielding $(n + 1)^{n-1}$ by Cayley's formula. Elizalde introduced the *quasi-Stirling polynomials* as follows

$$\overline{Q}_n(t) = \sum_{\pi \in \overline{\mathcal{Q}}_n} t^{\text{des}(\pi)}. \quad (5.36)$$

The first few values of $\overline{Q}_n(t)$ are given as follows, for $1 \leq n \leq 6$,

$$\begin{aligned} \overline{Q}_1(t) &= 1, \\ \overline{Q}_2(t) &= 1 + 3t, \\ \overline{Q}_3(t) &= 1 + 13t + 16t^2, \\ \overline{Q}_4(t) &= 1 + 39t + 171t^2 + 125t^3, \\ \overline{Q}_5(t) &= 1 + 101t + 1091t^2 + 2551t^3 + 1296t^4, \\ \overline{Q}_6(t) &= 1 + 243t + 5498t^2 + 28838t^3 + 43653t^4 + 16807t^5. \end{aligned}$$

The following result is a reformulation of Elizalde's formula [Eli21, Theorem 2.5].

Theorem 5.9 ([Eli21]). For $n \geq 1$, we have

$$\sum_{m \geq 0} \binom{m+1+n}{m+1} \frac{(m+1)^n}{n+1} t^m = \frac{\overline{Q}_n(t)}{(1-t)^{2n+1}}. \quad (5.37)$$

Consider a multiset $\mathcal{M} = \{1^{k_1}, 2^{k_2}, \dots, n^{k_n}\}$ and put $K = k_1 + k_2 + \dots + k_n$, where $k_i \geq 1$ is the multiplicity of i in $\mathcal{M}$. Let $\mathfrak{S}_{\mathcal{M}}$ denote the set of all multipermutations of $\mathcal{M}$. The *multiset Eulerian polynomials* $A_{\mathcal{M}}(t)$ are defined by

$$A_{\mathcal{M}}(t) := \sum_{\pi \in \mathfrak{S}_{\mathcal{M}}} t^{\text{des}(\pi)}. \quad (5.38)$$

MacMahon [Mac04, Volume 2, p.211] proved the factorial generating function formula

$$\frac{A_{\mathcal{M}}(t)}{(1-t)^{K+1}} = \sum_{m \geq 0} \binom{k_1+m}{m} \dots \binom{k_n+m}{m} t^m, \quad (5.39)$$

where $\mathcal{M} = \{1^{k_1}, 2^{k_2}, \dots, n^{k_n}\}$ and $K = k_1 + k_2 + \dots + k_n$.

³The restatement of the conjecture in [Eli21] mentions permutations with n descents, since their definition considers the last position $2n$ to be a descent.

Generalizing quasi-Stirling permutations to arbitrary multisets, we define multiset quasi-Stirling permutations as follows. A permutation π of a multiset $\mathcal{M}$ is called a *multiset quasi-Stirling permutation* if it avoids the patterns 1212 and 2121. Denote by $\overline{\mathcal{Q}}_{\mathcal{M}}$ the set of all multiset quasi-Stirling permutations of $\mathcal{M}$. Note that when $\mathcal{M} = \{1^2, 2^2, \dots, n^2\}$ and $\mathcal{M}' = \{1, 2, \dots, n\}$, we have $\overline{\mathcal{Q}}_{\mathcal{M}} = \overline{\mathcal{Q}}_n$ and $\overline{\mathcal{Q}}_{\mathcal{M}'} = \mathfrak{S}_n$. A descent of $\pi \in \overline{\mathcal{Q}}_{\mathcal{M}}$ is an index $i \in [K-1]$ if $\pi_i > \pi_{i+1}$. The *multiset quasi-Stirling polynomials* are defined by

$$\overline{Q}_{\mathcal{M}}(t) := \sum_{\pi \in \overline{\mathcal{Q}}_{\mathcal{M}}} t^{\text{des}(\pi)}. \quad (5.40)$$

The following result is a reformulation of Yan and Zhu [YZ22, Theorem 1.2].

Theorem 5.10 ([YZ22]). *Let $\mathcal{M} = \{1^{k_1}, 2^{k_2}, \dots, n^{k_n}\}$ and $K = k_1 + k_2 + \dots + k_n$ with $k_i \geq 1$. We have*

$$\frac{\overline{Q}_{\mathcal{M}}(t)}{(1-t)^{K+1}} = \sum_{m \geq 0} \frac{(m+1)^n}{K-n+1} \binom{K-n+m+1}{m+1} t^m. \quad (5.41)$$

Using the following binomial identity

$$\frac{m}{n+1} \binom{m+n}{m} = \binom{m+n}{m-1},$$

Eq. (5.41) can be rewritten as

$$\frac{\overline{Q}_{\mathcal{M}}(t)}{(1-t)^{K+1}} = \sum_{m \geq 0} \binom{K-n+m+1}{m} (m+1)^{n-1} t^m. \quad (5.42)$$

Comparing both right hand sides of (5.42) and (5.39), we derive the following result.

Proposition 5.2.1. *Let $\mathcal{M} = \{1^{k_1}, 2^{k_2}, \dots, n^{k_n}\}$ and $K = k_1 + k_2 + \dots + k_n$ with $k_i \geq 1$. We have*

$$\overline{Q}_{\mathcal{M}}(t) = A_{\mathcal{M}'}(t), \quad (5.43)$$

where $\mathcal{M}'$ is any element in the set

$$\{ \{1^{K-n+1}, 2, 3, \dots, n\}, \{1, 2^{K-n+1}, 3, \dots, n\}, \dots, \{1, 2, 3, \dots, n^{K-n+1}\} \}.$$

We recall the *bi-Stirling Eulerian polynomials*, defined by

$$E_n(t | \alpha, \beta) := \sum_{\pi \in \mathfrak{S}_{n+1}} t^{\text{des}(\pi)} \alpha^{\text{lrmin}(\pi)-1} \beta^{\text{rlmin}(\pi)-1}. \quad (5.44)$$

These polynomials have the following rational generating function

$$\frac{E_n(t | \alpha, \beta)}{(1-t)^{n+\alpha+\beta}} = \sum_{m \geq 0} \binom{\alpha + \beta + m - 1}{m} (m + \beta)^n t^m. \quad (5.45)$$

When $\beta = 0$, the summation for $E_n(t | \alpha, \beta)$ reduces to the permutations $\pi \in \mathfrak{S}_{n+1}$ with $\pi_{n+1} = 1$. Let $\pi'_i = \pi_i - 1$ for $i \in [n]$, then $\pi' \in \mathfrak{S}_n$ and

$$\text{des}(\pi) = \text{des}(\pi') + 1 \text{ and } \text{lmin}(\pi) = \text{lmin}(\pi') + 1.$$

Thus, we have

$$\sum_{\substack{\pi \in \mathfrak{S}_{n+1} \\ \text{rmin}(\sigma)=1}} t^{\text{des}(\pi)} \alpha^{\text{lmin}(\pi)-1} = \sum_{\pi' \in \mathfrak{S}_n} t^{\text{des}(\pi')+1} \alpha^{\text{lmin}(\pi)}. \quad (5.46)$$

It follows that from (5.45) with $\beta = 0$ and (5.46),

$$\frac{\sum_{\pi \in \mathfrak{S}_n} t^{\text{des}(\pi)} \alpha^{\text{lmin}(\pi)}}{(1-t)^{n+\alpha}} = \sum_{m \geq 0} \binom{\alpha+m}{m+1} (m+1)^n t^m. \quad (5.47)$$

Assuming $\alpha \neq 0$, dividing both sides of (5.47) by α and using

$$\frac{1}{\alpha} \binom{\alpha+m}{m+1} = \frac{1}{m+1} \binom{\alpha+m}{m},$$

we obtain

$$\frac{\sum_{\pi \in \mathfrak{S}_n} t^{\text{des}(\pi)} \alpha^{\text{lmin}(\pi)-1}}{(1-t)^{n+\alpha}} = \sum_{m \geq 0} \binom{\alpha+m}{m} (m+1)^{n-1} t^m. \quad (5.48)$$

Comparing both right hand sides of (5.48) and (5.42) yields the following result.

Theorem 5.11. *For $n \geq 0$, we have*

$$\sum_{\pi \in \mathfrak{S}_n} t^{\text{des}(\pi)} (K-n+1)^{\text{lmin}(\pi)-1} = \overline{Q}_{\mathcal{M}}(t), \quad (5.49)$$

where $\mathcal{M} = \{1^{k_1}, 2^{k_2}, \dots, n^{k_n}\}$ and $K = k_1 + k_2 + \dots + k_n$.

Corollary 5.2.1. The number of $\pi \in \overline{\mathcal{Q}}_{\mathcal{M}}$ with $\text{des}(\pi) = n-1$ is equal to $((k-1)n+1)^{n-1}$ when $\mathcal{M} = \{1^k, 2^k, \dots, n^k\}$.

Proof. Let $\mathcal{M} = \{1^k, 2^k, \dots, n^k\}$. Then $K = nk$, and hence $K-n+1 = (k-1)n+1$. By Theorem 5.11, we have

$$\overline{Q}_{\mathcal{M}}(t) = \sum_{\pi \in \mathfrak{S}_n} t^{\text{des}(\pi)} ((k-1)n+1)^{\text{lmin}(\pi)-1}.$$

The coefficient of t^{n-1} on the right-hand side comes only from the unique permutation $\pi = n(n-1) \cdots 1$, since it is the only permutation in $\mathfrak{S}_n$ with $\text{des}(\pi) = n-1$. For this permutation, every entry is a left-to-right minimum, so $\text{lmin}(\pi) = n$. Therefore

$$[t^{n-1}] \overline{Q}_{\mathcal{M}}(t) = ((k-1)n+1)^{n-1},$$

where $[t^n]F(t)$ denotes the coefficient of t^n in $F(t)$. Since $[t^{n-1}] \overline{Q}_{\mathcal{M}}(t)$ counts the number of $\pi \in \overline{\mathcal{Q}}_{\mathcal{M}}$ with $\text{des}(\pi) = n-1$, the result follows. $\square$

Remark 5.12. When $k = 2$, Corollary 5.2.1 covers Conjecture 5.8.

Building upon the generalized cluster complex introduced by Fomin and Reading [FR05], Douvropoulos and Josuat-Vergès [DJV25] defined the notion of cluster parking functions.

Let $n \geq 1$ and $m \geq 1$ be integers. Consider the generalized cluster complex of type A_{n-1} , denoted by $\Delta_n^{(m)}$, which is combinatorially described as the complex of dissections of a convex $(mn+2)$ -gon, where each inner polygon in the dissection is an $(mk+2)$ -gon for some integer $k \geq 1$. An *m-cluster parking function* of type A_{n-1} , as defined by Douvropoulos and Josuat-Vergès, is a face $f \in \Delta_n^{(m)}$ equipped with a labeling that satisfies the following conditions:

- Each inner $(mk+2)$ -gon defined by the dissection f is assigned a set of exactly k unordered labels.

- Every integer $i \in \{1, 2, \dots, n\}$ appears exactly once as a label across all inner polygons of the dissection.

Let $\text{CPF}_n^{(m)}$ denote the m -cluster parking function complex of type A_{n-1} . In the classical case where $m = 1$, we drop the superscript and simply write CPF_n .

We briefly recall the notions of *f-vector* and *h-vector* of a simplicial complex. Let Γ be an abstract simplicial complex of dimension $d - 1$. Its *f-vector* is

$$f(\Gamma) = (f_{-1}, f_0, \dots, f_{d-1}), \quad (5.50)$$

where f_i denotes the number of i -dimensional faces of Γ , and by convention $f_{-1} = 1$, corresponding to the empty face. The *f-polynomial* is defined by

$$f(\Gamma; t) = \sum_{i=-1}^{d-1} f_i t^{i+1}. \quad (5.51)$$

The *h-polynomial* is obtained from the *f-polynomial* by the transformation

$$h(\Gamma; t) = (1 - t)^d f\left(\Gamma; \frac{t}{1 - t}\right). \quad (5.52)$$

Equivalently, expanding this relation yields $h(\Gamma; t) = \sum_{i=0}^d h_i t^i$, and the coefficient sequence $h(\Gamma) = (h_0, h_1, \dots, h_d)$ is called the *h-vector* of Γ .

The following result by Doupvopoulos and Josuat-Vergès [DJV25] establishes a connection between the *h-polynomial* of the cluster parking function complex and the quasi-Stirling polynomials $\overline{Q}_n(t)$.

Theorem 5.13 ([DJV25, Proposition 7.5]). *For $n \geq 1$, we have*

$$h(\text{CPF}_n; t) = \overline{Q}_n(t). \quad (5.53)$$

Combining this with Proposition 5.2.1, Theorem 5.11 in the special case $k_1 = k_2 = \dots = k_n = 2$, we obtain the following chain of equalities:

$$\sum_{\pi \in \mathfrak{S}_n} t^{\text{des}(\pi)} (n+1)^{\text{lmin}(\pi)-1} = \overline{Q}_n(t) = A_{\mathcal{M}'}(t) = h(\text{CPF}_n; t), \quad (5.54)$$

where $\mathcal{M}'$ is any element in the set

$$\{1^{n+1}, 2, 3, \dots, n\}, \{1, 2^{n+1}, 3, \dots, n\}, \dots, \{1, 2, 3, \dots, n^{n+1}\}.$$

Problem 5. Find a direct combinatorial or topological explanation of the identity (5.54). Specifically, does there exist a specific shelling order for the cluster parking function complex CPF_n such that the dimension of the restriction faces naturally tracks the joint distribution of the statistics (des, lmin) on the symmetric group $\mathfrak{S}_n$?

Problem 6. Extend the identity (5.54) to the generalized Fuss parameter $m \geq 1$.

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Combinatoire des (p, q) -analogues des polynômes eulériens de type bi-Stirling

Résumé: Cette thèse porte sur les polynômes eulériens généralisés et les statistiques de permutations. Nous introduisons d'abord une généralisation à neuf variables des polynômes eulériens, définie à partir de statistiques de descentes, et nous établissons une formule de connexion avec les polynômes eulériens cycliques (α, t) de Carlitz et Scoville. On en déduit des fonctions génératrices exponentielles explicites ainsi que plusieurs développements γ -positifs, accompagnés d'une interprétation combinatoire des γ -vecteurs normalisés en termes de permutations web, ou, de manière équivalente, de permutations d'André cycliques. Nous établissons ensuite des développements en fractions continues de type Jacobi pour des (p, q) -analogues de polynômes eulériens généralisés, où (p, q) encode soit les motifs vinculaire $(31-2, 2-13)$, soit les statistiques de croisements et d'emboîtements. Ces développements conduisent à de nouveaux résultats sur la positivité totale et la γ -positivité. De plus, nous donnons une interprétation combinatoire d'une sous-famille des nombres eulériens remixed introduits par Nadeau et Tewari, en les réalisant comme des polynômes générateurs suivant les minima de gauche à droite, les minima de droite à gauche, les descentes et un indice majeur mixte. Ces polynômes généralisent simultanément les polynômes bi-Stirling-eulériens de Carlitz-Scoville et les polynômes de Stirling-Euler-Mahoniens de Butler. Enfin, des fonctions génératrices sont obtenues pour des statistiques raffinées de descentes selon la parité sur les permutations colorées $G(r, n)$, ce qui étend et unifie des résultats concernant les types de Coxeter classiques et permet d'évaluer les polynômes de descentes alternées signées en termes de polynômes eulériens.

Mots clés: polynômes eulériens, γ -positivité, statistiques sur les permutations, fractions continues, (p, q) -analogues, permutations colorées, groupes de Coxeter.

Image en couverture : Dessin de Franz Kafka. Crédit image : Franz Kafka, dessin du 17 décembre 1911.

